

MEASUREMENTS OF TRANSVERSE SPIN
DEPENDENT $\pi^+\pi^-$ AZIMUTHAL
CORRELATION ASYMMETRY AND
UNPOLARIZED $\pi^+\pi^-$ CROSS-SECTION IN pp
COLLISIONS AT $\sqrt{s} = 200$ GeV AT STAR

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ABSTRACT

The transversity distribution function, $h_1^q(x)$, where x is the longitudinal momentum fraction of the proton carried by quark q , encodes the proton's transverse spin structure at leading twist. Difficulties arise when extracting $h_1^q(x)$ due to its chiral-odd nature. However, it can be coupled with a spin-dependent interference fragmentation function (FF), $H_1^{\lessdot, h_1 h_2}$, in a dihadron ($h_1 h_2$) production channel in polarized proton-proton ($p^\uparrow p$) collisions. The coupling of $h_1^q(x)$ and $H_1^{\lessdot, h_1 h_2}$ produces an experimentally measurable azimuthal correlation asymmetry, A_{UT} , between the spin of the fragmenting quark and the final state dihadron. A model-independent extraction of transversity from these measurements relies on the knowledge of dihadron FFs, namely the unpolarized dihadron FFs, $D_1^{h_1 h_2/q(g)}$ for quarks, q (gluons, g). Extraction of these FFs requires measurements of the unpolarized dihadron cross-section in pp collisions, which are urgently needed. In pp collisions, the unpolarized cross-section measurement provides access to the $D_1^{h_1 h_2}$ for both quarks and gluons. This thesis outlines the measurements of the $\pi^+\pi^-$ azimuthal correlation asymmetry in the forward ($\eta > 0$) and backward ($\eta < 0$) pseudorapidity regions with respect to the polarized beam using the RHIC Run 2015 polarized pp data and the measurement of the unpolarized $\pi^+\pi^-$ cross-section in the invariant mass bins in the mid-pseudorapidity ($|\eta| < 1$) region using the RHIC Run 2012 pp data at $\sqrt{s} = 200$ GeV. These data sets were collected at the STAR experiment. The STAR Time Projection Chamber (TPC), Barrel Electromagnetic Calorimeter (BEMC), and Time-of-Flight Detector

(TOF) were used in conjunction to measure outgoing particle energy, tracking, and identification.

Dedicated To My Family.

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CHAPTER 1

INTRODUCTION

1.1 Overview

Humans have always been curious about the composition of matter around them. However, the reality is that all entities in the universe are built upon the same foundation - electrons, protons, and neutrons. The protons and neutrons are commonly called nucleons. This leads us to wonder why objects display various properties. Thankfully, a lot of these queries have already been resolved. By comprehending the traits and configurations of these fundamental components, we can understand the formation of atoms, molecules from atoms, and how molecules combine to create matter in our surroundings.

To understand the properties of matter around us, it's essential to comprehend the nature of its building blocks. One pertinent question is whether these "building blocks" are breakable or contain substructures. Unfortunately, these particles are incredibly small, making it difficult to examine them for their properties. The only way to gain insight into their composition is by colliding them with high energy and observing the resulting debris. These experiments, which involve the collision of leptons

(spin-1/2 fermions), nucleons (composite structures containing quarks), and heavy ions, have provided crucial early information. Many sophisticated experiments are underway, answering several unanswered questions while uncovering new mysteries.

We currently understand that electrons are fundamental particles. On the other hand, protons and neutrons comprise smaller components called quarks and gluons, which bind them together. Quarks are fermions with fractional electromagnetic charge, while gluons are neutral charge (electromagnetic) spin-1 intermediate bosons that act as force carriers between quarks within nucleons by the strong force. Hadrons, which consist of three quarks, are classified as mesons or baryons based on their quark content¹. The unique properties of nucleons arise from the interactions between quarks and gluons, commonly called partons. Therefore, uncovering the parton structure of nucleons is a critical area of study in high-energy physics.

There has been a development of theory that successfully explains the bound structure of nucleons. A theory that explains electromagnetic interaction between electromagnetically charged particles via virtual photon exchange is called quantum electrodynamics (QED). The QED, combined with the theory of weak decay of particles via the intermediate vector bosons (z , $W^\pm$) by Winberg and Salam, is now commonly called an electroweak theory. The theory of strong interaction, called quantum chromodynamics (QCD), describes the strong interaction between quarks via the exchange of color-charged-gluons.

After the discovery of the EMC collaboration in 1989, suggesting that the valence quarks contribute little to none to the nucleon's spin, the spin structure of the nucleon remains an unsolved problem thus far, which is widely known as "proton spin puzzle". The unpolarized nucleon structure is known to much higher precision; however, the polarized structure is still not fully understood, specifically the contribution of gluon

¹mesons contain a quark and anti-quark pair and baryons contain three quarks.

polarization. It is important to understand how the quarks and gluons dynamics and their spin contribute to the total nucleon property given by the parton distribution functions (PDFs). At leading twist, three PDFs are enough to describe the nucleon structure: unpolarized (or helicity averaged), helicity PDF, and transversity PDF. Spin asymmetry measurements in polarized semi-inclusive deep inelastic scattering (SIDIS) and polarized proton-proton collision (pp) have served as crucial tools to understand the (polarized) nucleon structure. The transversity PDF, which is less understood than other PDFs, is lately gaining more attention for its connection to the tensor charge. This thesis specifically outlines the measurement of two crucial observables, azimuthal correlation asymmetry and the unpolarized cross-section, via $\pi^+\pi^-$ channel necessary for the precision measurement of transversity PDF.

This thesis outline is as follows. In this chapter, I will first provide an overview of elementary particles and a model that describes them. I will also provide the basics of the QCD theory and relevant terminology that will be used frequently hereafter. In the second chapter, I will be more specific about the proton's structure, providing current experimental and theoretical progress, specifically on the transverse momentum-dependent structure of the proton. Chapter 3 will provide experimental aspects: the relativistic heavy ion collider (RHIC) and the solenoidal tracker at RHIC (STAR). Chapters 4 and 5 provide measurement details and results for the azimuthal correlation asymmetry and unpolarized $\pi^+\pi^-$ cross-section. Finally, in Chapter 6, I will summarize the thesis with the relevant outcomes of the measurements. More details, which are not appropriate in the main body of this thesis, will be provided in the appendices. I will refer to the respective appendix whenever relevant in the main body.

1.2 Review of Elementary Particles

An elementary particle is defined as the fundamental building block of matter, yet this designation is subject to revision as experimental techniques and theoretical frameworks evolve. Whether a particle is elementary or composite is contingent upon the sensitivity of experimental methods and the extent of scientific understanding at a given time. Thus, the characterization of particles as elementary may not be absolute but reflects our current knowledge and technological capability.

Initially, the electron, proton, and neutron were considered the fundamental particles, unbreakable without substructures, until the discovery confirmed that the proton is not fundamental: it has something more fundamental structure. As Rutherford's scattering experiment led to the discovery of the atomic nucleus and, eventually, protons and neutrons, the deep inelastic scattering experiment (DIS) led to the discovery of the internal structure of the proton, the quark, as shown in Figure 1.1.

Studying the inner workings of a nucleon and its quarks is a complex process that requires advanced detector technologies and particle colliders. These tools allow us to collide particles with enough energy to reveal their inner structure. The wavelength of the probe used in a particle collider is crucial, as it must be smaller than the de-Broglie wavelength² to probe the nucleon effectively.

It's worth noting the discovery of elementary particles. Heavier mass particles are produced as the energy of smashing particles increases. Lighter mass particles were discovered first, followed by heavier ones, due to the advancement of particle colliders that achieved higher energy limits.

The development of particle discovery started in the 19th century when the modern atomic theory was established. At that time, the exploitable size is $\sim 10^{-10}$ m,

² $\lambda = h/p$, where h = Planck's constant, p = momentum

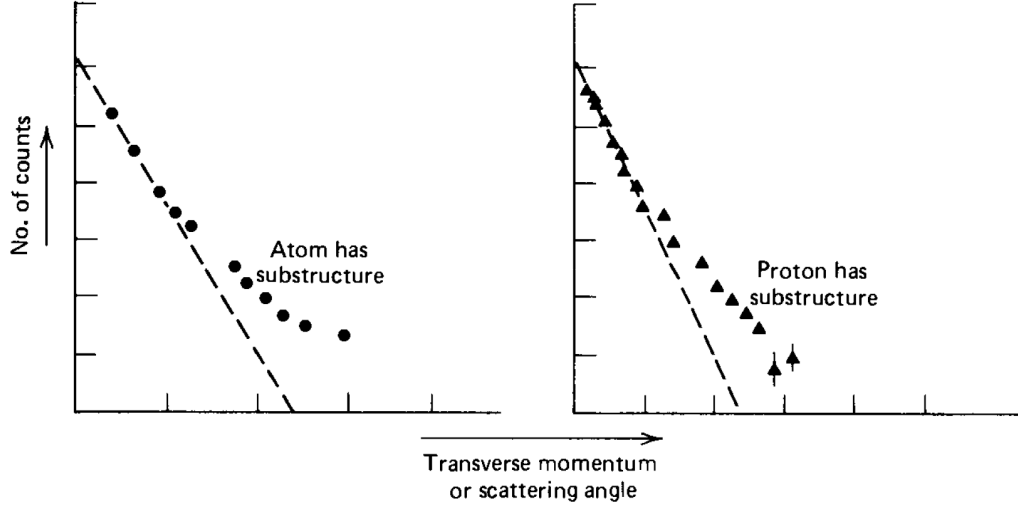


Figure 1.1: Left: In Rutherford scattering [1], the number of particles deflected through large angles indicates that the atom has an internal structure. Right: In deep inelastic scattering [2], the number of particles scattered through large angles indicates that the proton has an internal structure (quarks). The dashed line represents the expected distribution if the positive charge is uniformly distributed in an atom (left) and proton (right). Figure extracted from reference [3].

which is the size of an atom. In 1897, J. J. Thomson discovered an electron and its charge-to-mass ratio by studying the deflection of cathode rays in an external magnetic field [4]. However, the charge neutrality and large mass of an atom beg speculation that there must be something more massive than an electron and positive in charge that compensates for the electron’s charge. This was later confirmed by Rutherford’s α -scattering experiment [5], where he passed an α -particles³ on a gold foil and observed the spectrum of the scattered particles. He observed large deflection in a few of the α -particles; however, most remained undisturbed, suggesting that a hard and positive object is in charge at the center and a vast space all around. Rutherford coined the name “proton” for the nucleus of a hydrogen atom. In 1914, Neils Bhor proposed a model of a hydrogen atom as a negatively charged electron encircling around a positively charged proton at its center.

³Helium nucleus consisting of two protons and two neutrons with +2 net charge.

There was a problem with Bohr’s model, as it failed to explain the atomic structure of higher atomic number elements, for example, a helium atom with two electrons and two protons but four times as massive as hydrogen. It took about two decades until Chadwick discovered the neutron—a electrically neutral particle with a mass similar to that of the proton— in 1932 [6]. How positively charged protons reside within the nucleus of size ($\sim 10^{-15}$ m) was still puzzling. There should be a stronger force between nucleons that compensates for the repulsive force and is strong enough to hold protons and neutrons. Yukuwa proposed a theory in 1934 known as the meson theory [7]. He suggested that there should be a force carrier particle, like a photon in electromagnetic interaction, which holds nucleons together, which he called a “meson”. He calculated the mass of the meson to be intermediate between the mass of an electron and a proton. The Yukuwa meson was later discovered as a π –meson (pion) in 1947. The μ –meson (muon) was also discovered simultaneously with the pion by Powel and co-workers. The muon is the lighter version of an electron and has nothing to do with the strong force.

The discovery of antiparticles began with Paul Dirac’s groundbreaking work in relativistic quantum mechanics. In 1928, Dirac formulated the Dirac equation, a relativistic quantum equation that described the behavior of electrons while taking into account both quantum mechanics and special relativity [8] given by,

$$E^2 = \vec{p}^2 c^2 + m^2 c^4, \tag{1.1}$$

where E is the energy, $\vec{p}$ is the three momentum vector, m is the mass of the particle, and c is the speed of light, the solution of which suggested the existence of antiparticles from the negative energy solution, which was not well accepted at that time. This prediction led to the discovery of the first antiparticle, the positron, by Carl D.

Anderson in 1932 [9] when he observed tracks in a cloud chamber that were consistent with positively charged electrons. This discovery marked the birth of the concept of antimatter, revealing that for each particle in the universe, there exists an antiparticle with an opposite charge but the same mass. Later, in 1955 and 1956, antiparticles of proton and neutron, respectively, were discovered.

The discovery of a neutrino follows back to the 1930s when Wolfgang Pauli formulated the theory of beta decay, a type of radioactive decay in which a nucleus emits an electron (or positron) and a neutrino (or antineutrino). The problem Pauli faced was that the energy and momentum did not seem to be conserved in beta decay reactions, leading to violations of fundamental physical principles. To address this issue, Pauli postulated the existence of a new, nearly mass-less, and electrically neutral particle that was difficult to detect. Pauli originally suggested that this particle might be called a “neutron”, but this term was already in use to describe a different, massive particle found in atomic nuclei. The name “neutrino” was actually coined by Enrico Fermi when he developed a theory of beta decay that incorporated Pauli’s idea.

Enrico Fermi expanded on Pauli’s earlier work and formulated a theory known as the “Fermi theory of beta decay” [10]. This theory provided a quantitative explanation for the process in which a neutron decays into a proton, an electron (beta particle), and a neutrino. Fermi’s mathematical framework allowed physicists to calculate probabilities and rates for various β -decay reactions. The experimental verification of the neutrino’s existence and properties came after Fermi’s contributions through experiments like the Cowan-Reines experiment in the 1950s [11] and later experiments involving neutrino oscillations [12], which established that neutrinos have mass and can change between different types (flavors) as they travel through space.

In late 1947, the neutral kaon (K^0) was discovered by Rochester and Butler [13] looking at the cloud chamber photograph where it decays to the charge secondaries

π^+ and π^- . The charged kaon (K^+) was discovered in 1949 by Powel. In time, many more particles were found – the η , the ϕ , the ω , the ρ 's. Heavier particles, Λ 's, Ξ 's, Σ 's, and Δ 's, were discovered in successive years, which are collectively called strange particles as they produced strangely in the time scale of $\sim 10^{-23}$ seconds and decay relatively slower ($\sim 10^{-10}$ s), produced by a strong force, and decayed by a weak force. Most of the elementary particles were discovered by the year 1960's making particle physics complete chaos with a zoo of particles.

To handle this chaos of particles, Murray Gell-Mann proposed the “Eightfold Way” of classifying (anti-)particles in 1961 [14], where he arranged particles in different geometrical patterns (e.g., hexagon to represent meson octet⁴ and a triangular array to represent baryons decuplet⁵) based on their charge and strangeness. This way of classifying particles led to the discovery of the omega-minus (Ω^-), the missing spot in Gell-Mann's baryon decuplet, and he boldly predicted this particle to exist. The “Eightfold Way” of classifying particles served as a periodic table and initiated the modern era of particle physics.

Later, in 1964, the “Quark Model” (QM) was proposed by Gell-Mann [15] and independently by Zweig [16], which suggests that the hadrons⁶ comprise more fundamental substructures, which Gell-Mann named “quarks”. (Anti-)Baryons are composed of three (anti-)quarks, and mesons are composed of a pair of a quark and an antiquark. Quarks come in three flavors; up (u), down (d), strange (s), and three corresponding anti-quarks; anti-up ($\bar{u}$), anti-down ($\bar{d}$), and anti-strange ($\bar{s}$). All the quarks have a fractional charge; u has $\frac{2}{3}e$, d and s have $-\frac{1}{3}e$, and their antiparticles have the same magnitude charge, but of opposite sign. Combining these three fundamental quarks, all baryons and mesons could be reconstructed successfully.

⁴eight member family

⁵ten member family

⁶Baryons and mesons

Besides the intriguing description of the QM, it failed to explain some obvious questions about the fractional charges of (anti-)quarks, unobserved free (anti-)quarks, and violating the Pauli exclusion principle. Even though the quarks are confined within the hadrons, their existence as a point particle has been observed by the inclusive deep-inelastic scattering experiment (DIS), similar to the Rutherford scattering experiment, where most probed particles pass unaffected. In contrast, some are hard-deflected, as shown in figure 1.1, which strongly supports the QM. In 1964, O. W. Greenberg suggested the concept of the “color charge” to the quark to address the QM skepticism. He suggested that each quark, in addition to the fractional electromagnetic charge, has a “color charge”, which comes in three colors, “red”, “blue” and “green”. A hadron with quarks of different colors (that form a colorless state) is allowed by the exclusion principle. A colorless state is a combination of color and anti-color or a combination of all primary colors in equal proportion. This also explains why quarks are always present within the hadrons and do not exist individually.

In 1969, Richard Feynmann proposed a “Parton Model” [17] successfully describing the high energy SIDIS result from SLAC [18], where the nucleon constituents are considered as point-like objects called “partons” which interact via strong force. In the Feynman parton model, partons (quarks and gluons) are treated as free and independent particles within the hadron, allowing for a convenient way to calculate cross-sections and understand the dynamics of high-energy particle collisions. While the model simplifies the complex dynamics of QCD, it has been foundational in our understanding of particle interactions and remains a cornerstone of modern theoretical physics.

In 1974, the revolutionary J/ψ meson was discovered at Brookhaven National Laboratory (BNL) [19] and Stanford Linear Accelerator (SLAC) [20] simultaneously. The J/ψ is a bound state of the charm and anti-charm quarks ($c\bar{c}$) [21], which is

about three times heavier than the proton and has a much longer lifetime ($\sim 10^{20}$ s). The charm quark was suggested by Bjorken and Glashow, speculating the parallel between already discovered four types of leptons (e, μ, ν_e, ν_μ) and only three quarks (u, d , and s). In the successive years, more charmed baryons were discovered (e.g., $\Lambda_c^+(udc)$, $\Sigma_c^+(uuc)$, $D^0(c\bar{u})$, $D^+(c\bar{d})$), which established the J/ψ as $c\bar{c}$. Moreover, the discovery of the “tau” (τ) in 1975 [22] and “upsilon” ($\Upsilon(b\bar{b})$) in 1977 [23] adds two leptons (presuming ν_τ) and a “bottom” or “beauty” quark. After discovering the “top” quark in 1995 [24, 25], the quark-lepton symmetry seems to be restored with the six leptons and six quarks.

1.3 Fundamental Forces, Gauge Bosons and The Standard Model

1.3.1 Fundamental Forces and Gauge Bosons

The universe is governed by four fundamental forces of nature: gravity, strong, weak, and electromagnetic forces.

1.3.1.1 Gravitational Force

The General Theory of Relativity [26], proposed by Albert Einstein, revolutionized our understanding of gravity by describing it as the curvature of space-time caused by the presence of mass and energy. In this theory, massive objects, such as planets and stars, warp the fabric of space-time around them, causing other objects to move along curved paths. The theory predicts phenomena like the bending of light around massive objects (gravitational lensing) and the existence of gravitational waves. In situations where gravitational fields are weak, velocities are much smaller than the speed of light,

and energies are relatively low, the predictions of General Relativity closely converge with those of Newtonian gravity. This convergence means that General Relativity's effects are most apparent in extreme conditions, such as near black holes or during high-energy cosmic events, while in everyday scenarios, its predictions align closely with the familiar results from Newtonian physics.

1.3.1.2 Electromagnetic Force

The nature of electromagnetic force is similar to the gravitational force, except that it is associated with electromagnetically charged particles. It also varies as an inverse square of the distance between the charged particles. It is attractive for the unlike charges and repulsive for the like charges. It is the decisive force that binds electrons and the nucleus to form an atom, atoms together to form molecules, and molecules to form matter that we see in our everyday lives. It is also a long-range force, which is $\sim 10^{42}$ times stronger than the gravitational force and a thousand times weaker than the strong force. The charged particles, the electromagnetic field, and their interactions are described by a mathematical framework called Quantum Electrodynamics (QED).

The gravitational and electromagnetic forces share a common nature: they are long-range forces that vary as an inverse power of distance, $F \sim r^{-2}$. These forces can be felt in everyday life, such as the mutual attraction between the sun and the planets, and the high and low tides on Earth are due to the gravitational. A tiny magnet aligning north-south, the aurora borealis in the polar region, to name a few, are electromagnetic phenomena due to the Earth's magnetic field.

1.3.1.3 Strong Force

The strong nuclear force, often referred to simply as the strong force or strong interaction, is one of the fundamental forces that govern the behavior of subatomic

particles. It is responsible for binding quarks together to form protons, neutrons, and other hadrons, and it also binds these particles together within atomic nuclei. Unlike the electromagnetic force, which can be both attractive and repulsive, the strong force behaves differently. It acts between particles with color charge (quarks and gluons), mediated by the exchange of gluons – the force-carrying particles of the strong interaction. One of the most intriguing features of the strong force is its property of confinement, which means that quarks and gluons are never found in isolation but are always bound within larger particles due to their increasing force as they move apart. This phenomenon is a result of the way the strong force increases with distance, preventing the separation of quarks into free particles. Additionally, the strong force exhibits asymptotic freedom, meaning that at extremely short distances, quarks and gluons behave almost as if they were non-interacting, allowing for the perturbative calculations used in the theory of QCD. The interplay of confinement at larger scales and asymptotic freedom at smaller scales is a defining characteristic of the strong force and has profound implications for our understanding of the behavior of particles at the subatomic level.

1.3.1.4 Weak Force

The weak force, also known as the weak interaction, is one of the four fundamental forces of nature, which plays a pivotal role in processes involving the decay of subatomic particles, nuclear reactions in stars, and the fundamental interactions responsible for the behavior of fundamental particles. What distinguishes the weak force from other fundamental forces is its incredibly short range, typically on the order of 10^{-18} m, making it relevant only at the subatomic scale. Moreover, the weak force is unique in its ability to change one type of elementary particle (a quark or a lepton) into another through processes called weak interactions, a feature known as

“flavor-changing.”

One of the most significant observations related to the weak force is its role in β -decay. The discovery of neutrinos and the weak force’s involvement in beta decay were pivotal in establishing the existence of this force. Furthermore, the experimental observation of neutral current interactions in the Gargamelle bubble chamber at CERN provided important evidence for its existence [27], as predicted by the electroweak theory. The discovery of weak neutral currents played a crucial role in confirming the unification of the electromagnetic and weak forces within the electroweak theory. This pioneering work was recognized with the 1980 Nobel Prize in Physics, jointly awarded to Sheldon Glashow, Abdus Salam, and Steven Weinberg for their contributions to the electroweak theory [28, 29, 30], which successfully unified the weak force and the electromagnetic force under a single theoretical framework.

1.3.2 The Standard Model

The Standard Model (SM) of particle physics is a theoretical framework describing the universe’s fundamental particles and forces. It provides a comprehensive understanding of three fundamental forces: the electromagnetic force, the weak nuclear force, and the strong nuclear force. Gravity, the fourth fundamental force, is not included in the Standard Model and is described by a separate theory called general relativity.

SM classifies elementary particles into three categories: quarks and leptons, as shown in figure 1.2. Quarks are the building blocks of protons and neutrons, which make up atomic nuclei. There are six types of quarks: up, down, charm, strange, top, and bottom. Leptons include the electron, muon, and tau particles and their corresponding neutrinos. Each particle has its antiparticle counterpart, with opposite charge and other properties.

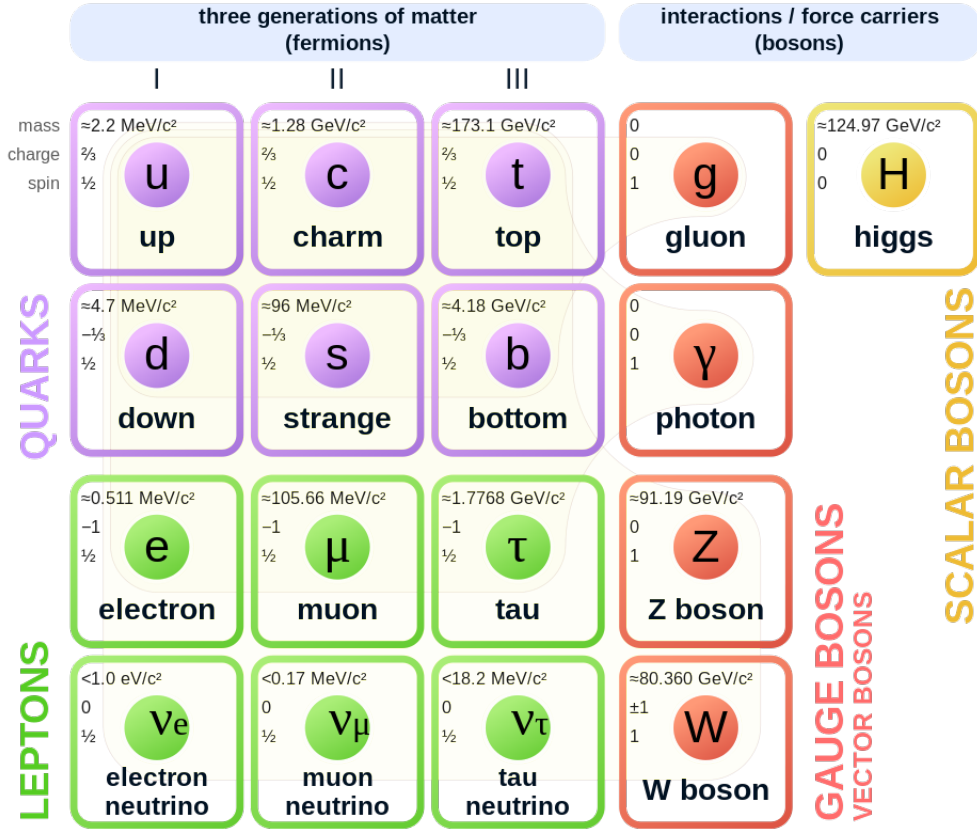


Figure 1.2: The Standard Model of elementary particles [31].

The Standard Model describes the fundamental forces as exchanges of particles called gauge bosons. The photon mediates the electromagnetic force, the $W^\pm$, and Z^0 bosons mediate the weak nuclear force, and the gluons mediate the strong nuclear force. These gauge bosons carry the forces between particles and govern their interactions.

The Higgs boson is a particle that was discovered in experiments at the Large Hadron Collider (LHC) in 2012 [32, 33]. It is associated with the Higgs field, which permeates all of space. The Higgs field gives mass to fundamental particles such as leptons, W 's, and Z through their interactions with it. The discovery of the Higgs boson confirmed the mechanism responsible for particle mass in the Standard Model.

The mass of a proton, however, is primarily a result of the strong force interactions among its constituent quarks and the gluons that mediate the strong force. This differs from the mechanism through which particles acquire mass from the Higgs mechanism. It's important to understand that the mass of a proton (or any hadron) is not simply the sum of the masses of its constituent quarks. Instead, it arises predominantly from the energy stored in the fields between quarks and gluons. This complexity makes the calculation of hadron masses a challenging task within the framework of QCD [34].

1.4 Strong Interaction and Quantum Chromodynamics

Quantum chromodynamics (QCD) is a gauge field theory that describes the strong interactions of colored quarks and gluons. At the heart of QCD lies the $SU(3)$ group theory, a mathematical framework that plays a central role in describing the strong interaction. The $SU(3)$ group symmetry, a special unitary group, governs the interactions and transformations of quarks and gluons based on their color charges, which are red, green, and blue.

The QCD lagrangian describes how quarks and gluons interact through the exchange of gluons, mediating the strong force, given by [35]

$$\mathcal{L} = \sum_q \bar{\psi}_{q,a} \left(i\gamma^\mu \partial_\mu \delta_{ab} - g_s \gamma^\mu t_{ab}^C \mathcal{A}_\mu^C - m_q \delta_{ab} \right) \psi_{q,b} - \frac{1}{4} F_{\mu\nu}^A F^{A\mu\nu}, \quad (1.2)$$

where repeated indices are summed over. The γ^μ are Dirac γ -matrices. The $\psi_{q,a}$ are quark field spinors for a quark flavor q and mass m_q , with color index a . The color index a runs from 1 to 3, corresponding to the number of color charges in the

QCD ($N_c = 3$). The gluon field is represented by $\mathcal{A}^C$, where C runs from 1 to 8, representing eight types of gluons, which transform under the adjoint representation of the SU(3) color group. The t_{ab}^C are the generators of the SU(3) group in the form of eight 3×3 Gell-Mann matrices, which encodes the fact that gluon's interaction with the quark rotates the quark's color in SU(3) space. The $g_s (= \frac{\alpha_s^2}{4\pi})$ is a QCD coupling constant, which is the fundamental parameter of the QCD. The $F_{\mu\nu}^A$ is a field tensor given by

$$F_{\mu\nu}^A = \partial_\mu \mathcal{A}_\nu^A - \partial_\nu \mathcal{A}_\mu^A - g_s f_{ABC} \mathcal{A}_\mu^B \mathcal{A}_\nu^C, \quad (1.3)$$

with $[t^A, t^B] = if_{ABC} t^C$, where f_{ABC} are the structure constants of the SU(3) group.

The QCD theory applies when the strong coupling constant α_s is small, at which quarks are essentially “free” within the nucleon. This regime corresponds to the very high energy limit (short distance), where QCD perturbation theory successfully describes the strong interaction between partons. However, the self-gluon interaction makes the g_s to be varied with the scale of the process, which causes g_s to be small in the short distance (high energy) and large in the longer distance (small distance), leading to the phenomena called “asymptotic freedom”. At the leading order, the strong coupling constant α_s can be expressed as [36]

$$\alpha_s(Q^2) = \frac{4\pi}{(11 - 2n_f/3) \ln \left(\frac{Q^2}{\Lambda^2} \right)}, \quad (1.4)$$

where n_f is the number of quark flavors and Λ is a free scale parameter. The logarithmic dependence of the α_s is because the anti-screening of the effective color charge is because of the gluon loops, as shown in figure 1.3. When $(11 - 2n_f/3) > 0$, the anti-screening due to the gluon self-interaction dominates, and as a consequence, α_s decreases with increasing Q^2 and becomes relatively weak at shorter distances. This groundbreaking discovery of “asymptotic freedom” illuminated the behavior of quarks

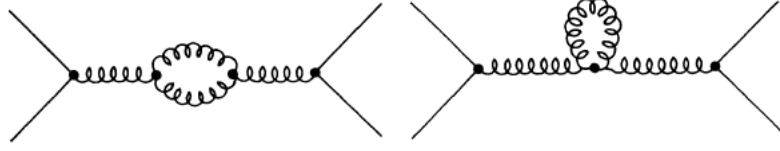


Figure 1.3: Gluon self-interactions are responsible for anti-screening of the color charge.

and gluons, paving the way for our understanding of the strong force. In recognition of their contributions [37, 38], David J. Gross, Frank Wilczek, and H. David Politzer was jointly awarded the Nobel Prize in Physics in 2004. The experimental measurement of $\alpha_s(Q^2)$ as a function of Q from world data is shown in figure 1.4.

When Q^2 is small (at longer distances), the coupling becomes so strong that the perturbative theory is no longer applicable; this region corresponds to the non-perturbative regime. The self-interaction of gluons forms the “glue tube” by squeezing the lines of forces together, which implies the quark-antiquark potential of the form $V(r) = -\frac{4}{3} \frac{\alpha_s}{r} + \lambda r$. Because of this linear potential, it requires infinite energy to separate quark and antiquark pairs, resulting in the phenomenon called “confinement”. The “confinement” suggests that quarks and gluons are confined within the hadrons.

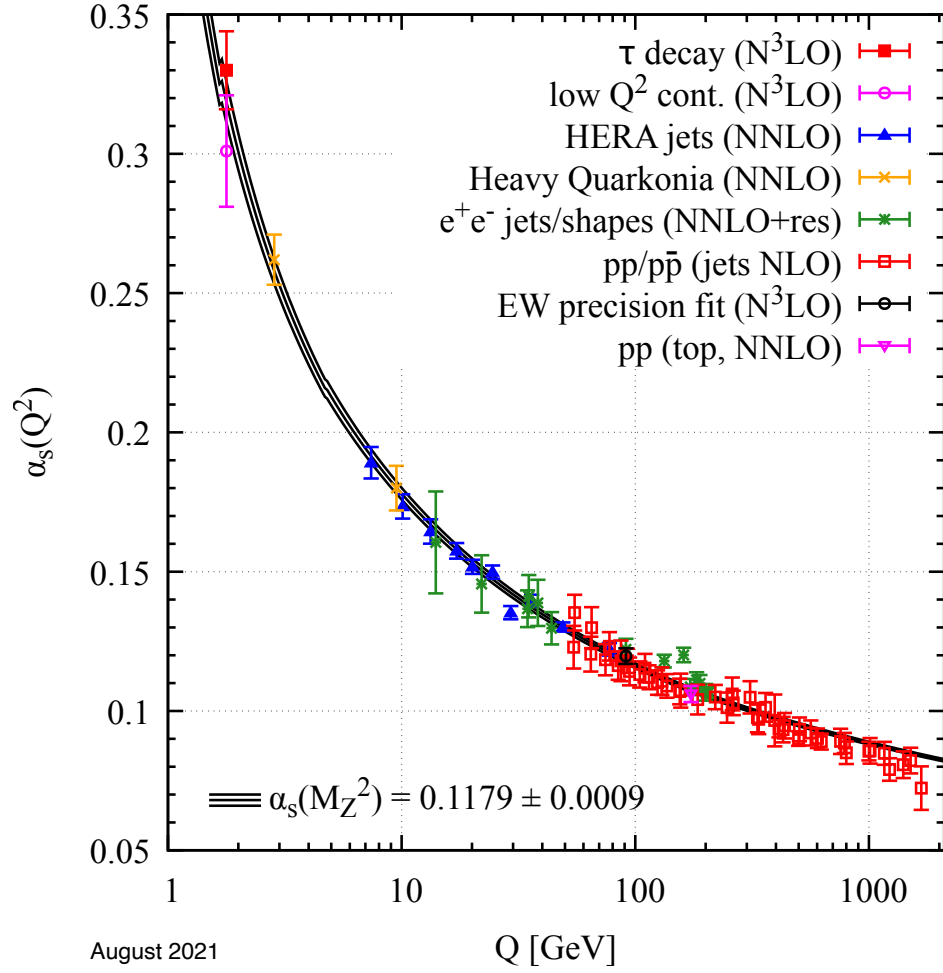


Figure 1.4: The running coupling constant from experiments as a function of the energy scale Q [39].

CHAPTER 2

SPIN STRUCTURE OF PROTON

2.1 Introduction

A proton is a complex composite structure comprising three quarks: two up (u) and one down (d). Most of its spin structure comes from sources other than its valence quarks. Understanding the proton structure has become challenging for the particle physicist community after the compelling discovery of the EMC collaboration [40]. While the proton constituents are well understood, their contribution to the proton spin is still puzzling. Despite tremendous development in the experiment and theory of (non-)perturbative QCD, we are not yet able to fully understand how quarks and gluons spin and orbital angular momentum contribute to the proton's spin.

The knowledge of the proton spin structure is based on annihilation and scattering experiments. In the annihilation experiments, electrons (e^-) and positrons (e^+) beams are used, which provides clean access to the light-quark FFs. When highly energetic e^- and e^+ beams collide, they annihilate to form a highly energetic virtual photon (as leptons undergo electromagnetic interaction) enough to generate a quark and antiquark pair rapidly flying apart. As they fly, they are pulled so strongly to

store more energy, and, eventually, they break apart radiating gluons, which further fragment into quark and antiquark pairs. This is called the fragmentation process, defined by the fragmentation function (FF) discussed in section 2.4. This process continues until the energy is compensated for by producing streams of particles forming jets. In the process, (anti-)quarks combine to form a stable colorless state of baryons and mesons called hadronization. The early e^+e^- experiments at SPEAR and PEP at SLAC in the US, TRISTAN at KEK in Japan, DORIS and PETRA at DESY in Germany, and LEP at CERN, to name a few, have been instrumental in providing precision FFs measurements.

On the other hand, the scattering experiments used either lepton-proton collision or proton-proton (pp) collision. In both cases, enough energy is supplied in the colliding beams such that a quark in the proton gets knocked out and undergoes fragmentation and hadronization process as in the annihilation process producing jets, which serve as a proxy for the partons producing it.

The proton's parton structure is characterized by a parton distribution function (PDF). These functions provide insight into the distribution of momentum carried by the partons within the proton. Assuming the probed nucleon moves along the $\hat{z}$ direction, the longitudinal momentum-dependent PDF can be extended in the momentum or coordinate space. The transverse momentum extension to the longitudinal momentum dependence PDF is called the transverse momentum dependent (TMD) PDFs[41, 42, 43, 44]. In contrast, the extension in the coordinate space is called the generalized parton distributions (GPDs)[45, 46, 47]. The longitudinal momentum-dependent PDFs, TMDs, and GPDs enable three-dimensional tomography of the proton structure, offering a way to understand the spatial and momentum distributions of partons within the proton.

2.2 Leading Twist Transverse Momentum Dependent Distribution Functions

The PDFs depend on the polarization state of the nucleon and the parton within the nucleon. Eight leading-order TMDs exist, whose physical description is listed in Figure 2.1. $f_1(x, k_T)$ is the simplest unpolarized TMD, which describes a probability of finding a quark carrying longitudinal momentum x of the nucleon and a transverse momentum $k_T = |\vec{k}_T|$ in a fast-moving nucleon. The Boer-Mulder function, $h_1^\perp(x, k_T)$, gives the transverse quark distribution for an unpolarized nucleon. Suppose the nucleon is polarized longitudinally to the momentum direction. In that case, the longitudinal quark distribution is given by the helicity PDF, $g_{1L}(x, k_T)$, and a transverse distribution is given by the Worm Gear function, $h_{1L}(x, k_T)$. The Sivers function, $f_{1T}^\perp(x, k_T)$, is introduced to describe spin-independent quark distribution in a transversely polarized nucleon. Furthermore, for a transversely polarized nucleon, the longitudinal quark polarization is given by a Worm Gear function, $g_{1T}(x, k_T)$, and a transverse quark polarization is given by transversity $h_{1T}(x, k_T)$ or the pretzelosity $h_{1T}^\perp(x, k_T)$. The Sivers, Boer-Mulders, and pretzelosity require orbital angular momentum in the nucleon since they involve a transition between initial and final nucleon states whose orbital angular momentum differs by ± 1 (Sivers and Boer-Mulder) or by ± 2 (pretzelosity). The Worm-gears link two perpendicular spin directions and are connected to the quark orbital motion inside nucleons [48].

		Quark polarization		
		Unpolarized (U)	Longitudinally polarized (L)	Transversely polarized (T)
Nucleon polarization	U	$f_1 = \text{Nucleon spin} \uparrow$		$h_1^\perp = \text{Boer-Mulder}$
	L		$g_1 = \text{Helicity}$	$h_{1L}^\perp = \text{Helicity}$
	T	$f_{1T}^\perp = \text{Sivers}$	$g_{1T}^\perp = \text{Transversity}$	$h_{1T}^\perp = \text{Transversity}$



Nucleon spin



Quark spin

Figure 2.1: Leading twist TMDs classified according to the quark polarizations (f , g , h) and nucleon (U, L, T) [49]. A similar classification of TMD distributions exists for gluons.

The three TMD distributions: $f_1(x, k_T)$, $g_{1L}(x, k_T)$, and $h_{1T}(x, k_T)$, survives upon integration over the k_T as they are even under the exchange of $k_T \rightarrow -k_T$, leading to three collinear PDFs. All the other TMD distributions are k_T -odd, which vanishes upon k_T integration. These k_T -odd TMDs are beyond the scope of this thesis; Only the leading order TMD distribution functions will be discussed, focusing on the transversity PDF.

The quest for transverse spin provides a unique way to unravel the proton spin structure. The spin of the proton perpendicular to the proton's momentum direction is a vector that correlates with the transverse component of the parton (TMD) or the transverse position of the parton (GPD). The first observation of the single spin asymmetry (SSA) was in 1976 from polarized pp data at $\sqrt{s} = 6$ GeV and 12 GeV for the charged pions at Argonne National Laboratory Zero Gradient Synchrotron [50]. This observation was supported by similar experiments [51, 52, 53, 54]. Similar asymmetries have been observed in polarized pp by the RHIC where the asymmetry

reaches up to 40% [55], lepton-nucleon experiment in at HERMES, COMPASS, and JLab. The observed asymmetry in the hadron production off the polarized nucleon tells us about the spin-orbit coupling in the nucleon and/or in the fragmentation process [48].

The quantitative understanding of PDFs has been achieved through QCD analysis of an extensive dataset, encompassing experiments such as Deep-Inelastic Scattering (DIS), Drell-Yan lepto-production, and inclusive processes in pp collisions [56, 57, 58]. At leading twist, the nucleon structure is fully described by three Parton Distribution Functions (PDFs): the unpolarized PDF, $f_1(x)$, the helicity PDF, $g_1(x)$, and the transversity PDF, $h_1^q(x)$, where x is the nucleon momentum fraction carried by patrons. Although $f_1(x)$ and $g_1(x)$ are reasonably well constrained by experimental data [59, 60], the knowledge of $h_1^q(x)$ is limited [61, 62, 63] to semi-inclusive deep inelastic scattering (SIDIS) [64, 65, 66, 67, 68, 69], e^+e^- [70, 71] and $p^\uparrow p$ [72] data. This is because $h_1^q(x)$ is a chiral-odd object, and it needs to be coupled with another chiral-odd partner to form an experimentally observable chiral-even cross-section.

2.3 Current Understanding of the Collinear PDFs

2.3.1 Unpolarized PDF

Unpolarized PDFs, $f_1(x)$, describe the probability of finding an unpolarized parton within an unpolarized proton as a function of x . $f_1(x)$ is well constrained by the global QCD analysis of a vast amount of experimental data from the ATLAS, CMS, and LHCb data from the Large Hadron Collider (LHC), HERA I+II combined DIS data, and measurements from the fixed-target data from Fermilab Tevatron $p\bar{p}$ collider by CTEQ [73, 56], MSHT [74, 75], and NNPDF [76] groups. Figure 2.2 shows the most

recent global QCD analysis of $f_1(x)$ as a function of x for the different quark flavors and the gluon from CTEQ collaboration. The higher x region is dominated by valence quarks, u and d , and gluons. The $f_1(x)$ for gluon is scaled down by a factor of 5, indicating that gluons contribute as much as the valence quarks in the valence region and completely dominate the lower x region. Although the high- x region is well constrained ($0.1 < x$), the low- x region is poorly understood, requiring much higher center-of-mass energy collision data, and very limited experimental data is available to probe in this region.

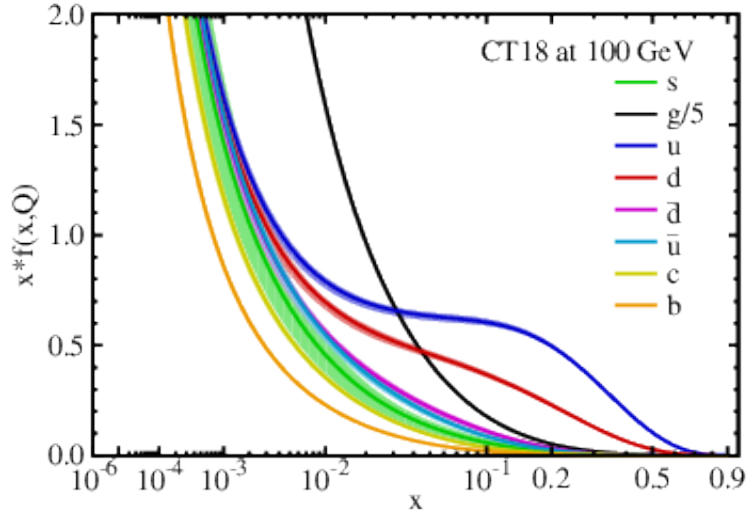


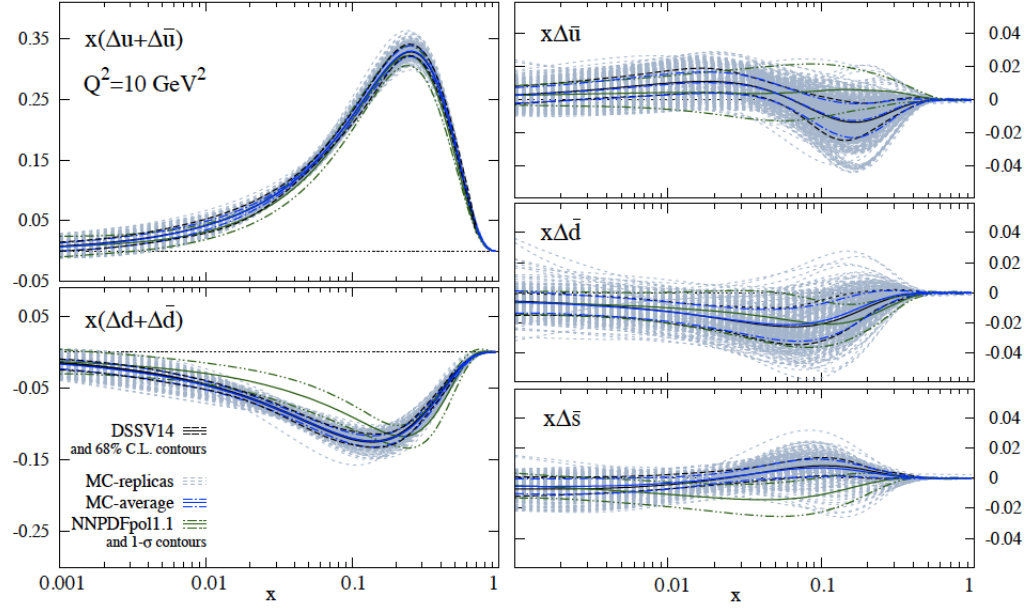
Figure 2.2: Unpolarized PDFs at $Q = 100$ GeV for u , $\bar{u}$, d , $\bar{d}$, $s = \bar{s}$, and g . The gluon PDF has been scaled down as $g(x, Q)/5$ [56].

2.3.2 Helicity PDF

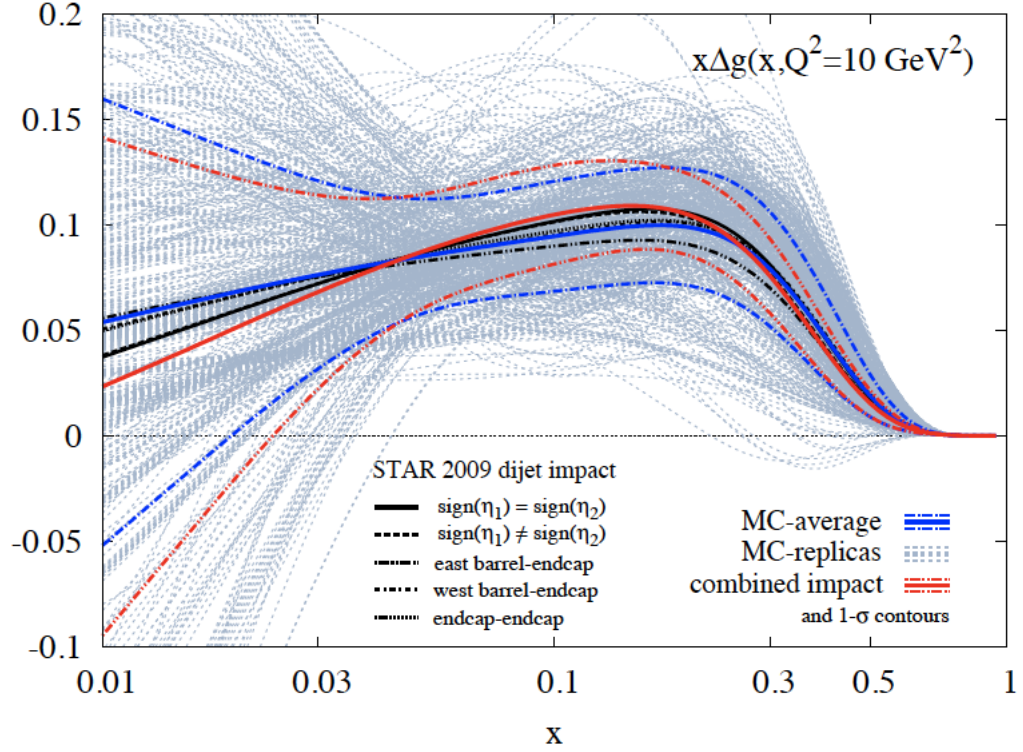
The helicity PDF, $g_1(x)$, is the probability of finding a longitudinally polarized parton in a longitudinally polarized nucleon. DIS data mainly constrain the $g_1(x)$ for quarks. However, DIS data are less sensitive to gluons. Polarized pp collision data from RHIC is crucial to probe gluon polarization. The first evidence of the gluon polarization is established by the global QCD analysis from DSSV14 [77] and confirmed by NNPDF

[76], and further constrained by DSSV14 [57] utilizing the di-jet measurements from the RHIC [78, 79].

Figure 2.3a shows $g_1(x)$ from DSSV14 for different quark and antiquark flavors, compared to the NNPDFpol1.1. The u -quark has a positive helicity, and d -quark has negative helicity, meaning that the u -quark tends to align its spin along the nucleon polarization, whereas the d -quark tends anti-align to the nucleon spin. Helicity PDF for quarks is approaching the precision of the unpolarized. However, the gluon helicity is far from reaching the quark precision level, shown in Figure 2.3b.



(a) Helicity PDFs for the quark and antiquark at $Q^2 = 10 \text{ GeV}^2$.



(b) Gluon helicity distribution.

Figure 2.3: Helicity PDFs [57].

2.3.3 Transversity PDF

At the leading twist, the transversity PDF, denoted as $h_1^q(x)$, is defined as the probability of finding a transversely polarized quark in a transversely polarized nucleon. This concept was initially introduced by Ralston and Soper [80]. Notably, this distribution, $h_1^q(x)$, is a property specific to quarks and holds no relevance for gluons [81]. Its significance has grown in recent years due to its role in accessing the nucleon tensor charge, denoted as g_T . This tensor charge, mathematically defined as the integral involving the transversity of valence quarks, takes the form:

$$g_T = \delta u - \delta d, \quad \delta u = \int_0^1 dx \left(h_1^{u_v}(x) - h_1^{\bar{u}_v}(x) \right), \quad \delta d = \int_0^1 dx \left(h_1^{d_v}(x) - h_1^{\bar{d}_v}(x) \right) \quad (2.1)$$

In this context, q_v signifies the valence quark of flavor q . The increasing attention towards g_T stems from its fundamental role in comprehending the spin structure of the nucleon. g_T can be calculated in the lattice QCD from the first principle, allowing a direct comparison between experimentally measured transversity distributions and relatively precise lattice QCD calculation.

The present understanding of $h_1(x)$ emerges from comprehensive analyses of data obtained from SIDIS experiments carried out by HERMES [82, 64, 83] and COMPASS [66, 67, 68], e^+e^- collision data provided by Belle [71, 84, 85] and BABAR [70], and proton-proton (pp) collision data gathered by STAR [72, 86]. The extraction of the phenomenological $h_1(x)$ from these datasets can be found in Ref. [62, 63, 58].

Recently, the Jefferson Lab Angular Momentum (JAM) group released a comprehensive report on the simultaneous global QCD analysis of the $\pi^+\pi^-$ FFs and transversity PDFs for u , d , and antiquarks from e^+e^- annihilation, SIDIS, and pp data for the first time [87]. This report incorporates data from the Belle cross-section

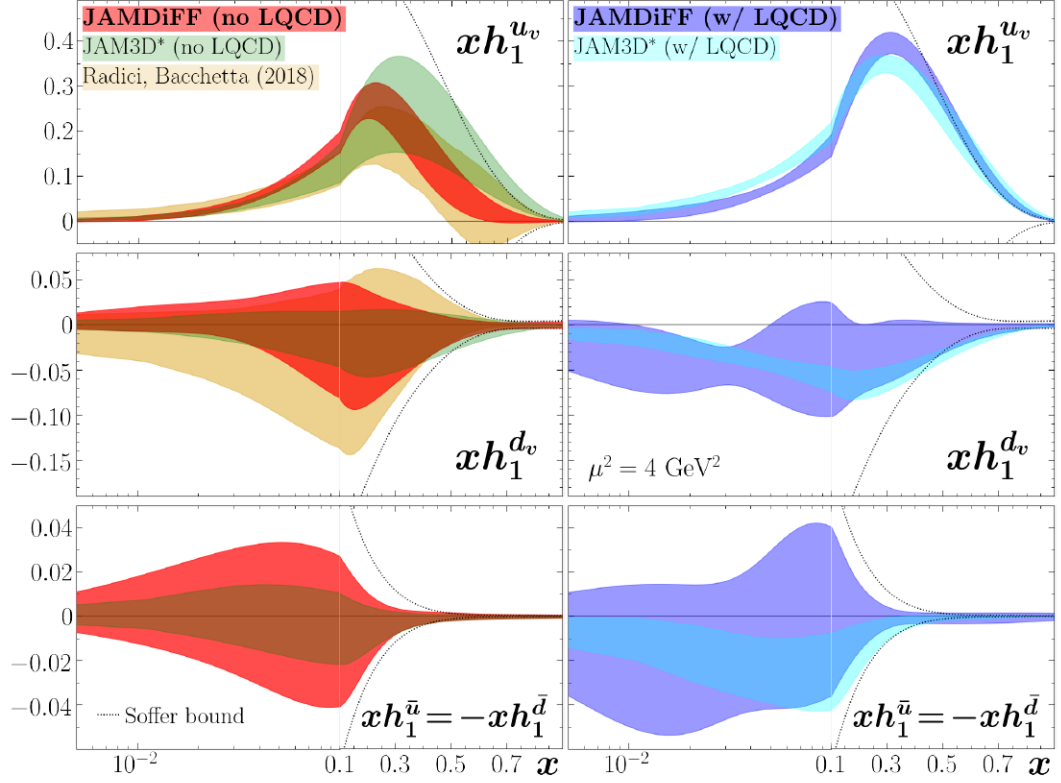


Figure 2.4: Transversity PDFs $xh_1^{u_v}$ (top row), $xh_1^{d_v}$ (middle row), and $xh_1^{\bar{u}}$ (bottom row) plotted as a function of x at the scale $\mu^2 = 4 \text{ GeV}^2$ from the JAM collaboration (JAMDiFF) [87]. The left panel shows JAMDiFF results without including lattice QCD data, whereas the right panel shows corresponding results, including lattice QCD data. The JAMDiFF result is compared with the JAM3D* with (cyan) and without (green) lattice QCD [88] and RB18 (gold) [63]. The dashed lines indicate the Soffer bound.

[85], STAR $\pi^+\pi^-$ asymmetry [72, 86], as well as lattice QCD (LQCD) results from ETMC [89] and PNDME [90]. It covers all relevant kinematic variable binnings for the processes under consideration, enabling a realistic extraction of quark and gluon FFs.

Furthermore, by incorporating theoretical constraints at both lower and higher values of x , where experimental data are limited, the JAM group found good compatibility between different techniques for extracting tensor charges, including LQCD data, while maintaining a good description of experimental measurements. Additionally, they observed a close agreement between tensor charges and the x dependence of transversity measured in single-hadron and dihadron cases, providing the first demonstration of the universality of transversity and tensor charges.

Figure 2.4 illustrates the recent global extraction of $h_1(x)$ by the JAM collaboration [87], in comparison to the extraction from Ref. [63] (left panel), along with the incorporation of lattice QCD data (right panel) [91, 92]. The behavior of $h_1(x)$ for the valence quarks follows a similar trend as that of $g_1(x)$, though with less constraint, particularly in the valence quark region. A positive value of xh_1^{uv} indicates that u -quarks favor the alignment of nucleon spin, while d -quarks tend to anti-align. However, the latter tends to be less constrained due to limited experimental data sensitive to it. In addition, the first extraction of antiquark transversity is shown in the bottom panel for the dihadron production. It's worth noting that including Lattice QCD data substantially enhances the constraints on these distributions.

2.4 Fragmentation Functions

Fragmentation functions (FFs) usually appear as a non-perturbative ingredient in the factorization formula when the specific particle species are observed in the final

state. FF is defined as the probability of finding color-neutral particles inside partons. In other words, it describes how color-neutral particles are formed from the colored quarks and gluons.

FF can be studied in single-inclusive hadron production in e^+e^- (SIA), in SIDIS and pp collision processes, where FF frequently appears in the QCD factorized cross-section:

- SIA: $\sigma^{e^+e^- \rightarrow hX} = \hat{\sigma} \otimes FF$
- SIDIS: $\sigma^{lN \rightarrow lhX} = \hat{\sigma} \otimes PDF \otimes FF$
- pp : $\sigma^{pp \rightarrow hX} = \hat{\sigma} \otimes PDF \otimes PDF \otimes FF$

Here, $\hat{\sigma}$ represents the process-dependent hard scattering cross-section, which is perturbatively calculable, and only the leading order factors are shown. This factorization scheme holds only if the hard scale is present in each of the processes, such as the center-of-mass energy, $\sqrt{s}$, serves as a hard scale in the SIA, the virtual momentum transfer between lepton and hadron, Q^2 , serves as a hard scale on the SIDIS. The transverse momentum of the final state hadron relative to the proton's momentum, $p_{h\perp}$, serves as a hard scale.

The application of FF was first introduced in the 1970s to explain the observation of large transverse momentum of hadrons in the hadronic collisions from CERN, which could not be explained without including gluonic subprocesses [93]. Several FFs appear in the process depending on the inclusion of spin of the hadron and/or parton, transverse momentum of hadron relative to the parton, higher-twist effects, and fragmentation to more than one hadrons, which are summarized in Table 2.1.

H \ q(g)	U	L (Circ)	T (Lin)
U	$D_1^{h/q(g)}$		$H_1^{\perp h/q(g)}$
L		$G_1^{h/q(g)}$	$H_{1L}^{\perp h/q(g)}$
T	$D_{1T}^{\perp h/q(g)}$	$G_{1T}^{h/q(g)}$	$H_1^{h/q(g)} H_{1T}^{\perp h/q(g)}$

Table 2.1: Interpretation of TMD FF for quarks (gluons).

The $D_1^{h/q(g)}$ is the unpolarized FF, which describes the number density of unpolarized hadrons in an unpolarized quark (gluon). The density of longitudinally polarized hadrons in longitudinally(circularly) polarized quark(gluon) is given by $G_1^{h/q(g)}$. The $H_1^{h/q(g)}$ describes the density of transversely polarized hadron in transversely (linearly) polarized quark(gluon). One should remember that if the parton polarization is involved, then the FF gives the difference in densities, such as $D_1^{h/q(g)}$ has a simple number density interpretation; however, $G_1^{h/q(g)}$ represents the difference in densities of two opposite parton polarization. $H_1^{\perp h}$ is the well-known Collins FF [94], as it provides access to the transversity in the SIDIS process.

The dihadron FFs (DiFFs) appear in the factorization formula of the processes, where a parton fragments in two hadrons. It is gaining more attention lately because it can be an analyzer for the transversity in the collinear framework, unlike the TMD Collins FF. The DiFFs measure the probability of finding two color-neutral hadrons in a parton. $D_1^{h_1 h_2}$ represents the measure of such probability for two hadrons h_1 and h_2 in an unpolarized parton. Including the transverse momentum of parton k_T and the polarization, the probability density of finding two unpolarized hadrons in a transversely polarized target is given by two DiFFs: $H_1^{\triangleleft, h_1 h_2}$ and $H_1^{\perp, h_1 h_2}$ [93]. The $H_1^{\triangleleft, h_1 h_2}$ is a well-studied DiFF than other DiFFs, as it provides access to the transversity, for example, in polarized pp collisions. It is often called the Interference Fragmentation Function (IFF) because it requires interference between different dihadron production channels for its survival [95, 96, 97].

2.5 Accessing Transversity Via Dihadron Channel: Theoretical Description

Accessing the transversity PDF, which is chiral-odd, can be quite challenging. This must be combined with another chiral-odd partner to yield a chiral-even cross-section. As a result, it cannot be accessed in inclusive processes. Furthermore, to access transversity, knowledge of another chiral-odd object that needs to appear together is required. This difficulty makes it less straightforward to access than other leading-order parton distribution functions, making it the least understood among other leading-order PDFs.

Transversity can be accessed through a single hadron production, either by exploiting TMD factorization (e.g., Collins effect [94] in SIDIS) or collinear sub-leading twist (twist 3) factorization in hadronic collisions [98, 99]. Alternatively, one can consider the unpolarized dihadron production at leading twist collinear factorization, where transversity couples with the interference fragmentation function, $H_1^\triangleleft$ (IFF) [100, 95, 96, 97]. The choice of identified hadron pair, such as $\pi\pi$, KK , and πK , provides additional sensitivity to quark flavor. The collinearity is preserved in this channel as the dihadron center of mass travels along the jet axis, and the IFF survives even after integrating over the transverse momentum of the hadron pair. Moreover, this channel doesn't require a jet reconstruction, devoid of systematic uncertainty associated with the jet reconstruction.

The discussion will focus solely on accessing transversity through the dihadron channel in $p^\uparrow p$ collision. This specific channel is used to measure the azimuthal

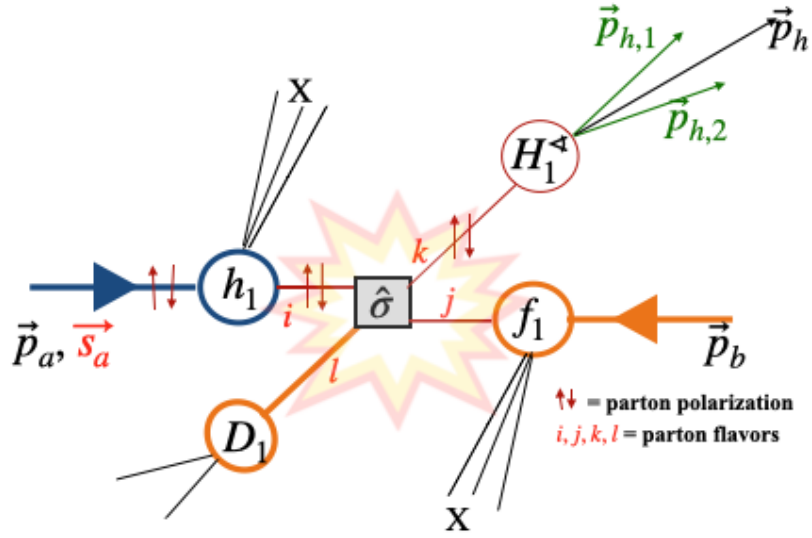


Figure 2.5: Dihadron channel in $p^\uparrow p$ showing the fragmentation of polarized parton and final state unpolarized dihadron pair. The respective PDFs and FFs are indicated at where they appear in the process.

correlation asymmetry in this analysis, which can be expressed as,

$$p_a^\uparrow(\mathbf{S}_a, \mathbf{P}_a) + p_b(\mathbf{P}_b) \rightarrow (h_1 h_2)_{jet}(\mathbf{P}_h) + X, \quad (2.2)$$

where $p_a^\uparrow(p_b)$ is a transversely polarized (unpolarized) proton, h_1 and h_2 are the final state unpolarized charged hadrons within a jet, and X represent undetected final state particles. The boldface quantities in the equation are regular three vectors, where $\mathbf{S}$ is the polarization vector and $\mathbf{P}$ is the momentum vector. The final state hadrons could be protons, kaons, or pions.

The dihadron channel is depicted schematically in Figure 2.5. The polarized (unpolarized) proton originates from the left (right), and the associated kinematic variables are indicated with subscripts a (b). The transversity PDF of a transversely polarized proton a is denoted by h_1 , while the unpolarized PDF of proton b is de-

noted by f_1 . The hard scattering occurs between a transversely polarized parton i from proton a and an unpolarized parton j from proton b . The cross-section of the hard scattering, denoted as $\hat{\sigma}$, is perturbatively calculable. This scattering occurs at an extremely small distance ($\lesssim 10^{-15}$ m), where the partons are considered asymptotically free. Subsequently, the hard scattered partons undergo fragmentation and hadronizes into color-neutral stable particles. The fragmentation of the polarized scattered parton k is described by the spin-dependent fragmentation function $H_1^\perp$, while the fragmentation of the unpolarized parton l is described by D_1 . At the leading twist, these fragmentation functions also possess a probabilistic interpretation, representing the probability of finding color-neutral hadrons within the parton. In the final state, we observe hadrons emerging from the polarized parton k , either in the form of jets or exclusively, which grants access to the quark transversity coupled with the IFF.

Measuring two hadrons in the final state provides two vectors associated with the hadron pair, offering a convenient theoretical framework [101]. If $\mathbf{P}_{\mathbf{h}_1}$ and $\mathbf{P}_{\mathbf{h}_2}$ represent the momenta of the two hadrons, these vectors can be defined as follows:

$$\mathbf{P}_{\mathbf{h}} = \mathbf{P}_{\mathbf{h}_1} + \mathbf{P}_{\mathbf{h}_2}, \text{ and } \mathbf{R} = \frac{\mathbf{P}_{\mathbf{h}_1} - \mathbf{P}_{\mathbf{h}_2}}{2}, \quad (2.3)$$

Here, $\mathbf{P}_{\mathbf{h}}$ denotes the resultant momentum of the hadrons, and $\mathbf{R}$ represents the relative momentum vector. Due to the presence of the relative momentum vector $\mathbf{R}$, the IFF can be integrated over the transverse component of $\mathbf{P}_{\mathbf{h}}$ with respect to the jet axis. Additionally, the transverse component of the fragmenting quark can be expressed in terms of the transverse component of the relative momentum, denoted as $\mathbf{R}_{\mathbf{T}}$, through the mixed product $\mathbf{S} \cdot \mathbf{R} \times \mathbf{P}_{\mathbf{h}}$. In this channel, the center of mass of the hadron pair moves collinearly with the jet axis, thereby preserving the concept of

collinear factorization; hence, the jet reconstruction is not required. For a dihadron channel given by equation 2.2, only one type of asymmetry, A_{UT} , exists, which is given by,

$$A_{UT} = \frac{d\sigma^\uparrow - d\sigma^\downarrow}{d\sigma^\uparrow + d\sigma^\downarrow} \equiv \frac{d\sigma_{UT}}{d\sigma_{UU}}, \quad (2.4)$$

where $d\sigma_{UT}$ is the difference between the cross-section when the polarized proton a is polarized up ($\uparrow$) and down ($\downarrow$), whereas $d\sigma_{UU}$ is the unpolarized cross-section. The polarized and unpolarized part of the cross-section can be expressed as [97],

$$d\sigma_{UT} = 2|\mathbf{P}_{\mathbf{h}\perp}| \sum_{i,j,k,l} |\mathbf{S}_{\mathbf{a}}| \sin(\phi_S - \phi_R) \int \frac{dx_a dx_b}{8\pi^2 z_h} f_1^{j/b}(x_b) h_1^{i/a}(x_a) \frac{d\Delta\sigma^{i^\uparrow j \rightarrow k^\uparrow l}}{d\hat{t}} \times \frac{\mathbf{R}}{M_h} \sin\theta H_1^{\triangleleft, h_1 h_2/k}(z_h, \cos\theta, M_h^2) \quad (2.5)$$

$$d\sigma_{UU} = 2|\mathbf{P}_{\mathbf{h}\perp}| \sum_{i,j,k,l} \int \frac{dx_a dx_b}{8\pi^2 z_h} f_1^{i/a}(x_a) f_1^{j/b}(x_b) \frac{d\Delta\sigma^{ij \rightarrow kl}}{d\hat{t}} \times D_1^{h_1 h_2/l}(z_h, \cos\theta, M_h^2) \quad (2.6)$$

In the equations above, $|\mathbf{P}_{\mathbf{h}\perp}|$ represents the magnitude of the transverse component of $\mathbf{P}_{\mathbf{h}}$ with respect to the beam direction. This component serves as a hard scale for the process. The PDFs f_1 and h_1 correspond to the unpolarized and transversity PDFs, respectively. $H_1^{\triangleleft, h_1 h_2}$ denotes the IFF, while $D_1^{h_1 h_2}$ represents the unpolarized counterpart of the IFF.

The definitions of the azimuthal angles ϕ_S and ϕ_R are derived from Figure 2.6 and are given by Equations 2.7-2.8. These angles are defined in the center of mass of the hadronic system. Specifically, ϕ_S denotes the angle between the scattering plane formed by $\mathbf{P}_{\mathbf{a}}$ and $\mathbf{P}_{\mathbf{h}}$, and the polarization vector $\mathbf{S}_{\mathbf{a}}$. On the other hand, ϕ_R

represents the angle between the scattering plane and the dihadron plane formed by the momenta $\mathbf{p}_{h1}$ and $\mathbf{p}_{h2}$.

The hard scale is assumed to significantly exceed the dihadron system's proton mass and the invariant mass, denoted as M_h . The validity of the assumptions holds only at leading twist (leading order in $1/|\mathbf{P}_{h\perp}|$).

$$\cos \phi_S = \frac{\hat{P}_a \times \mathbf{P}_h}{|\hat{P}_a \times \mathbf{P}_h|} \cdot \frac{\hat{P}_a \times \mathbf{S}_a}{|\hat{P}_a \times \mathbf{S}_a|}, \quad \cos \phi_R = \frac{\hat{P}_h \times \mathbf{P}_a}{|\hat{P}_h \times \mathbf{P}_a|} \cdot \frac{\hat{P}_h \times \mathbf{R}}{|\hat{P}_h \times \mathbf{R}|} \quad (2.7)$$

$$\sin \phi_S = \frac{(\mathbf{P}_h \times \mathbf{S}_a) \cdot \hat{P}_a}{|\hat{P}_a \times \mathbf{P}_h| |\hat{P}_a \times \mathbf{S}_a|}, \quad \sin \phi_R = \frac{(\mathbf{P}_a \times \mathbf{R}) \cdot \hat{P}_h}{|\hat{P}_h \times \mathbf{P}_a| |\hat{P}_h \times \mathbf{R}|} \quad (2.8)$$

The polarized cross-section exhibits a sinusoidal modulation of the azimuthal angle ϕ_{RS} (defined as $\phi_S - \phi_R$), while such modulation is suppressed in the unpolarized cross-section. This observation indicates a bias in the dihadron yields within the specific ϕ_{RS} bin, resulting in an asymmetry that arises solely from polarization effects. By utilizing Equations 2.5 and 2.6, Equation 2.4 can be expressed as follows:

$$A_{UT}^{\sin \phi_{RS}} = \frac{1}{|\mathbf{S}_a|} \frac{d\sigma^\uparrow - d\sigma^\downarrow}{d\sigma^\uparrow + d\sigma^\downarrow}, \quad (2.9)$$

This equation isolates A_{UT} as the amplitude of the "sine" modulation of dihadron yields in ϕ_{RS} at the leading twist. To represent the asymmetry associated with the "sine" modulation in ϕ_{RS} , it will henceforth be denoted as $A_{UT}^{\sin(\phi_{RS})}$. Experimental measurement of $A_{UT}^{\sin(\phi_{RS})}$ involves counting the number of dihadrons in ϕ_{RS} bins for both beam polarizations ($\uparrow$ and $\downarrow$), which is sensitive to the product $h_1 H_1^\triangleleft$:

$$A_{UT} \propto \sum_{i,j,k,l} \frac{f_1^{j/b}(x_b) h_1^{i/a}(x_a) H_1^{\triangleleft h_1 h_2/k}(z_h, M_h^2)}{f_1^{j/b}(x_b) f_1^{i/a}(x_a) D_1^{h_1 h_2/l}(z_h, M_h^2)} \quad (2.10)$$

Here, all components are non-perturbative objects, and precise experimental knowl-

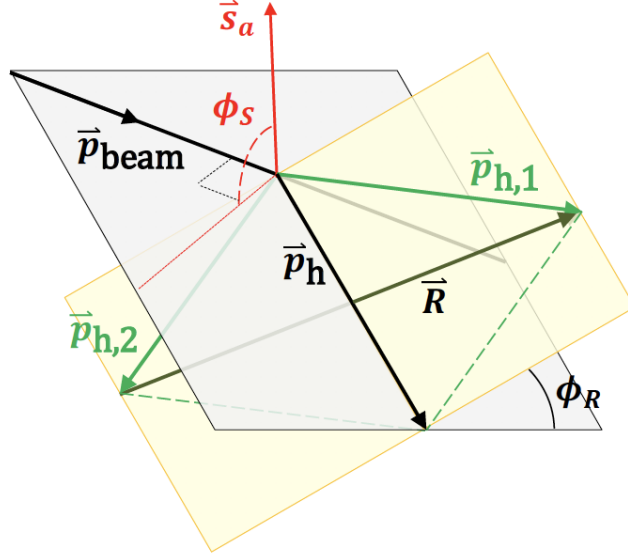


Figure 2.6: The azimuthal angle definitions in the dihadron channel.

edge is required to extract transversity. The unpolarized PDF is known to have higher precision based on SIDIS data [102], and the IFF has limited knowledge from the e^+e^- experiments. However, the unpolarized Fragmentation Function (FF) $D_1^{h_1 h_2}$ lacks experimental constraints, and the global extraction of transversity relies on simulations, leading to large model-dependent uncertainties. To constrain D_1 and, consequently, transversity, it is crucial to experimentally measure $d\sigma_{UU}$, an observable discussed in Chapter 5 that is of great importance.

The relative partial wave expansion of FFs in the cross-sections $d\sigma_{UT}$ and $d\sigma_{UU}$ using the basis of Legendre polynomials through the $\cos \theta$ dependence provides intriguing physics insight [103]. Keeping up to $L = 1$, we have,

$$D_1(z, \cos \theta, M_h^2) \rightarrow D_{1,oo}(z, M_h^2) + D_{1,ol}(z, M_h^2) \cos \theta + D_{1,ul}(z, M_h^2) \frac{1}{4}(\cos^2 \theta - 1) \quad (2.11)$$

$$\sin \theta H_1^\triangleleft(z, \cos \theta, M_h^2) \rightarrow H_{1,ot}^\triangleleft(z, M_h^2) \sin \theta + H_{1,lt}^\triangleleft(z, M_h^2) \sin \theta \cos \theta \quad (2.12)$$

In this context, the double-index notation is used to describe the polarization state in the center of mass of the dihadron for each specific probability amplitude. For instance, $D_{1,oo}$ corresponds to the decay into a $L = 0$ pair (s-wave). In contrast, $D_{1,ol}$ refers to the interference between the decay amplitudes of a $L = 0$ pair and a $L = 1$ pair (s-p interference) polarized longitudinally along the center of mass of $\mathbf{P}_h$. Similarly, $H_{1,ot}^\triangleleft$ represents the interference between the decay amplitudes of a $L = 0$ pair and a transversely polarized $L = 1$ pair. Furthermore, $H_{1,lt}$ and $D_{1,ll}$ denote the interference between decay amplitudes at different polarization states of $L = 1$ pairs.

The p -wave contribution arises from the spin-1 resonance in the dihadron production channel. As a result, the p -wave FFs (ll or lt terms) are expected to exhibit a characteristic mass dependence typical of the resonance. Furthermore, the $D_{1,oo}$ contains both s and p -wave contributions. On the other hand, the interference terms (ol or ot terms) require the presence of both s and p -wave channels. Therefore, the intrinsic FFs, $D_{1,ol}$ and $H_{1,ot}^\triangleleft$, are expected to be sizeable in the vicinity of the resonance region [95, 103, 97], for example, around the ρ -mass in the $\pi^+\pi^-$ case.

2.6 Experimental Data for DiFFs

From the discussion in 2.5, it is clear that independent measurement of $H_1^{\triangleleft h_1 h_2}$ and $D_1^{h_1 h_2}$ with z and M_h dependence is required to decouple transversity from the $h_1 H_1^\triangleleft$.

The recent Belle measurements of $\pi^+\pi^-$ cross-section differential in M_h and z [85] provide crucial observable for the $D_1^{h_1 h_2}$ for quarks. However, these measurements are less sensitive to gluon fragmentation. The measurement from the pp data is required to probe gluon fragmentation. The unpolarized $\pi^+\pi^-$ cross-section measurement in pp collision data from the STAR experiment (Chapter 5) will fulfill the dire need of this observable for the first time. Belle also measured back-to-back dihadron single

spin asymmetry (SSA) sensitive to the spin-dependent IFF [84]. The measured SSA is proportional to the product $H_1^\triangleleft \bar{H}_1^\triangleleft$.

The SSA to probe the transverse spin-dependent IFF was measured in the transversely polarized proton target by the HERMES experiment for the first time [64] followed by the COMPASS experiment with the polarized proton and deuteron target [104] in SIDIS. The COMPASS SSA is of higher precision than the HERMES, showing enhancement around the ρ -meson mass ($\sim 0.8 \text{ GeV}/c^2$).

In polarized pp , the STAR experiment reported the first observation of SSA [72], followed by higher precision measurement [105] at $\sqrt{s} = 200 \text{ GeV}$, which is sensitive to the product $h_1 H_1^\triangleleft$. These measurements also depict the distinct resonance feature consistent with the SIDIS measurements. Similar measurements from STAR at $\sqrt{s} = 500 \text{ GeV}$ [86] observed significant asymmetry peaked around the ρ -mass, implying the universality of the physical process involved in the resulting SSA.

CHAPTER 3

EXPERIMENTAL ASPECT

3.1 Relativistic Heavy Ion Collider (RHIC)

This dissertation is a part of research conducted at the relativistic heavy ion collider (RHIC) at Brookhaven National Lab (BNL), located at Upton, New York, United States. RHIC is the world's first collider, which can collide polarized proton beams up to center-of-mass energy of $\sqrt{s} = 510$ GeV. It can also collide heavy ion particles of varying energy up to $\sqrt{s} = 200$ GeV. RHIC comprises two 3834 m long rings that carry particles in opposite directions, known as the “blue beam” and the “yellow beam”. This terminology is essential during data analysis for distinct beams. The blue beam circulates clockwise, while the yellow beam circulates counterclockwise when observed from above. The collider features six interaction points at the 12, 2, 4, 6, and 8 o'clock positions, where particle collisions can occur. Presently, only the STAR and sPHENIX (upgraded PHENIX) detectors, situated at the 6 and 8 o'clock positions, respectively, are actively collecting data. A comprehensive overview of the RHIC complex and its components can be found in reference [106]. This section focuses on the pertinent components of RHIC that directly influence the analysis at

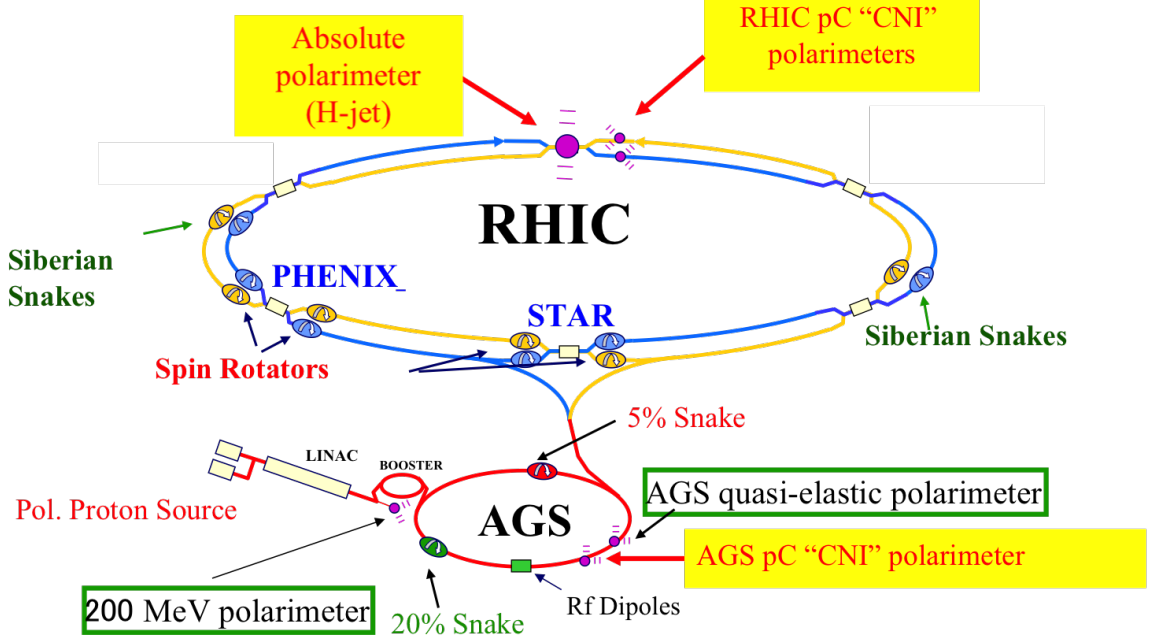


Figure 3.1: Relativistic heavy ion collider complex [107].

hand.

3.1.1 Siberian Snakes and Spin Dynamics

In high-energy particle colliders, it is important to understand the evolution of the spin of the highly accelerated colliding beam and the tools that control it. For a circular accelerator as RHIC, the spin evolution in the external magnetic field is given by the Thomas-BMT equation [108, 109],

$$\frac{d\vec{P}}{dt} = - \left(\frac{e}{\gamma m} \right) \left[G\gamma \vec{B}_\perp + (1 + G)\vec{B}_\parallel \right] \times \vec{P}, \quad (3.1)$$

where $\vec{P}$ is the polarization vector expressed in the frame that moves with the particle. $G(= 1.7928)$ is the anomalous magnetic moment of the proton and $\gamma = E/m$. $G\gamma$ is called the spin tune ν_{sp} , which gives the number of full spin precessions in every full revolution. The value of the ν_{sp} can reach up to 478 at the RHIC top beam energy

of 250 GeV.

Maintaining the polarization of the highly accelerated beam in the circular path is difficult. Two resonances cause the depolarization; one is due to the misalignment and magnet errors called “imperfection resonances”, and the other is called “intrinsic resonances,” caused by the focusing magnetic fields. The imperfection resonances occur when $\nu_{sp} = G\gamma = n$, where n is an integer. In contrast, the intrinsic resonance occurs when $\nu_{sp} = G\gamma = kP \pm \nu_y$, where k is an integer, ν_y is the vertical betatron tune, and P is the superperiodicity. These resonances cause the spin of the polarized beam to be fully flipped (highly accelerated beam) or partially flipped (slowly accelerated beam), which are traditionally overcome using a betatron jump. At RHIC, “Siberian Snake” is used to maintain the stable beam polarization, which can produce a 180° spin rotation about the horizontal axis, which is greater than the spin rotation due to the depolarization resonances. These Siberian snakes are constructed either by using a solenoidal magnet or a sequence of interleaved horizontal and vertical dipole magnets producing local orbit distortion. A full snake is used; however, for a low-energy accelerator, a partial snake suffices to maintain stable beam polarization, overcoming the depolarization resonances.

3.1.2 Polarized Proton Injector

The RHIC polarized beam requires a bunch intensity of 2×10^{11} protons per bunch. To meet this requirement, the alternating gradient synchrotron (AGS) needs to reach bunch intensity of 4×10^{11} to allow room for some losses while transferring it to the RHIC rings. The optically pumped polarized ion source (OPPIS) produces a polarized H^- . It produces $500 \mu A$ in a single $300 \mu s$ pulse, which corresponds to the 9×10^{11} polarized H^- . The polarized H^- ions are accelerated to 200 MeV with a radio frequency quadrupole (RFQ) and 200 MHz LINAC with an efficiency of about

50%. The pulse of H^- ions is strip-injected and captured in the AGS Booster as a single bunch, containing 4×10^{11} polarized protons. Each polarized proton bunch is accelerated to the Booster to 1.5 GeV kinetic energy and transferred to the AGS, which is accelerated to 25 GeV. Finally, the beam bunches are handed to the RHIC rings for further acceleration to 100 or 250 GeV.

3.1.3 Acceleration in the Booster and the AGS

The polarized proton bunch in the AGS may lose its polarization due to the depolarization resonances. This is overcome by installing a partial (5%) Siberian snake at one of the straight sections of the AGS, which is sufficient to avoid any depolarization due to imperfection resonances. The acceleration and beam transfer efficiency is greater than 50%, giving 2×10^{11} protons per bunch. This process is repeated until all 120 bunches of the RHIC ring are filled.

3.1.4 RHIC Rings General Layout

Two full Siberian Snakes on the opposite side of each RHIC ring serve to avoid depolarization from imperfection and intrinsic depolarization resonances up to the top energy of 250 GeV. These Siberian Snakes introduce a 180° spin rotation around the horizontal axis without generating a net orbit distortion. In addition, the two axes of the two Snakes in each ring must be perpendicular. Moreover, Snakes are installed in the exact opposite direction in the ring; in fact, the beam direction in one snake should be exactly opposite. Two pairs of Snakes, one pair in each ring, are installed at the 3 and 9 o'clock sections of the RHIC as shown in figure 3.1.

To achieve the desired beam polarization at the interaction region, a set of two pairs of spin rotators, two sets of that one for the STAR and the other for the PHINIX,

are installed right before and after the interaction region. The equilibrium position of the beam polarization is the transverse direction. Whenever a longitudinal polarization is needed, the spin rotators before the interaction rotates the beam polarization to the longitudinal direction, and the rotators beyond the interaction point bring it back to the transverse direction. Having its own set of spin rotators for the STAR and PHENIX experiments allowed an independent choice of beam polarization simultaneously.

3.1.5 Polarimetry

Measuring the beam polarization with higher precision at RHIC is crucial, as the polarization value normalizes many spin physics analyses. There are two proton-carbon (pC) Coulomb Nuclear Interference (CNI) [110], one in each ring, for the relative polarization measurement, and one hydrogen gas jet (H-jet) polarimeter [111] at one of the interaction points for the absolute polarization.

The pC CNI polarimeter uses a very thin ribbon of carbon targets developed by IUCF, which is simpler and easy to handle in the vacuum than the H-jet CNI. The analyzing power of pC CNI is about 3-5% for an elastic scattering in the small angle CNI region with a larger cross-section being weakly energy dependent over the RHIC energy range from 24 GeV/ c to 250 GeV/ c —the carbon target results in the high luminosity required for fast polarization measurements. The sizeable analyzing power, larger cross-section, and the advantages of the solid carbon target make this process ideal for a fast primary RHIC polarimetry.

The H-jet CNI polarimeter is located at the IP-12 in the RHIC ring, which is close to the pC CNI polarimeter, as shown in figure 3.1. It is used for the absolute polarization measurement based on the elastic pp scattering in the CNI region. It consists of three parts: the polarized atomic beam source (ABS), the scattering chamber, and

the Breit-Rabi polarimeter [112]. With the state-of-art of ABS, the thickness of the H-jet target is $\sim 10^{12}$ atoms/cm². The polarimeter target is free from the atomic beam, which crosses the RHIC beam in the vertical direction. Because of the particle identity, the beam polarization can be directly expressed in terms of the proton target polarization, which is directly measured by the Breit-Rabi polarimeter. Since it is remotely located, there is minimal interference with the other experiments, and it excludes the background generations for the other experiments.

3.2 STAR Detector and Sub-Systems

The solenoidal tracker at RHIC (STAR) is one of the two large detectors constructed at the six o'clock position in the RHIC complex at Brookhaven National Laboratory (BNL). It is constructed to investigate the behavior of strongly interacting matter at high energy density and to search for the signatures of quark-gluon plasma formation. Moreover, the polarized proton-proton collision program opened up multiple opportunities for the spin physics study of the quarks and gluon in a proton. STAR can measure a hadron production over a large solid angle, featuring detector systems for high-precision tracking, momentum analysis, and particle identification. In addition, large acceptance of the STAR allows event-by-event characterizations of heavy ion collisions and detection of a hadron in jets.

Figure 3.2 displays the STAR detector layout with a cutaway revealing the inner detector systems. The STAR detector system, cylindrical in shape, includes multiple detectors working in unison. The interaction point is at the center of the detector system, where the closest detector to it is the Silicon Vertex Tracker (SVT) [114]. The SVT spans $|\eta| \leq 1$ in pseudorapidity with full azimuthal coverage ($\delta\phi = 2\pi$), which provides charged particle tracking, precision localization of the primary vertex,

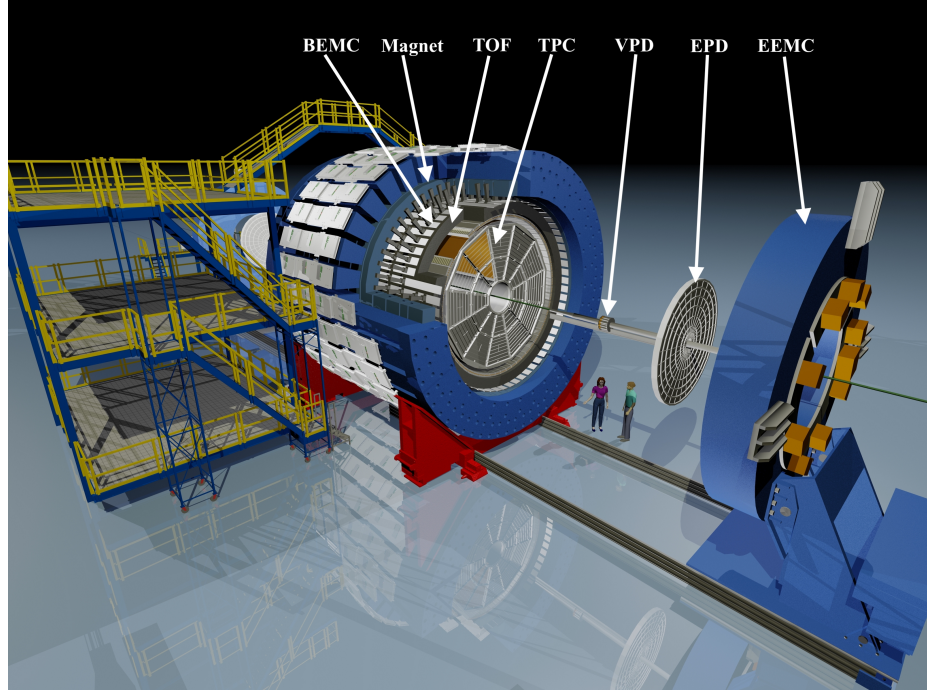


Figure 3.2: Perspective view of the STAR detector, with a cutaway for viewing inner detector systems.

and identification of secondary vertices from the weak decays, for example, λ , Ξ , and Ω s. The Time Projection Chamber [115], which is the heart of the STAR detector system, provides charged particle tracking, Particle Identification (PID), and momentum reconstruction using a 0.5T field from the solenoidal magnet [116]. The details of the TPC are discussed in section 3.2.1. In addition, a Time of Flight (TOF) [117] detector vastly improves STAR PID by providing the timing information with the same coverage as the TPC in η and ϕ , which is discussed in section 3.2.3. The STAR event triggering system is based on the transverse energy of the events measured by the electromagnetic calorimeter in the barrel region (BEMC) [118] and in the end-cap region (EMC)[119]. The BEMC and EMC have a $-1 < \eta < 2$ coverage in pseudorapidity and whole 2π azimuthal angle.

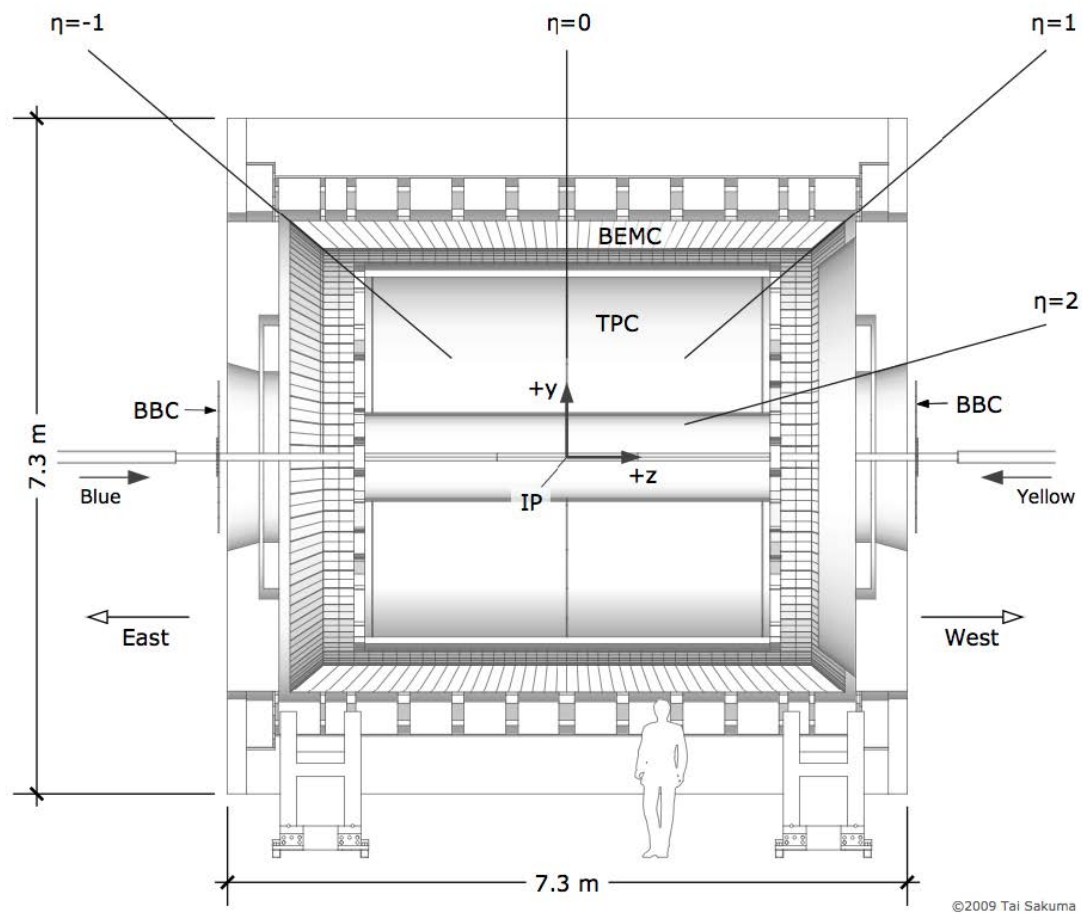


Figure 3.3: Cutaway side view of the STAR detector [113].

3.2.1 TPC and Magnet

Figure 3.4 shows the schematic view of the STAR TPC. It's cylindrical, measuring 4 meters long and 4.2 meters in diameter. Its acceptance covers $|\eta| \leq 1.8$ in pseudorapidity and full 2π coverage in azimuth over a full range of multiplicities, which makes it one of the largest operating TPC in the world. It sits in a large solenoidal magnet that operates in a 0.5 T magnetic field. It consists of a thin conductive Central Membrane (CM) at its center, which is $70\text{ }\mu\text{m}$ thick carbon-loaded Kapton film with a surface resistance of $250\text{ }\Omega$ per square that operates at 28 kV potential, with the end caps at the ground providing a uniform electric field of 135 V/cm. The proper boundary conditions between the CM, end caps, and field cage cylinders establish the uniformity of the electric field. The field cage cylinders, inner field cage, and outer field cage, shown in figure 3.4, provide a series of equipotential rings that divide the space between the CM and the anode planes into 182 equally spaced segments. One ring is common to both sides, where the CM sits. The uniform gradient between the CM and the grounded end caps is maintained by biasing these rings with a chain of precision resistors of $2\text{ M}\Omega$.

The empty volume between the field cages is filled by the P10 gas, which is 10% methane and 90% argon regulated at 2 millibars above atmospheric pressure [120]. The P10 gas attributes the fast drift velocity, peaking at a low electric field, which provides the stable drift velocity of the particles insensitive to the variation in the atmospheric conditions, temperature, and pressure.

The STAR experiment uses the TPC as the primary tracking device, which records particle tracks, measures particle momenta, and identifies the particles by measuring their ionization energy loss (dE/dx). It effectively separates pions and protons up to 1.2 GeV/c; however, separating high-energy particles is difficult as the dE/dx

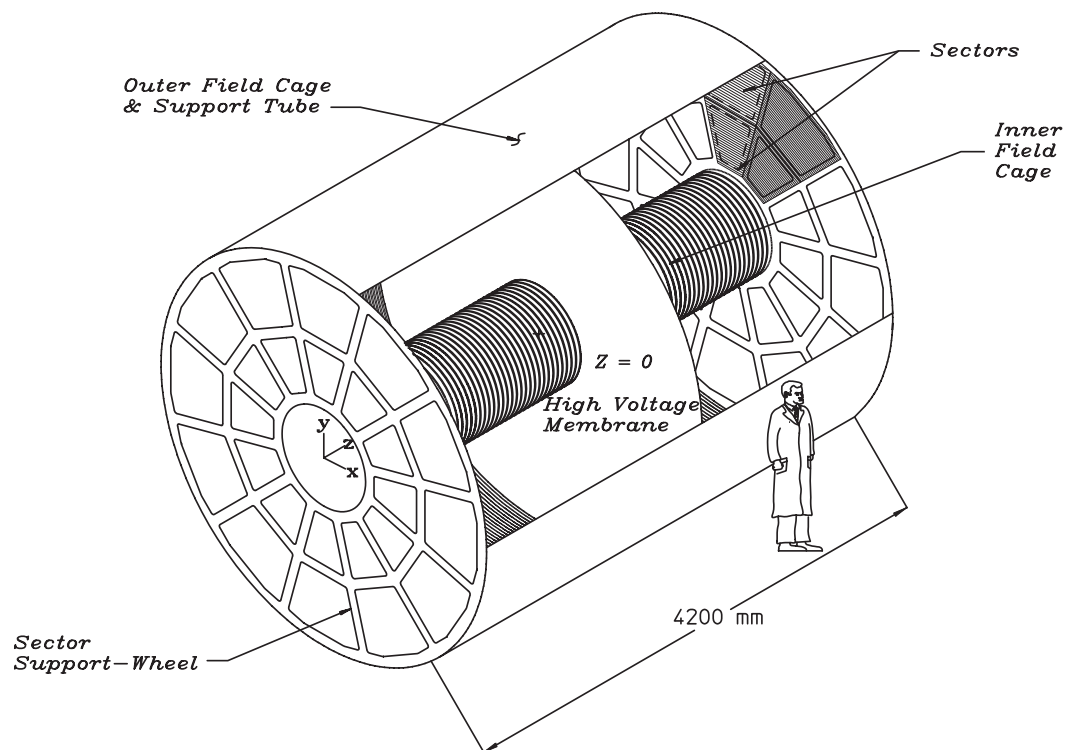


Figure 3.4: STAR time projection chamber.

becomes less mass-dependent at higher energies. As the primary ionizing particles enter the TPC gas volume, their paths are reconstructed with high precision from the released secondary electrons, which drift in the electric field to the readout end caps at the end of the chamber. The TPC performance relies heavily on the magnetic field as electrons may diffuse in the transverse direction, directly impacting particle track reconstruction. The momentum resolution for most of the tracks in the TPC reaches the value of $\delta p/p = 0.02$. The relative momentum resolution increases for a low-momentum track with more hit points along the track.

The electric and magnetic fields in the TPC are uniform all around, radially and longitudinally. However, the non-uniformities and global misalignment of the electric and magnetic fields can cause distortions in the position of the secondary electrons reaching the readout pads. One of the causes of the field distortion in TPC is the space charge, which occurs due to the piling-up of slow drifting positively charged ions in the gas from the standard operation of the TPC. The effect is propagated to the recorded tracks, corrected by the well-established space charge and grid leak calibration procedure [121].

3.2.2 Barrel Electromagnetic Calorimeter (BEMC)

The schematic view of the Barrel Electromagnetic Calorimeter (BEMC)[118] is shown in figures 3.5 (cross-section view). The BEMC sits underneath the aluminum coil of the solenoid, covering the $|\eta| \leq 1$ and 2π in azimuth at a radial distance of about 220 cm from the beam axis matching the coverage of the TPC acceptance.

The BEMC design includes 120 calorimeter modules placed 60 in ϕ and 2 in η . Each module measures about 26 cm wide and 293 cm long with an active depth of 23.5 cm, each subtending 6° in $\Delta\phi$ and 1 unit in $\Delta\eta$. Each module is divided into 40 towers, 20 in η and 2 in ϕ , with each tower subtending 0.05×0.05 in $\Delta\phi \times \Delta\eta$. Thus,

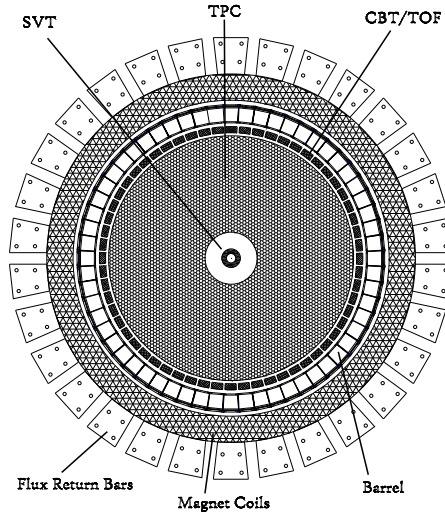


Figure 3.5: BEMC Cross-Section view

the BEMC comprises 4800 calorimeter towers spanning a full 2π coverage in azimuth. Each of the towers is constructed so that they point to the interaction diamond at the center. This projective nature of the tower in a module is shown in figure 3.6.

The core of the calorimeter consists of a stacked layer of 5 mm thick lead and 5 mm plastic scintillator alternately and a Shower Max Detector (SMD) situated roughly at five radiation lengths from the front of the stack. There are 21 active scintillating layers and 20 layers of lead absorber plates. The plastic scintillator, called “megatile”, consists of 40 optically isolated tiles. The mechanical structure of a calorimeter module with the compression components and the rail mounting system is shown in figure 3.7.

3.2.3 Time of Flight (TOF)

The STAR Time of Flight (TOF) complements the particle identification capability of the TPC by using the timing information in conjunction with the vertex position

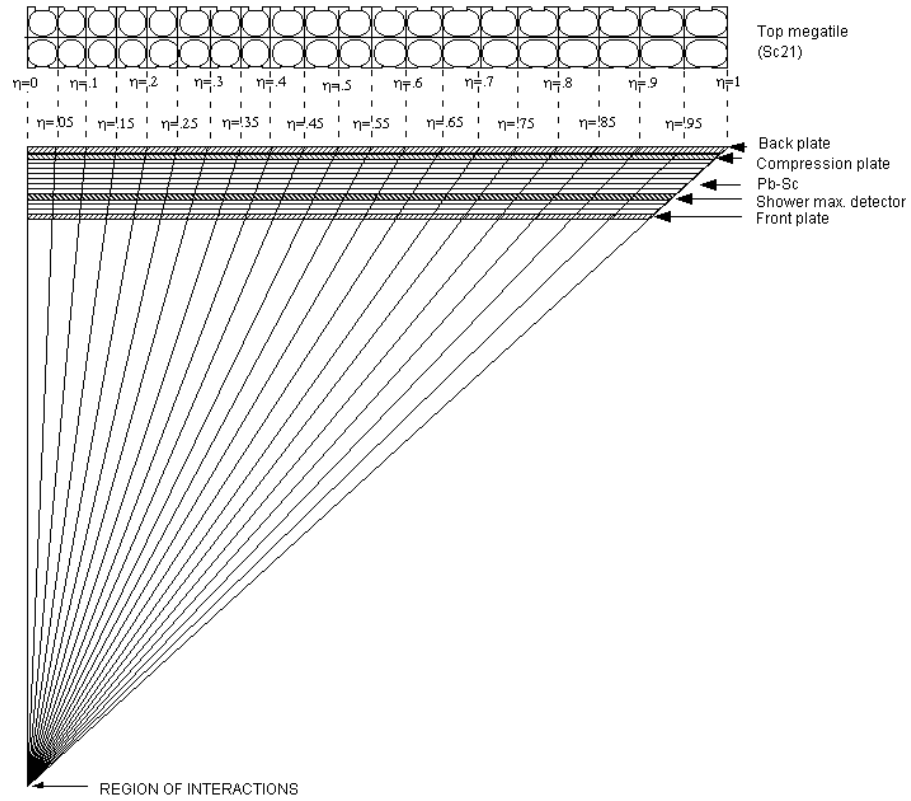


Figure 3.6: Projective nature of the BEMC module.

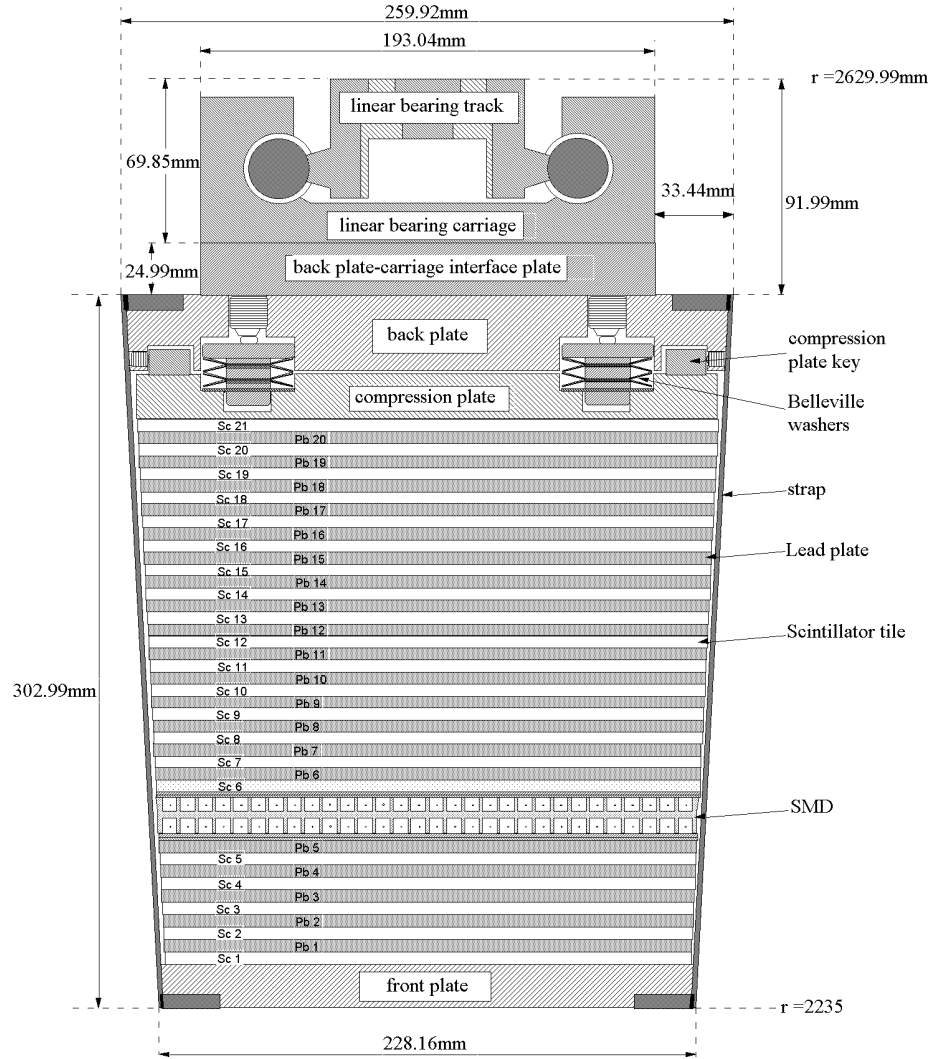


Figure 3.7: Side view of the BEMC module.

detector (VPD). TOF sits outside the TPC, wrapping all around and spanning the same coverage as the TPC. The TOF system is designed based on the multi-gap resistive plate chamber (MRPC), first developed by the CERN ALICE group [122]. The side view of the MRPC module for the STAR TOF system is shown in Figure 3.8. The long edge of the module is shown at the top, whereas the bottom figure shows the short edge. The MRPC module consists of a series of 0.54 mm thick glass plates kept parallel by using 0.22 mm diameter nylon fishing line as a spacer. A mixture of 90% C₂H₂F₄, 5% isobutane, and 5% SF₆ fills the gaps between the plates. The resistivity of these glass plates is of the order of $10^{13} \Omega/\text{cm}$, thus being transparent to the electrical charges. These plates are electrically floating, and a strong electric field is generated in each sub-gap of gas with the electrodes applied on the outer surface of the outer plate. Since the glass plates and electrodes are charge transparent, the induced signal picked up by the copper readout pad is the sum of all the avalanche of charges induced in the gas gaps.

The TOF system measures time intervals for a particle track using two detectors; one is the VPD (see section 3.2.4 for details), which provides an event start time, and the TOF MRPC provides the stop time for reconstructed charged tracks that point back to the event vertex. This time interval, Δt , for reconstructed tracks, with the momentum, p , and total path length, s , allows us to calculate the inverse velocity, $1/\beta$, for each track as,

$$\frac{1}{\beta} = \frac{c \Delta t}{s}, \quad (3.2)$$

where c is the speed of light. With $1/\beta$ and p , the particle mass, m , can be calculated as,

$$m = p \sqrt{\frac{1}{\beta^2} - 1} \quad (3.3)$$

β , as calculated above using the TOF system timing information, versus particle

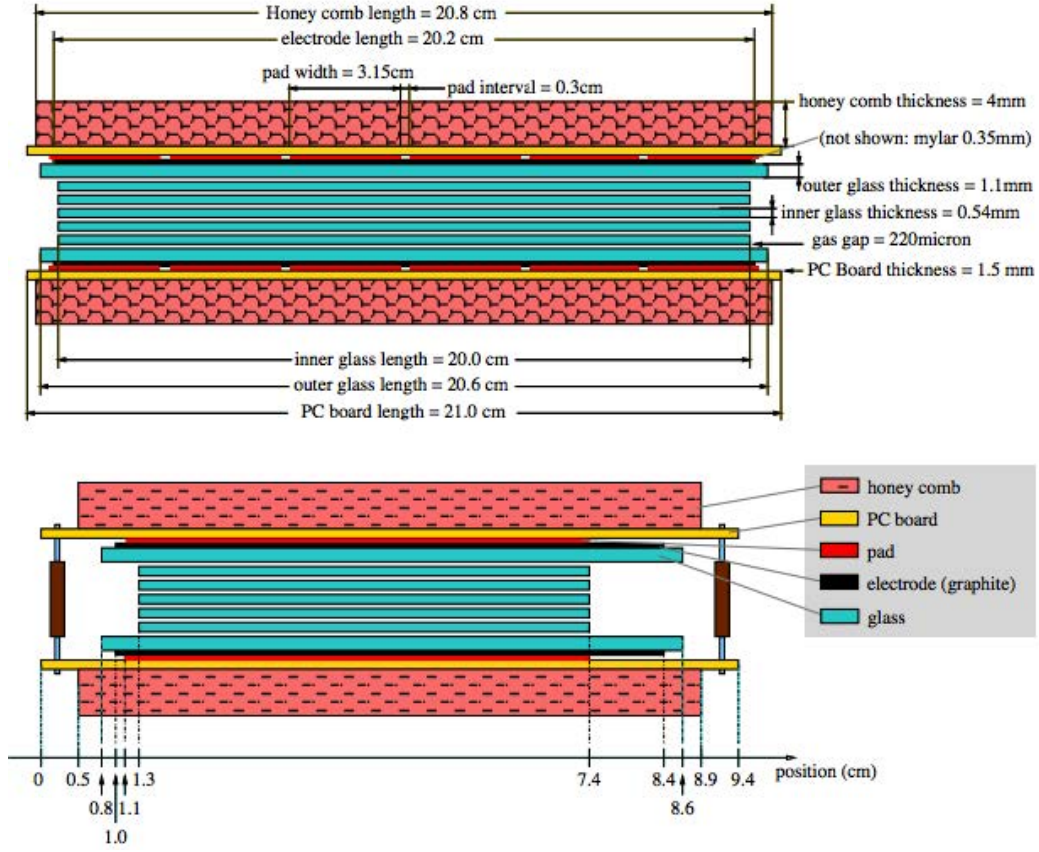


Figure 3.8: Side view of an MRPC module for the long (top) and short (bottom) sides.

momentum provides a basis for the TOF particle identification.

In recent studies, it has been observed that for an event vertex with the z-vertex position difference identified by the TPC ($V_{z,tpc}$) and VPD ($V_{z,vpd}$) greater 5 cm, the VPD timing information may be incorrect. A new method was developed called “StartLessTOF”, which addresses this issue by calculating the start time starting at the stop time and moving backward for a track without relying on the VPD timing. In the IFF asymmetry analysis, discussed in chapter 4, the StartLessTOF method of PID is used, and more details will be discussed in the respective section.

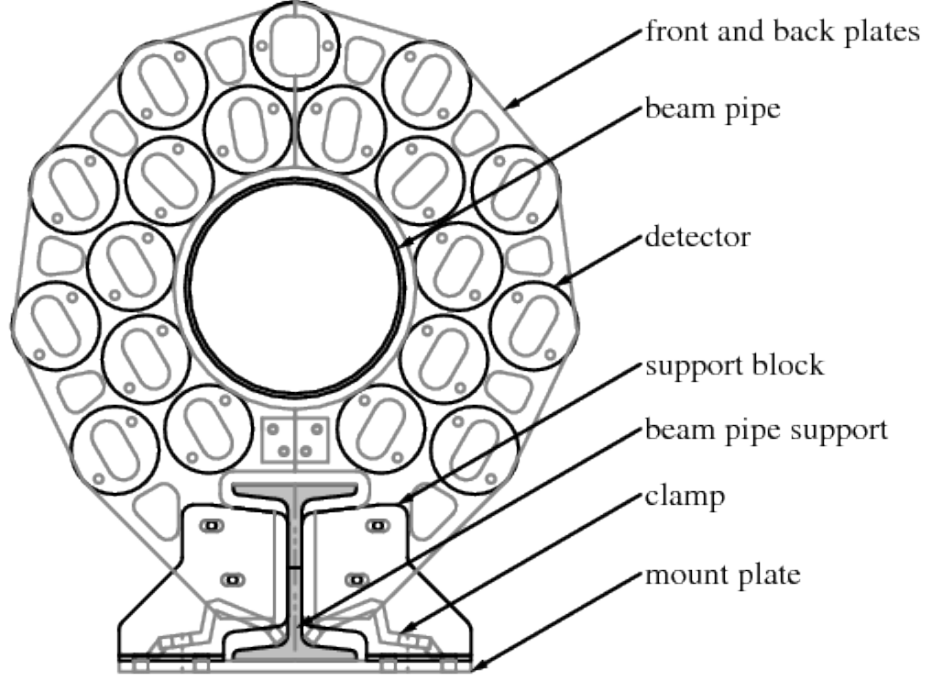


Figure 3.9: VPD front view.

3.2.4 Vertex Position Detector (VPD)

There are two identical Vertex Position Detector (VPD) [123] assemblies, one at the east side and the other at the west side, located symmetrically surrounding the beam pipe at a distance of 5.7 m from the TPC center. Each VPD assembly consists of nineteen detectors, each composed of a lead (Pb) converter followed by a plastic scintillator read out by a photo-multiplier tube. The side view of this detector assembly is shown in figure 3.10. The front view of the VPD with all nineteen detectors assembled in two circles is shown in figure 3.9.

The VPD can provide timing information of an event, which is required for event triggering and for a fast detector as the event start time, for example, TOF. It measures the vertex z -position. It provides the timing information by measuring photon arrival time produced by the decay of π^0 in the very forward direction, close to the beam pipe.

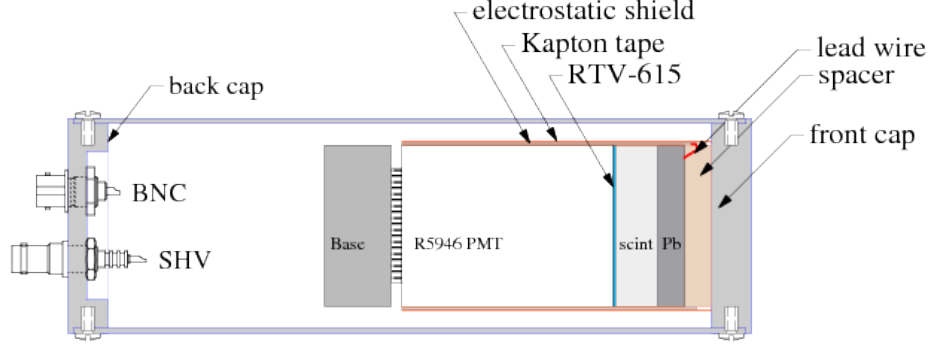


Figure 3.10: VPD side view.

Besides serving as the primary detector input for the minimum bias trigger in A+A collisions, each VPD assembly measures up to nineteen times in each event, which is used to measure the location of the vertex along the beam direction, Z_{vtx} , given by,

$$Z_{vtx} = c \frac{T_{east} - T_{west}}{2}, \quad (3.4)$$

where T_{east} and T_{west} are the times measured by the two VPD assemblies. Moreover, the start time used by the TOF for the particle identification, T_{start} is given by,

$$T_{start} = \frac{T_{east} + T_{west}}{2} - \frac{L}{c}, \quad (3.5)$$

where L is the distance of either assembly from the center of the STAR. The experimental resolution of the VPD timing, T_{east} or T_{west} , depends on the number of channels used in the assembly N , which improves as $\delta T \propto 1/\sqrt{N}$, where T is T_{east} or T_{west} . Which follows the Z_{vtx} resolution of equation 3.4 given by,

$$\delta Z_{vtx} = \frac{c}{2} \delta \Delta T = \frac{c}{\sqrt{2}} \frac{\sigma_0}{\sqrt{N}}, \quad (3.6)$$

where σ_0 is the time resolution of the single readout detector and $\delta \Delta T$ is the resolution of the time difference ($T_{east} - T_{west}$). Thus, the large number of readout channels allows

averaging over more lit channels, improving the overall VPD performance.

CHAPTER 4

$\pi^+\pi^-$ AZIMUTHAL CORRELATION ASYMMETRY

In this chapter, we will explore the precision measurement of the dihadron azimuthal correlation asymmetry $A_{UT}^{\sin(\phi_{RS})}$ for $\pi^+\pi^-$ production in polarized proton-proton ($p^\uparrow p$) collisions based on the theoretical framework discussed in section 2.5. This measurement is based on the STAR polarized pp data from Run 2015, which corresponds to an integrated luminosity, L_{int} , of $\sim 52 \text{ pb}^{-1}$, which provides the most statistical precision in the asymmetry in the valence quark region ($0.1 < x < 0.3$). I will discuss asymmetry extraction methods, followed by a brief review of previous STAR measurements in the same framework. The rest of the chapter will provide various details of the analysis.

4.1 $A_{UT}^{\sin(\phi_{RS})}$ Extraction Methods

In the STAR experiment, the asymmetry $A_{UT}^{\sin(\phi_{RS})}$ can be extracted in $p^\uparrow p$ collisions by counting the number of dihadron pairs in different ϕ_{RS} bins for various beam polar-

ization conditions (up and down). Two commonly employed methods for extracting $A_{UT}^{\sin(\phi_{RS})}$ are discussed below.

4.1.1 Relative Luminosity Method

The relative luminosity formula is given by

$$A_{UT} \cdot \sin \phi_{RS} = \frac{1}{P} \cdot \frac{N^{\uparrow}(\phi_{RS}) - r \cdot N^{\downarrow}(\phi_{RS})}{N^{\uparrow}(\phi_{RS}) + r \cdot N^{\downarrow}(\phi_{RS})}, \quad (4.1)$$

Where $N^{\uparrow(\downarrow)}$ represents the number of dihadron pairs in each ϕ_{RS} bin when the beam is polarized up (down). The parameter r corresponds to the ratio of the luminosities $L^{\uparrow}$ and $L^{\downarrow}$, while P represents the absolute beam polarization.

This equation incorporates relative luminosities, a quantity measured by the detector and subject to associated systematic uncertainties. In contrast, the cross-ratio method, free from any reliance on detector-related parameters, has gained wide acceptance within the STAR community due to its elegant nature. Notably, the consistency between these two methods has been rigorously verified and confirmed in previous work [72]. Given this established validation, we adopted the cross-ratio method, as it aligns with our pursuit of reliable results.

4.1.2 Cross-Ratio Method

The cross-ratio method, which eliminates the need for detector-related quantities, is extensively discussed in reference [124]. In the following, we outline the derivation of the cross-ratio formula for the spin- $\frac{1}{2}$ analyzing power in a symmetric detector system.

The cross-section for a polarized spin- $\frac{1}{2}$ beam can be written as,

$$I(\theta, \phi) = I_0(\theta) [1 + p_y A_y(\theta)], \quad (4.2)$$

where θ is the scattering angle, I_0 is the unpolarized cross-section at an scattering angle θ , p_y is y -component of the beam polarization, and $A_y(\theta)$ is the analyzing power. In terms of $p^\uparrow p$ to dihadron channel, the above equation can be expressed as,

$$I(\theta, \phi_{RS}) = I_{0,\phi_{RS}}(\theta)[1 + P A_{UT} \sin \phi_{RS}], \quad (4.3)$$

where P is the absolute polarization of the transversely polarized beam. The number of dihadrons recorded in particular ϕ_{RS} bin when the beam is polarized up $N_{\phi_{RS}}^\uparrow$ is given by,

$$N^\uparrow(\phi_{RS}) = L^\uparrow I_{0,\phi_{RS}}[1 + A_{UT} P \sin \phi_{RS}], \quad (4.4)$$

where $L^\uparrow$ is the luminosity when the beam is polarized up. In the same ϕ_{RS} bin, the dihadron counts when the polarization is flipped $N_{\phi_{RS}}^\downarrow$ is given by,

$$N^\downarrow(\phi_{RS}) = L^\downarrow I_{0,\phi_{RS}}[1 - A_{UT} P \sin \phi_{RS}], \quad (4.5)$$

where the negative sign accounts for the opposite spin flip.

It is expected that, by rotational symmetry in a perfectly symmetric detector, the observed counts in a ϕ_{RS} bin when the beam polarization is $\uparrow$ should be similar in the opposite ϕ_{RS} bin shifted by π , i.e., $\phi_{RS} + \pi$ when the beam is polarized $\downarrow$. So, the counts in the $\phi_{RS} + \pi$ bin when beam polarization is up, $N^\uparrow(\phi_{RS} + \pi)$, is given by,

$$N^\uparrow(\phi_{RS} + \pi) = L^\uparrow I_{0,\phi_{RS}+\pi}[1 - A_{UT} P \sin \phi_{RS}], \quad (4.6)$$

where the negative sign is from $\sin(\phi_{RS} + \pi) = -\sin \phi_{RS}$. Again, the counts in the $\phi_{RS} + \pi$ bin when spin is flipped, $N^\downarrow(\phi_{RS} + \pi)$, is given by,

$$N^\downarrow(\phi_{RS} + \pi) = L^\downarrow I_{0,\phi_{RS}+\pi} [1 + A_{UT} P \sin \phi_{RS}], \quad (4.7)$$

where the sign is positive because of an extra negative sign from the spin-flip. Now, forming a geometric mean of counts from equivalent bins as,

$$\sqrt{N^\uparrow(\phi_{RS})N^\downarrow(\phi_{RS} + \pi)} = \sqrt{L^\uparrow L^\downarrow I_{0,\phi_{RS}} I_{0,\phi_{RS}+\pi}} [1 + A_{UT} P \sin \phi_{RS}], \quad (4.8)$$

and,

$$\sqrt{N^\downarrow(\phi_{RS})N^\uparrow(\phi_{RS} + \pi)} = \sqrt{L^\uparrow L^\downarrow I_{0,\phi_{RS}} I_{0,\phi_{RS}+\pi}} [1 - A_{UT} P \sin \phi_{RS}] \quad (4.9)$$

Solving equations 4.8 and 4.9 for $A_{UT} \sin \phi_{RS}$, we get,

$$A_{UT} \sin \phi_{RS} = \frac{1}{P} \cdot \frac{\sqrt{N^\uparrow(\phi_{RS})N^\downarrow(\phi_{RS} + \pi)} - \sqrt{N^\downarrow(\phi_{RS})N^\uparrow(\phi_{RS} + \pi)}}{\sqrt{N^\uparrow(\phi_{RS})N^\downarrow(\phi_{RS} + \pi)} + \sqrt{N^\downarrow(\phi_{RS})N^\uparrow(\phi_{RS} + \pi)}} \quad (4.10)$$

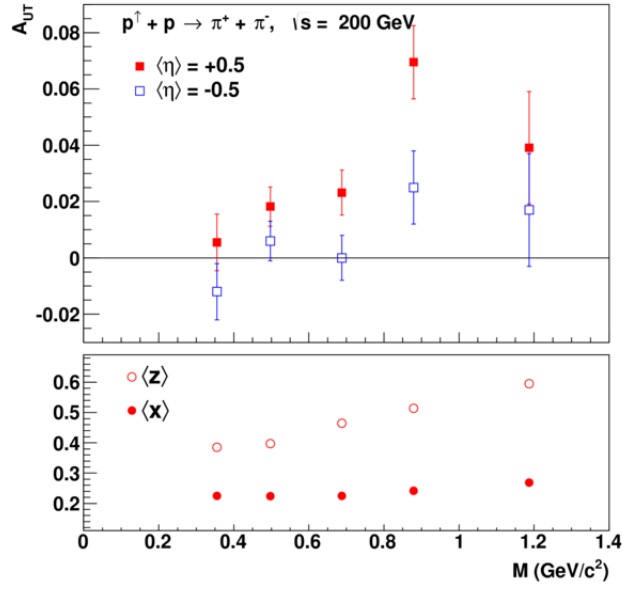
Equation 4.10 is the cross-ratio formula, also called the “geometric mean formula”, which only involves the numbers without the relative luminosity. This formula has a simple and elegant structure, which is why it has been widely used by STAR measurements since the 2006 analysis-the first asymmetry analysis via dihadron channel in $p^\uparrow p$.

4.2 STAR $A_{UT}^{\sin(\phi_{RS})}$: Proof-of-principle And Global Impact

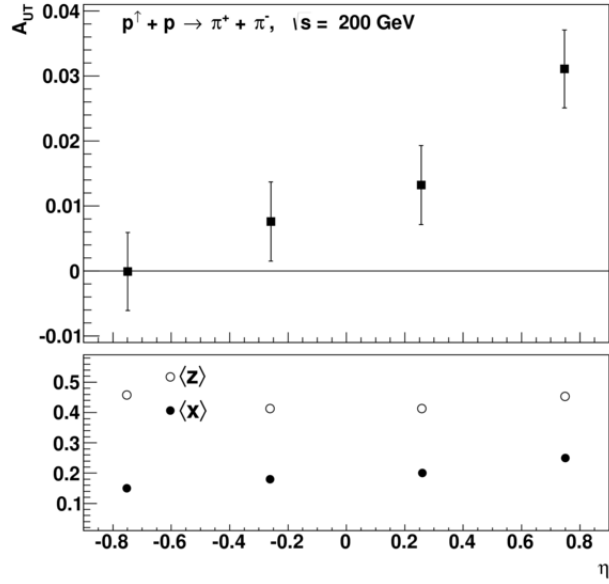
STAR reported the first measurement of $A_{UT}^{\sin(\phi_{RS})}$ for $\pi^+\pi^-$ in $p^\uparrow p$ collisions at $\sqrt{s} = 200$ GeV using Run 2006 data [72]. This was followed by a measurement using Run 2011 data at $\sqrt{s} = 500$ GeV [86]. The 2006 dataset corresponds to an integrated luminosity of $L = 1.8 \text{ pb}^{-1}$ and an average beam polarization of approximately 60%, allowing for the exploration of the valence quark transversity in the range $0.15 < x < 0.30$. The 2011 data correspond to an integrated luminosity of 25 pb^{-1} with an average beam polarization of 53% and covers x -range of about $0.05 < x < 0.2$.

Figure 4.1 shows the STAR $A_{UT}^{\sin(\phi_{RS})}$ results at $\sqrt{s} = 200$ GeV [72]. Figure 4.1a depicts $A_{UT}^{\sin(\phi_{RS})}$ as a function of the invariant mass ($M_{inv}^{\pi^+\pi^-}$) in two pseudorapidity bins, $\eta^{\pi^+\pi^-} > 0$ and $\eta^{\pi^+\pi^-} < 0$ in the top panel, whereas the bottom panel shows the partonic variables x and z (fractional energy of fragmenting quark carried by the $\pi^+\pi^-$) from simulation in the same $M_{inv}^{\pi^+\pi^-}$ bins. Figure 4.1b shows $A_{UT}^{\sin(\phi_{RS})}$ as a function of $\eta^{\pi^+\pi^-}$ integrating over $M_{inv}^{\pi^+\pi^-}$ and $p_T^{\pi^+\pi^-}$ in the top panel and x and z in the bottom panel. The STAR $A_{UT}^{\sin(\phi_{RS})}$ results at $\sqrt{s} = 500$ GeV is shown in figure 4.2. The signal differential in mass in the forward $\eta^{\pi^+\pi^-}$ region is displayed together with the STAR 2006. The theoretical prediction is shown in Figure 4.2a. Figure 4.2b shows the $A_{UT}^{\sin(\phi_{RS})}$ signal differential in $\eta^{\pi^+\pi^-}$ (top panel) and x and z in the corresponding $\eta^{\pi^+\pi^-}$ bins (bottom panel).

The measured $A_{UT}^{\sin(\phi_{RS})}$ signal exhibits significant magnitudes in the $\eta^{\pi^+\pi^-} > 0$ (forward) region, corresponding to the higher x range where transversity is expected to be sizable. Conversely, the observed signal is small in the $\eta^{\pi^+\pi^-} < 0$ (backward) region, which probes low x values. This observation suggests a dependence of transversity



(a)



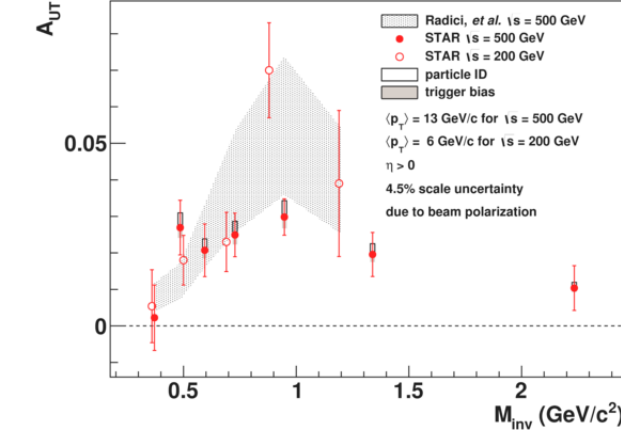
(b)

Figure 4.1: STAR A_{UT} at $\sqrt{s} = 200$ GeV [72]. (a) A_{UT} as a function of $M_{inv}^{\pi^+\pi^-}$ (upper panel) and corresponding partonic variables x and z (lower panel) from the simulation. (b) A_{UT} as a function of $\eta^{\pi^+\pi^-}$ (upper panel) and x and z in the lower panel.

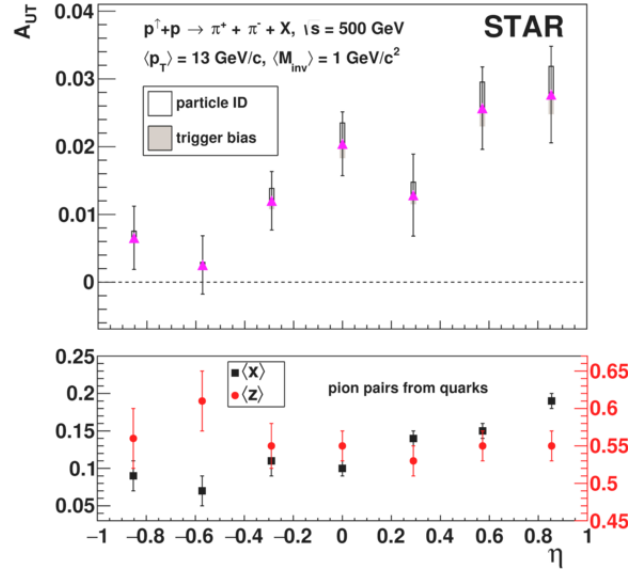
on x . Additionally, the forward A_{UT} signal exhibits a peak around the ρ -meson mass ($\sim 0.77 \text{ GeV}/c^2$). This characteristic is anticipated due to the interference between pions originating from different production channels, as discussed in section 2.5. Figure 4.2a compares the STAR 2006 and 2011 measurements with theoretical calculations in $M_{inv}^{\pi^+\pi^-}$ bins. The measured asymmetries and the resonance peak align well with theoretical predictions.

The STAR 2006 $A_{UT}^{\sin(\phi_{RS})}$ data points obtained at $\sqrt{s} = 200 \text{ GeV}$ have been incorporated into a global analysis for transversity extraction, alongside the SIDIS and e^+e^- data [63]. Figure 4.3 illustrates the behavior of xh_1 with and without the inclusion of the STAR data. Specifically, Fig. 4.3a depicts the valence u -quark transversity (xh_1^{uv}), while Fig. 4.3b displays the valence d -quark transversity (xh_1^{dv}) as functions of x . The dark band in Fig. 4.3a and the light band in Fig. 4.3b shows the transversity with the inclusion of the STAR data, which is compared against the same result without the STAR data [125]. Despite the statistical limitations, the impact of the STAR data is significant and results in notable improvements in constraining transversity in the valence quark region.

The measurement conducted using the Run 2015 dataset (this measurement) provides the most precise determination of $A_{UT}^{\sin(\phi_{RS})}$ at $\sqrt{s} = 200 \text{ GeV}$. Consequently, this measurement is expected to enhance the precision in constraining transversity in the valence quark region substantially.

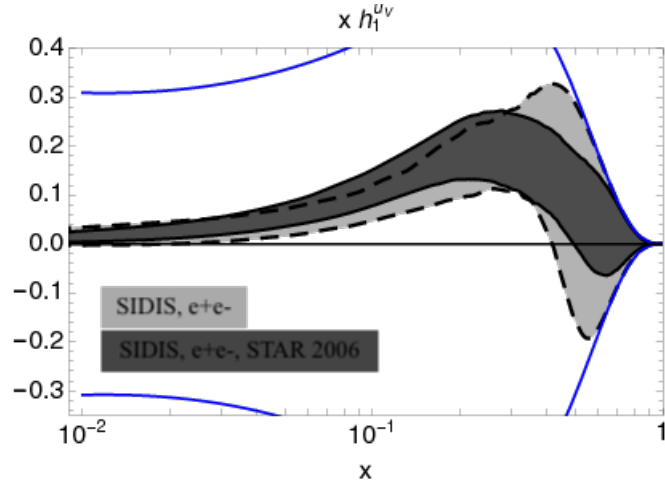


(a)

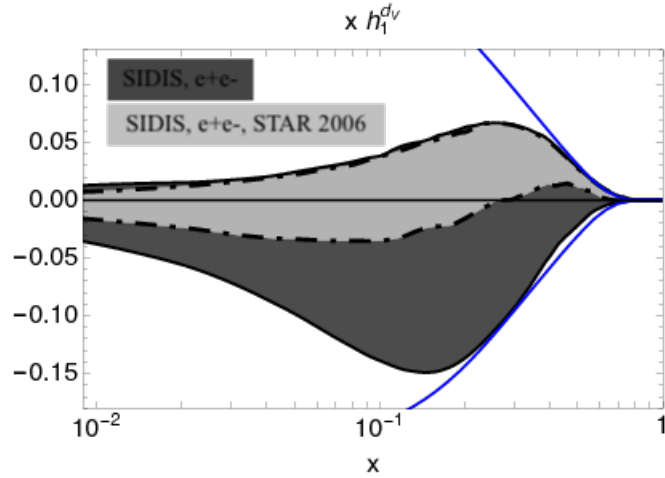


(b)

Figure 4.2: STAR A_{UT} at $\sqrt{s} = 500$ GeV [86]. (a) A_{UT} as a function of $M_{inv}^{\pi^+\pi^-}$ compared with the theory predictions from the SIDIS and e^+e^- data [126] and STAR 2006 result. (b) A_{UT} as a function of $\eta^{\pi^+\pi^-}$ for $\langle p_T^{\pi^+\pi^-} \rangle = 13$ GeV/c (top panel), and x and z in corresponding $\eta^{\pi^+\pi^-}$ bins (lower panel).



(a)



(b)

Figure 4.3: Transversity xh_1 as a function of x at $Q^2 = 2.4 \text{ GeV}^2$. The dark (blue) lines represent Soffer bounds. **(a)** Valence u -quark transversity. The dark band includes STAR pp , SIDIS, and e^+e^- , whereas the lighter band excludes the STAR data. **(b)** Valence d -quark transversity. The dark band only includes SIDIS and e^+e^- data, and the lighter band is after including the STAR data. Significant improvement in the $xh_1^{q_v}$ because of the STAR data can be observed.

4.3 STAR Run 2015 IFF Analysis

4.3.1 Dataset and Quality Analysis

Transversely polarized proton-proton collision data from the STAR run 2015 at $\sqrt{s} = 200$ GeV is utilized to measure the IFF asymmetry. The description of the data set is as follows:

- Collision: proton-proton
- Center-of-mass energy ($\sqrt{s}$): 200 GeV
- Beam polarization: transverse
- Production year: year2015
- Trigger Sets: production_pp200trans_2015
- Production tag: P16id
- Production library: SL16d
- Integrated luminosity (L_{int}): $\sim 52 \text{ pb}^{-1}$

The data analyzed in this study comprises collision events triggered by the jet patch triggers. Our focus is directed towards events triggered by JP1 and JP2, denoted as “JP”, constituting approximately 90% of the total events. Specifically, events are selected using offline trigger IDs: JP1 (480404, 480414) and JP2 (480401, 480411). It is worth noting that two distinct offline trigger IDs were employed for the JP triggers due to a trigger ID modification after collecting one-third of the data during the data-taking process. Data quality assessment (QA) is conducted at the event level, examining the average values of the physical quantities for each Run. Runs

with a duration of less than three minutes are omitted. Besides Run time, another key criterion for a Run is the presence of JP-triggered events. Additionally, due to the polarization sensitivity of this analysis, two fills with unfavorable polarization patterns (fills 18749 and 18764) are excluded.

For QA purposes, the following event and track quality cuts are applied,

- Z-Vertex (V_z): $|V_z| < 60$ cm
- Vertex ranking ($V_{z,rank}$): $V_{z,rank} > 1e6$
- Track pseudorapidity (η): $|\eta| < 1$
- Track transverse momentum (p_T): $p_T > 1.5$ GeV/ c
- Track distance of closest approach (dca): $dca < 1$ cm

$$dca = \begin{cases} 2 \text{ cm} & \text{if } p_T < 0.5 \text{ GeV}, \\ 2.5 \text{ cm} - p_T(1 \text{ cm/GeV}) & \text{if } 0.5 < p_T < 1.5 \text{ GeV}, \\ 1 \text{ cm} & \text{if } p_T \geq 1.5 \text{ GeV} \end{cases}$$

- Number of fit points ($nHitsFit$): $nHitsFit > 25$
- Fit points ratio ($nHitsFit/nHitsFitPoss$): $nHitsFit/nHitsFitPoss > 0.51$
- Pion selection: $-1 < n\sigma_\pi < 2$

The analysis focuses on primary tracks originating from primary vertices. To eliminate pileup vertices cuts on V_z and $V_{z,rank}$ are implemented. Tracks with more TPC hits exhibit better precision than those with fewer fit points. The track- η cut confines the track within the TPC acceptance in the central region. A cut on the fit points ratio prevents duplicating tracks with different segments reconstructed as

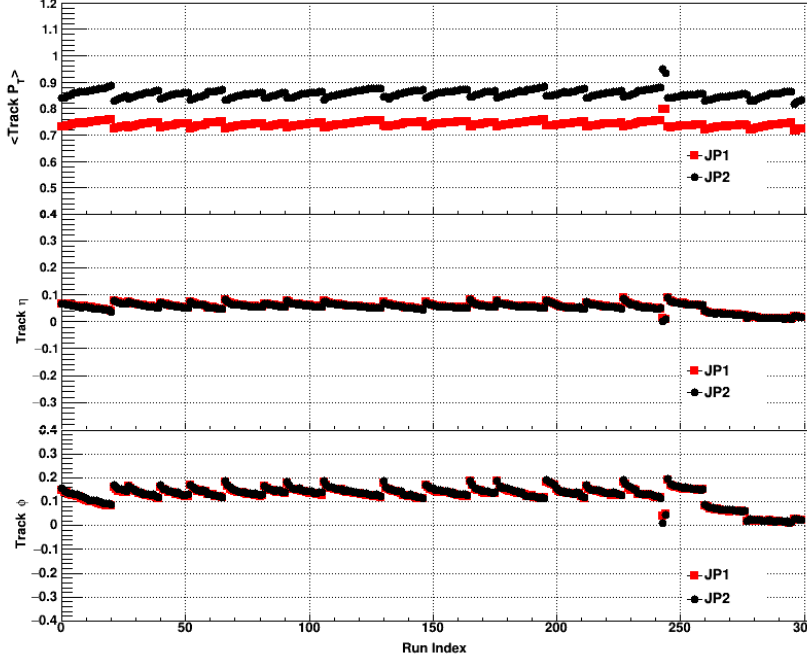


Figure 4.4: Average track η , ϕ , and p_T per run.

distinct entities. The *dca* cut removes tracks arising from secondary decays. These selection cuts are tailored for the QA analysis, while cuts for the physics analysis will be detailed in the respective section.

The mean event and track parameters per Run, including V_z , η , ϕ , p_T , etc., are graphed against the Run index. Ideally, the average per Run should remain consistent over time. However, variations due to environmental and experimental conditions can render certain event records unsuitable for physics analysis. Any Run with an average value deviating by more than 3σ from the mean is excluded.

Figure 4.4 illustrates the post-QA average p_T , η , and ϕ of tracks for JP1 and JP2 triggers as a function of the Run index for the initial 300 runs. The depicted average values exhibit dependence on the data collection period, with each band corresponding to a specific fill. This Run dependence within the fill is averaged over the data-taking period and does not impact the physics analysis. Following QA,

749 Runs are identified as suitable for subsequent physics analysis. Appendix A.4 provides a comprehensive list of these runs.

4.3.2 Simulation and Comparison with Data

This analysis necessitates a Monte Carlo simulation sample to evaluate systematic uncertainties stemming from event triggering. The simulated events are generated utilizing PYTHIA 6.4.28 [127], employing the Perugia 2012 tune [128]. In the Perugia 2012 tune, the parameter PARP(90), governing the energy dependence of underlying event processes and the low- p_T cut-off, is tuned from 0.24 to 0.213. This adjustment aligns with STAR-specific measurements of $\pi^\pm$ cross-sections in 200 GeV pp collisions [129, 130]. This alteration has yielded a satisfactory depiction of jet and underlying event characteristics in pp collisions for both 200 [131, 132, 133], and 510 [134] GeV energies.

Subsequently, the simulated events undergo comprehensive detector simulations aligned with the 2015 Run configurations, following the implementation in GEANT 3 [135]. The GEANT 3 simulation precisely emulates the STAR 2012 detector setup. However, for the 2015 Run, additional material originating from the Heavy-Flavor Tracker (HFT) [136] is accounted for in the region $z \lesssim -30$ cm. To accommodate this supplementary material effect, the GEANT simulation employs the 2012 detector setup while utilizing digitization routines consistent with the 2015 detector readout procedures [137].

To encompass event pileup and beam background effects, simulated events are embedded into Zero-bias events collected during the 2015 Runs. These zero-bias events are triggered at clock triggers unrelated to collisions. Furthermore, the online trigger algorithm is implemented within the embedding sample, permitting trigger selection.

The simulation software records hard scattered partons, final state particles from the fragmentation and hadronization of partons, and the detector’s response to the final state particles. The final state particles are referred to as “particle level”, while their response to the detector is called “detector level”. The detector responses are recorded in the same format as those generated in real data events.

To directly apply the embedding-driven measurement to data, it’s essential that the detector-level quantities obtained through embedding accurately replicate real data characteristics. To ensure this, data and embedding are compared for all relevant physics quantities pertaining to JP1 and JP2 triggers.

The embedding sample is tailored to possess an increasing partonic p_T ($\hat{p}_T$), aligning with triggered data statistics to facilitate meaningful statistical comparisons. For merging of $\hat{p}_T$ bins, a weight proportional to the number of events generated in each specific $\hat{p}_T$ bin is employed. This weight is essentially the inverse of the luminosity for every $\hat{p}_T$, expressed as $\mathcal{L}_i = N_i/\hat{\sigma}_i$, where N_i represents the total generated events in $\hat{p}_T$ bin i , and $\hat{\sigma}$ is the corresponding partonic cross-section. Furthermore, the luminosity is normalized by the luminosity of the highest $\hat{p}_T$ bin. Consequently, the weight equals one for the highest bin. This method of recombining $\hat{p}_T$ bins is fortified by the comprehensive $\hat{p}_T$ spectrum, which is anticipated to exhibit a smooth distribution across the entire range. Figure 4.5 showcases the weighted $\hat{p}_T$ combining all partonic bins. The smoothness of the curve substantiates the efficacy of the weighting scheme, providing confidence in the reliable recombination of $\hat{p}_T$ bins.

Figure 4.6 and 4.9 depict a comparison of $p_T^{\pi^+}$ and $p_T^{\pi^-}$ between JP1 (left panel) and JP2 (right panel) triggers for both data and embedding results. The close agreement observed across a wide range of p_T values demonstrates the reliability of the embedding process in describing the data. A comprehensive set of comparison plots for various event and track-level quantities can be found in Appendix A.2.

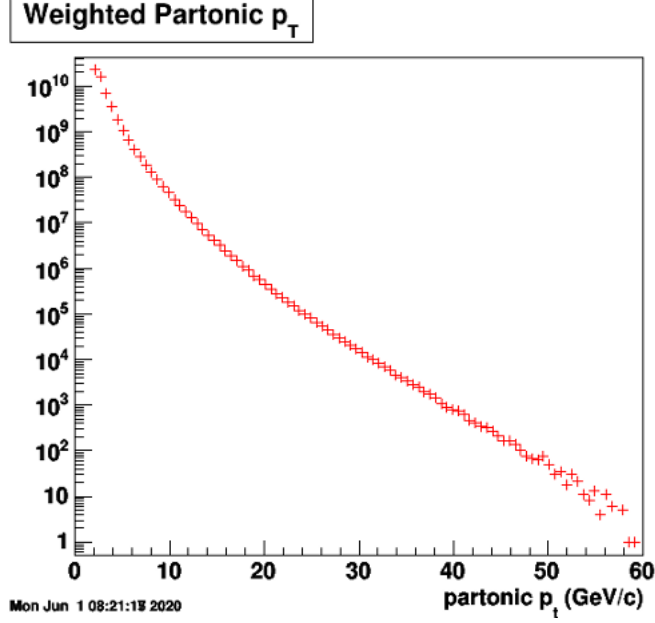


Figure 4.5: Weighted partonic p_T distribution.

While combining distributions in the embedding, the JP1 and JP2 triggers are merged using the prescale values obtained from the data. For this combination, a prescale value of 5.887 (from the database) is applied to the JP1 trigger while combining with the JP2 trigger, with a prescale value of one.

4.3.3 Events and Track Selection Cuts

This analysis considers only high-quality tracks originating from event vertices within 60 cm along the beam direction from the nominal TPC center. Each event must be triggered by JP and should contain at least two tracks meeting quality criteria. A track is deemed valid if it possesses a transverse momentum p_T greater than 1.5 GeV/ c and a distance of closest approach (dca) to the event vertex of less than 1 cm.

Charged pions are identified based on their ionization energy loss, denoted as dE/dx . The selection process involves imposing a cut on the standard deviations of the measured dE/dx relative to the anticipated pion energy loss. Specifically, a

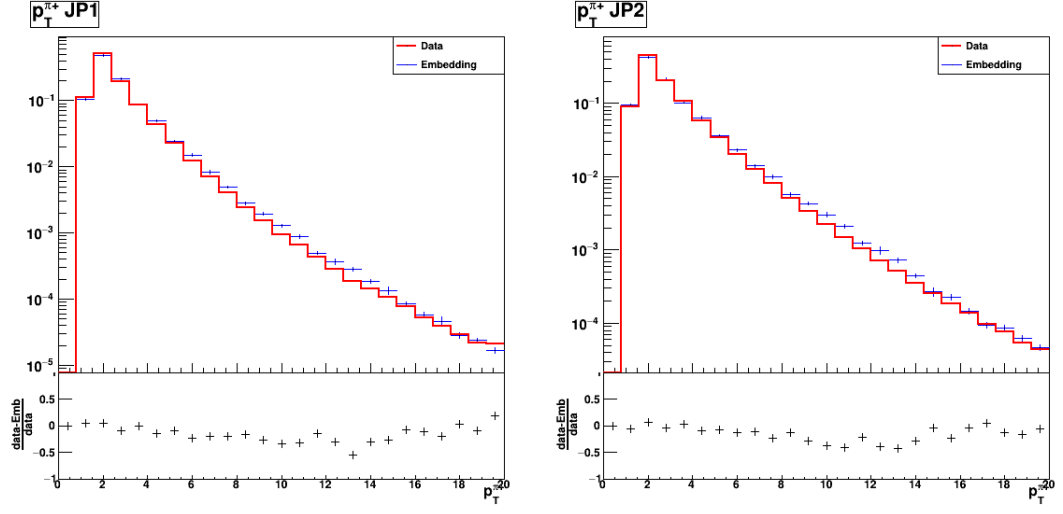


Figure 4.6: $p_T^{\pi^+}$ for JP1 and JP2.

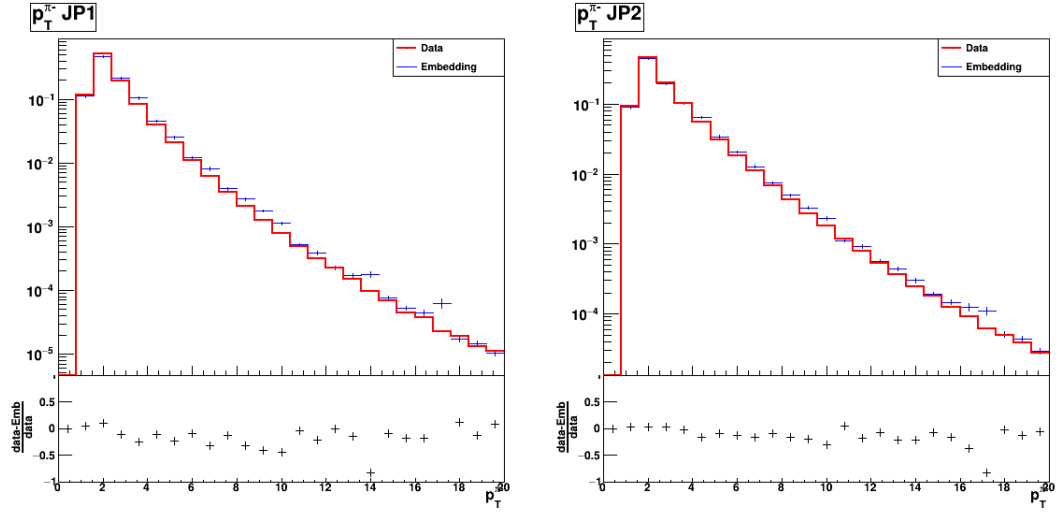


Figure 4.7: $p_T^{\pi^-}$ for JP1 and JP2.

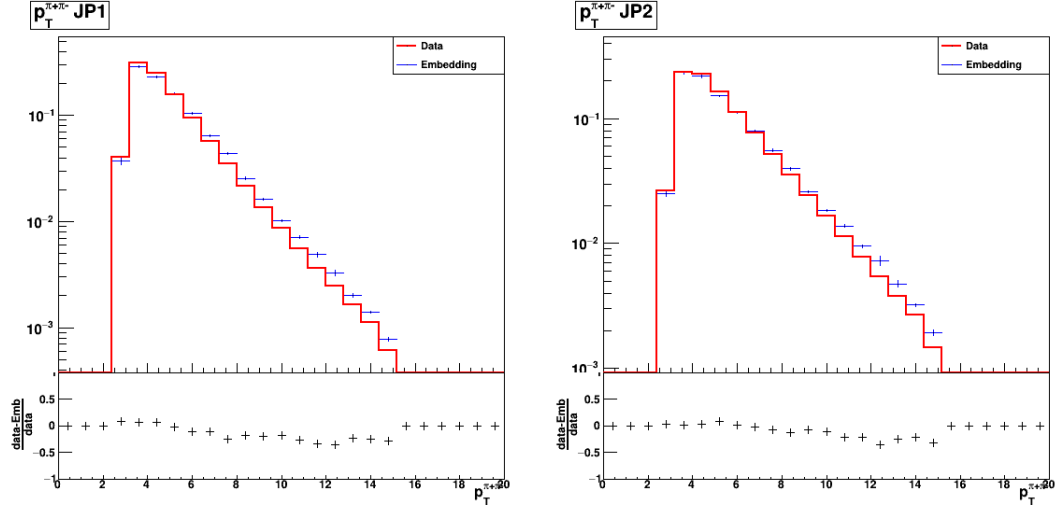


Figure 4.8: $p_T^{\pi^+\pi^-}$ for JP1 and JP2.

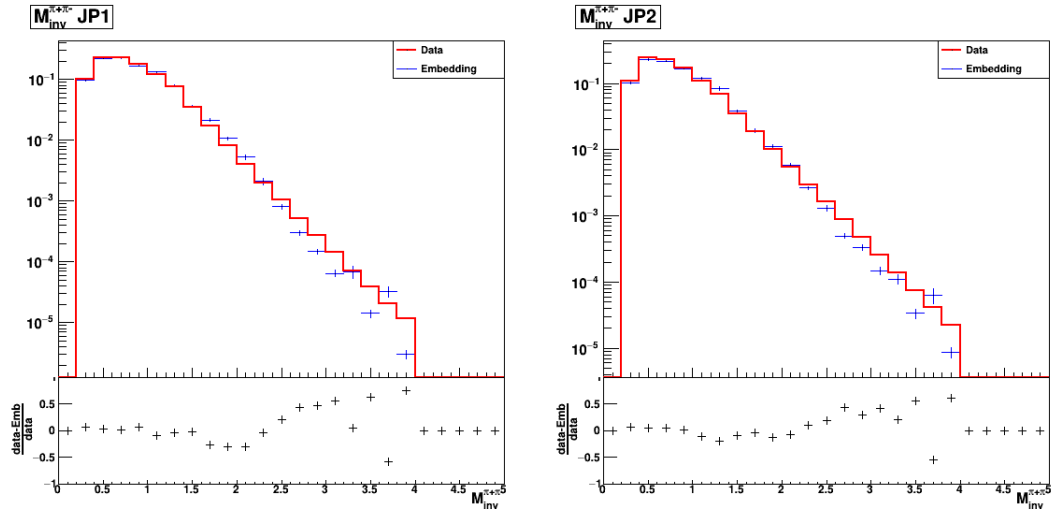


Figure 4.9: $M_{inv}^{\pi^+\pi^-}$ for JP1 and JP2.

condition of $-1 < n\sigma_\pi < 2$ is applied, where $n\sigma_\pi$ is defined as:

$$n\sigma_\pi = \frac{1}{\sigma_{\text{exp}}} \ln \left(\frac{dE/dx_{\text{obs}}}{dE/dx_{\pi, \text{calc}}} \right), \quad (4.11)$$

In this equation, dE/dx_{obs} represents the observed dE/dx value for the particle track, $dE/dx_{\pi, \text{calc}}$ is the expected mean dE/dx for pions at the given momentum, and σ_{exp} stands for the dE/dx resolution of the TPC.

Final Selection Cuts (Analysis)	
Z-Vertex	$< 60 \text{ cm}$
Vertex Ranking	$> 1e6$
Triggers	JP1, JP2
Tracks	Primary
track dca	$< 1 \text{ cm}$
track η	$ \eta < 1$
track p_T	$> 1.5 \text{ GeV}/c$
track nHitsFit	> 15
Pion Selection	$-1 < n\sigma_\pi < 2$
Fit ratio	> 0.51
cone ($\pi^+\pi^-$)	< 0.7
$M_{inv}^{\pi^+\pi^-}$	$0.2 < M_{inv} < 4 \text{ (GeV}/c^2\text{)}$
$p_T^{\pi^+\pi^-}$	$2.50 < p_T < 15 \text{ (GeV}/c\text{)}$
$\eta^{\pi^+\pi^-}$	$-1 < \eta^{\pi^+\pi^-} < 1$

Table 4.1: Final selection cuts for the asymmetry analysis.

Terminology definitions:

- Fit ratio = nHitsFit/nHitsFitPossible.

- $\eta = -\ln \tan \left(\frac{\theta}{2} \right)$, where θ is the angle between the particle momentum and the positive direction of the beam axis.
- cone: Opening angle between π^+ and π^- in $\eta - \phi$ space

$$\text{cone} = \sqrt{(\eta^{\pi^+} - \eta^{\pi^-})^2 + (\phi^{\pi^+} - \phi^{\pi^-})^2}$$
- $M_{inv}^{\pi^+\pi^-}$: Invariant mass of $\pi^+\pi^-$ pair

$$M_{inv}^{\pi^+\pi^-} = \sqrt{2M_\pi^2 + 2(E^{\pi^+}E^{\pi^-} - \vec{p}^{\pi^+} \cdot \vec{p}^{\pi^-})}$$
- $p_T^{\pi^+\pi^-}$: Transverse momentum of $\pi^+\pi^-$ pair relative to the beam direction

$$p_T^{\pi^+\pi^-} = \sqrt{p_x^{\pi^+\pi^-2} + p_y^{\pi^+\pi^-2}}$$
- $\eta^{\pi^+\pi^-}$: $\pi^+\pi^-$ pseudorapidity, $\eta^{\pi^+\pi^-} = \arcsin \left(\frac{p_z^{\pi^+\pi^-}}{p_T^{\pi^+\pi^-}} \right)$

4.3.4 $\pi^+\pi^-$ Pair Reconstruction and Binning

The $\pi^+\pi^-$ pairs are formed by selecting pion tracks with opposite charges. The corresponding azimuthal angles, ϕ_S and ϕ_R , and kinematic characteristics are established using the geometry outlined in Figure 2.6. All possible combinations of opposite pairs are formed for each event, excluding repetitions (e.g., $\pi^+\pi^-$ and $\pi^-\pi^+$ pairs are considered the same and are avoided). The π^+ and π^- tracks must be close in $\eta - \phi$ space, adhering to the criterion $\text{cone} < 0.7$. This cone cut ensures that the $\pi^+\pi^-$ pair remains within a jet. Jet reconstruction is not required in this channel; the cone cut guarantees that the pair falls within a jet.

The kinematic variables for the pair are formulated using the expressions listed in Section 4.3.3. Key quantities at the pair level include the invariant mass of the $\pi^+\pi^-$ pair, denoted as $M_{inv}^{\pi^+\pi^-}$, the transverse momentum relative to the beam direction, denoted as $p_T^{\pi^+\pi^-}$, and the pseudorapidity, denoted as $\eta^{\pi^+\pi^-}$. The definition of “forward” and “backward” in $\eta^{\pi^+\pi^-}$ is relative to the direction of the polarized beam

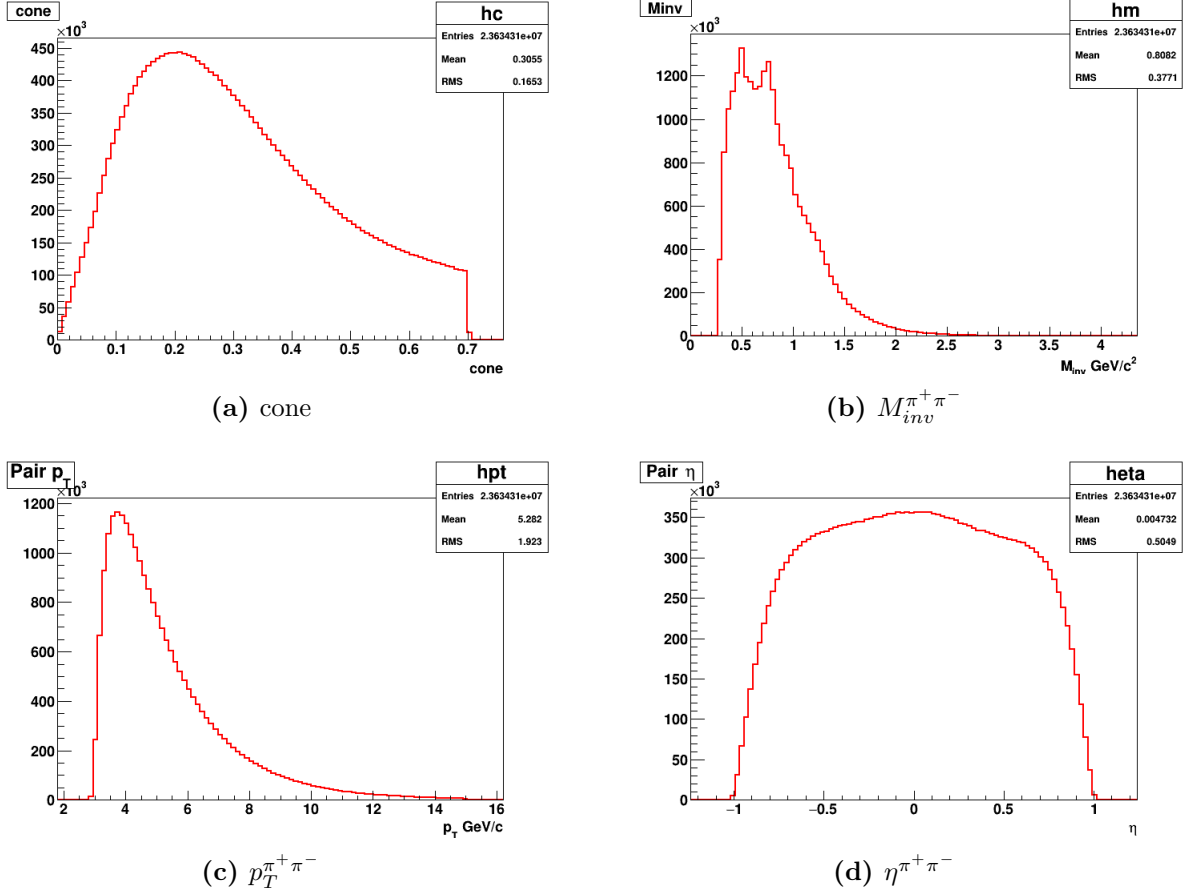
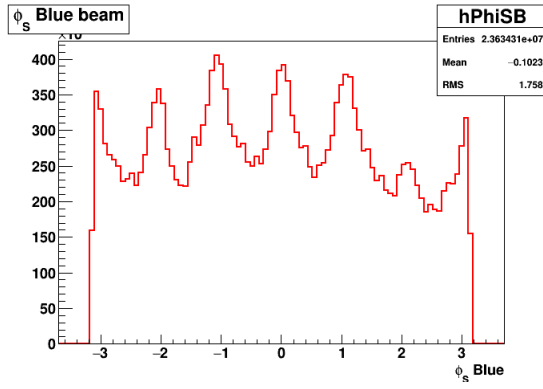


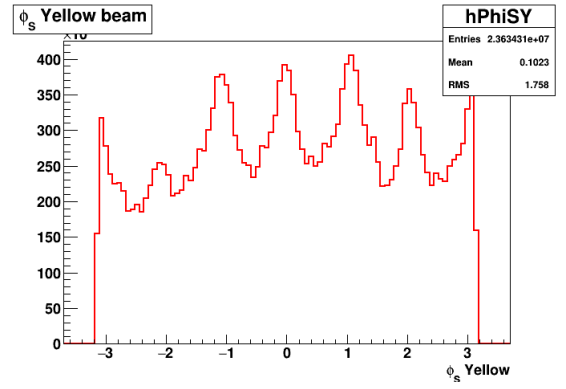
Figure 4.10: $\pi^+\pi^-$ Kinematic distributions.

under consideration. At STAR, the $+z$ region corresponds to $+\eta$ and the $-z$ region corresponds to $-\eta$. So, for the Blue beam, the “forward” $\eta^{\pi^+\pi^-}$ corresponds to the $+z$ side; however, the same region is considered “backward” for the Yellow beam. These quantities are crucial for the analysis of $\pi^+\pi^-$ asymmetry. Reconstructed distributions for pair kinematic properties are illustrated in Figures 4.10 through 4.13.

The reconstruction of azimuthal angles adheres to the STAR coordinate system. In this system, the blue beam direction aligns with the positive z axis, while the yellow beam corresponds to the negative z axis. As a default configuration for pair reconstruction, the polarization direction for both beams is fixed along the positive y axis. The determination of the positive x axis is achieved through the application of

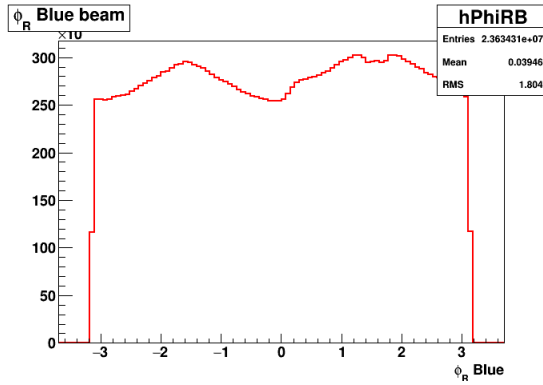


(a) ϕ_S Blue

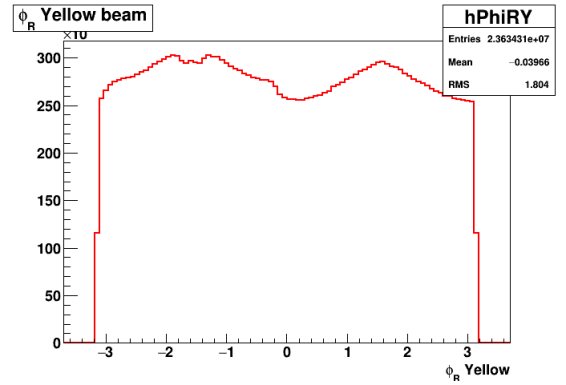


(b) ϕ_S Yellow

Figure 4.11: ϕ_S distributions for blue and yellow beam.



(a) ϕ_R Blue



(b) ϕ_R Yellow

Figure 4.12: ϕ_R distributions for blue and yellow beam.

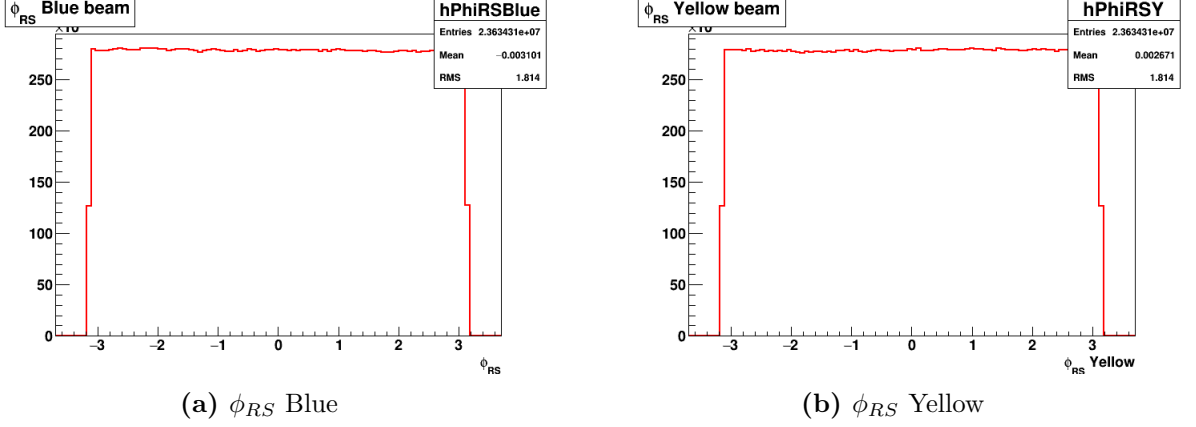


Figure 4.13: $\phi_{RS}(= (\phi_S - \phi_R))$ distributions for blue and yellow beam.

the right-hand rule. Notably, a disparity exists between the blue and yellow beams in the η direction. Specifically, the blue beam spans from $-\eta$ to $+\eta$, while the yellow beam operates in the opposite direction. To account for this distinction, a sign inversion is applied to the $\eta^{\pi^+\pi^-}$ values for $\pi^+\pi^-$ events from the yellow beam. Consequently, $\eta^{\pi^+\pi^-}$ ($-\eta^{\pi^+\pi^-}$) is modified to $-\eta^{\pi^+\pi^-}(\eta^{\pi^+\pi^-})$, enabling the direct combination of results from both beams.

The distribution of the azimuthal angle ϕ_{RS} ($\phi_{RS} = \phi_S - \phi_R$) is illustrated in Figure 4.13, with its construction based on Equations 2.7 and 2.8. In these equations, ϕ_S represents the angle between the polarization vector $\vec{s}_a$ and the scattering plane. The scattering plane is formed by the beam momentum vector $\vec{p}_{beam}$ and the combined vector of the $\pi^+\pi^-$ momenta, $\vec{p}_h = \vec{p}_{h,1} + \vec{p}_{h,2}$. The angle ϕ_R is defined as the angle between the scattering plane and the $\pi^+\pi^-$ plane, which is formed by the momenta of the two pions, $\vec{p}_{h,1}$ and $\vec{p}_{h,2}$. The relative momentum vector of the $\pi^+\pi^-$ system is formulated as $\vec{R} = \frac{1}{2}(\vec{p}_{h,1} - \vec{p}_{h,2})$. Importantly, the designation of $\vec{p}_{h,1}$ for π^+ and $\vec{p}_{h,2}$ for π^- is pivotal, as it ensures the preservation of the direction of $\vec{R}$. Without this specific charge ordering, the direction of $\vec{R}$ becomes randomized, leading to a diluted asymmetry (see Appendix A.5.7).

The $\pi^+\pi^-$ yields are categorized based on the beam polarization direction ($\uparrow / \downarrow$). In the STAR 2015 data context, both beams are polarized transversely. This advantageous situation permits separate treatment of each beam to measure asymmetry. Notably, the spin patterns during the beam fill adhere to a fixed order in beam bunches and are flipped between fills systematically, such that when the polarization states are integrated, the resultant net polarization is close to zero. This unique feature of the spin patterns of polarized beams at STAR enables the execution of physics analyses that involve unpolarized beams, as exemplified in the discussion of the unpolarized cross-section measurement in Chapter 5.

For the asymmetry measurement, the blue and yellow beams are independently analyzed. When analyzing the yellow beam, the blue beam is treated as unpolarized, integrating over all possible spin states. The 8-bit spin pattern, which corresponds to the decimal values listed below, is employed to designate a specific beam polarization state (refer to Appendix A.3):

- Blue $\uparrow$, Yellow $\uparrow \equiv 51$ (8-bit = 0011 0011)
- Blue $\downarrow$, Yellow $\downarrow \equiv 85$ (8-bit = 0101 0101)
- Blue $\uparrow$, Yellow $\downarrow \equiv 53$ (8-bit = 0011 0101)
- Blue $\downarrow$, Yellow $\uparrow \equiv 83$ (8-bit = 0101 0011)

Consequently, it follows that the $\pi^+\pi^-$ yields from the blue beam in the $\uparrow$ state are indicated by spin states 51 and 53. On the other hand, yields from the blue beam in the $\downarrow$ state are reflected by spin states 83 and 85. Similarly, the $\uparrow$ yields for the yellow beam correspond to spin states 51 and 83, while the $\downarrow$ yields are associated with spin states 53 and 85. These numerical values effectively label the yields based on specific spin states at the analysis level.

In this analysis, the measurement of $A_{UT}^{\sin(\phi_{RS})}$ is carried out with respect to three crucial kinematic variables: $M_{inv}^{\pi^+\pi^-}$, $p_T^{\pi^+\pi^-}$, and $\eta^{\pi^+\pi^-}$, to prevent potential signal dilution [138]. Notably, $\eta^{\pi^+\pi^-}$ serves as a proxy for the partonic variable x , where, within the STAR coordinate system, $\eta^{\pi^+\pi^-} > 0$ (< 0) corresponds to the high (low)- x region. Furthermore, the extraction of asymmetry through multi-dimensional binning in both $M_{inv}^{\pi^+\pi^-}$ and $p_T^{\pi^+\pi^-}$ within two $\eta^{\pi^+\pi^-}$ regions provides deeper insight into the kinematic dependence of the asymmetry, allowing for more refined kinematic granularity in the transversity extraction process.

4.3.5 Polarization Correction

The beam polarization is recorded at the beginning of the fill and followed by subsequent measurements during the beam fill. The beam polarization decays linearly because of the depolarization, as discussed in section 3.1.1. The slope of the linear polarization decay ($\frac{dP}{dt}$) is provided for each beam fill along with the initial polarization (P_0) at the beginning of the fill (t_0). At any time, t , the polarization value is given by,

$$P(t) = P_0 + \frac{dP}{dt}(t - t_0) \quad (4.12)$$

The actual polarization value is calculated for every event using the equation 4.12 at the event time t . The corrected polarization values for all events are filled in a histogram for every run for the blue and yellow beams. Figure 4.14 shows the event-by-event corrected polarization distribution for a full data sample for blue and yellow beams, and corrected polarization as a function of the run index is shown in figure 4.15. From figure 4.14, the corrected polarization are found to be,

- $P_B = 57.11\% \pm 3.75\%$ for the blue beam.

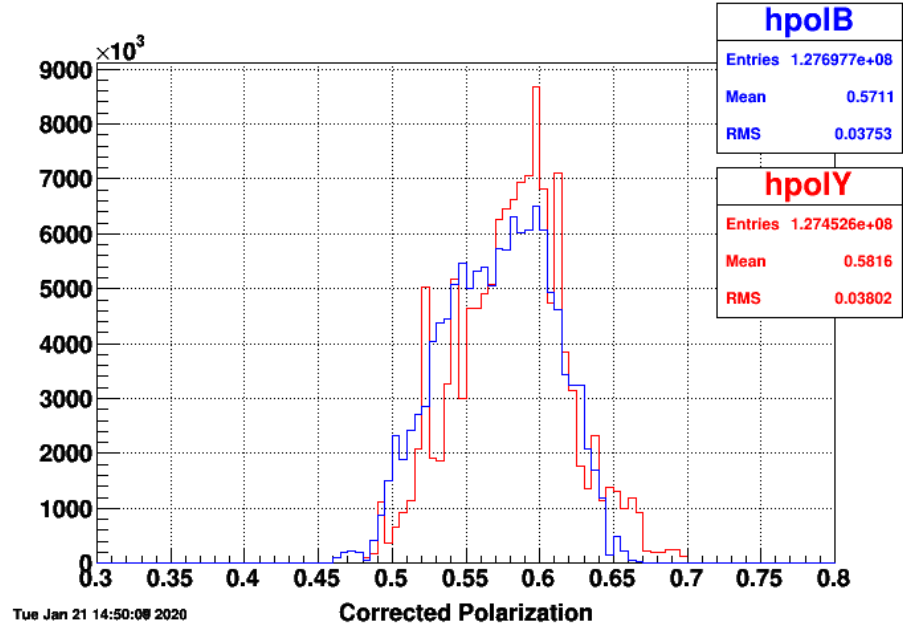


Figure 4.14: Event-by-event polarization corrected values for blue and yellow beams.

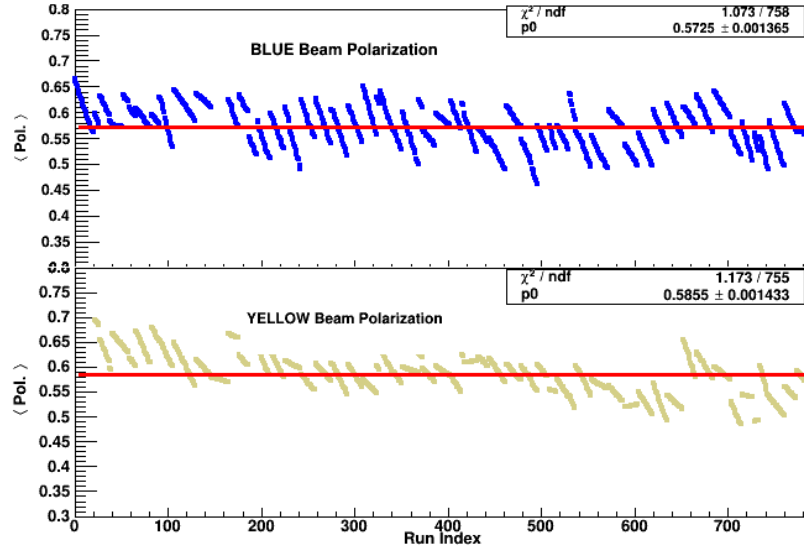


Figure 4.15: Event-by-event polarization corrected values vs. run index for blue and yellow beams.

- $P_Y = 58.16\% \pm 3.80\%$ for the yellow beam.

The P_B and P_Y values and associated errors are used in the asymmetry calculation. The 3% systematic uncertainty from the beam polarization measurement is not included in the asymmetry extraction, which will be mentioned separately with the result.

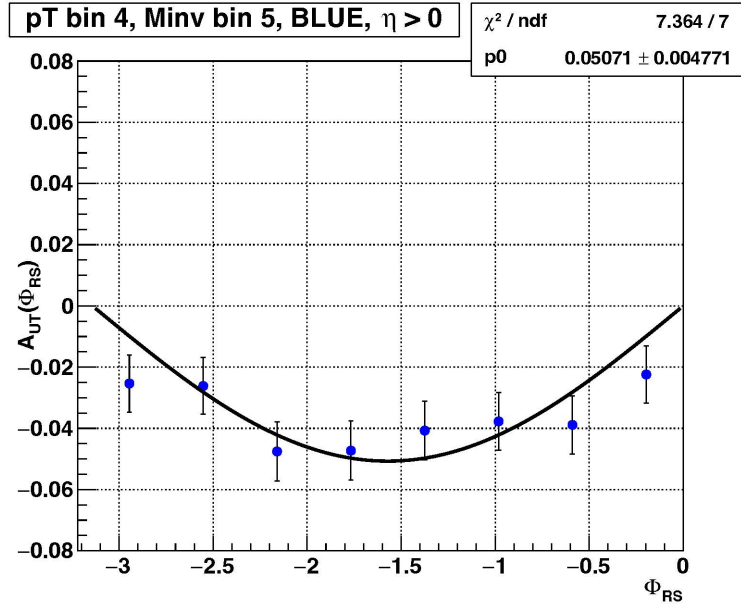
4.3.6 $A_{UT}^{\sin(\phi_{RS})}$ Extraction in $p^\uparrow p$

Since the $\pi^+\pi^-$ azimuthal correlation asymmetry arises due to the convolution of $h_1(x)$ and $H_1^{\leq, \pi^+\pi^-}(z, M^2)$, it inherits the dependence on the partonic variable x from $h_1(x)$ and the final state variables z and M , from $H_1^{\leq, \pi^+\pi^-}(z, M^2)$. Therefore, $A_{UT}^{\sin(\phi_{RS})}$ must be extracted in $M_{inv}^{\pi^+\pi^-}$, $p_T^{\pi^+\pi^-}$, x , and z bins with the binning scheme discussed in section 4.3.4. The x and z are the partonic variables associated with the final state $\pi^+\pi^-$, which are not accessible in the data and should be estimated from the embedding sample. Mathematically, these partonic variables can be expressed as,

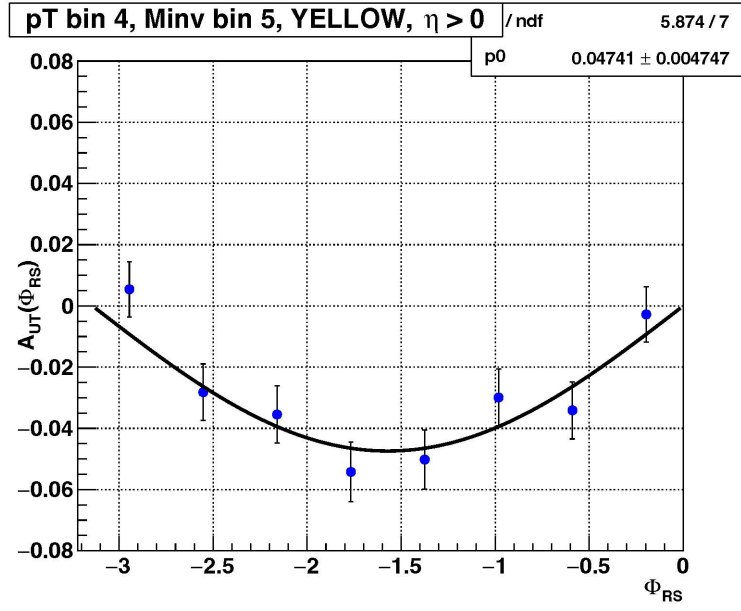
$$x = \frac{p^{quark}}{p^{proton}}, \quad z = \frac{E^{\pi^+\pi^-}}{E^{quark}} \quad (4.13)$$

where p and E refer to the momentum and energy, respectively. These variables are estimated by finding an underlying parton associated with the final state $\pi^+\pi^-$ imposing a matching condition requiring that the $\pi^+\pi^-$ should be closest to the initial state parton set by the condition $\Delta R = \sqrt{(\eta^{\pi^+\pi^-} - \eta^{parton})^2 + (\phi^{\pi^+\pi^-} - \phi^{parton})^2} < 0.5$.

The cross ratio formula, given by equation 4.10, is used for the $A_{UT}^{\sin(\phi_{RS})}$ extraction. Within each asymmetry bin, the right side of the equation 4.10 is evaluated for the $8-\phi_{RS}$ bins in the range $[-\pi, 0]$. This is possible by dividing the whole ϕ_{RS} range into 16 bins of equal width and counting $\pi^+\pi^-$ yields in each bin when the



(a)



(b)

S

Figure 4.16: “sine” modulation and fits for (a) Blue (b) Yellow beams in $\eta > 0$ in the $p_T^{\pi^+\pi^-}$ bin $([6.4 - 15] \text{ GeV}/c)$ and $M_{inv}^{\pi^+\pi^-}$ bin $([0.77, 0.88] \text{ GeV}/c^2)$, where the asymmetry signal is largest.

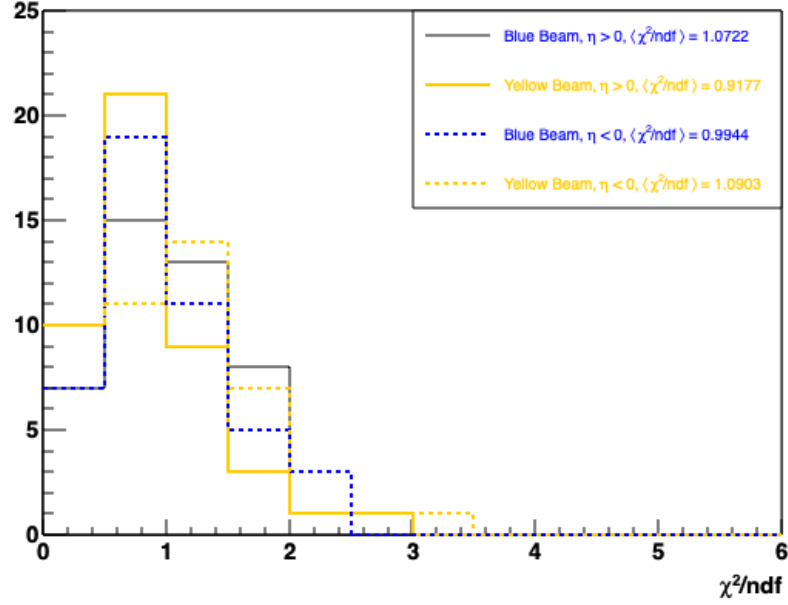


Figure 4.17: χ^2/ndf of the “sine” fit in all kinematic bins. The quality of the “sine” fit is monitored by the χ^2/ndf ratio.

beam is polarized up or down. Then, these eight data points are fitted with the “sine” function, allowing the amplitude to float freely. The amplitude of the fit and corresponding fit error is the $A_{UT}^{\sin(\phi_{RS})}$ signal and associated statistical uncertainty in the particular kinematic bin. An example of the fit is shown in figure 4.16 in the kinematic bin $(p_T^{\pi^+\pi^-} \equiv [6.4, 15) \text{ GeV}/c, M_{inv}^{\pi^+\pi^-} \equiv [0.8823, 1.031) \text{ GeV}/c^2$, and $\eta^{\pi^+\pi^-} > 0$), where the asymmetry signal is largest. Figure 4.16a is for the blue, and figure (4.16b) is for the yellow beam. Displayed fits are for the kinematic bin, where the asymmetry signal is the largest. The central values in the fit are the numerical values from the equation 4.10 evaluated in $8\text{-}\phi_{RS}$ bins in $[-\pi, 0]$, and the error bars are the statistical uncertainty calculated by the error propagation method. The fit provides a very good description, as indicated by the χ^2/ndf of the fit. Figure 4.17 shows the χ^2/ndf distribution from the “sine” fit for blue and yellow beam asymmetry in forward and backward pseudorapidity regions associated with the results shown in

figure 4.30. The same level of fit quality is observed in all kinematic bins.

4.3.7 Systematic Uncertainties

4.3.7.1 Pion Contamination Fraction

This measurement uses $\pi^+\pi^-$ pairs for asymmetry measurement; however, any of the hadron pairs, such as $\pi^+\pi^-$, K^+K^- , or $p\bar{p}$, can be used to achieve the same physics motivation, probing different underlying quark sensitivity. Different QCD mechanisms may involve different hadron species, resulting in different asymmetry signals. Thus, knowing the hadron purity in the kinematic bin where the asymmetry is measured is important. The STAR TPC measures the charged particle ionization loss, dE/dx , with which we can identify charged particle species based on the track momentum and charge sign with the direction of curvature in the uniform magnetic field. The TPC, however, can identify particles of momentum $p < 1$ GeV/ c with better precision, and the TPC capability degrades for higher momentum particles. The STAR timing detector, TOF, complements the TPC PID, providing better resolution up to $p < 1.6$ GeV/ c ; however, this measurement uses only the TPC information for the particle fraction calculation at the preliminary level. In appendix A.1, we discuss the results of the after-preliminary PID analysis, which includes TOF. The findings show a significant improvement in the PID systematic uncertainty.

Figure 4.18 displays dE/dx as a function of particle track momentum in a logarithmic scale, measured by TPC, alongside Bichsel’s prediction [139]. Each band corresponds to a distinct charged particle species. In the lower momentum range of $p < 1$ GeV/ c , clear separation of dE/dx for pions, kaons, protons, and electrons is evident, allowing for precise particle identification at lower momenta.

In the $1 \leq p < 1.3$ GeV/ c region, the dE/dx bands cross, making particle iden-

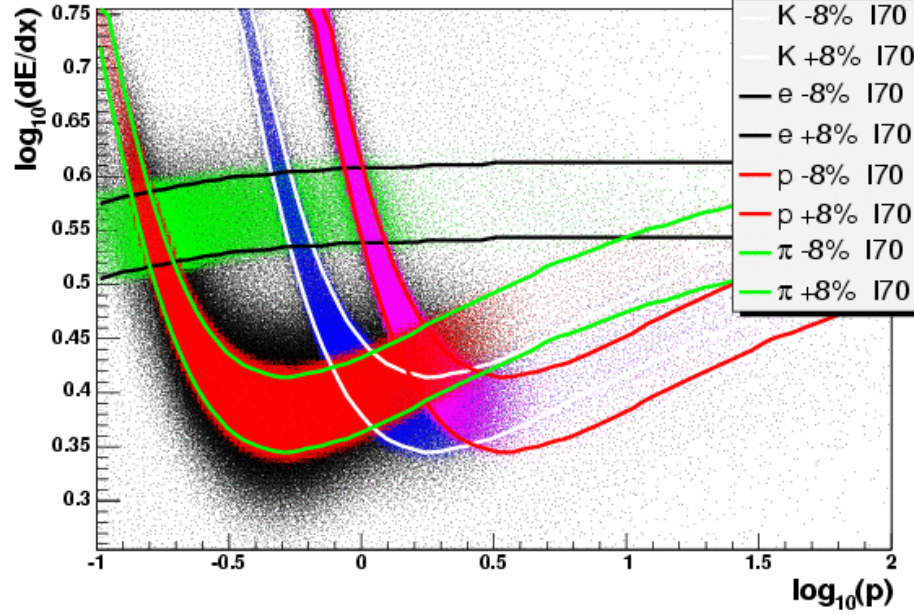


Figure 4.18: Distribution of $\log_{10}(dE/dx)$ as a function of $\log_{10}(p)$ for pion, kaon, electron, proton, and their antiparticles. The units of dE/dx and p are KeV/cm and GeV/c, respectively. The color band represents the $\pm 1\sigma$ band of the dE/dx resolution. 170 means Bichsel's prediction for 30% truncated dE/dx mean [139].

tification challenging. Beyond $p > 1.3$ GeV/c, the relativistic rise of dE/dx and decreased yields of electrons and kaons lead to a 15% difference in dE/dx between pions and kaons, which is even more pronounced for protons. This enables pion identification from kaons and protons in the higher momentum range using TPC alone, while identifying kaons and protons becomes difficult due to overlapping dE/dx bands.

This analysis focuses on pion tracks with $p_T \geq 1.5$ GeV/c, effectively avoiding the dE/dx crossover region. The relativistic rise effect facilitates pion isolation from protons and kaons, with electrons distinctly separating from pions. A standard approach calculates the fraction of other particle species ($p's, K's, e's$) present within the pion region, using the $n\sigma_\pi$ distribution.

Figure 4.19 illustrates the $n\sigma_\pi$ distribution for positive charge tracks in the lowest $p_T^{\pi^+\pi^-}$ bin within the $\eta^{\pi^+\pi^-} > 0$ region. The distribution features a distinct pion peak centered around zero. Protons and kaons manifest on the left side (p+k region), while

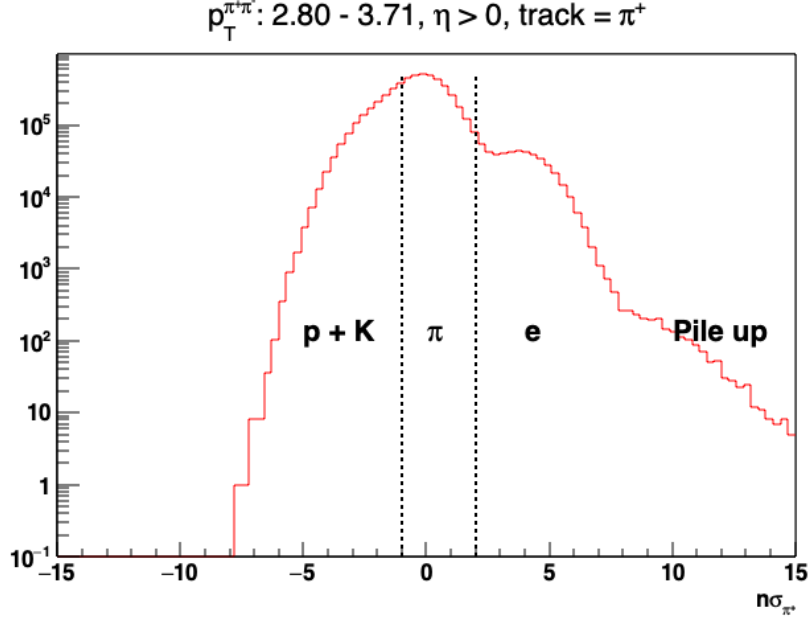


Figure 4.19: $n\sigma_{\pi^+}$ distribution in the lowest $p_T^{\pi^+\pi^-}$ bin (2.8 - 3.71 GeV/ c) in $\eta^{\pi^+\pi^-} > 0$.

electrons appear to the right of the pion peak. The selected pion region is demarcated by the two vertical lines, defined by the $n\sigma_\pi$ cut $-1 < n\sigma_\pi < 2$. Given the proximity of the p+k region to the pion region, the primary background sources within the pion signal region originate from protons and kaons. Electrons, exhibiting a distinct peak further away from the pion, are readily identifiable and minimally contribute to the pion signal region. Tracks labeled "Pile up" on the far right are disregarded, as they do not contribute to the pion signal region.

A multi-gaussian fit method is employed to quantify contributions from other particle species in the pion signal region. The initial step involves determining the peak position for each particle species in the $n\sigma_\pi$ distribution to construct a comprehensive multi-gaussian fit for the entire distribution. This is achieved by generating a two-dimensional (2D) distribution, with $n\sigma_\pi$ along the x -axis and $n\sigma_{p,K,e}$ along the y -axis. Figure 4.20 displays sample 2D plots of $n\sigma_p$ vs $n\sigma_{\pi^+}$ across five $p_T^{\pi^+\pi^-}$ bins in the $\eta^{\pi^+\pi^-} > 0$ region. Profiling the 2D histogram through a third-order polynomial fit

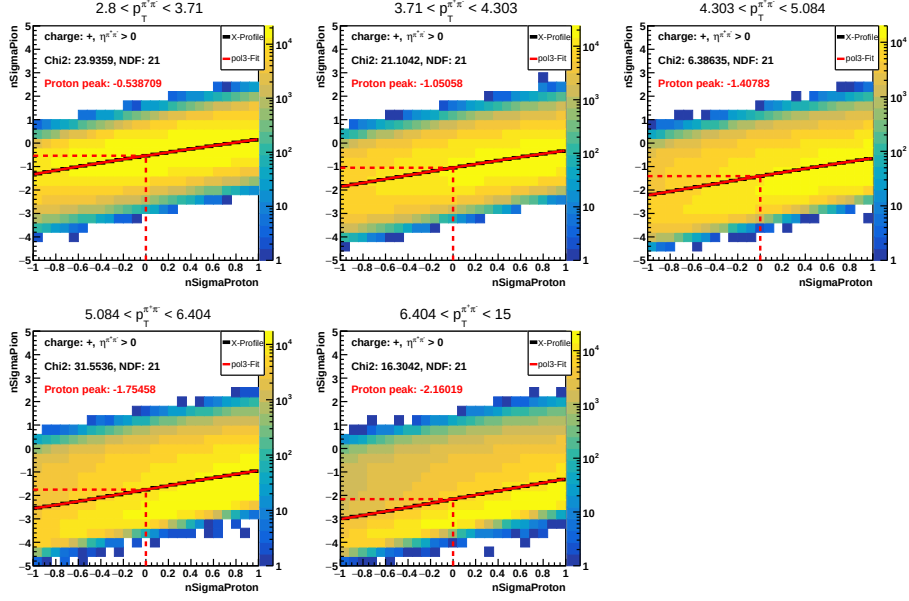


Figure 4.20: Track $n\sigma_{\pi^+}$ vs $n\sigma_p$ in five invariant mass bins.

(pol3) provides the mean y -values for each x -bin, allowing the identification of peak positions along the y -axis corresponding to the pion mean position at zero along the x -axis. Figure 4.20 showcases the proton peak positions across various $p_T^{\pi^+\pi^-}$ bins. A similar approach is used to identify peak positions for kaons and electrons relative to the pion peak at zero.

Next, a multi-gaussian fit is constructed, comprising five Gaussian fits – one for each particle species (protons, electrons, kaons) and one for “Pile up”. The mean of the Gaussian fits for protons, electrons, and kaons is set to their respective peak positions, with varying width. The mean of the Gaussian fit for pions is fixed at zero. The “Pile up” Gaussian fit is determined by the mean and width that best characterize the overall distribution. Figure 4.21 visualizes the resulting multi-gaussian fit for positive charge tracks across five $p_T^{\pi^+\pi^-}$ bins in the $\eta^{\pi^+\pi^-} > 0$ region. Notably, in this kinematic region, protons and kaons are closely aligned and infiltrate the pion region from the left, while electrons contribute from the right.

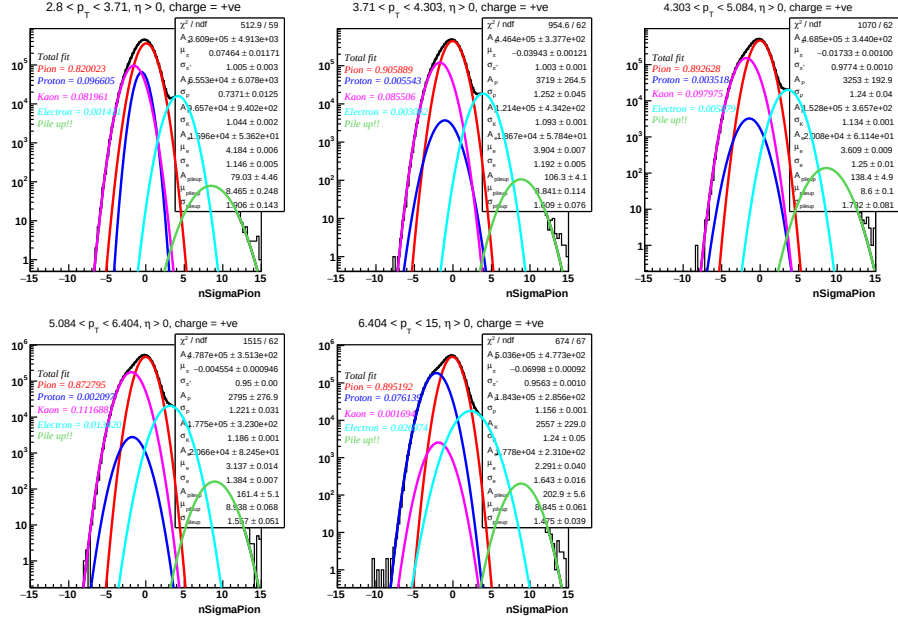


Figure 4.21: Track $n\sigma_{\pi^+}$ multi-gaussian fit in five invariant mass bins.

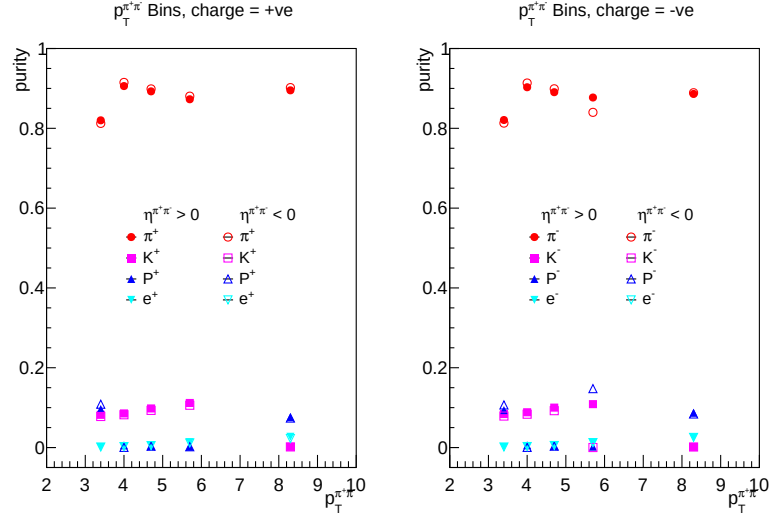


Figure 4.22: Particle fractions at the track level in $p_T^{\pi^+\pi^-}$ bins

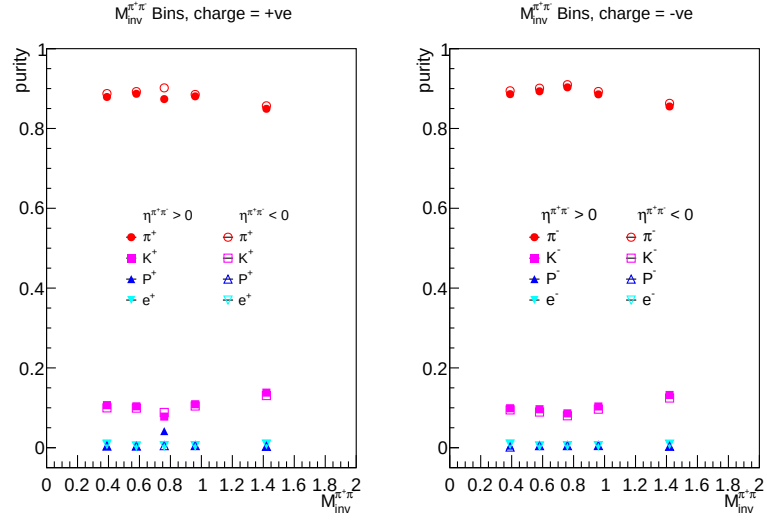


Figure 4.23: Particle fractions at the track level in $M_{inv}^{\pi^+\pi^-}$ bins.

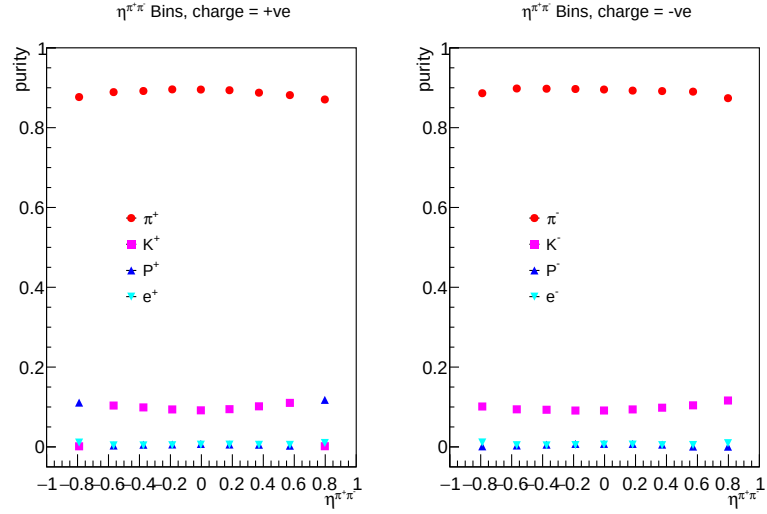
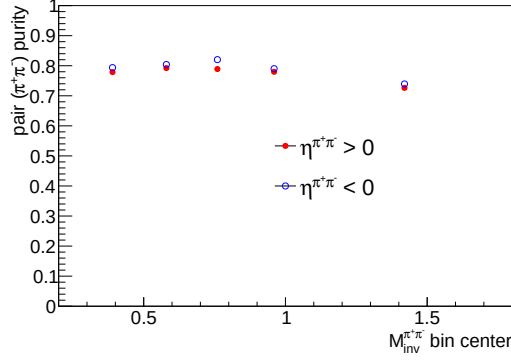
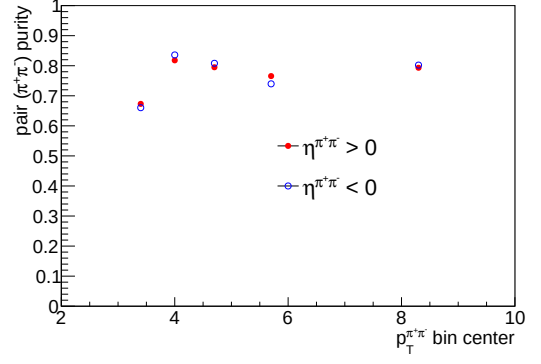


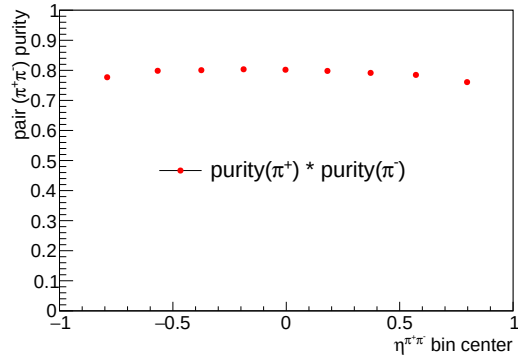
Figure 4.24: Particle fractions at the track level in $\eta^{\pi^+\pi^-}$ bins.



(a) $\pi^+\pi^-$ fraction in $M_{inv}^{\pi^+\pi^-}$ bins.



(b) $\pi^+\pi^-$ fraction in $p_T^{\pi^+\pi^-}$ bins.



(c) $\pi^+\pi^-$ fraction in $\eta^{\pi^+\pi^-}$ bins.

Figure 4.25: $\pi^+\pi^-$ purity fractions in the asymmetry bins.

Subsequently, the fractions of p , K , and e that contribute to the pion signal region are calculated by evaluating the ratio of the Gaussian tail area for each particle species to the total area under the combined fit within the pion signal region ($-1 < n\sigma_\pi < 2$). Mathematically:

$$f_{\pi,p,K,e} = \frac{I_{\pi,p,K,e}(-1, 2)}{I_{\text{total}}(-1, 2)}, \quad (4.14)$$

where $I(-1, 2)$ denotes the Gaussian integral over the range $-1 < n\sigma_\pi < 2$. This computation provides the pion purity at the track level. The track-level particle fractions are depicted in figures 4.22-4.24. Meanwhile, the pion purity at the pair level, representing the likelihood that both tracks are pions, is obtained by multiplying f_{π^+} and f_{π^-} . Figure 4.25 displays the pair-level pion purity fractions. The pair-level purity varies across $p_T^{\pi^+\pi^-}$ bins, with the lowest $p_T^{\pi^+\pi^-}$ bin nearing the crossover region in the dE/dx band, leading to a lower purity. Conversely, the higher $p_T^{\pi^+\pi^-}$ bins exhibit better pion purity due to the separation of the pion band from the kaon and proton bands. Specifically, pion purity is approximately 85% at higher $p_T^{\pi^+\pi^-}$ and around 75% in the lowest bin. Particle fractions are similarly computed for $M_{inv}^{\pi^+\pi^-}$ and $\eta^{\pi^+\pi^-}$ bins. However, due to momentum dilution, these bins do not exhibit clear particle fraction dependencies, yielding an average $\pi^+\pi^-$ purity of about 80%.

The pion impurity fraction ($1 - f_{\pi^+\pi^-}$), mainly driven by protons and kaons, contributes to a systematic uncertainty estimation. It is assumed that the kaon and proton asymmetry signals are either the same size as the pion signal or smaller. The systematic uncertainty magnitude is calculated as the product of pion impurity and either the asymmetry signal or the statistical uncertainty, based on whichever is larger:

$$\delta_{\text{sys,pid}} = (1 - f_{\pi^+\pi^-}) \times \text{Max}(A_{\text{UT}}, \delta_{\text{A}_{\text{UT}}}) \quad (4.15)$$

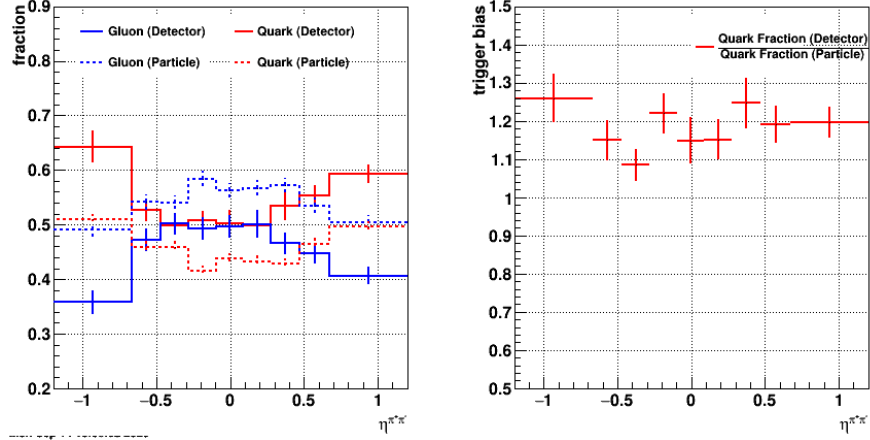


Figure 4.26: Trigger bias in $\eta^{\pi^+\pi^-}$ bins.

4.3.7.2 Trigger Bias

The event-triggering algorithm employed at STAR can introduce a bias at the parton level, favoring quark-initiated events over gluon-initiated events. Given that parton-level information remains inaccessible in the data, the extent of this bias can solely be approximated through simulations. This involves determining the fraction of quark events at the detector level compared to the fraction derived from the particle-level events – those devoid of event-triggering effects. The size of the trigger bias is ascertained by calculating the ratio between these fractions.

The bias fraction, f_{bias} , can be written as,

$$f_{bias} = \frac{f_q^{detector}}{f_q^{particle}} = \frac{(N_q/N_{q+g})^{detector}}{(N_q/N_{q+g})^{particle}}, \quad (4.16)$$

where q and g refer to quark and gluon, respectively.

As in the data, identical event and track-level cuts are applied at the detector and particle levels. The track η and p_T cuts are retained at the particle level. Assigning a parton to each $\pi^+\pi^-$ is a more complex task since jet reconstruction is not performed.

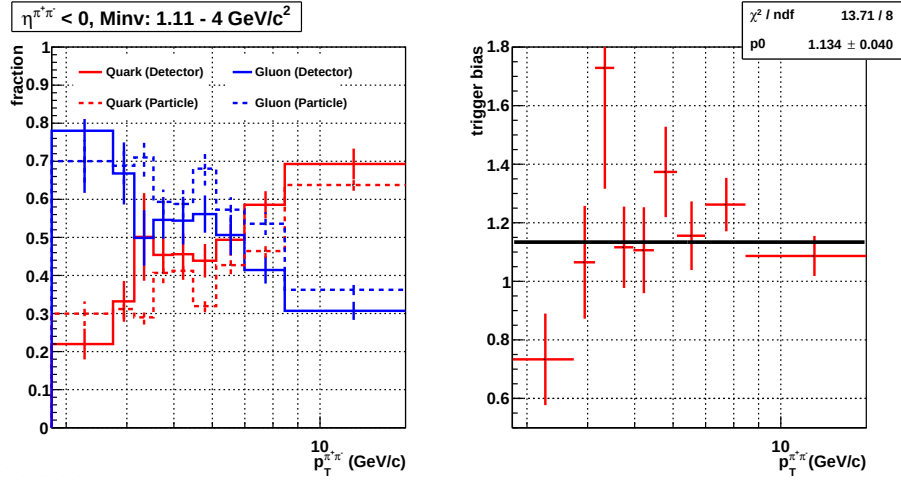


Figure 4.27: Trigger bias in the highest $p_T^{\pi^+\pi^-}$ bin.

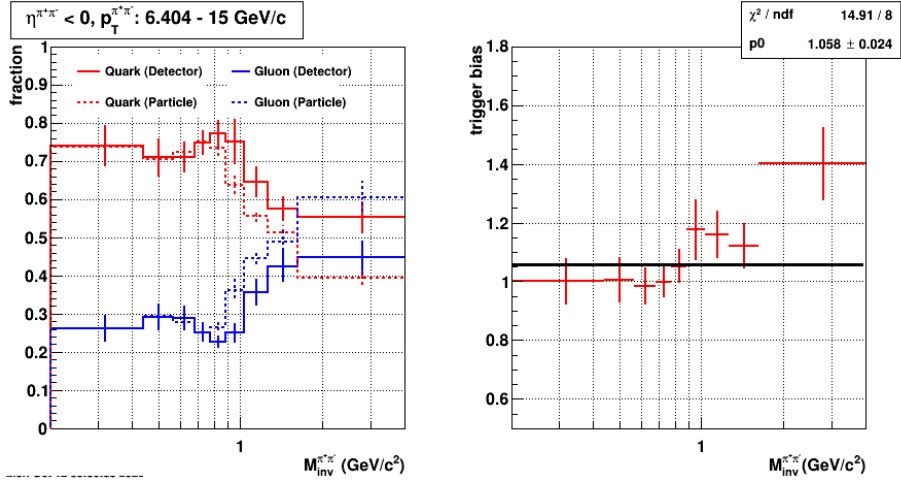


Figure 4.28: Trigger bias in the highest $M_{inv}^{\pi^+\pi^-}$ bin.

As an alternative, we replicate an approach of assigning a parton to each $\pi^+\pi^-$ as it would be done with jets. The assignment of a parton to each $\pi^+\pi^-$ involves calculating an angle between two outgoing partons in $\eta-\phi$ space. The parton closest to the $\pi^+\pi^-$ is designated as the source parton, subject to a constraint that the opening angle is less than 0.5 ($\Delta R = \sqrt{(\eta_{\text{parton}} - \eta_{\text{pair}})^2 + (\phi_{\text{parton}} - \phi_{\text{pair}})^2} < 0.5$). This guarantees that a $\pi^+\pi^-$ meets the criterion of being within the jet. Subsequently, histograms are generated for each parton flavor with $\pi^+\pi^-$ kinematic bins, following the same binning as for asymmetry. The bin content of a parton's histogram is incremented by one for every $\pi^+\pi^-$ event tagged with the associated parton. This process yields histograms for each parton flavor at the detector and particle levels. Dividing the histogram for a quark by the respective parton's histogram yields a quark fraction histogram. Ultimately, the actual trigger bias fraction is derived by dividing the quark fraction histograms at the detector and particle levels.

Figures 4.26-4.28 illustrate the quark fraction (in red) and gluon fraction (in blue) at both the detector and particle levels in the left panel. The right panel displays the bias fraction in $\eta^{\pi^+\pi^-}$, the highest $M_{inv}^{\pi^+\pi^-}$ bin as a function of $p_T^{\pi^+\pi^-}$ in $\eta^{\pi^+\pi^-} < 0$, and the highest $p_T^{\pi^+\pi^-}$ bin as a function of $M_{inv}^{\pi^+\pi^-}$ in $\eta^{\pi^+\pi^-} < 0$. Solid lines represent detector-level fractions, while dashed lines denote particle-level fractions. The bias fractions do not display distinct trends across all kinematic bins, which is similar in the $\eta^{\pi^+\pi^-} > 0$ region. Consequently, a constant fit is applied to the bias fraction, yielding a single value to estimate the trigger bias. This procedure is replicated to estimate the bias fraction for all asymmetry bins. Given a bias fraction of f_{bias} , the systematic uncertainty due to event triggering bias, $\delta_{\text{sys,bias}}$, is calculated as:

$$\delta_{\text{sys,bias}} = (1 - f_{\text{bias}}) \times \text{Max}(A_{\text{UT}}, \delta_{A_{\text{UT}}}) \quad (4.17)$$

4.3.7.3 Systematic Error Propagation

Systematic uncertainties are evaluated for each of the asymmetry bins. The overall systematic uncertainty, denoted as δ_{sys} , is obtained through the quadratic summation of the systematic errors stemming from trigger bias and pion contamination:

$$\delta_{\text{sys}} = \sqrt{\delta_{\text{sys,bias}}^2 + \delta_{\text{sys,pid}}^2} \quad (4.18)$$

The total systematic uncertainty is symmetrically assigned to the signal and is reported together with the statistical uncertainty, δ_{stat} , as $A_{UT}^{\sin(\phi_{RS})} \pm \delta_{\text{stat}} \pm \delta_{\text{sys}}$.

4.3.8 Results

4.3.8.1 $A_{UT}^{\sin(\phi_{RS})}$ vs $\eta^{\pi^+\pi^-}$

Figure 4.29 displays $A_{UT}^{\sin(\phi_{RS})}$ as a function of $\eta^{\pi^+\pi^-}$, integrated across $M_{inv}^{\pi^+\pi^-}$ and $p_T^{\pi^+\pi^-}$ (*upper panel*). The vertical lines denote the statistical uncertainty, while the box illustrates the total systematic uncertainty. The bottom panel presents the average fraction of proton momentum carried by a fragmenting quark, $\langle x \rangle$, and the average fraction of fragmenting quark energy carried by $\pi^+\pi^-$, $\langle z \rangle$, in the same $\eta^{\pi^+\pi^-}$ bins, obtained from simulation. Within the considered $\eta^{\pi^+\pi^-}$ bins, the value of x ranges from 0.1 to 0.22, moving from the backward to the forward region. On the other hand, z does not exhibit a clear dependence on $\eta^{\pi^+\pi^-}$, maintaining an average value of approximately 0.46.

A substantial $A_{UT}^{\sin(\phi_{RS})}$ signal is evident in the forward region, increasing from $\eta^{\pi^+\pi^-} \sim 0$ towards the forward region. This region corresponds to the probing of valence quarks with higher x values. The x -dependent behavior of transversity and the substantial asymmetry signal indicate a significant transversity signal in the for-

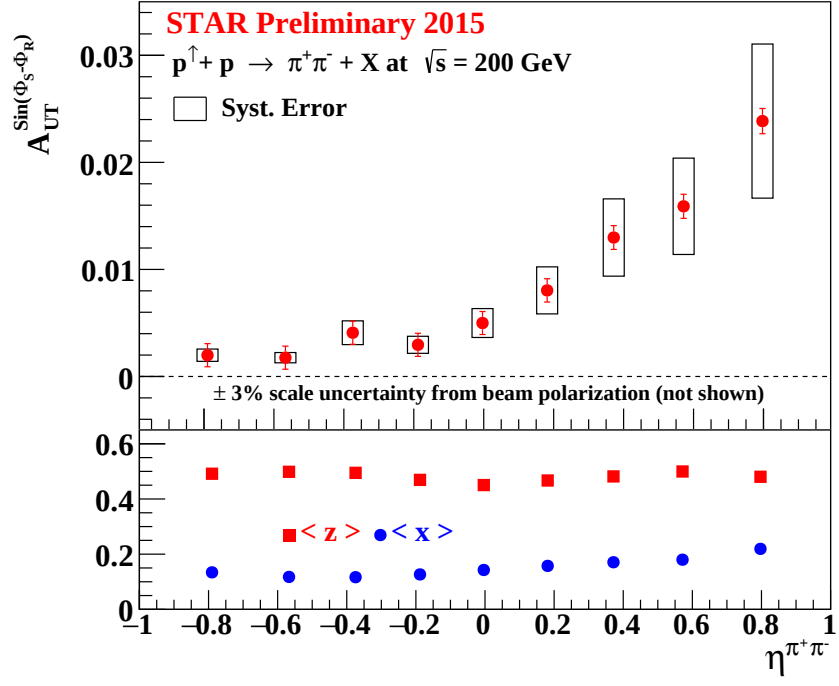


Figure 4.29: $A_{UT}^{\sin(\phi_{RS})}$ as a function of $\eta^{\pi^+\pi^-}$ integrated over $M_{inv}^{\pi^+\pi^-}$ and $p_T^{\pi^+\pi^-}$ (*top panel*). The quark $\langle z \rangle$ and $\langle x \rangle$, in the corresponding $\eta^{\pi^+\pi^-}$ bins, are shown in the *bottom panel*.

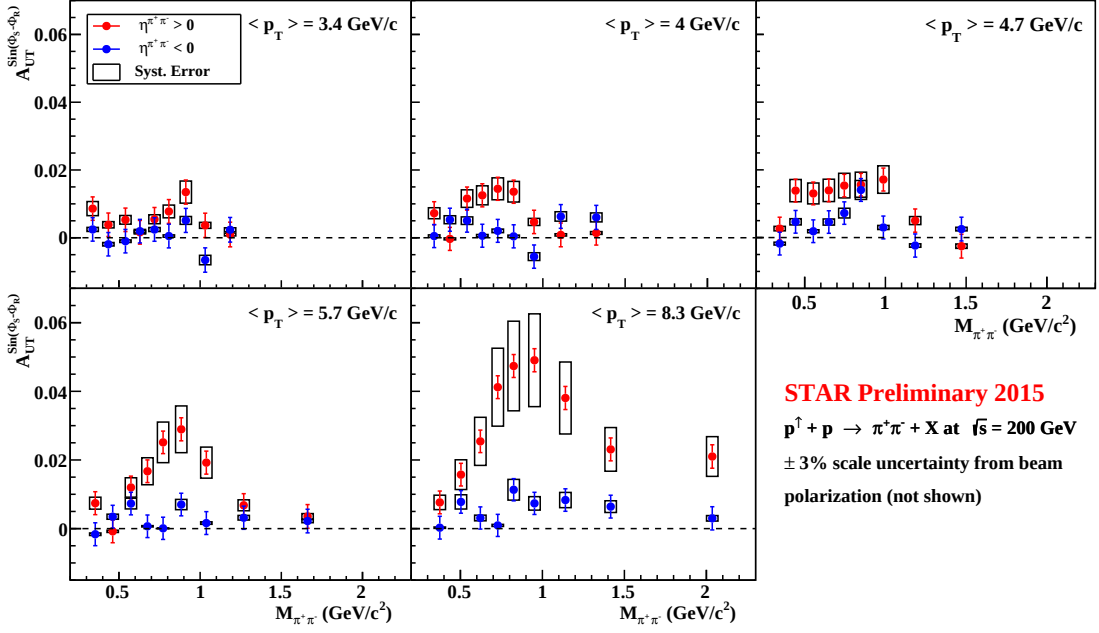


Figure 4.30: $A_{UT}^{\sin(\phi_{RS})}$ measured in five $p_T^{\pi^+\pi^-}$ bins in $\eta^{\pi^+\pi^-} > 0$ (in red) and $\eta^{\pi^+\pi^-} < 0$ (in blue) regions. The value of $\langle p_T^{\pi^+\pi^-} \rangle$ for each transverse momentum bin is shown at each panel's top right corner.

ward region. As anticipated, the backward region, representing the low- x domain, showcases a smaller asymmetry signal.

4.3.8.2 $A_{UT}^{\sin(\phi_{RS})}$ vs $M_{inv}^{\pi^+\pi^-}$

Figure 4.30 illustrates $A_{UT}^{\sin(\phi_{RS})}$ as a function of $M_{inv}^{\pi^+\pi^-}$ across five $p_T^{\pi^+\pi^-}$ bins. For each $p_T^{\pi^+\pi^-}$ bin, $A_{UT}^{\sin(\phi_{RS})}$ is extracted in both the forward and backward $\eta^{\pi^+\pi^-}$ regions. Each panel corresponds to a distinct $p_T^{\pi^+\pi^-}$ bin, increasing its range from left to right and top to bottom. The average $p_T^{\pi^+\pi^-}$ value for each bin is displayed in the top-right corner of the respective panel. This binning approach for the asymmetry signal provides enhanced insights into the characterization of transversity, enabling the simultaneous investigation of $M_{inv}^{\pi^+\pi^-}$ and $p_T^{\pi^+\pi^-}$ dependencies.

The forward asymmetries are mostly positive and exhibit a distinct resonance

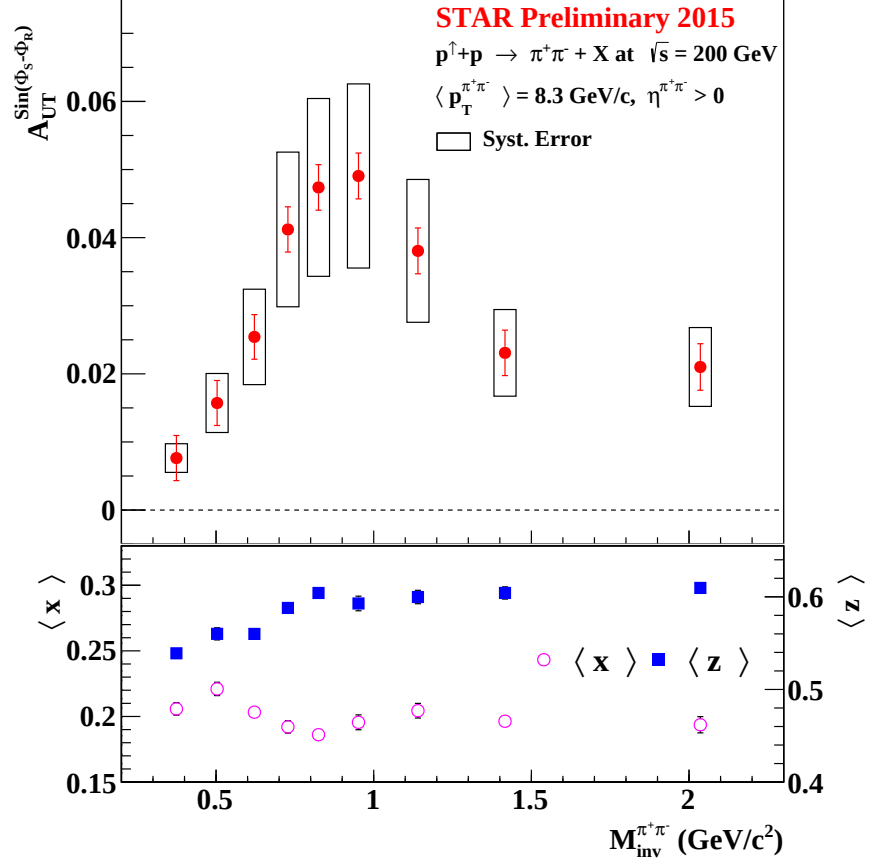


Figure 4.31: $A_{UT}^{\sin(\phi_{RS})}$ as a function of $M_{inv}^{\pi^+ \pi^-}$, in highest $p_T^{\pi^+ \pi^-}$ bin and in $\eta^{\pi^+ \pi^-} > 0$ region (top panel). The bottom panel shows the $\langle x \rangle$ and $\langle z \rangle$ in the corresponding mass bins.

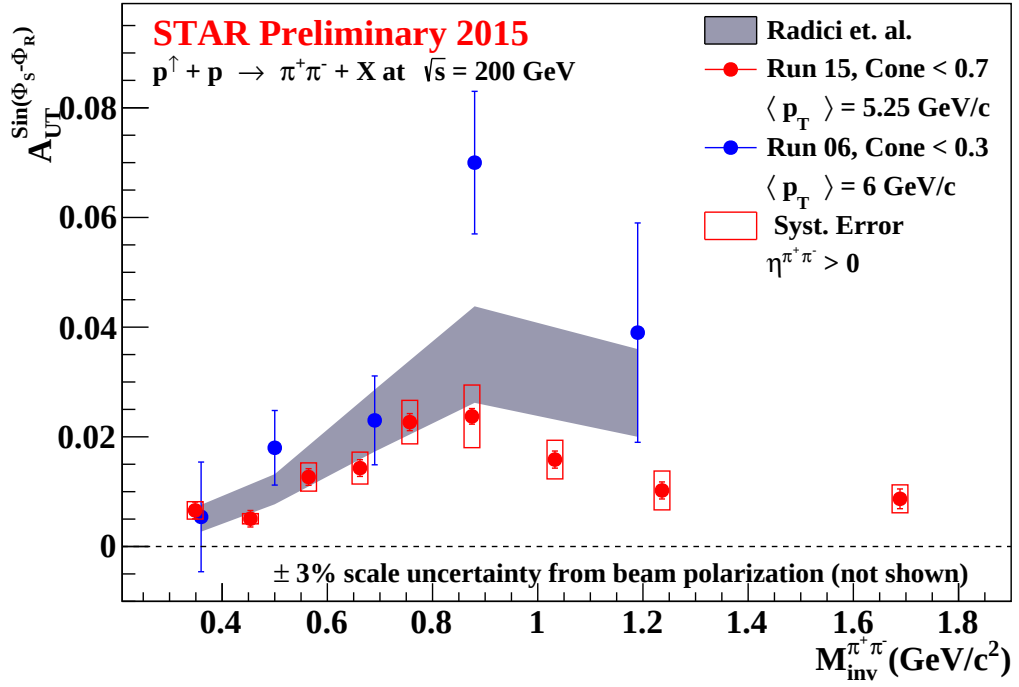


Figure 4.32: $A_{UT}^{\sin(\phi_{RS})}$ as a function of $M_{inv}^{\pi^+\pi^-}$, in $\eta^{\pi^+\pi^-} > 0$ region, compared with the theoretical calculation from Ref. [140].

peak at $M_{inv}^{\pi^+\pi^-} \sim 0.8 \text{ GeV}/c^2$, close to the ρ -meson mass. The prominence of this peak increases with higher $p_T^{\pi^+\pi^-}$ bins. This resonance peak aligns with expectations from the IFF model calculations, attributed to the interference of vector meson decays in a relative p -wave with non-resonant background processes in a relative s -wave [141, 96, 97]. The observed resonance peak in this measurement agrees with prior STAR observations [72, 86]. Backward asymmetries, while small, are non-zero. The backward region is predominantly influenced by scattered quarks from the unpolarized beam and low- x quarks from the polarized beam, which carries lesser spin information. Consequently, a smaller asymmetry signal is expected in this region. The kinematic coverage of x and z in the highest $p_T^{\pi^+\pi^-}$ bin within the forward region, where the asymmetry signal is most pronounced, is displayed in the bottom panel of figure 4.31. Here, z increases with mass, from 0.25 to slightly above 0.3, while $\langle x \rangle \sim 0.2$.

Figure 4.32 presents the STAR 2015 $A_{UT}^{\sin(\phi_{RS})}$ as a function of $M_{inv}^{\pi^+\pi^-}$, integrated over $p_T^{\pi^+\pi^-}$ within the forward region. A comparison is made with the STAR 2006 and a theoretical curve (*gray band*) that is fitted to the SIDIS, e^+e^- , and STAR 2006 data [140]. The uncertainty band, generated via the bootstrap method with 600 replicas at a 90% confidence level, incorporates the systematic theoretical uncertainty linked to gluon fragmentation into $\pi^+\pi^-$, which remains currently undetermined. The STAR measurements exhibit concordance with the theoretical model, showcasing a significant resonance peak at $M_{inv}^{\pi^+\pi^-} \sim M_\rho$.

4.3.8.3 $A_{UT}^{\sin(\phi_{RS})}$ vs $p_T^{\pi^+\pi^-}$

Figure 4.33 presents $A_{UT}^{\sin(\phi_{RS})}$ as a function of $p_T^{\pi^+\pi^-}$ across five $M_{inv}^{\pi^+\pi^-}$ bins, indicated by red (*forward region*) and blue (*backward region*) curves. The asymmetry signal exhibits an increasing trend with rising $p_T^{\pi^+\pi^-}$ in the forward region, with this trend being notably more pronounced in the invariant mass bin when $\langle M_{inv}^{\pi^+\pi^-} \rangle \sim M_\rho$.

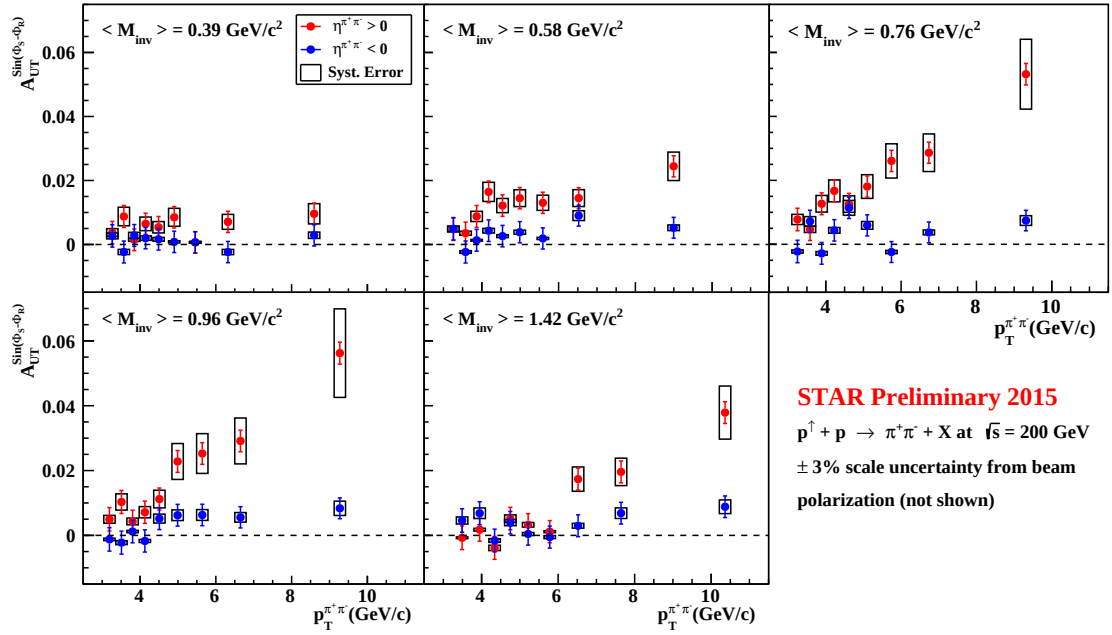


Figure 4.33: $A_{UT}^{\sin(\phi_{RS})}$ vs $p_T^{\pi^+\pi^-}$, in five $M_{inv}^{\pi^+\pi^-}$ bins in $\eta^{\pi^+\pi^-} > 0$ (in red) and $\eta^{\pi^+\pi^-} < 0$ (in blue) regions. The value of $\langle M_{inv}^{\pi^+\pi^-} \rangle$ for each of the invariant mass bins is shown at the top left corner of each panel.

The backward asymmetry signal showcases a similar trend to the forward region, albeit with a relatively smaller amplitude. This observation aligns with expectations, as higher asymmetries are anticipated in the ρ -resonance region ($\eta^{\pi^+\pi^-} > 0$) corresponding to the high- x quark regime, while smaller asymmetries are expected in the low- x region. This further suggests the sizeable transversity in the valence quark region.

CHAPTER 5

UNPOLARIZED $\pi^+\pi^-$

CROSS-SECTION

5.1 Introduction

This chapter presents the first measurement of the unpolarized $\pi^+\pi^-$ cross-section in pp collisions at center-of-mass energy ($\sqrt{s}$) of 200 GeV using data from the STAR Run 2012. The data set is dominated by the jet patch triggered with a lower energy threshold than the Run 2015 dataset. It is a suitable choice for measuring the cross-section as it enhances the sensitivity to gluons.

5.2 Analysis

5.2.1 Dataset

The STAR run 12 pp collision data at $\sqrt{s} = 200$ GeV is used in this analysis. This dataset has been widely used previously for multiple physics analyses by STAR [142,

143, 144, 145, 146]. The final set of 601 runs after quality assurance [147] is used in this analysis, corresponding to an integrated luminosity of 14 pb^{-1} .

At RHIC, both beams are polarized, with stable polarization in the transverse direction to the collider plane. The polarization direction alternates in subsequent bunches, and the polarization pattern is modified from fill to fill to minimize systematic uncertainty. While both beams maintain transverse polarization, integrating over all polarization states enables the determination of the unpolarized cross-section, effectively reducing the net beam polarization to nearly zero.

Jet Patch Trigger	Offline ID	ADC	Threshold (GeV)
JP0	370601	20	3.5
JP1	370611	28	5.4
JP2	370621	36	7.3

Table 5.1: Energy threshold for Jet Patch Triggers used in this dataset.

5.2.2 Simulation and Comparison with the Data

The experimental data’s cross-section is affected by detector inefficiencies, including event triggering and particle tracking efficiency. To ensure direct comparison with the theoretical measurements at the particle level, it’s essential for the measured results to be corrected for these detector effects, requiring the use of simulations.

For the simulation of particle-level events in proton-proton collisions at a center-of-mass energy of $\sqrt{s} = 200 \text{ GeV}$, the PYTHIA 6 Monte Carlo event generator [148] is employed, utilizing the Perugia 12 tune [128] with the modified parameter (PARP(90)=0.213) [147]. This setup incorporates the CTEQ6 PDF sets [149].

Subsequently, the response of the STAR detector is simulated using GEANT3 [150]. Before reconstruction, the raw simulated detector responses are combined event-by-event with actual detector responses from the zero bias trigger samples

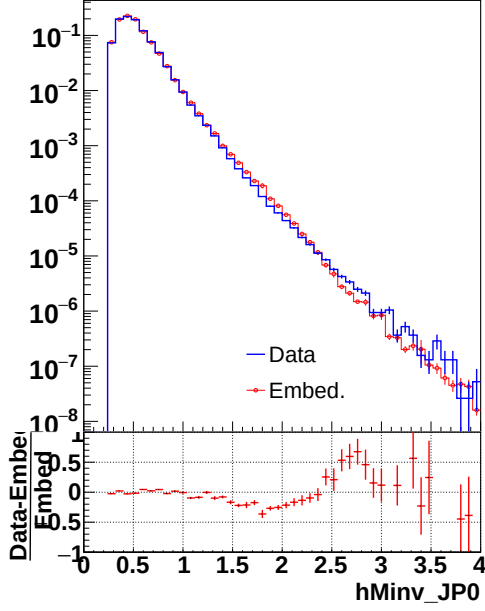
through an embedding process. This process exclusively utilizes events triggered by the zero bias trigger, representing random detector states. The comprehensive details of the embedding sample’s production are discussed in the STAR thesis [151].

Applying the embedding-driven results to the experimental measurements requires a good agreement between the embedding sample and the actual data. This is accomplished by directly comparing the normalized reconstructed distributions of combined partonic $\hat{p}_T$ from the embedding with those from the data. The procedure for determining weights to combine distributions from all partonic $\hat{p}_T$ bins is similar to the method discussed in 4.3.2.

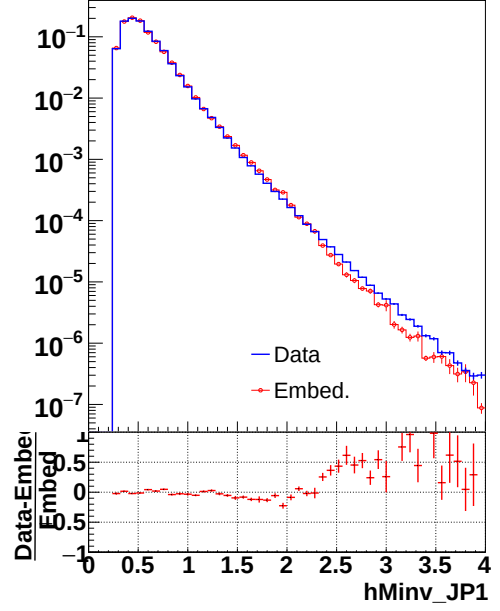
Figure 5.1 displays the data-embedding comparison for $M_{inv}^{\pi^+\pi^-}$ distributions for all three triggers (top panel). For all triggers, the distribution shows good agreement over a wide range of $M_{inv}^{\pi^+\pi^-}$ (the relative difference is shown in the bottom panels). The disagreement in the higher mass region, which is statistically limited, is because of poor reconstruction of higher momentum tracks in the simulation. For a more comprehensive comparison, refer to the appendix B.1.

5.2.3 Event Selection

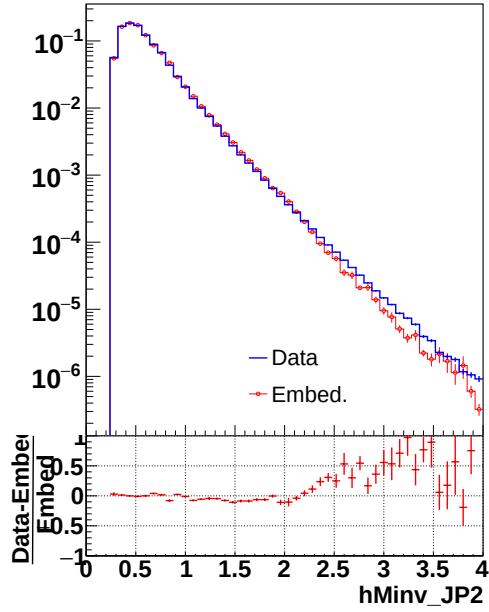
This analysis adopts the standard event and track-level quality cuts established by the STAR Spin community. The primary vertex with the highest rank ($V_{z,rank} > 10^6$) is required to be within 60 cm from the TPC’s center along the beam direction. Moreover, only events satisfying the JP0, JP1, or JP2 triggering criteria are considered, as they constitute the dominant triggers in the data sample. These triggers are identified using offline IDs 370601, 370611, and 370621 for JP0, JP1, and JP2, respectively. Primary tracks originating from the selected primary vertices and fulfilling track-level quality cuts are employed for pion identification. The analysis uses the range of $-1 < n\sigma_\pi < 2$ as the default pion selection cut.



(a) $M_{inv}^{\pi^+\pi^-}$ comparison for JP0 trigger.



(b) $M_{inv}^{\pi^+\pi^-}$ comparison for JP1 trigger.



(c) $M_{inv}^{\pi^+\pi^-}$ comparison for JP2 trigger

Figure 5.1: $M_{inv}^{\pi^+\pi^-}$ distribution comparison for (a) JP0, (b) JP1, and (c) JP2 triggers.

The construction of all-inclusive $\pi^+\pi^-$ events involves pairs of oppositely charged pion tracks within a collision event. $\pi^+\pi^-$ events with an opening angle in $\eta - \phi$ space (cone) exceeding 0.7 or falling below 0.02 are excluded. The upper cone cut is intended to ensure that the $\pi^+\pi^-$ pairs are within the jet. Note that jet reconstruction is not required in this channel. The minimum cone cut is determined through a study of cone resolution using embedding, which eliminates $\pi^+\pi^-$ events composed of tracks too close for separate identification by the detector. Figure 5.2 illustrates the cone resolution, the difference in $\pi^+\pi^-$ opening angles between particle and detector levels, for various upper cone cuts. This resolution's Gaussian width, found to be 0.018 GeV/c^2 , leads to a minimum cone cut of 0.02 GeV/c^2 applied consistently in this analysis. Furthermore, hard cuts are enforced on the pair p_T ($1 < p_T^{\pi^+\pi^-} < 15 \text{ GeV}/c$) and pair mass ($0.27 < M_{inv}^{\pi^+\pi^-} < 4 \text{ GeV}/c^2$) to further refine the $\pi^+\pi^-$ event selection.

A comprehensive list of all selection cuts employed in this analysis is provided in table 5.2. In the table, the terms “Detector”, “particle”, and “PYTHIA” refer to reconstructed quantities, corresponding Monte-Carlo generated PYTHIA quantities associated with detector quantities, and pure PYTHIA particles, respectively.

The dca has the following p_T dependence:

$$dca = \begin{cases} < 2 \text{ cm}, & \text{if } p_T < 0.5 \text{ GeV}/c \\ < 2.5 - p_T \cdot (\text{cm}/\text{GeV}), & \text{if } 0.5 \leq p_T < 1.5 \text{ GeV}/c \\ < 1 \text{ cm}, & \text{if } p_T \geq 1.5 \text{ GeV}/c \end{cases}$$

The measurement of the differential unpolarized cross-section is performed in the $M_{inv}^{\pi^+\pi^-}$ bins. The width of these bins must exceed the invariant mass resolution, determined from the individual momenta of the pion tracks and their energies in each

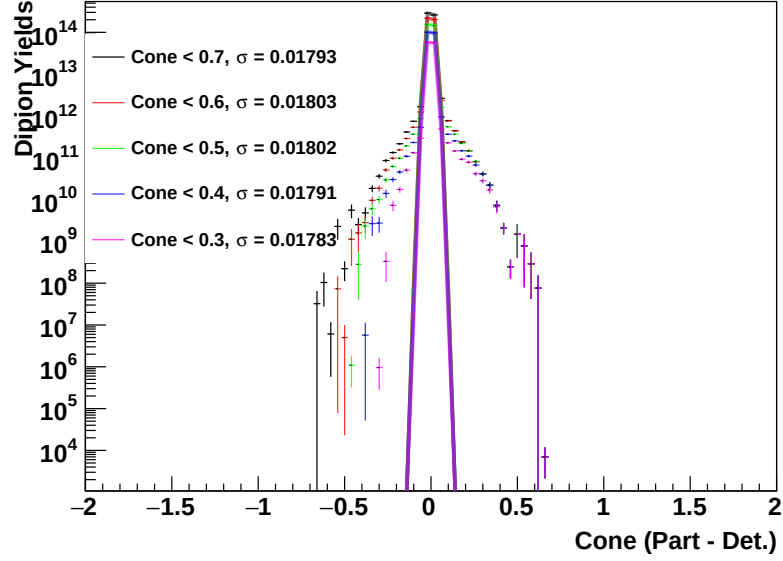


Figure 5.2: Difference between generated and reconstructed $\pi^+\pi^-$ invariant mass for different upper cone cut.

Quantities	Selection Cuts		
	Data	Detector	Particle (PYTHIA)
Z-Vertex(V_z , cm)	< 60	< 60	
$V_{z,rank}$	$> 1e6$	$> 1e6$	
Trigger	JP0, JP1, JP2	JP0, JP1, JP2	
$p_{T,track}$ (GeV/c)	$0.5 < p_T < 15$	$0.5 < p_T < 15$	$0.5 < p_T < 15$
$ \eta_{track} $	< 1	< 1	< 1
dca (cm)	p_T dependence	p_T dependence	
nHitsFit	> 15	> 15	
nHits Ratio	> 0.51	> 0.51	
Pion Selection	$-1 < n\sigma_\pi < 2$	$-1 < n\sigma_\pi < 2$	GePcID:8,9 (PDG: ± 211)
Cone	$0.02 < Cone < 0.7$	$0.02 < Cone < 0.7$	$0.02 < Cone < 0.7$
$\eta^{\pi^+\pi^-}$	$ \eta^{\pi^+\pi^-} < 1$	$ \eta^{\pi^+\pi^-} < 1$	$ \eta^{\pi^+\pi^-} < 1$
$p_T^{\pi^+\pi^-}$ (GeV/c)	$1 < p_T^{\pi^+\pi^-} < 15.0$	$1 < p_T^{\pi^+\pi^-} < 15.0$	$1 < p_T^{\pi^+\pi^-} < 15.0$
$M_{inv}^{\pi^+\pi^-}$ (GeV/ c^2)	$0.27 < M_{inv}^{\pi^+\pi^-} < 4.0$	$0.27 < M_{inv}^{\pi^+\pi^-} < 4.0$	$0.27 < M_{inv}^{\pi^+\pi^-} < 4.0$
$cone = \sqrt{(\eta^{\pi^+} - \eta^{\pi^-})^2 + (\phi^{\pi^+} - \phi^{\pi^-})^2}$			

Table 5.2: Event and track selection cuts

pair. The energy of the identified pions is calculated using the known pion mass ($M^{\pi^\pm} = 0.1396 \text{ GeV}/c^2$) along with the pion's momentum obtained from the TPC.

For a thorough resolution study, an embedding sample is employed that closely matches the data. The discrepancy between the $\pi^+\pi^-$ mass at the detector level and particle level is taken as the mass resolution. This is studied in 12 mass bins, each with approximately equal widths of $0.3 \text{ GeV}/c^2$. Figure 5.3 depicts the differences between the detector and particle masses for each bin, with the Gaussian width of each distribution representing the mass resolution. These bin resolutions are plotted as vertical error bars in relation to the average mass, as shown in Figure 5.4. It's evident that the resolution is better in the lower mass region (< 0.03) compared to the higher mass region. This dependence of mass resolution on the invariant mass aligns with the (transverse) momentum resolution of the particle tracks, given the linear relationship between the invariant mass and the transverse momentum of the $\pi^+\pi^-$.

5.2.4 Cross-Section Binning

The $M_{inv}^{\pi^+\pi^-}$ bin width for the cross-section measurement should not be smaller than the $M_{inv}^{\pi^+\pi^-}$ resolution. While the bin resolution varies with the mass, a minimum bin width is chosen to be greater than the absolute resolution, which is $0.054 \text{ GeV}/c^2$. This ensures the bin width is large enough to avoid smearing effects on the measured cross-section.

The $\pi^+\pi^-$ yields for each trigger are shown in Figure 5.5. The dominant statistics come from the JP1-triggered events, followed by JP2 and JP0. The yields are lower at higher masses for JP0 than JP1 and JP2 due to the trigger threshold energy. The same mass binning is used for all triggers, with the width varied to account for the decreased yields at higher masses. The selected mass bins for the cross-section and

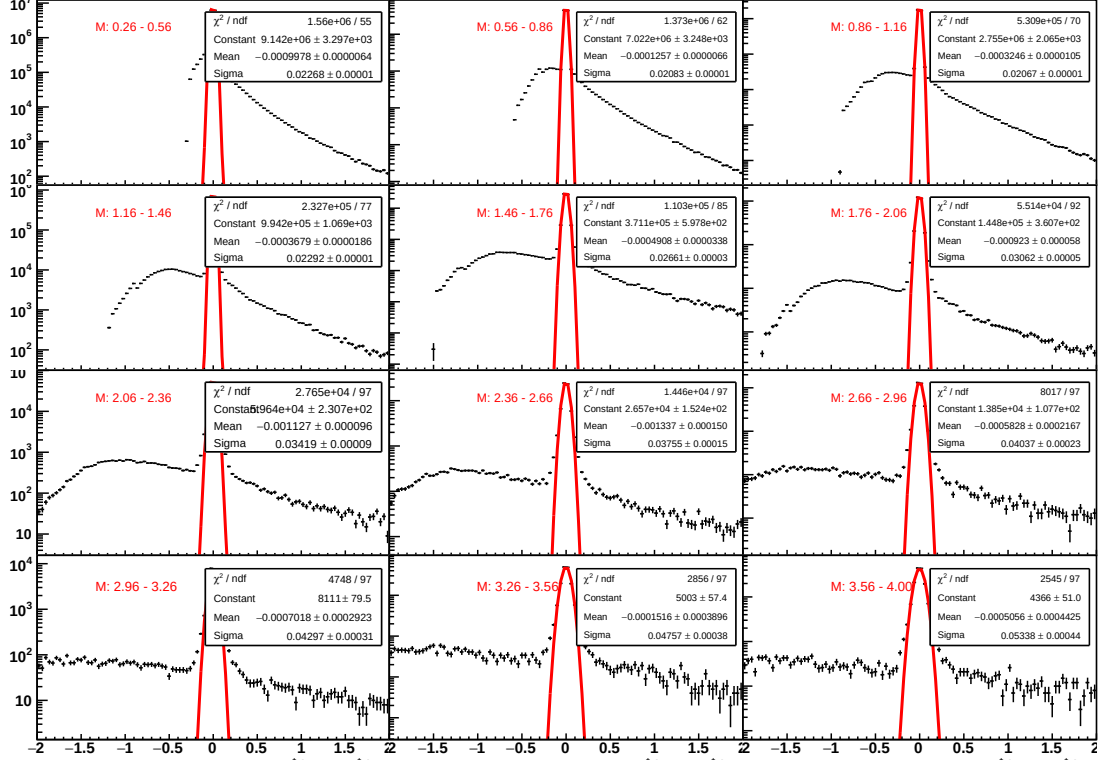


Figure 5.3: $M_{gen}^+ \pi^- - M_{rec}^+ \pi^-$ distributions in 12 mass bins fitted with the Gaussian distribution. The bin range is displayed in each panel.

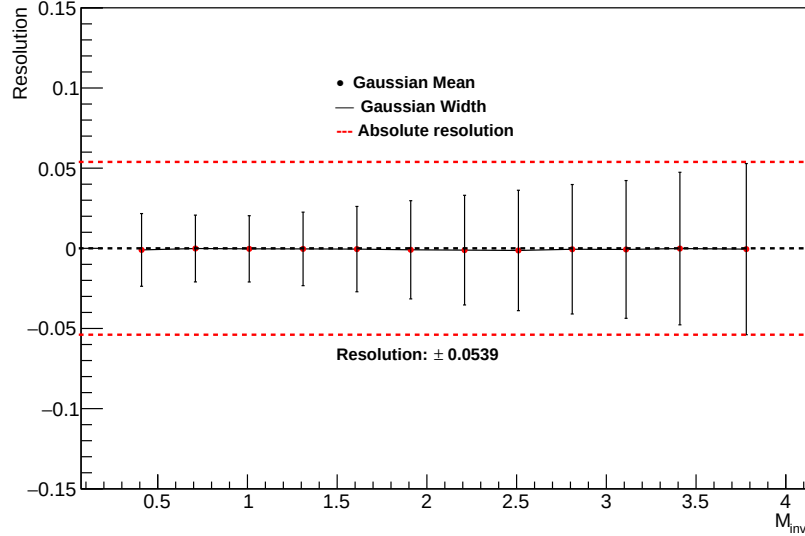


Figure 5.4: Bin resolution (Gaussian widths from figure 5.3) as a function of $M_{inv}^+ \pi^-$. The envelope of the vertical error bars gives the absolute resolution (bin resolution when mass integrated.)

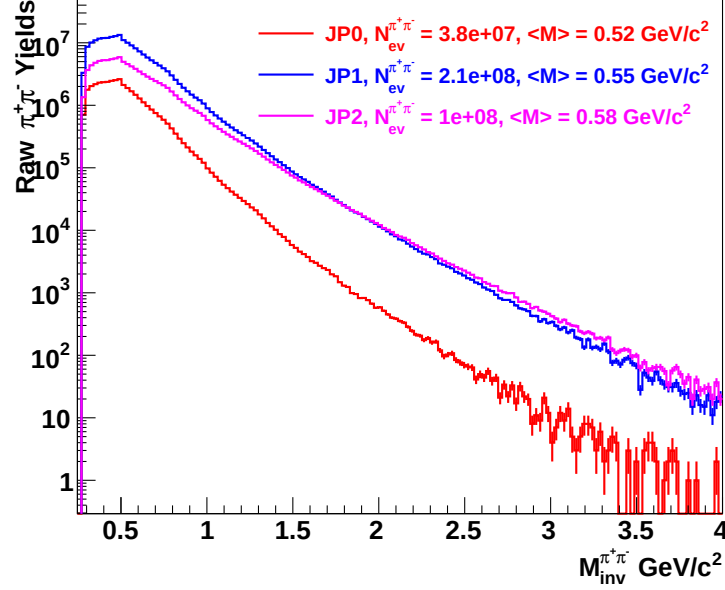


Figure 5.5: Raw yields for JP0, JP1, and JP2 triggers.

other relevant kinematics are shown in Table 5.4.

5.2.5 Luminosity

The STAR experiment selects events based on specific triggering conditions to meet particular physics goals. The three JP triggers, each with a different trigger threshold (table 5.1), have different luminosity per trigger for the data sample used, different from the overall integrated luminosity reported with the dataset. The luminosities per trigger are recorded during the data-taking process. They are available online (JP0, JP1, JP2). The total sum of luminosity per run per trigger for all 601 runs gives the luminosity per trigger. Also, the average prescale¹ value for each trigger is calculated similarly. The prescale values are useful for trigger-specific analysis in the embedding.

¹prescale value for a particular trigger represents how often the triggered event is recorded. For example, if the prescale for a JP0 trigger is 118, only one event for every 118 events is kept.

Triggers	JP0	JP1	JP2	$L_{JP} = 26.64 \text{ pb}^{-1}$
Trig. ID	370601	370611	371621	
$L(\text{pb}^{-1})$	0.16	7.68	18.80	
Prescale	118.335	2.514	1	

Table 5.3: Total luminosity and prescale value per trigger.

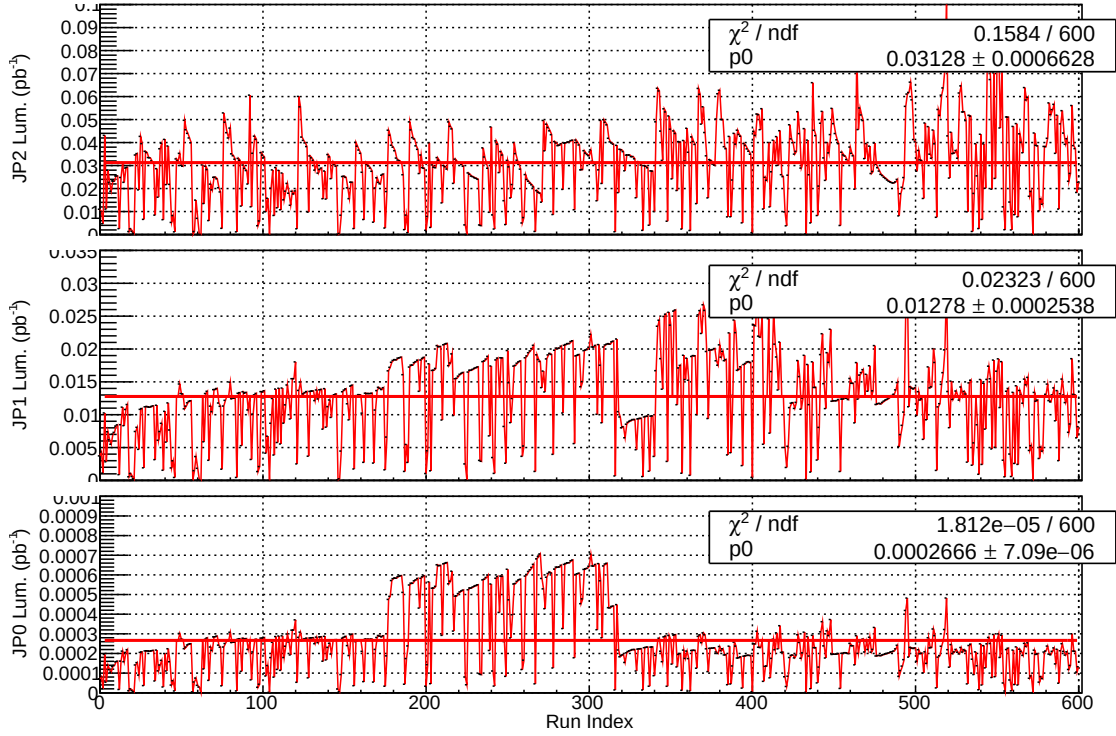


Figure 5.6: Luminosity as a function of the Run index, where the Run index is assigned by sorting Run numbers in increasing order, for JP0 (*bottom*), JP1 (*middle*), and JP2 (*top*). The fit parameter p0 displays the average luminosity per trigger for the full data set.

Table 5.3 shows the total luminosity and average prescale values for each trigger for the data sample used in this analysis. Figure 5.6 displays the recorded luminosity as a function of the Run index for JP0 (bottom), JP1 (middle), and JP2 (top) triggers. The saw-tooth pattern in the plot depicts the fill dependence, where the recorded luminosity falls off with time within the fill.

5.2.6 Unfolding Analysis

5.2.6.1 Why Unfolding?

The $\pi^+\pi^-$ pairs are reconstructed from the measured pion tracks from the detector. The detector effects, such as detector inefficiency in the particle tracking and event triggering, impact these reconstructed $\pi^+\pi^-$ pairs. Consequently, an event could be reconstructed in different kinematic phase spaces. Moreover, the data has active sources of backgrounds (pile-up, beam backgrounds, etc.), which could suppress an event of interest. The unfolding process addresses such issues using the embedding sample. We have used the TUnfoldDensity algorithm[152] for cross-section unfolding.

We are interested in the $\pi^+\pi^-$ cross-section measurement differentiated in thirteen $M_{inv}^{\pi^+\pi^-}$ bins, which needs to be corrected for the detector effects. The unfolding problem can be expressed as

$$y_i = \sum_{j=1}^m A_{ij}x_j, \quad 1 \leq i \leq n, \quad (5.1)$$

where, x_j represents the true counts in j bins, A_{ij} is a matrix of probabilities describing the migrations from bin j to any of the n bins, also called a response matrix, and y_i is the detector counts in the bin i . If the background events, b_i , are present in the

Bin Id	M-Bin	BW	$\langle M_{inv} \rangle$	$\langle p_T^{\pi^+\pi^-} \rangle$	$\langle \eta^{\pi^+\pi^-} \rangle$
1	0.27 - 0.35	0.08	0.319	1.74	-0.0168
2	0.35 - 0.45	0.1	0.401	1.87	-0.0173
3	0.45 - 0.6	0.15	0.517	2.06	-0.0176
4	0.6 - 0.75	0.15	0.666	2.64	-0.0197
5	0.75 - 0.95	0.2	0.83	3.42	-0.0217
6	0.95 - 1.15	0.2	1.03	4.3	-0.0233
7	1.15 - 1.35	0.2	1.24	5.22	-0.0243
8	1.35 - 1.6	0.25	1.45	6.16	-0.0264
9	1.6 - 1.9	0.3	1.72	7.24	-0.0252
10	1.9 - 2.2	0.3	2.02	8.33	-0.0273
11	2.2 - 2.6	0.4	2.36	9.33	-0.0289
12	2.6 - 3.2	0.6	2.81	10.5	-0.0337
13	3.2 - 4	0.8	3.45	11.9	-0.0149

Table 5.4: Binning and kinematics for the $\pi^+\pi^-$ cross-section.

bin i , these can be included in the problem as

$$y_i = \sum_{j=1}^m A_{ij}x_j + b_i, \quad 1 \leq i \leq n \quad (5.2)$$

The solution to the above problem might be subject to instabilities because of statistical fluctuation in y_i . This issue is addressed by applying a smoothness condition on x_j , called regularization. This unfolding algorithm uses a procedure to estimate x_j using a least square method with Tikhonov regularization [153]. For the best unfolding output for the least square minimization, the bins in y_i should be greater or equal to those in x_j , i.e., $n \geq m$.

5.2.6.2 Binning Consideration for the Unfolding

The response matrix is reconstructed with $\pi^+\pi^-$ yields binned in $M_{inv}^{\pi^+\pi^-}$ with $n = 160$ uniform bins (detector bins along the x-axis) and $m = 13$ variable-width bins (true bins along the y-axis), which correspond to the cross-section bins (see details in sec-

tion 5.2.4) onto which the data spectrum is unfolded. The input is the $\pi^+\pi^-$ yields binned in $M_{inv}^{\pi^+\pi^-}$ from the data with the same binning as the detector axis in the response matrix. This binning scheme meets the requirement for the TUnfoldingDensity algorithm.

The cross-section binning and relevant kinematic values in those bins are presented in the table 5.4. The optimum value of n is selected after several iterations for a reasonable unfolding output. The response matrices for JP0, JP1, and JP2 triggered events are shown in the top-left panel of figures 5.8–5.10, respectively, where the y-axis contains “true” values and the x-axis contains “reconstructed” values of $M_{inv}^{\pi^+\pi^-}$. The response matrix shows a good correlation between the true and reconstructed masses, as most entries fall along the diagonal.

5.2.6.3 Background Definition and Normalization Scale

The embedding records information about the generated (“generated” and “true” refer to the same PYTHIA generated quantities recorded by GEANT, which are used interchangeably in this document) and the reconstructed events and tracks quantities. Some particle tracks not associated with any generated tracks may have been reconstructed and stored. Those are the background tracks, which can be identified via the idTruth of a reconstructed track, which gives information about the association between the generated and reconstructed tracks. Any reconstructed tracks with the zero idTruth value are not associated with the generated tracks and should be treated as a background. In addition, the matching condition is imposed between the idTruth of the reconstructed and the track ID of the generated for the certainty of the association. Thus, to consider a signal track, it must match both conditions, otherwise considered a background track. The definition of a background at the $\pi^+\pi^-$ level is based on whether a pair consists of the background track. In a $\pi^+\pi^-$ pair, if

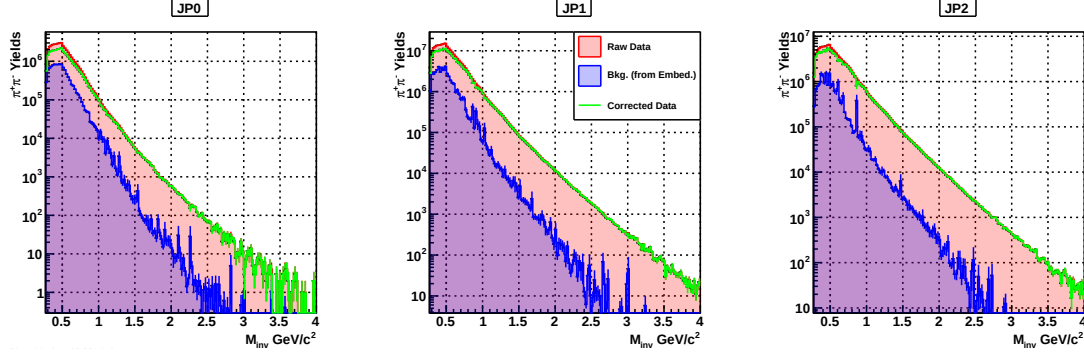


Figure 5.7: Total $\pi^+\pi^-$ yields from the data (“Raw Data” in red), $\pi^+\pi^-$ backgrounds from embedding (“Bkg.” in blue), and background subtracted data (“Corrected Data” in green) as a function of $\pi^+\pi^-$ invariant mass, $M_{inv}^{\pi^+\pi^-}$, for the JP0, JP1, and JP2 triggers from left to right, respectively. Backgrounds are multiplied by the scale factor S_{bkg} .

any or both tracks are background, those pairs are considered background pairs.

Figure 5.7 shows the raw $\pi^+\pi^-$ yields from the data (shown in red), backgrounds from the embedding (shown in blue), and background-subtracted data (shown in green) for JP0, JP1, and JP2 triggers from left to right, respectively. The binning scheme for the background is the same as in the data spectrum, so bin-by-bin background subtraction is possible. The $\pi^+\pi^-$ background yield from the embedding, which contributes only in the lower mass region, is multiplied by a scale factor so that it can be directly subtracted from the data. The scale factor, S_{bkg} , is determined from the total $\pi^+\pi^-$ yields in the data (I_{dat}), and the total $\pi^+\pi^-$ yields in the embedding, I_{emb} , as,

$$S_{bkg} = \frac{I_{dat}}{I_{emb}} \quad (5.3)$$

5.2.6.4 Unfolding Using TUnfoldDensity Algorithm

The unfolding algorithm determines the stationary point in the Lagrangian [152]

$$\mathcal{L}(x, \lambda) = (\mathbf{y} - \mathbf{A}\mathbf{x})^T \mathbf{V}_{\mathbf{y}\mathbf{y}}^{-1} (\mathbf{y} - \mathbf{A}\mathbf{x}) + \tau^2 (x - f_b x_0)^T (\mathbf{L}^T \mathbf{L}) (x - f_b x_0) + \lambda (Y - e^T x) \quad (5.4)$$

where $Y = \sum_i y_i$ and $e_j = \sum_i A_{ij}$. The first term is expected from the least square minimization, where $\mathbf{V}_{\mathbf{y}\mathbf{y}}$ of $\mathbf{y}$ is the covariance matrix which holds the square of the uncertainties, and all other vector definitions follow from Eq. 5.2. The second term describes the regularization condition, which damps fluctuations in $\mathbf{x}$. The regularization strength, τ^2 , is considered constant while determining the stationary point of $\mathcal{L}$, which is determined using the L-Curve method [154]. The third term is the optional area constant with the Lagrangian parameter λ .

The TUnfoldDensity algorithm effectively handles background through the SubtractBackground() method, which inputs the background histogram and normalization scale factor. Unfolding is performed with the background subtraction for each trigger individually, allowing us to measure the cross-section for each trigger. The $\pi^+\pi^-$ invariant mass, which is to be unfolded, and background distributions are shown in figure 5.7. The number of bins in both cases is kept the same, and the subtraction is performed bin-by-bin using the histograms before the actual unfolding method is called.

Figures 5.8–5.10 show the unfolding analysis details for triggers JP0, JP1, and JP2, respectively. The top right panel displays the L-curve, a scanned region for the regularization parameter, τ , and the asterisk shows the minimum curvature of the scanned region, which is used to calculate τ . The L-curve should have an approximate L-shape with the kink somewhere on the scanned region but not too close to the edges for good unfolding results. The bottom left panel presents the bin correlation matrix

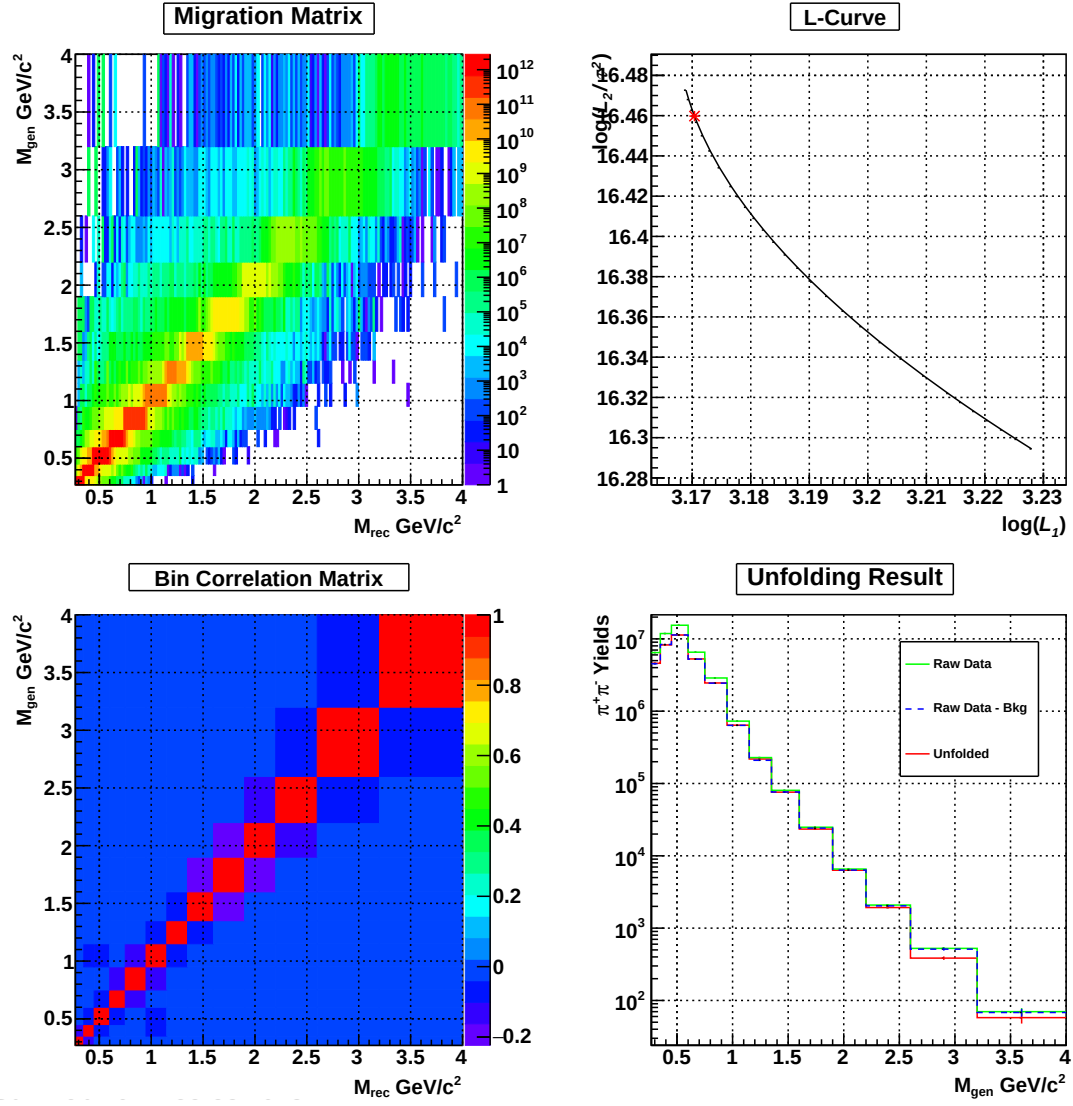


Figure 5.8: Unfolding for the JP0 triggered cross-section.

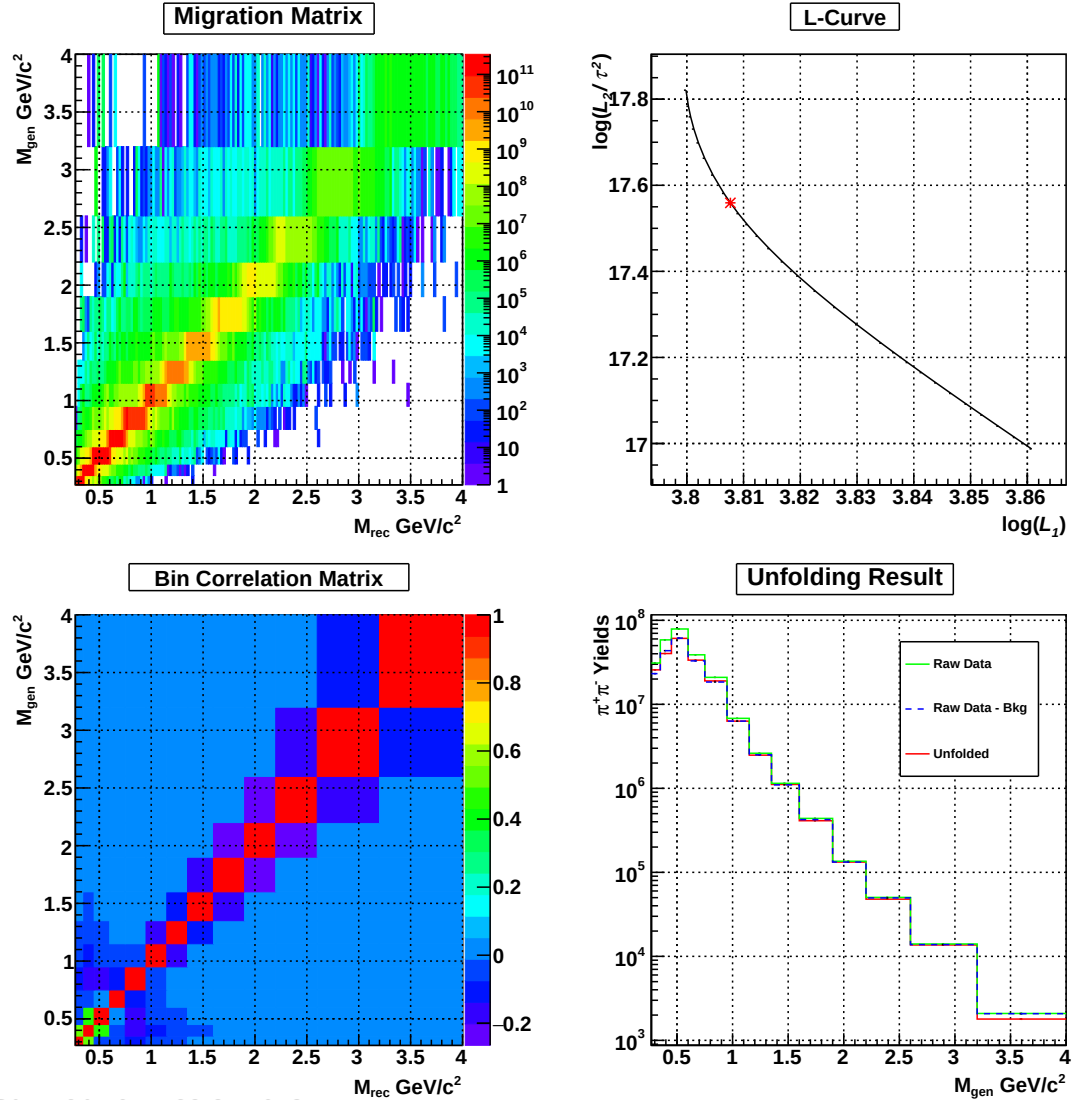


Figure 5.9: Unfolding for the JP1 triggered cross-section.

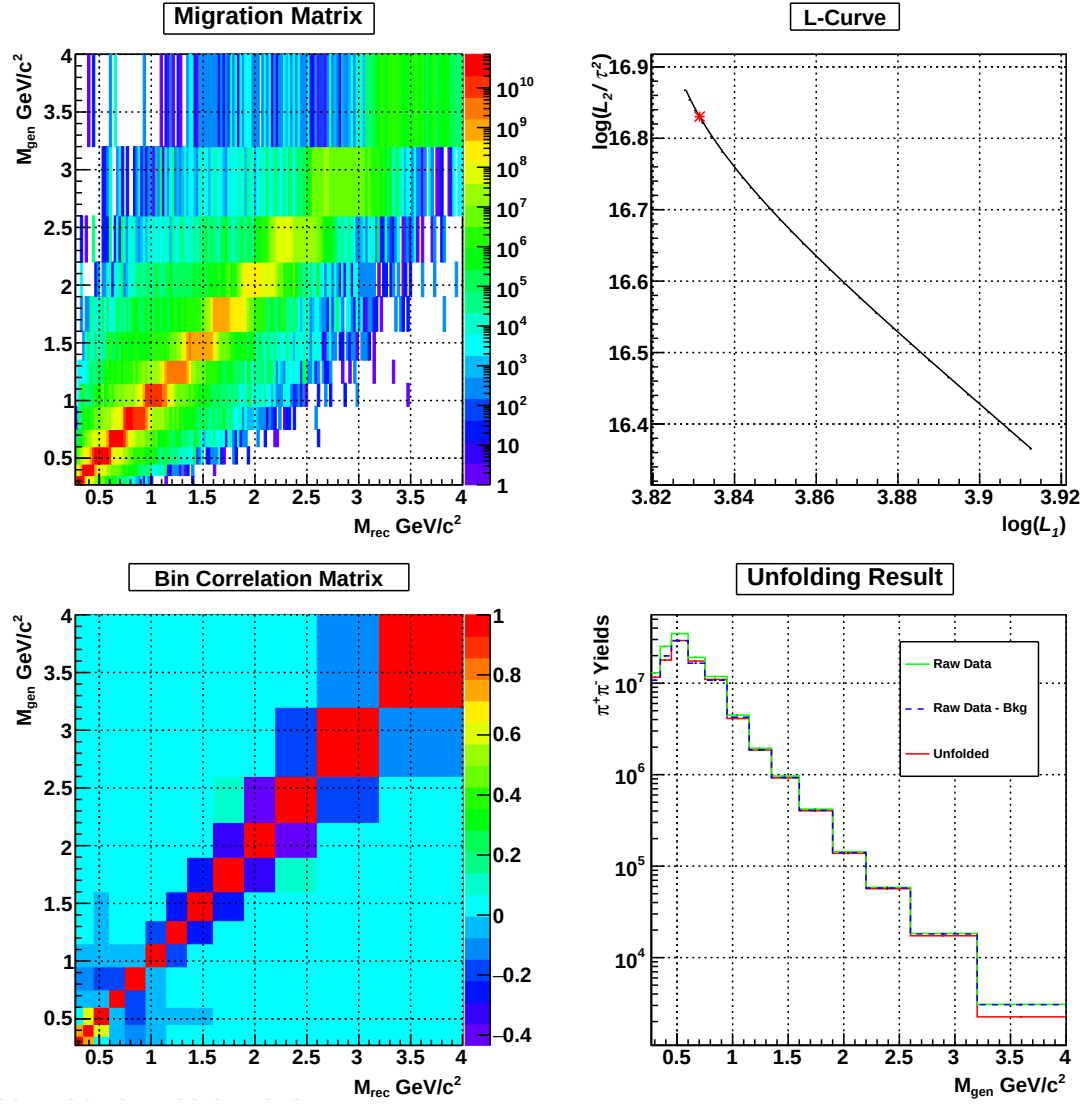


Figure 5.10: Unfolding for the JP2 triggered cross-section.

between the generated (y-axis) and reconstructed (x-axis) bins.

The bottom right panel depicts the raw data (solid green), the background-subtracted data (dashed blue), and the unfolded distribution (solid red) into 13 cross-section bins. The input data histograms for the unfolding have a different number of bins than those shown here; specifically, the data input consists of 160 reconstructed bins of uniform width, while the output bins are 13 cross-section bins at the truth level. As expected, the unfolded output differs slightly from the input, with yields from the higher mass bins migrating to the lower mass region. This final unfolded distribution is devoid of the bin migration effect and background and is at the true level. However, the true level yields still require corrections from the trigger and particle tracking efficiency to remove detector effects and corrections for the misidentified and loss of pions because of the adopted hard particle selection cuts.

5.2.7 Efficiency Studies

Two types of efficiencies appear in this analysis; tracking and triggering efficiencies. In high-energy physics, these analyses are performed using the embedding sample as it has generated and reconstructed event information. Each of these analyses is done separately in the following sub-sections. The detector level cuts listed in table 5.2 are applied for all efficiency analyses. Only the p_T and η cuts are retained at the particle level for both kinematic and acceptance consistency.

5.2.7.1 Tracking Efficiency (ϵ_{trk})

Mathematically, the tracking efficiency (ϵ_{trk}) is defined as the ratio of the number of reconstructed tracks to the number of generated tracks;

$$\epsilon_{trk}^{\pi} = \frac{N_{gen,rec}^{\pi}}{N_{gen}^{\pi}}, \quad (5.5)$$

where N_{gen} is the number of generated tracks and $N_{gen,rec}$ is the number of generated tracks that passed all the detector level cuts called reconstructed tracks. Both generated and reconstructed tracks are subject to common event-level selection cuts of $|V_z| < 60$ cm, $V_{z,rank} > 1e6$ and additional cuts on track p_T and η . In addition, the track is considered reconstructed if it passes all the detector cuts listed in table 5.2 and is part of a pair. The tracking efficiency is calculated for each of the cross-section bins.

Figure 5.11 illustrates the tracking efficiency for π^+ , π^- , their product, and the $\pi^+\pi^-$ efficiency, as a function of the $\pi^+\pi^-$ invariant mass, for each trigger used. The top panel displays the tracking efficiency for π^+ (left) and π^- (right) in cross-section $M_{inv}^{\pi^+\pi^-}$ bins. Pion tracking efficiency increases with mass in the lower mass region but tends to decrease in the higher mass regions. The bottom left panel highlights the mass dependence of the tracking efficiency for a $\pi^+\pi^-$, which is the product of pion track efficiencies in a pair. The bottom right panel shows the tracking efficiency at the $\pi^+\pi^-$ level, which is consistent with the product-level efficiency and provides a consistency check of the two approaches. Efficiency at the product level is utilized to correct each bin in the cross-section.

5.2.7.2 Triggering Efficiency (ϵ_{trg})

Trigger efficiency correction is necessary for cross-section measurement analysis in particle physics because the cross-section measures the probability of a certain interaction occurring, and the trigger system plays a crucial role in selecting the events of interest. Given the high rate of events produced from collisions, the trigger system is designed to select only a subset of events of interest for a particular analysis. The trigger efficiency correction aims to account for these inefficiencies and ensure that the observed number of events in the selected sample accurately reflects the number

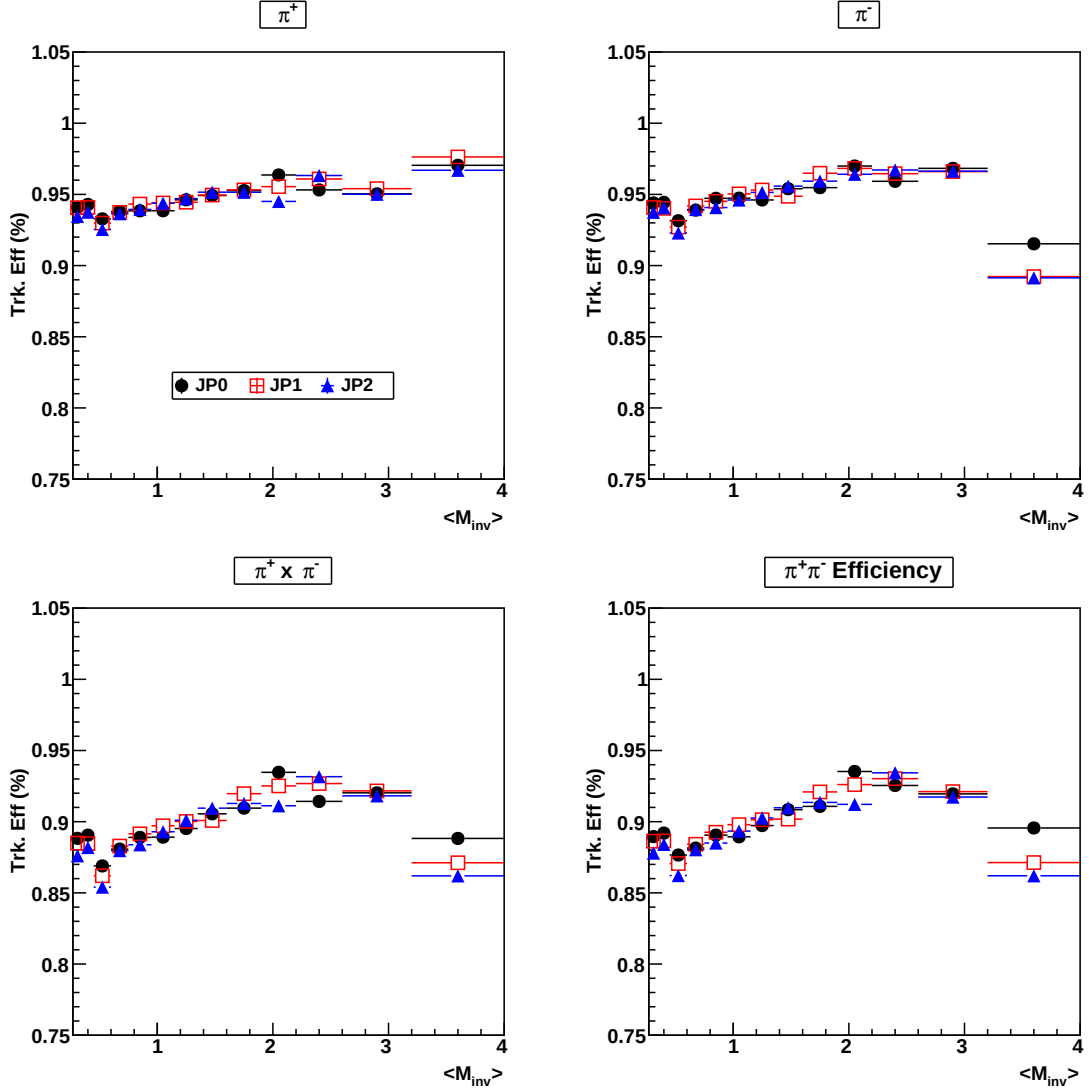


Figure 5.11: Tracking efficiencies in the cross-section $M_{inv}^{\pi^+\pi^-}$ bins are shown for JP0 (black circle), JP1 (red open square), and JP2 (blue triangle) triggers.

of events that would have been observed without the triggering condition.

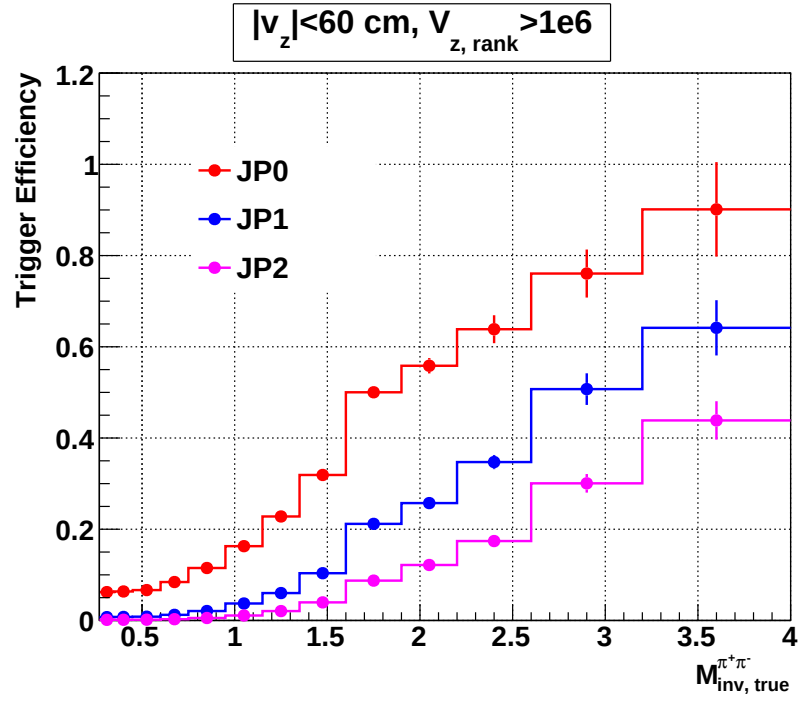
We use the embedding sample to estimate trigger efficiency, which does not know about event triggering unless a trigger simulator algorithm is invoked. By comparing event information with and without triggering effects, we can calculate the trigger efficiency, denoted as ϵ_{trg} , for a particular trigger of interest. Mathematically, ϵ_{trg} is the ratio of the number of events triggered to the number of unbiased events, as represented by the equation,

$$\epsilon_{trg} = \frac{\sum N_{triggered}^{\pi^+\pi^-}}{\sum N_{unbiased}^{\pi^+\pi^-}}, \quad (5.6)$$

where $N_{triggered}^{\pi^+\pi^-}$ represents the number of $\pi^+\pi^-$ events from a triggered event, while $N_{unbiased}^{\pi^+\pi^-}$ represents the number of $\pi^+\pi^-$ events without the application of a trigger algorithm, both at the truth level. The “truth level” event refers to the $\pi^+\pi^-$ event from Monte-Carlo tracks associated with reconstructed pion tracks, with all the selection cuts applied at the detector level. The trigger efficiency is calculated for each invariant mass bin for JP0, JP1, and JP2 triggers using Eq. 5.6, as shown in Figure 5.12. The triggering efficiency is lower in the lower mass region and higher in the higher mass region for all triggers. The JP0 trigger is activated more frequently, as it has the lowest energy threshold, while the JP1 and JP2 triggers are activated less often due to their higher energy threshold criteria. The trigger efficiency correction is applied for each of the cross-section bins.

5.2.8 Corrections for Misidentification and Loss of Pions

Usually, in STAR, the dE/dx -based data-driven approach is used to make corrections associated with the PID. In this analysis, we opted for the embedding-driven approach as there is a good description of $n\sigma_\pi$ from the embedding, which is used for the PID



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Figure 5.12: Trigger efficiency for JP0, JP1, and JP2 triggers as a function of $M_{\text{inv}}^{\pi^+\pi^-}$ in cross-section bins.

in data and embedding at the detector level (see section B.1).

Two corrections are considered associated with the PID. The first correction is for misidentifying protons and kaons as pions within the default $n\sigma_\pi$ cut, which accounts for the contribution from pions in the signal region. The second correction is for the loss of true pions due to the default $n\sigma_\pi$ cut, which needs to be maintained. This correction accounts for the inefficiency of the default pion selection adopted in this analysis. Both corrections are applied to the cross-section to improve the accuracy of the results.

5.2.8.1 Correction for the Fake Pion Contribution (f_{fake})

The f_{fake} is calculated as a fraction of true $\pi^+\pi^-$ pairs within the default $n\sigma_\pi$ cut, accounting for fake pions entering the signal region. Mathematically,

$$f_{fake} = \frac{N_{n\sigma\pi\&\&true}^{pair}}{N_{n\sigma\pi}^{pair}} \quad (5.7)$$

where the denominator is the $\pi^+\pi^-$ pair selected with the default $n\sigma_\pi$ cut at the detector level, and the numerator is the number of detected pairs that are true pairs (the GEANT particle ID identifies true pairs: 8 for π^+ and 9 for π^-).

Figure 5.13 shows the f_{fake} in $M_{inv}^{\pi^+\pi^-}$ cross-section bins for all three JP triggers. f_{fake} has a slight mass dependence, with the increase in the fraction as the mass increases. Also, the fraction has a trigger dependence, specifically in the region $M_{inv}^{\pi^+\pi^-} < 1.5 \text{ GeV}/c^2$; the fraction is lower for the trigger of a lower trigger threshold. The correction for fake contributions is applied at the triggered cross-section level on a bin-by-bin basis.

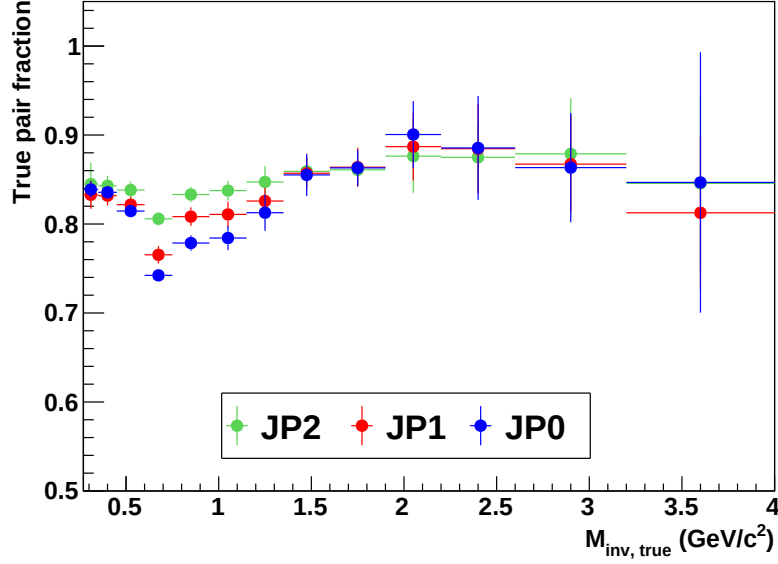


Figure 5.13: Fraction of true $\pi^+\pi^-$ pair.

5.2.8.2 Correction for the Pion Loss (f_{loss})

Some of the true $\pi^+\pi^-$ may be lost due to the restrictive PID cut. This loss of $\pi^+\pi^-$ events can be estimated by calculating the fraction;

$$f_{loss} = \frac{N_{true}}{N_{n\sigma_\pi \& true}} \quad (5.8)$$

where N_{true} is the number of true $\pi^+\pi^-$ pairs and $N_{n\sigma_\pi \& true}$ is the number of true $\pi^+\pi^-$ pairs which falls within the $-1 < n\sigma_\pi < 2$ range. This accounts for the inefficiency of the applied PID cut on identifying pion. This fraction is greater than one and increases the yields by this factor to account for the loss of $\pi^+\pi^-$ events.

Figure 5.14 shows f_{loss} as a function of $M_{inv}^{\pi^+\pi^-}$ in cross-section bins for all JP triggers. The f_{loss} is independent of the triggers. The loss is higher in the lower mass region, which decreases in the higher mass region. f_{loss} varies from 40% at the lower mass region to 30% at the higher mass regions.

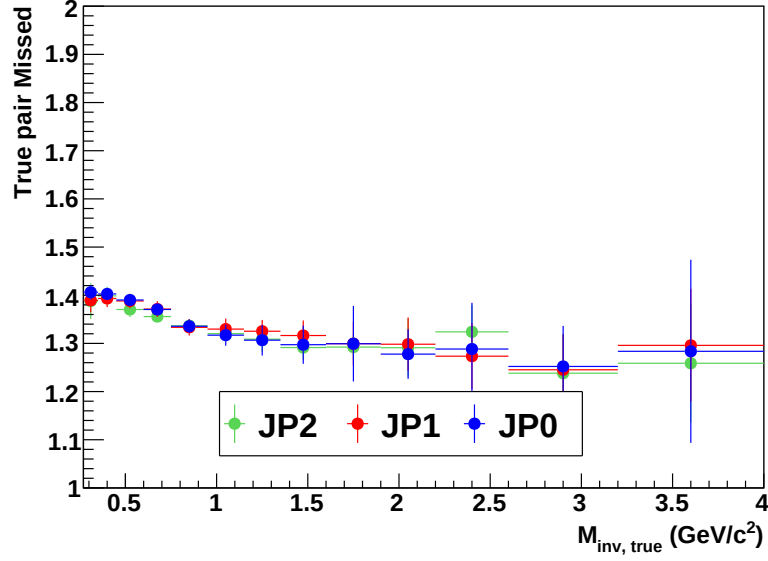


Figure 5.14: PID cut efficiency fraction, which accounts for the loss of true $\pi^+\pi^-$ pairs due to the detector level PID cut, $-1 < n\sigma_\pi < 2$.

5.2.9 Measured Cross-Section Per Trigger

With all the correction factors on hand as discussed above, the unpolarized $\pi^+\pi^-$ cross-section can be calculated using the formula;

$$\frac{d\sigma_{mes}^{\pi^+\pi^-}}{dM_{inv}^{\pi^+\pi^-}} = \frac{f_{fake} * f_{loss}}{L_{int} \cdot \epsilon_{trk}^{\pi^+\pi^-} \cdot \epsilon_{trg}^{\pi^+\pi^-}} \frac{dN_{true}^{\pi^+\pi^-}}{dM_{inv}^{\pi^+\pi^-}}, \quad (5.9)$$

where $dN_{true}^{\pi^+\pi^-}/dM_{inv}^{\pi^+\pi^-}$ is the unfolded yields normalized by the bin width. We use f_{fake} to correct for the fake pion contribution, as discussed in section 5.2.8.1, and f_{loss} to correct for the $n\sigma_\pi$ efficiency on the particle identification, as defined in section 5.2.8.2. L_{int} is the integrated luminosity in pb^{-1} (discussed in section 5.2.5). $\epsilon_{trk}^{\pi^+\pi^-}$ is the tracking efficiency correction factor, as discussed in section 5.2.7.1, and $\epsilon_{trg}^{\pi^+\pi^-}$ is the trigger efficiency correction factor, as discussed in section 5.2.7.2.

We measure the cross-section for each trigger and obtain the final cross-section by calculating the weighted average for three triggered cross-sections. The weight,

w_i , for combining the cross-sections is determined from the statistical uncertainties in the final cross-section for each trigger.

$$w_i = \frac{1}{\delta_{stat,i}^2}, i = JP0, JP1, JP2 \quad (5.10)$$

$$(d\sigma/dM)_{mes,w} = \frac{\sum_i w_i * (d\sigma/dM)_{mes,i}}{\sum_i w_i} \quad (5.11)$$

$$\delta_{stat,w} = \frac{1}{\sqrt{\sum w_i}} \quad (5.12)$$

Equation 5.11 calculates the final cross-section as a weighted average of triggered cross-sections, and Eq. 5.12 gives the weighted statistical uncertainty for the final cross-section.

Figure 5.15 displays the measured cross-sections for all three triggers and the weighted average (labeled as “Comb”) of all three triggered cross-sections (top panel), and the relative difference between the triggered and weighted average cross-sections in the bottom panel. We find good agreement between trigger-separated cross-sections; the small disagreement is treated as a systematic uncertainty. The relative difference between the triggered cross-sections and the weighted average cross-section is applied as a systematic uncertainty, discussed in section 5.2.7.2. Tables 5.5–5.7 present the numerical details for each triggered cross-section. The measured cross-section is reported in picobarns (pb).

5.2.10 Systematic Studies

The systematic uncertainties associated with this measurement are studied and quantified in the following sections:

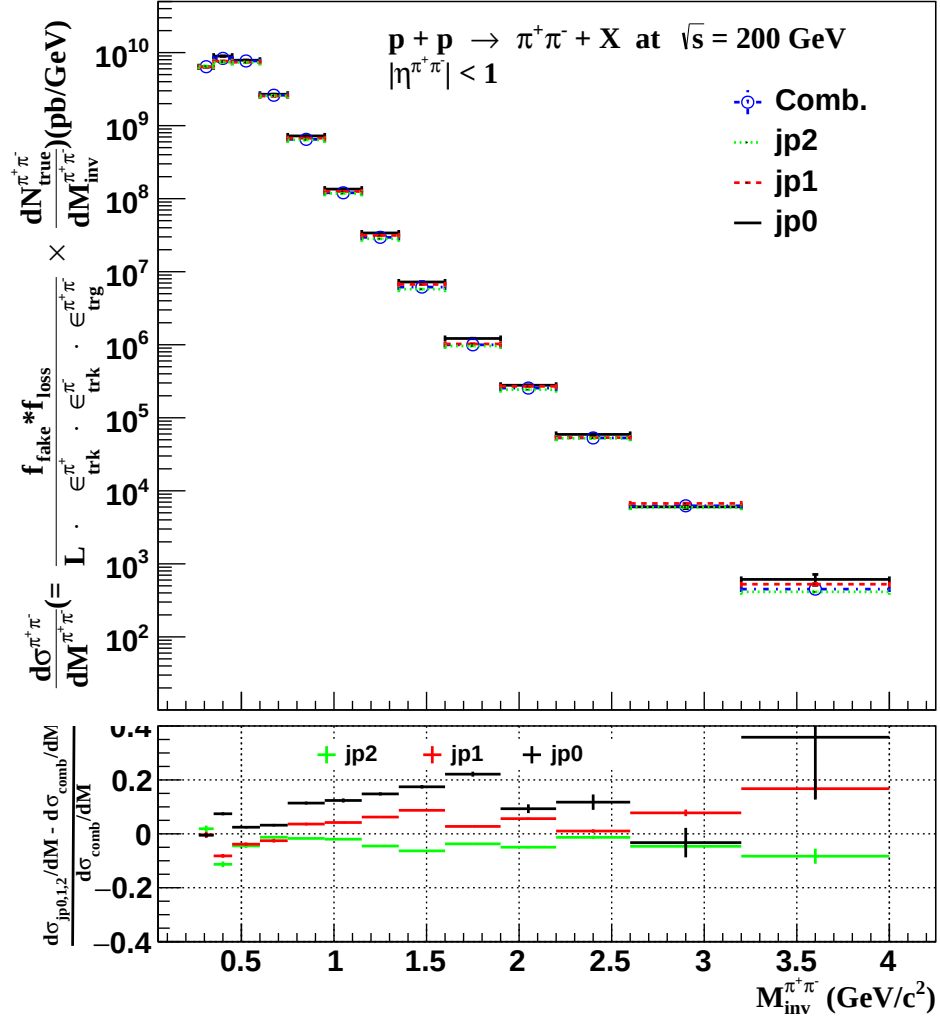


Figure 5.15: Top panel: Unpolarized $\pi^+\pi^-$ cross-section as a function of $M_{inv}^{\pi^+\pi^-}$ for the JP0, JP1, and JP2 triggers. The final cross-section (labeled “Comb”) is the weighted average of all three triggered cross-sections. Bottom panel: the relative difference between triggered cross-sections and final cross-section.

Bin No.	BW	$L(pb^{-1})$	ϵ_{trg}	ϵ_{trk}	f_{fake}	f_{loss}	dN_{un}/dM	δ_{un}	$d\sigma/dM(JP0)$	$\delta\sigma_{stat,JP0}$
0	0.08	0.16	0.0622	0.888	0.839	1.41	3.82e+06	2.33e+04	6.38e+09	3.89e+07
1	0.1	0.16	0.0636	0.891	0.836	1.4	6.92e+06	3.02e+04	8.95e+09	3.91e+07
2	0.15	0.16	0.0666	0.869	0.815	1.39	9.65e+06	3.33e+04	7.87e+09	2.72e+07
3	0.15	0.16	0.0843	0.881	0.742	1.37	4.71e+06	1.89e+04	2.69e+09	1.08e+07
4	0.2	0.16	0.115	0.889	0.779	1.34	2.28e+06	1.08e+04	7.23e+08	3.42e+06
5	0.2	0.16	0.163	0.889	0.784	1.32	6.07e+05	3.99e+03	1.35e+08	8.9e+05
6	0.2	0.16	0.228	0.895	0.813	1.31	2.08e+05	1.06e+03	3.39e+07	1.72e+05
7	0.25	0.16	0.319	0.906	0.858	1.3	7.51e+04	407	7.23e+06	3.92e+04
8	0.3	0.16	0.5	0.91	0.877	1.3	2.33e+04	173	1.22e+06	9.02e+03
9	0.3	0.16	0.558	0.935	0.901	1.28	6.08e+03	89.6	2.79e+05	4.11e+03
10	0.4	0.16	0.639	0.914	0.889	1.29	1.93e+03	49	5.92e+04	1.5e+03
11	0.6	0.16	0.761	0.92	0.863	1.25	375	21.1	6.03e+03	339
12	0.8	0.16	0.901	0.888	0.847	1.27	58.1	9.87	612	104

Table 5.5: Numerical details for the JP0 cross-section.

Bin No.	BW	$L(pb^{-1})$	ϵ_{trg}	ϵ_{trk}	f_{fake}	f_{loss}	dN_{un}/dM	δ_{un}	$d\sigma/dM(JP1)$	$\delta\sigma_{stat,JP1}$
0	0.08	7.68	0.00731	0.885	0.833	1.39	2.2e+07	2.27e+05	6.39e+09	6.6e+07
1	0.1	7.68	0.00774	0.885	0.832	1.39	3.47e+07	2.29e+05	7.65e+09	5.06e+07
2	0.15	7.68	0.0083	0.862	0.822	1.39	5.33e+07	3.27e+05	7.38e+09	4.53e+07
3	0.15	7.68	0.0123	0.883	0.765	1.37	3.04e+07	1.99e+05	2.54e+09	1.67e+07
4	0.2	7.68	0.0207	0.891	0.808	1.33	1.77e+07	8.24e+04	6.73e+08	3.13e+06
5	0.2	7.68	0.0372	0.897	0.811	1.33	5.97e+06	2.03e+04	1.25e+08	4.27e+05
6	0.2	7.68	0.0601	0.9	0.826	1.33	2.37e+06	5.07e+03	3.14e+07	6.7e+04
7	0.25	7.68	0.104	0.901	0.858	1.32	1.06e+06	2.3e+03	6.7e+06	1.45e+04
8	0.3	7.68	0.212	0.92	0.868	1.3	4.08e+05	997	1.02e+06	2.5e+03
9	0.3	7.68	0.257	0.925	0.887	1.3	1.28e+05	468	2.69e+05	982
10	0.4	7.68	0.347	0.927	0.885	1.27	4.71e+04	250	5.35e+04	284
11	0.6	7.68	0.507	0.922	0.867	1.25	1.34e+04	130	6.72e+03	65.2
12	0.8	7.68	0.642	0.871	0.813	1.28	1.73e+03	46.5	526	14.1

Table 5.6: Numerical details for the JP1 cross-section.

Bin No.	BW	$L(pb^{-1})$	ϵ_{trg}	ϵ_{trk}	f_{fake}	f_{loss}	dN_{un}/dM	δ_{un}	$d\sigma/dM(JP2)$	$\delta\sigma_{stat,JP2}$
0	0.08	18.8	0.00139	0.876	0.845	1.39	1.02e+07	1.04e+05	6.53e+09	6.68e+07
1	0.1	18.8	0.00151	0.882	0.843	1.4	1.57e+07	1.75e+05	7.39e+09	8.23e+07
2	0.15	18.8	0.00169	0.854	0.838	1.37	2.6e+07	1.97e+05	7.34e+09	5.56e+07
3	0.15	18.8	0.00277	0.88	0.806	1.36	1.62e+07	5.24e+04	2.57e+09	8.33e+06
4	0.2	18.8	0.00541	0.884	0.833	1.34	1.03e+07	1.82e+04	6.38e+08	1.13e+06
5	0.2	18.8	0.0109	0.893	0.838	1.32	3.89e+06	7.5e+03	1.18e+08	2.27e+05
6	0.2	18.8	0.0206	0.901	0.847	1.31	1.77e+06	3.08e+03	2.82e+07	4.91e+04
7	0.25	18.8	0.0397	0.909	0.86	1.29	8.81e+05	1.67e+03	5.77e+06	1.09e+04
8	0.3	18.8	0.0874	0.913	0.861	1.29	3.88e+05	926	9.6e+05	2.29e+03
9	0.3	18.8	0.122	0.911	0.876	1.29	1.34e+05	487	2.42e+05	882
10	0.4	18.8	0.174	0.932	0.875	1.32	5.53e+04	286	5.23e+04	270
11	0.6	18.8	0.301	0.918	0.879	1.24	1.7e+04	145	5.94e+03	50.7
12	0.8	18.8	0.439	0.862	0.846	1.26	2.21e+03	52.1	413	9.75

Table 5.7: Numerical details for the JP2 cross-section.

5.2.10.1 Trigger Bias (δ_{bias})

While collecting data, a triggering algorithm is applied to select high transverse energy (E_T) jets suitable for physics analysis, which is otherwise shadowed by garbage low E_T jets. However, selecting only higher E_T jets biased the events favoring the quarks jets over gluon jets as gluon jets are broader, which may leak the energy to neighboring calorimeter towers, failing to meet the triggering condition. The size of this effect, called trigger bias, can be estimated using an embedding sample calculating the ratio of the quark fraction at the detector level to the same fraction at the particle level. Here, the detector level is where we apply the trigger to select hard $\pi^+\pi^-$ events, and the particle level is at the level before the trigger is applied. Mathematically, trigger bias fraction, f_{bias} can be expressed as:

$$f_{bias} = \frac{\left(N_q^{\pi^+\pi^-}/N_{q+g}^{\pi^+\pi^-}\right)_{detector}}{\left(N_q^{\pi^+\pi^-}/N_{q+g}^{\pi^+\pi^-}\right)_{particle}} \quad (5.13)$$

$$\delta_{bias} = 1 - f_{bias} \quad (5.14)$$

where q and g referred to the quarks and gluons, respectively. The source parton, the parton associated with the $\pi^+\pi^-$, is identified by matching $\pi^+\pi^-$ with the closest parton in the $\eta - \phi$ space. Partons associated with each $\pi^+\pi^-$ event are counted at the detector and particle levels. PDG codes identify parton flavors and pions at the particle level.

The quark and gluon fraction calculated at the detector and particle level is shown in the left panel of figure 5.16. The quark fraction is shown in red, and the gluon fraction is shown in blue. The dashed line represents the detector level fraction, whereas the solid line represents the particle level fraction. The gluon-initiated events dominate the lower mass region, whereas the quark-initiated events dominate the

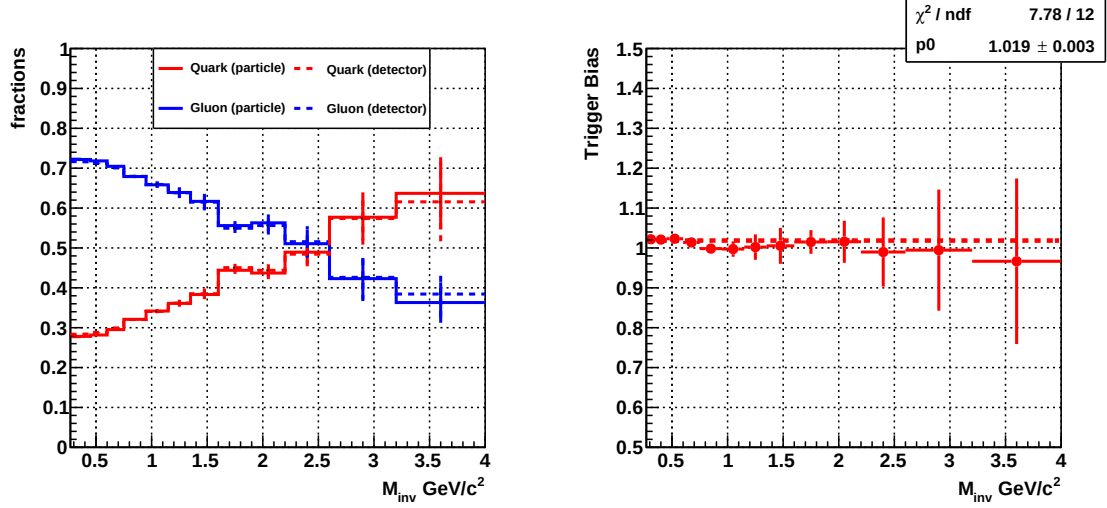


Figure 5.16: Trigger bias fraction as a function of invariant mass.

higher mass region, as expected. The calculated fraction, f_{bias} , is shown in the right panel of figure 5.16. The associated systematic uncertainty, δ_{bias} , is given by equation 5.14. The δ_{bias} is found to be very small over the range of invariant mass bins.

5.2.10.2 Systematic Uncertainty Associated with f_{fake} (δ_{fake})

The size of systematic uncertainty associated with the f_{fake} is estimated by varying the $n\sigma_\pi$ cut by 0.5 from both sides on the default PID cut $-1 < n\sigma_\pi < 2$. The difference between the ratio of the default fraction to the wider fraction ($-1.5 < n\sigma_\pi < 2.5$) and the ratio of the default fraction to the narrower fraction ($-0.5 < n\sigma_\pi < 1.5$) is assigned as a systematic uncertainty, δ_{fake} , as shown in figures 5.17–5.19 for each of the trigger. δ_{fake} is estimated for each cross-section bin.

5.2.10.3 Systematic Error Associated with f_{loss} (δ_{loss})

The systematic uncertainty associated with the f_{loss} is determined by studying the embedding description of the data. Figure 5.20 shows the reconstructed $n\sigma_{\pi^+}$ along the y-axis and $n\sigma_{\pi^-}$ along the x-axis of a true $\pi^+\pi^-$ pair from the embedding in

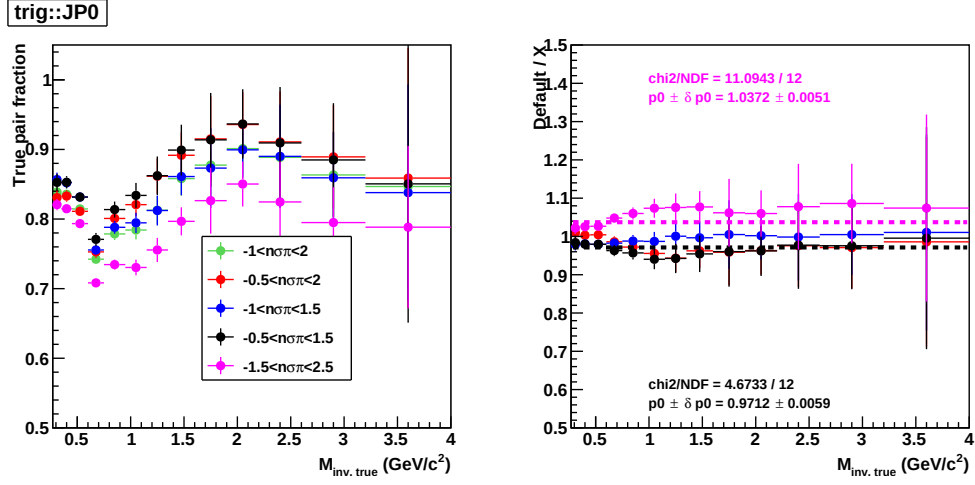


Figure 5.17: Systematic error associated to the f_{fake} for the JP0 trigger.

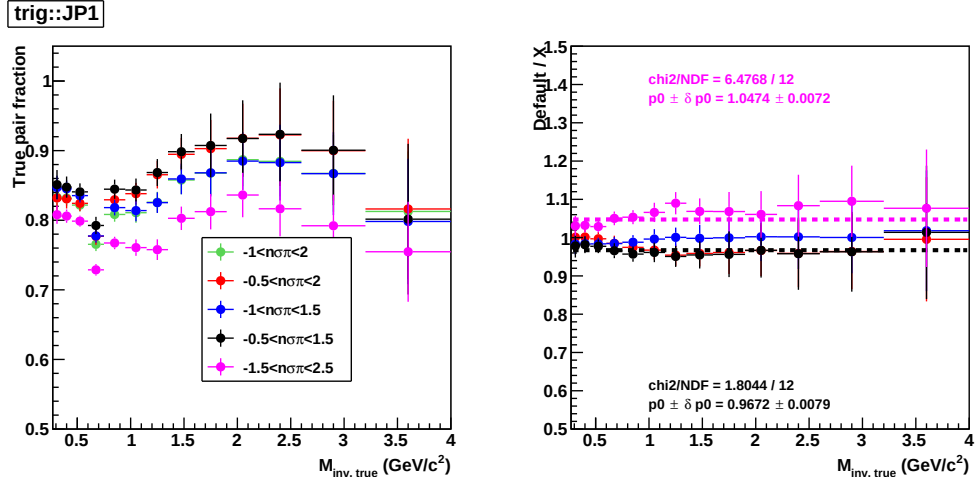


Figure 5.18: Systematic error associated to the f_{fake} for the JP1 trigger.

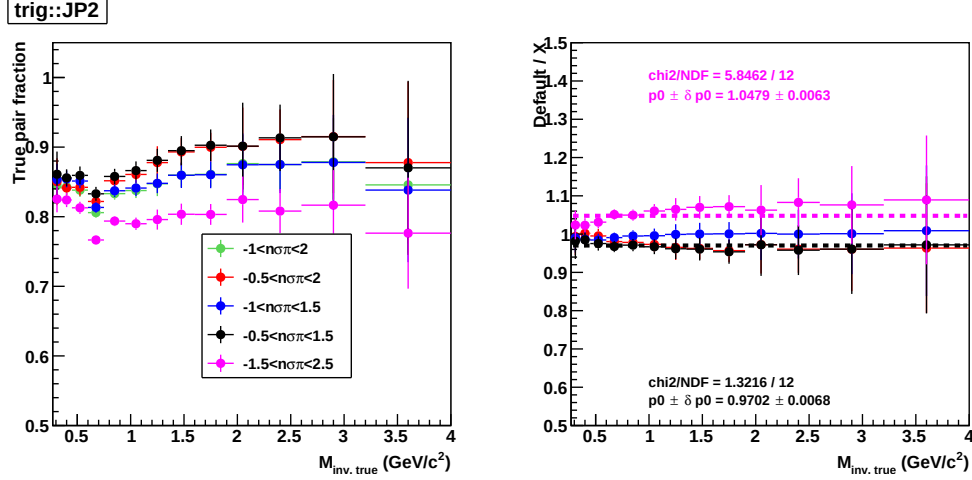


Figure 5.19: Systematic error associated to the f_{fake} for the JP2 trigger.

the lowest $M_{inv}^{\pi^+\pi^-}$ bin for JP0 trigger. This is just a representative plot to explain the general approach. All the data points displayed in the picture have a very good description of the data. However, embedding fails to describe the data in the region $n\sigma_\pi < -4$ (see figure B.6), and N_{true} fails to account for the $\pi^+\pi^-$ yields from this region.

To estimate the systematic uncertainty's size from f_{loss} (δ_{loss}), we examined the corresponding region on the opposite side ($n\sigma_\pi > 4$), where the embedding provides excellent data description. If the embedding accurately describes the entire data region, we expect similar trends in the $n\sigma_\pi < -4$ regions. The δ_{loss} is calculated as the fraction of the true pair in the $|n\sigma_\pi| > 4$ regions,

$$\delta_{loss} = \frac{N_{true, |n\sigma_\pi| > 4}}{N_{true}} \quad (5.15)$$

Figure 5.21 displays the δ_{loss} for JP0, JP1, and JP2 triggers. There is no trigger dependence in the δ_{loss} , which is about 6% in the overall mass region.

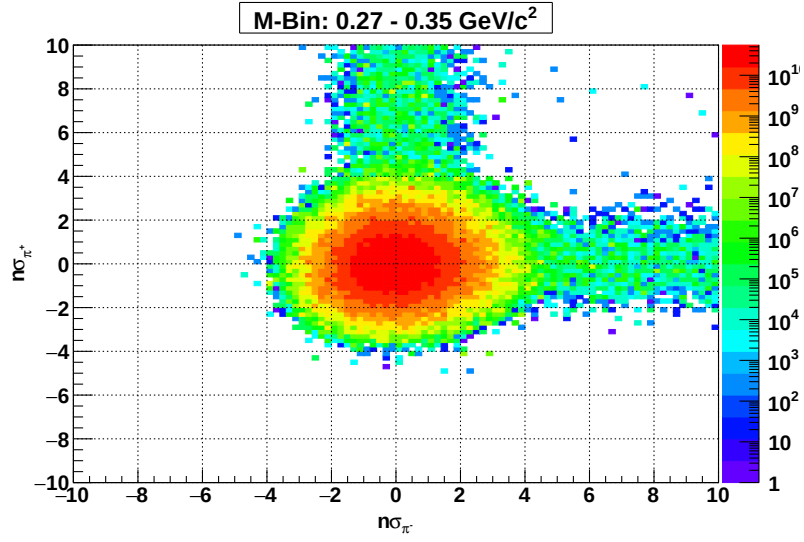


Figure 5.20: Reconstructed $n\sigma_{\pi^+}$ vs $n\sigma_{\pi^-}$ of a true $\pi^+\pi^-$ pair in the lowest invariant mass bin.

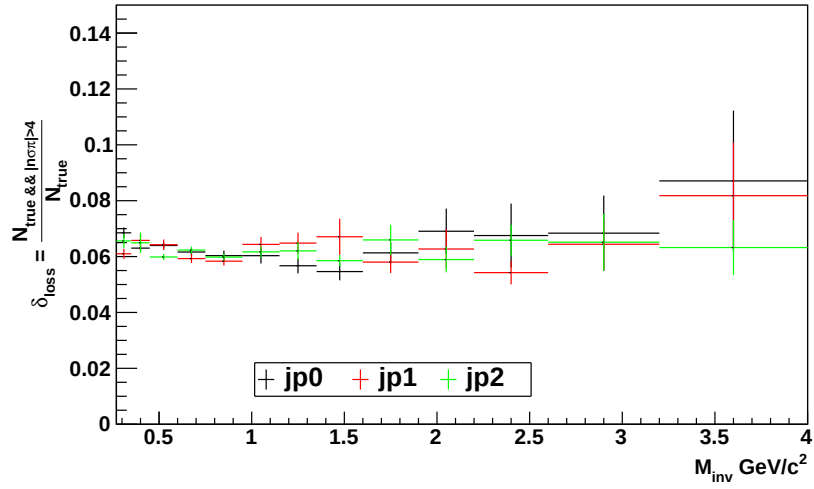


Figure 5.21: Systematic uncertainty associated with the f_{loss} .

5.2.10.4 Cross-Section Discrepancy Between Triggers (δ_{trg})

After correcting all the triggered cross-sections, a discrepancy remains, as shown in Figure 5.22, which displays the relative difference between the triggered and weighted average (final) cross-section. This difference represents a systematic uncertainty associated with the trigger effect. Instead of using a weighted average, we calculate the maximum offset between any triggered cross-section and the final cross-section as our systematic uncertainty.

It is reasonable to expect this effect to be more pronounced in the lower mass bins and less in the higher mass bins. However, we observe a larger difference in the higher mass bins, which we attribute to statistical fluctuations resulting from low statistics. To mitigate this, we combine the highest three bins for each trigger, as shown by the solid lines in figure 5.22. Then, we apply a single maximum offset value out of all triggers in those three bins, where the maximum offset comes from the JP0 trigger.

$$\delta_{trg} = \text{Max}(\delta_{trg,JP0}, \delta_{trg,JP1}, \delta_{trg,JP2}) \quad (5.16)$$

5.2.10.5 Embedding Sample Statistical Uncertainty ($\delta_{embstat}$)

The statistical fluctuation due to the embedding sample statistics is calculated from the covariance of the response matrix used for the unfolding. The diagonal element (N_i) for each cross-section bin i of the covariance matrix gives the statistical uncertainty of the embedding sample:

$$\delta_{N,i} = \sqrt{N_i} \quad (5.17)$$

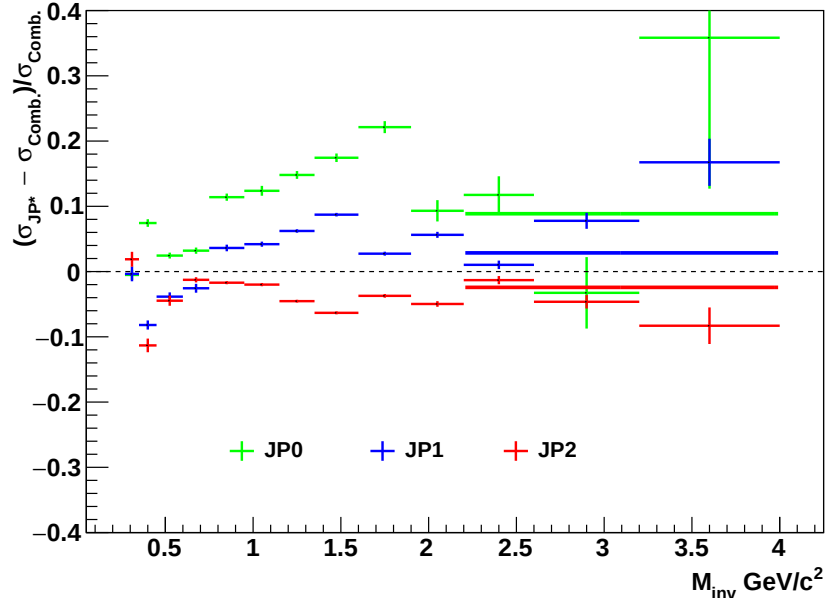


Figure 5.22: Trigger efficiency systematic uncertainty as a maximum relative difference between the triggered to the total cross-section. The last three bins are combined, and a single value of δ_{loss} is applied to those bins.

The $\delta_{N,i}$ is statistical fluctuation at the unfolding level, which is propagated to the cross-section level by multiplying with the same correction factor ($f_{xsec} = \frac{f_{fake} * f_{loss}}{L * BW * \epsilon_{trg} * \epsilon_{trk}}$) applied for the cross-section (equation 5.9):

$$\delta_{embstat,i} = f_{xsec,i} \times \delta_{N,i} \quad (5.18)$$

The $\delta_{embstat,i}$ is calculated for each trigger using the diagonal element of the covariance matrices, shown in figure 5.23. The weighted average of all $\delta_{embstat,i}$ values from all triggers is then applied as a systematic uncertainty due to the embedding sample statistics to the final cross-section result.

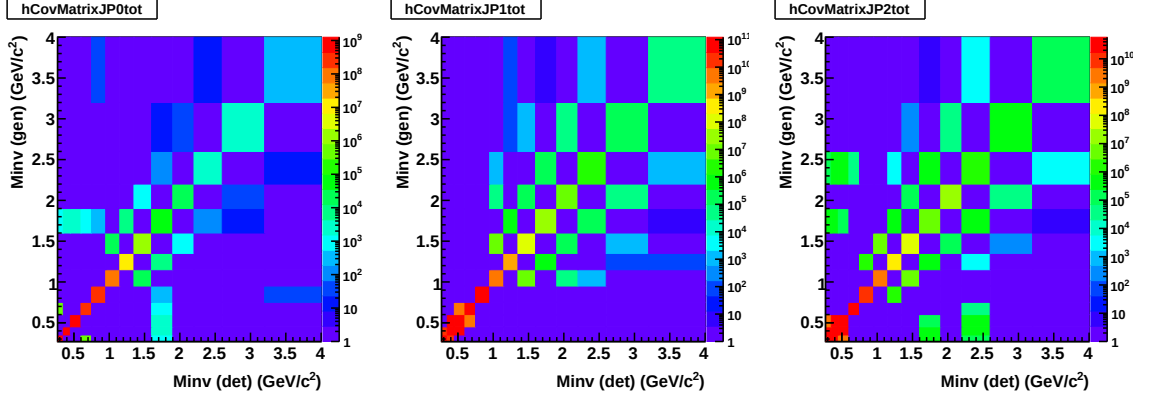


Figure 5.23: Covariance matrices for JP0, JP1, and JP2 triggers from left to right, respectively.

5.2.10.6 Total Systematic Uncertainty ($\delta\sigma_{sys}^{Syst.}$)

The total systematic uncertainties for the final cross-section is the quadrature sum of all the systematic uncertainties discussed above. The weighted averages of all triggered cases are the δ_{fake} and δ_{loss} . The δ_{bias} is calculated for the combined trigger, which directly applies to the final cross-section. The δ_{trg} is also directly applied to the final cross-section as it is calculated as the maximum discrepancy between the triggered and the final cross-section.

$$\delta\sigma_{mes}^{Syst.} = \sqrt{\delta_{trg}^2 + \delta_{fake}^2 + \delta_{loss}^2 + \delta_{bias}^2 + \delta_{embstat}^2} \quad (5.19)$$

Figure 5.24 displays a breakdown of each contributing systematic uncertainty to the final cross-section. The dominant systematic uncertainty comes from the trigger effect at the final cross-section. Table 5.8 presents numerical values of all the systematic uncertainties.

Bin No.	δ_{trg}	δ_{fake}	δ_{loss}	δ_{bias}	$\delta_{embstat}$	$\delta\sigma_{mes}^{Syst.}$
0	0.0189	0.0343	0.067	0.0218	0.00897	0.081
1	0.113	0.0392	0.0639	0.021	0.00626	0.138
2	0.0447	0.039	0.064	0.0231	0.00509	0.0904
3	0.032	0.0627	0.0613	0.0138	0.00682	0.0946
4	0.114	0.0789	0.0601	0.00169	0.0107	0.152
5	0.124	0.0992	0.0612	0.00281	0.012	0.17
6	0.148	0.108	0.0594	0.00214	0.0121	0.193
7	0.174	0.1	0.0589	0.00526	0.0104	0.21
8	0.221	0.0907	0.0599	0.0149	0.00896	0.247
9	0.0931	0.0841	0.0663	0.0155	0.0224	0.144
10	0.0131	0.0952	0.0613	0.0102	0.0244	0.117
11	0.0778	0.0969	0.0669	0.00551	0.0452	0.148
12	0.167	0.0573	0.0854	0.0334	0.12	0.232

Table 5.8: Final systematic uncertainties.

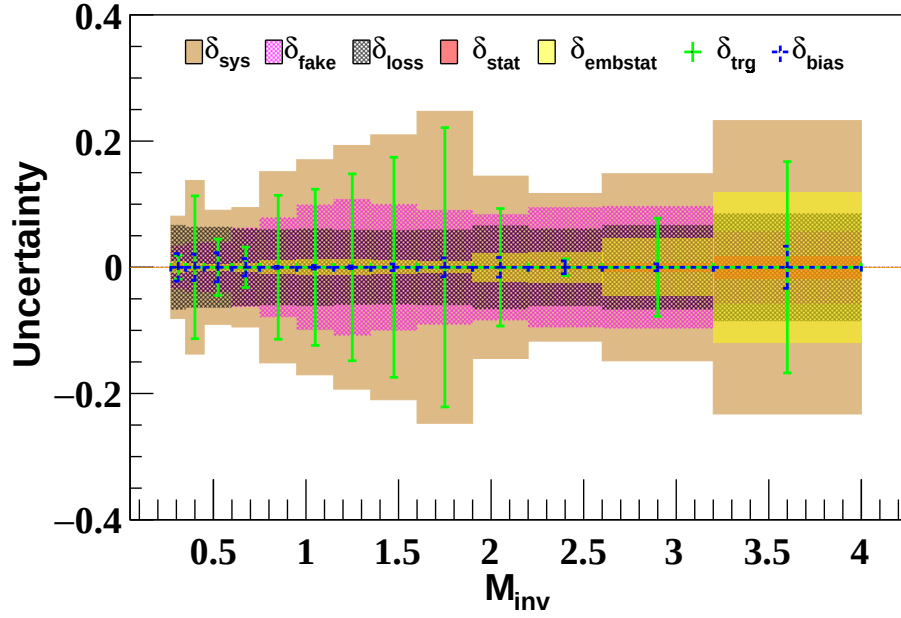


Figure 5.24: Breakdown of systematic uncertainties contributing to the final cross-section result.

5.3 Final Cross-Section Result

5.3.1 PYTHIA Cross-Section

The PYTHIA cross-section is calculated using standalone PYTHIA events from the embedding sample used in this analysis. All PYTHIA events are used, and the fiducial kinematic cuts on track p_T and η , as applied in the measured cross-section (see table 5.2), are retained at the PYTHIA level as well. The PYTHIA pions are selected using PDG codes (211 for π^+ and -211 for π^-). The $\pi^+\pi^-$ construction procedure is the same as in the measured construction, as discussed in section 5.2.3.

As the embedding sample is produced in the 13 partonic p_T ranges from 2 GeV to infinity, PYTHIA unpolarized $\pi^+\pi^-$ cross-section is calculated for each of the partonic p_T bins. The final cross-section combines cross-sections from all partonic bins. The total number of $\pi^+\pi^-$ events ($N_{p_T}^{\pi^+\pi^-}$) are counted and binned as a function of $M_{inv}^{\pi^+\pi^-}$ for each partonic p_T bins. The PYTHIA cross-section binning is the same as in the measured cross-section. The number of collision events processed ($N_{p_T}^{pro}$), which is smaller than the number of events tried or generated, and the partonic cross-section ($\hat{\sigma}_{p_T}$) is used to calculate luminosity (L_{p_T}) for each partonic p_T bin as,

$$L_{p_T} = \frac{N_{p_T}^{pro}}{\hat{\sigma}_{p_T}} \quad (5.20)$$

The unpolarized $\pi^+\pi^-$ cross-section for each p_T bin is then calculated as,

$$\frac{d\sigma_{p_T}^{\pi^+\pi^-}}{dM^{\pi^+\pi^-}} = \frac{N_{p_T}^{\pi^+\pi^-}}{L_{p_T} * \Delta M^{\pi^+\pi^-}} \quad (5.21)$$

where, $\Delta M^{\pi^+\pi^-}$ represents a bin width.

The final cross-section is the sum of all cross-sections from each of the partonic p_T

p_T Bin	$N_{p_T}^{tried}$	$N_{p_T}^{gen}$	$N_{p_T}^{pro}$	σ_{p_T}	$\delta\sigma_{p_T}$	L	δ L
2–3	1.77e7	3.69e6	3.33e6	9e+09	1.9e+08	0.00037	7.81e-06
3–4	1.43e7	3.69e6	3.31e6	1.46e+09	3.39e+07	0.00226	5.25e-05
4–5	1.52e7	3.69e6	3.3e6	3.54e+08	8.68e+06	0.0093	0.000228
5–7	1.79e7	3.69e6	3.28e6	1.52e+08	3.54e+06	0.0217	0.000506
7–9	1.91e7	3.69e6	3.29e6	2.49e+07	5.76e+05	0.132	0.00305
9–11	1.94e7	3.69e6	3.28e6	5.85e+06	1.37e+05	0.561	0.0131
11–15	2.2e7	3.69e6	3.28e6	2.3e+06	5.51e+04	1.42	0.0341
15–20	2.52e7	3.69e6	3.28e6	3.43e+05	8.14e+03	9.56	0.227
20–25	3.03e7	3.69e6	3.27e6	4.57e+04	1.09e+03	71.6	1.71
25–35	2.26e7	2.46e6	2.18e6	9.73e+03	275	224	6.34
35–45	2.35e7	2.46e6	2.18e6	470	12.1	4.65e+03	120
45–55	1.27e7	1.23e6	1.09e6	26.9	0.709	4.06e+04	1.07e+03
55–1	1.25e7	1.23e6	1.09e6	1.43	0.0398	7.61e+05	2.11e+04

Table 5.9: PYTHIA number of events and luminosity per partonic p_T bins. Cross-section (σ_{p_T}) is in picobarn (pb), luminosity (L) is in pb^{-1} , and partonic p_T is in GeV/c .

bin, which is shown (in black) in figure 5.15 (top panel), which follows a similar trend to the measured cross-section. The agreement of PYTHIA with the measured cross-section is within 10% up to $M_{inv}^{\pi^+\pi^-} \sim 2.25 \text{ GeV}/c^2$, above which the disagreement grows up to $\sim 50\%$.

5.3.2 Measured Cross-Section

The measured cross-section is a weighted average of JP0, JP1, and JP2 triggered cross-sections, depicted in red in the top panel of figure 5.25. The uncertainty shown is a quadrature sum of statistical and systematic uncertainties. The absolute PYTHIA cross-section and theoretical prediction from the JAM collaboration are also included for comparison. The agreement between the measured cross-section, PYTHIA, and theoretical prediction is excellent and falls within the uncertainty bounds. This consistency demonstrates the reliability of the analysis framework and its alignment with theoretical expectations. Notably, the measured cross-section encompasses $\pi^+\pi^-$ pro-

duction from quark and gluon fragmentation. Consequently, this marks a significant step toward constraining the gluon fragmentation function, complementing the results from the Belle experiment.

In the bottom panel, the green band represents the final systematic uncertainty, while the red band indicates the statistical uncertainty. The relative difference between the PYTHIA and measured cross-sections is shown as black closed circles. Additionally, the dashed blue lines depict the relative difference between the JAM DiFF prediction and the final cross-section. The 10% uncertainty from the luminosity measurement is not factored into the final uncertainty.

The theoretical prediction is sourced from the author of Ref. [58], which also details the fitting parameters and procedures. The JAM DiFF Prediction also includes quark and gluon fragmentation contributions to $\pi^+\pi^-$ production. The presented result is derived from an average of around 300 replicas, with the uncertainty being solely statistical. The theoretical curve is confined to the lower $M_{inv}^{\pi^+\pi^-}$ range (< 1.6 GeV/ c^2) due to the current framework's limitations in accurately predicting the higher mass region.

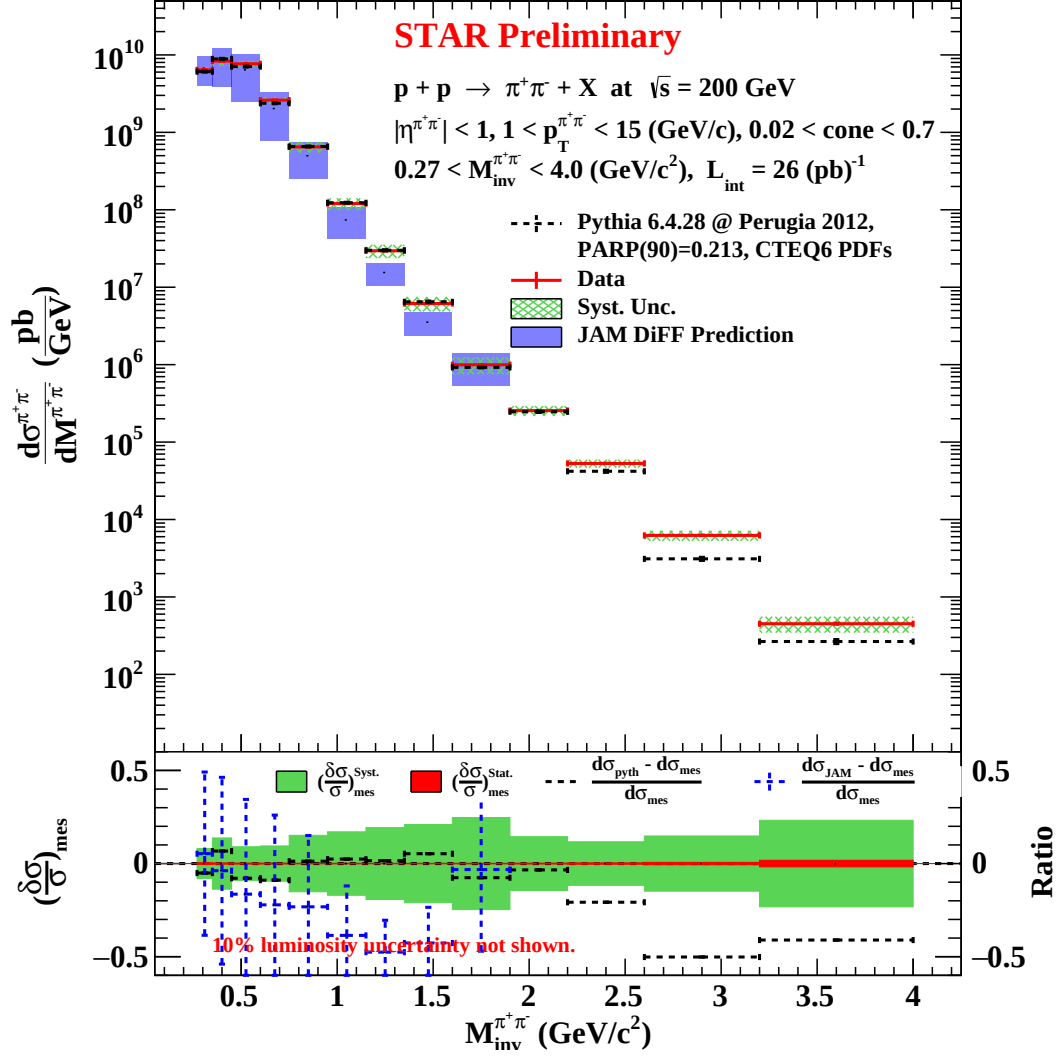


Figure 5.25: Top panel: measured unpolarized $\pi^+\pi^-$ cross-section, PYTHIA cross-section, and theory prediction from JAM collaboration as a function of $M_{\text{inv}}^{\pi^+\pi^-}$. Bottom panel: Relative difference between the measured and PYTHIA (dashed black) and JAMDiFF (dashed blue) cross-section on the right y-axis (Ratio = $\frac{d\sigma_{\text{pyth(JAMDiFF)}/dM - d\sigma_{\text{mes}}/dM}{d\sigma_{\text{mes}}/dM}$), and relative statistical ($(\frac{\delta\sigma}{\sigma})_{\text{mes}}^{\text{Stat.}}$) and systematic ($(\frac{\delta\sigma}{\sigma})_{\text{mes}}^{\text{Syst.}}$) errors on the left y-axis.

$M(GeV/c^2)$	$d\sigma_{mes}^{\pi^+\pi^-}(pb)$	$\delta\sigma_{mes}^{Stat.}(pb)$	$d\sigma_{pyth}^{\pi^+\pi^-}(pb)$	$\delta\sigma_{pyth}^{Stat.}(pb)$	$(\delta\sigma/d\sigma)_{mes}^{Syst.}$
0.27-0.35	6.41e+09	2.99e+07	6.09e+09	9.54e+07	0.081
0.35-0.45	8.33e+09	2.89e+07	8.89e+09	1.38e+08	0.138
0.45-0.6	7.68e+09	2.15e+07	7.07e+09	1.08e+08	0.0904
0.6-0.75	2.61e+09	6.13e+06	2.38e+09	3.41e+07	0.0946
0.75-0.95	6.49e+08	1.01e+06	6.57e+08	8.72e+06	0.152
0.95-1.15	1.2e+08	1.96e+05	1.23e+08	1.65e+06	0.17
1.15-1.35	2.96e+07	3.86e+04	3e+07	4.86e+05	0.193
1.35-1.6	6.16e+06	8.52e+03	6.48e+06	1.41e+05	0.21
1.6-1.9	9.98e+05	1.66e+03	9.22e+05	1.43e+04	0.247
1.9-2.2	2.55e+05	648	2.46e+05	5.02e+03	0.144
2.2-2.6	5.3e+04	194	4.2e+04	1.33e+03	0.117
2.6-3.2	6.23e+03	39.7	3.11e+03	144	0.148
3.2-4	451	8	266	16	0.232

Table 5.10: Final cross-section results.

CHAPTER 6

CONCLUSIONS AND OUTLOOK

The transversity distribution function is one of the leading order PDFs that provides transverse quark polarization in transversely polarized nucleons and is the least constrained PDF from experimental measurements. It can be accessed via the dihadron channel in polarized proton-proton collisions. In this channel, the transversity is coupled with the IFF, forming a collinear observable sensitive to the transversity PDF. To isolate the transversity, the IFF has to be measured independently, for example, in the e^+e^- process. In addition, the unpolarized fragmentation function is required. The Belle experiment has provided a cross-section that can provide the quark fragmentation; however, the knowledge of the gluon fragmentation has relied on various assumptions made in simulation (e.g., PYTHIA), resulting in large model-dependent uncertainty in the transversity. High-precision measurements of unpolarized dihadron cross-section in pp collisions can directly extract the unpolarized gluon FF, allowing for model-independent extraction of transversity.

The $\pi^+\pi^-$ azimuthal correlation asymmetry has been measured using polarized proton-proton collision data at the $\sqrt{s} = 200$ GeV from STAR Run 2015. This precision measurement is based on the proof-of-principle measurement in $p^\uparrow p$ collisions

from STAR Run 2006 [72]. This dataset provides the largest statistics for measurements of IFF asymmetry probing transversity in the valence quark region. The presented $\pi^+\pi^-$ asymmetry measurement agrees well with previous STAR measurements and is characterized by a mass peak at $M_{inv}^{\pi^+\pi^-} \sim 0.8 \text{ GeV}/c^2$ consistent with ρ -resonance, as predicted by the theory. A large asymmetry signal in the forward pseudorapidity region suggests a large transversity signal in the valence quark region. A smaller asymmetry signal in the backward region suggests a small transversity signal in the low- x region.

The unpolarized $\pi^+\pi^-$ cross-section differential in $M_{inv}^{\pi^+\pi^-}$ in a proton-proton collision at $\sqrt{s} = 200 \text{ GeV}$ has been measured for the first time using STAR Run 2012 data. The measured cross-section shows good agreement with the PYTHIA cross-section and the theoretical prediction from the JAM collaboration [58, 155]. The early work on the global extraction of transversity [62, 61] uses available SIDIS and e^+e^- data, which suffer from the large uncertainty associated with the unknown gluon fragmentation function. The inclusion of the STAR asymmetry results in $p^\uparrow p$ [72, 86] already shows a significant impact on the valence quark transversity [63, 58]. This unpolarized cross-section result in pp is much needed for constraining the gluon fragmentation function. The high-precision IFF asymmetries obtained from STAR Run 2015, combined with the unpolarized cross-section result from STAR Run 2012 in pp collision, complement SIDIS and e^+e^- data, providing a foundation for a model-independent extraction of transversity with greater precision than previously possible.

The significant systematic uncertainty observed in the IFF asymmetries, predominantly driven by the PID, undergoes further investigation following the preliminary release, which now includes TOF information. A thorough analysis is outlined in appendix A.1. The findings indicate that the background's contribution to the signal asymmetry is negligible, and this signal asymmetry remains consistent irrespective of

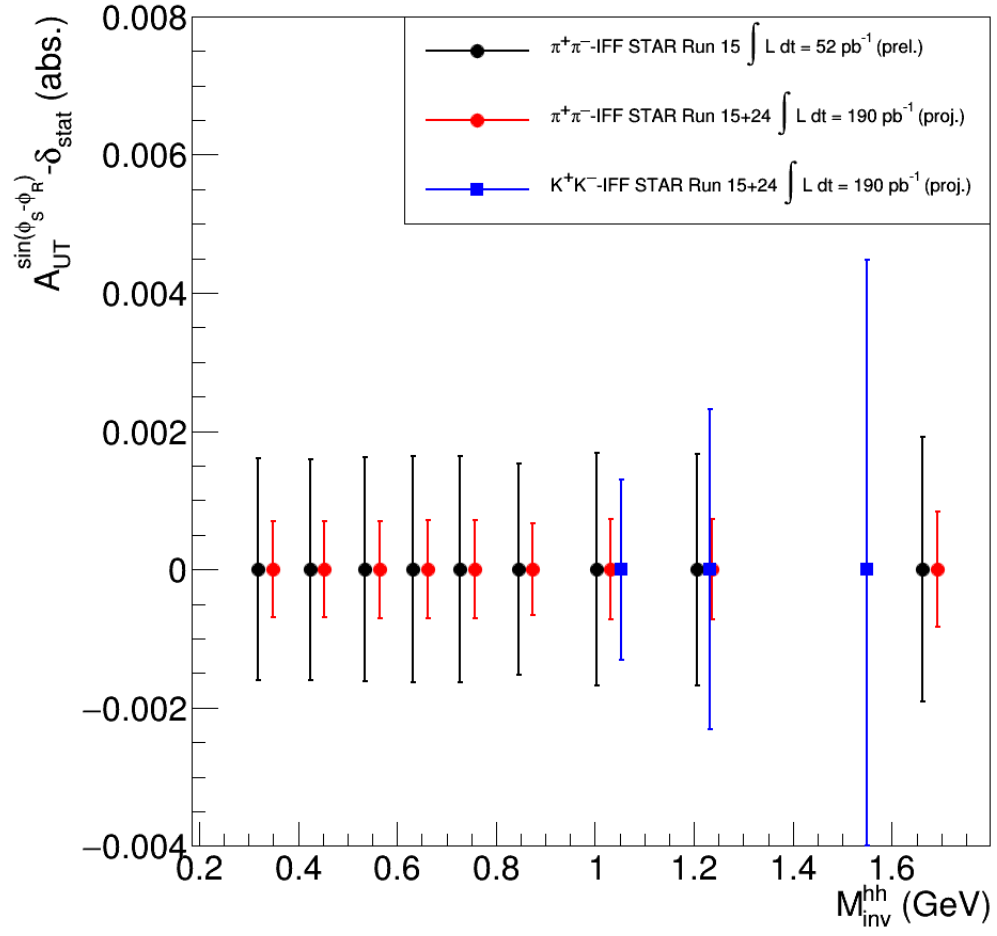


Figure 6.1: IFP asymmetries projection for $\pi^+\pi^-$ and K^+K^- for the Run 2015 and 2024 combined.

varying background sizes.

Hence, attributing systematic uncertainty by assuming the background asymmetry is akin to the signal's asymmetry is impractical. Instead, it can be estimated as the absolute difference between the asymmetries originating from significantly distinct background sizes. In the final publication, a considerably improved systematic uncertainty, closely aligned with the magnitude of the statistical uncertainty, is anticipated.

Exploring flavor-dependent transversity is of paramount importance, entailing the measurement of asymmetry and cross-section for various hadron pairs, including KK , $K\pi$, and $\pi\pi$, in polarized pp collisions. The STAR experiment has collected extensive data from Run 2022 at $\sqrt{s} = 510$ GeV and has a polarized pp Run scheduled at $\sqrt{s} = 200$ GeV in 2024. The combined dataset from Run 2015+2024 and Run 2017+2022, each at center-of-mass energies of 200 and 510 GeV, respectively, provides a robust statistical foundation for analysis. This encompasses the investigation of low multiplicity hadrons, such as kaons, which offer effective means to probe s -quark transversity.

Figure 6.1 estimates the anticipated statistical uncertainty for K^+K^- pair production utilizing the combined STAR Run 2015+2024 dataset. Given the comparatively lower kaon statistics, projections are generated using a reduced track p_T cut (> 0.2 GeV/ c) and a more stringent PID analysis, elaborated on in appendix A.5.8. Additionally, the figure provides a statistical projection for $\pi^+\pi^-$ pair production, employing the same kinematic binning as depicted in figure 4.1. We expect the high-statistics dataset from Run 2015+2024 will facilitate precise measurements of $\pi^+\pi^-$ production and mark the inaugural measurement of K^+K^- production in polarized pp collisions.

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APPENDIX A

RUN 2015 IFF ANALYSIS

A.1 PID Systematic Studies: After Preliminary

The PID dominates the large systematic uncertainty in the asymmetry results, estimated by a more traditional approach relying only on the TPC dE/dx . The size of systematic uncertainty is as expected, as the analysis sits in the momentum region (track $p_T > 1.5$ GeV/ c) where all hadronic species dE/dx bands almost overlap. Another PID detector, TOF, alone isn't useful as it cannot provide a good PID in this regime.

Alternatively, the simultaneous use of both PID detectors appears useful in understanding the systematic uncertainty associated with the PID. Similar to the $n\sigma$ in the TPC ($n\sigma_{\pi}^{TPC}$), TOF provides $n\sigma$ ($n\sigma_{\pi}^{TOF}$) information useful for the PID. The simultaneous use of the TPC and TOF PID requires a track to have both pieces of information. However, the TOF misses most of the time, which excludes many tracks leading to a reduction in the $\pi^+\pi^-$ events by more than 50% (in run 2015 sample).

To quantify PID systematic uncertainty alternatively, three different data samples based on the PID selection cuts applied are considered:

- Pure TPC sample (S^{TPC}): Preliminary result discussed in Chapter 4 is based on this sample where only $n\sigma_{\pi}^{TPC}$ is used for the PID. This sample has the highest background fraction.
- TPC and/or TOF sample ($S^{TPC\text{or}TOF}$): This sample uses the TOF PID and the TPC if TOF information is available. Otherwise, only TPC PID is used. This sample has a reduced background than the S^{TPC} sample, but still, significant background enters from the tracks that don't have TOF.
- TPC and TOF sample ($S^{TPC\&TOF}$): In this sample, all the tracks require both TPC and TOF information, and both PIDs are used to get high-purity pions. This sample removes up to $\sim 99\%$ of the background, and a high-purity sample is obtained at the expense of 55% of the total statistics.

Additionally, the modification in the multi-gaussian fit estimating particle fractions in the pion signal region is reconsidered. Based on the momentum dependence of the dE/dx , it is challenging to identify protons and kaons as these particles almost overlap in the higher momentum region $p > 1 \text{ GeV}/c$, where this analysis sits. This issue can be resolved by accommodating kaon and proton into a single Gaussian. This is a reasonable approach as these species are almost inseparable in this region, and there is no need for separate identification.

Furthermore, the different PID selection cut is employed for the tracks with TPC and TOF information. Figure A.1 shows a 2D plot of $n\sigma_{\pi}^{TPC}$ (along y -axis) and $n\sigma_{\pi}^{TOF}$ (along x -axis) for π^+ in varying $p_T^{\pi^+\pi^-}$ bins. The central blob represents the pion, and the blobs on the right are kaons and protons, respectively. The kaons and protons move closer as the $p_T^{\pi^+\pi^-}$ increases and become inseparable. The proton and kaon contribution can be removed significantly by employing a polynomial cut, removing those background regions. What we left off after removing these backgrounds

is displayed in figure A.2, where mostly the pion and electrons are present. As electrons are well separated, the multi-gaussian fit on $n\sigma_\pi$ would result in high purity pions.

Figure A.3 compares the $n\sigma_\pi$ distribution between the S^{TPC} and $S^{TPCorTOF}$ samples, where the effect of the polynomial cut is visible, which only reduces the proton and kaon backgrounds as intended. The multi-gaussian fit accounting proton+kaon as single species for $S^{TPCorTOF}$ sample is shown in figure A.4, where the pion fraction reaches up to 96% in the lowest $p_T^{\pi^+\pi^-}$ bin; However, the effect is small in the higher $p_T^{\pi^+\pi^-}$ region.

The multi-gaussian fit for $S^{TPC\&TOF}$ sample, corresponding to the figure A.2, is shown in figure A.5. In the lower $p_T^{\pi^+\pi^-}$ region, this approach successfully removes the proton+kaon background, and negligible background contribution appears in the higher region. The $\pi^+\pi^-$ level purity for $S^{TPCorTOF}$ and $S^{TPC\&TOF}$ samples in all asymmetry kinematic bins are shown in figure A.6.

We have used these three samples of significantly different pion purity to estimate the PID systematic uncertainty. First, the background asymmetry contribution is quantified by measuring and comparing asymmetries from all three samples. Figures A.8-A.10 show such asymmetries compared with the preliminary results (S^{TPC} sample). Surprisingly, albeit with different background sizes, asymmetries are consistent within the statistical uncertainty signaling that the background asymmetries are negligible. This strongly ruled out the assumption that the background asymmetries are “about the same size” as the signal asymmetries while assigning PID systematic uncertainty scaling the signal asymmetry linearly by the background fraction.

The smaller size of the background asymmetry is implied from the measurement in the signal region, which needs further proof. This is possible by measuring asymmetry

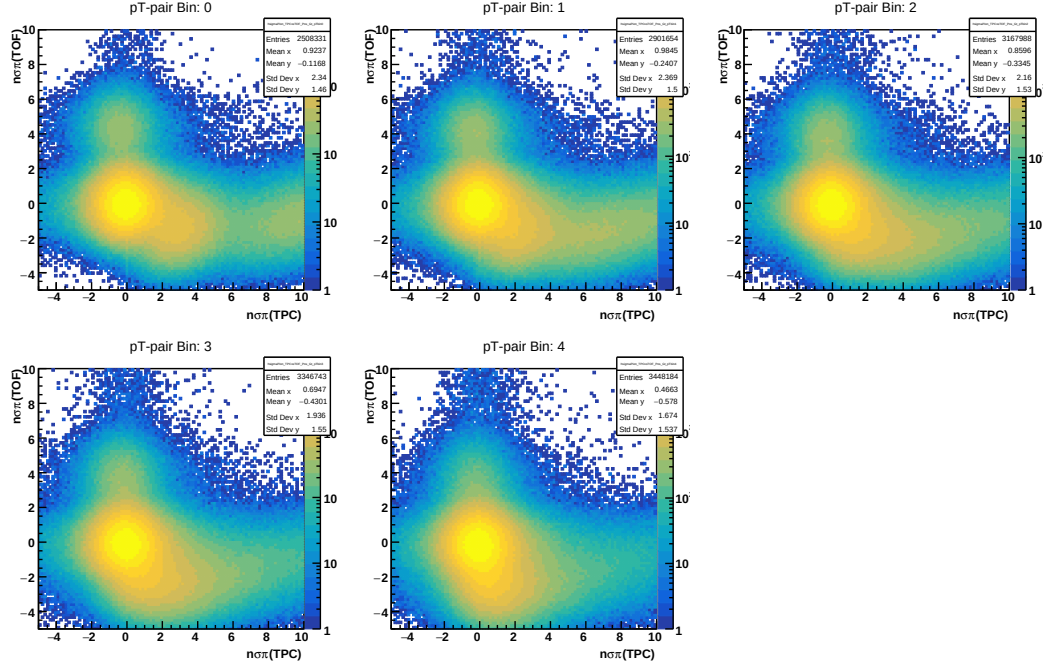


Figure A.1: $n\sigma_{\pi^+}(\text{TPC})$ vs $n\sigma_{\pi^+}(\text{TOF})$ in $\eta > 0$, in different p_T bins for $S^{\text{TPC}\&\text{TOF}}$ sample.

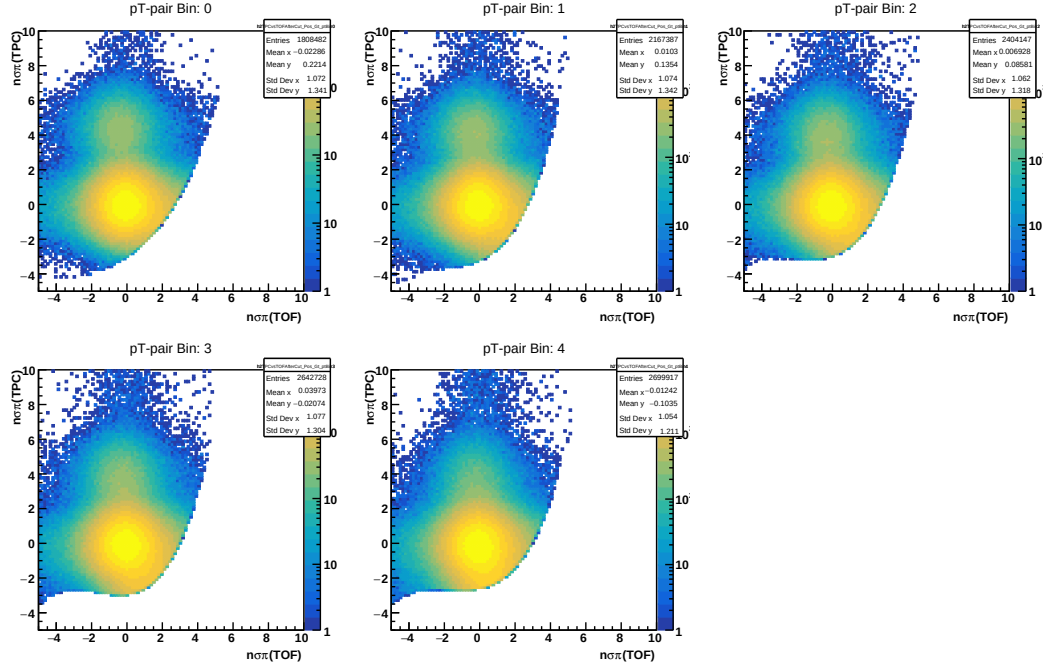


Figure A.2: $n\sigma_{\pi^+}(\text{TPC})$ vs $n\sigma_{\pi^+}(\text{TOF})$ in $\eta > 0$, in different p_T bins after polynomial PID cut for $S^{\text{TPC}\&\text{TOF}}$ sample.

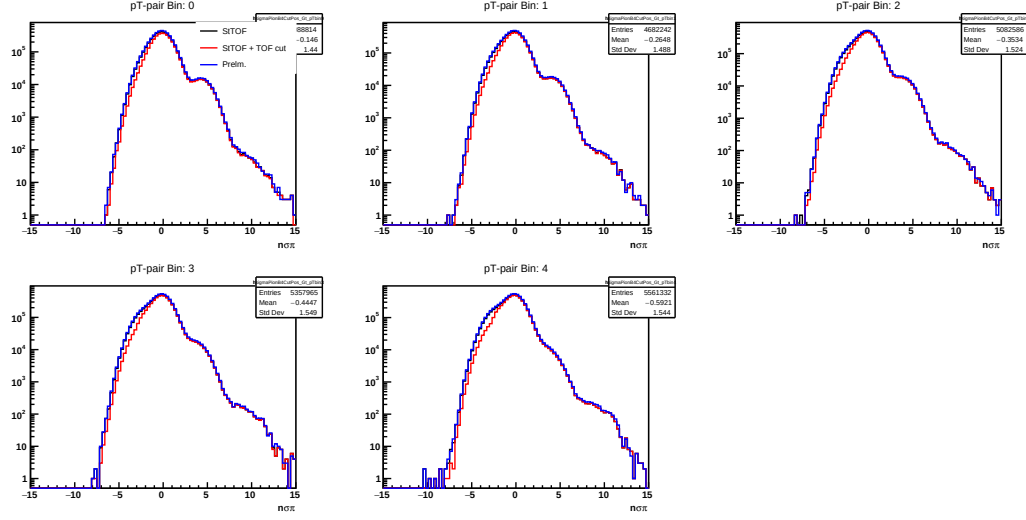


Figure A.3: $n\sigma_{\pi^+}$ before (black) and after (red) polynomial TOF cut. Blue curve is used for preliminary PID.

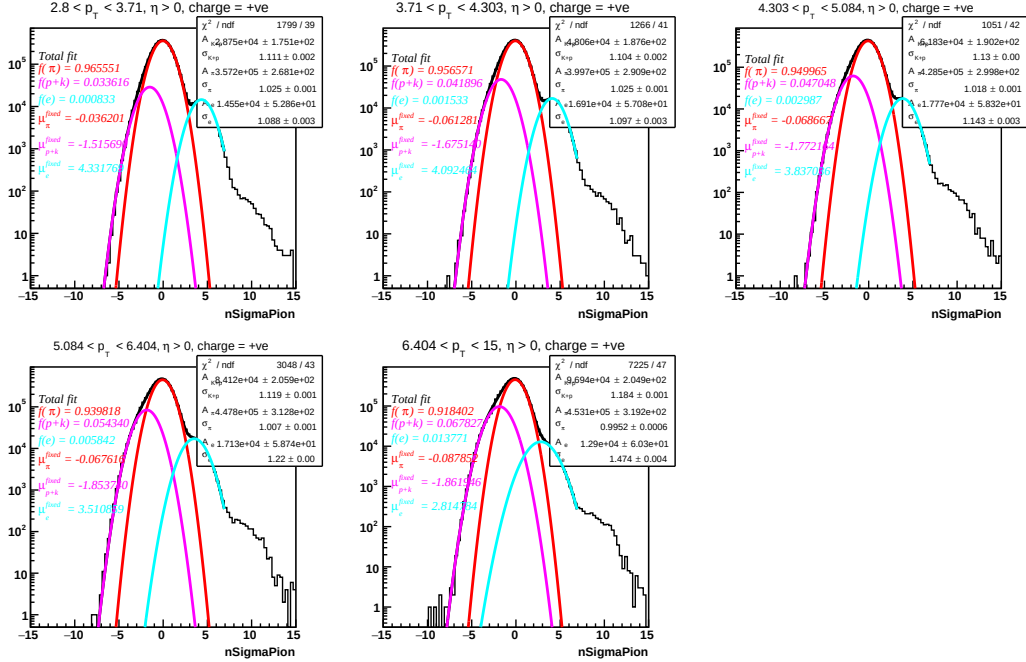


Figure A.4: $n\sigma_{\pi}$ multi-gaussian fit in p_T bins, for $charge > 0$, $\eta > 0$ from $S_{TPCorTOF}$ sample.

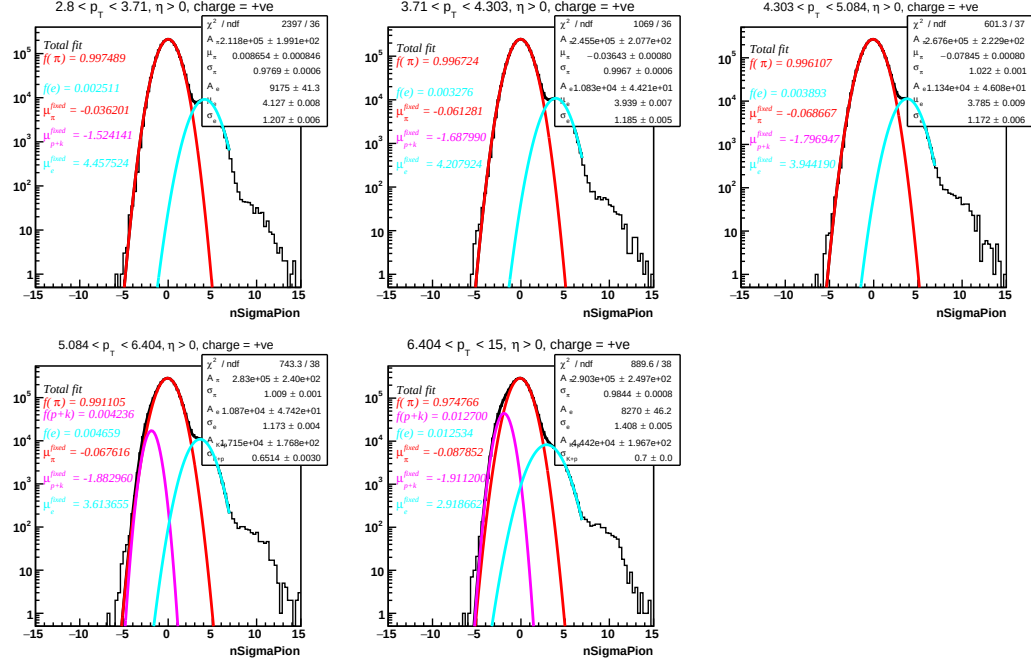


Figure A.5: $n\sigma_\pi$ multi-gaussian fit in p_T bins, for $charge > 0$, $\eta > 0$ from $S^{TPC\&TOF}$ sample.

for the background in the $S^{TPC\&TOF}$ sample. The defined signal and the background regions are depicted in figure A.11 depicts the defined signal and background regions. Provided limited statistics in the background region, asymmetry as a function of $M_{inv}^{\pi^+\pi^-}$ with $p_T^{\pi^+\pi^-}$ integrated and as a function of $p_T^{\pi^+\pi^-}$ with $M_{inv}^{\pi^+\pi^-}$ integrated is measured, shown in figure A.12. Within the statistical uncertainty, all the background asymmetries are consistent with zero.

Observing consistency between the signal asymmetries of different background sizes and zero background asymmetry suggests a different approach to quantifying PID systematic uncertainty. The background effect is thus quantified as the absolute difference between the $S^{TPC\&TOF}$ and $S^{TPC\&TOF}$ samples. Moreover, the asymmetry from $S^{TPC\&TOF}$ is considered a final asymmetry as it poses an improved pion fraction without compromising statistics. Figure A.13 shows such difference in $A_{UT}^{\sin(\phi_{RS})}$ in multi-binning differential in $p_T^{\pi^+\pi^-}$ and $M_{inv}^{\pi^+\pi^-}$, and the corresponding final result

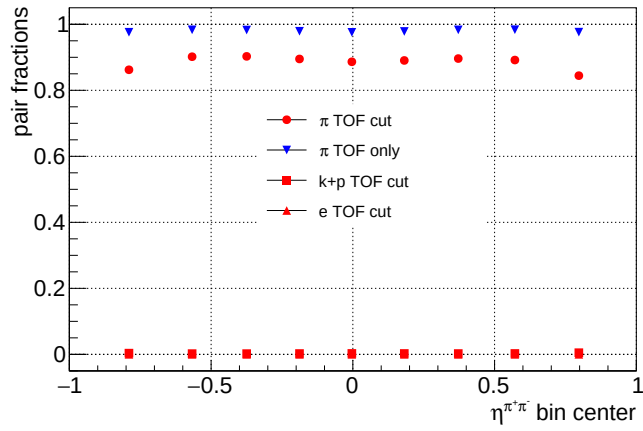
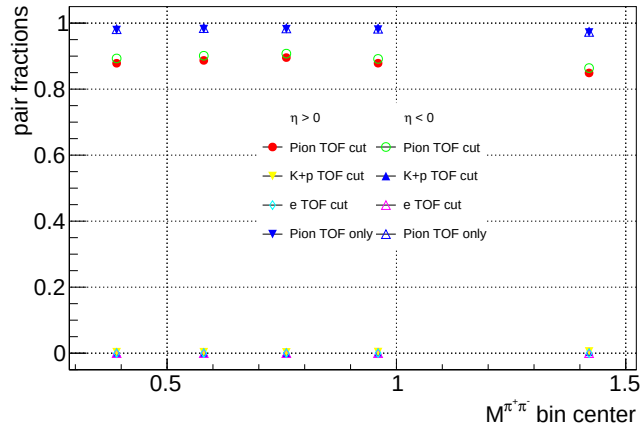
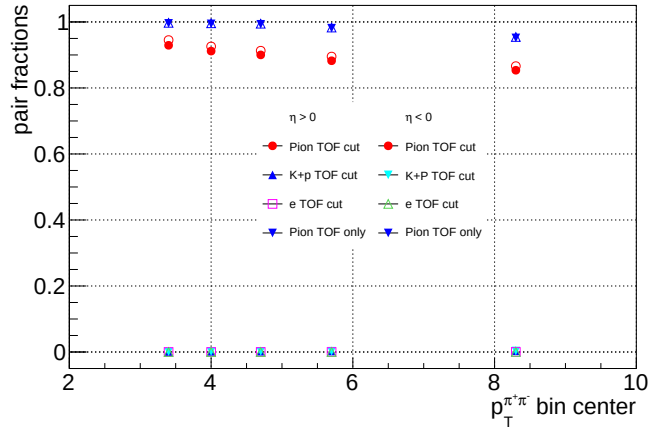


Figure A.6: $\pi^+\pi^-$ fractions for different purity samples.

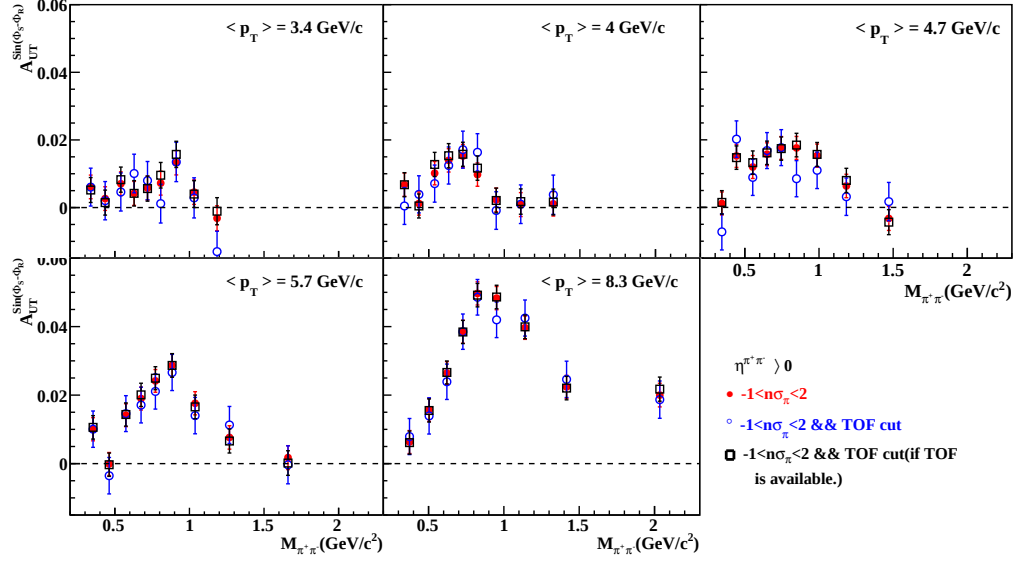


Figure A.7: $A_{UT}^{\sin(\phi_{RS})}$ vs $M_{inv}^{\pi^+\pi^-}$ for a varying pair purity samples. The purity is varied by applying PID cuts using TPC and TOF $n\sigma_\pi$.

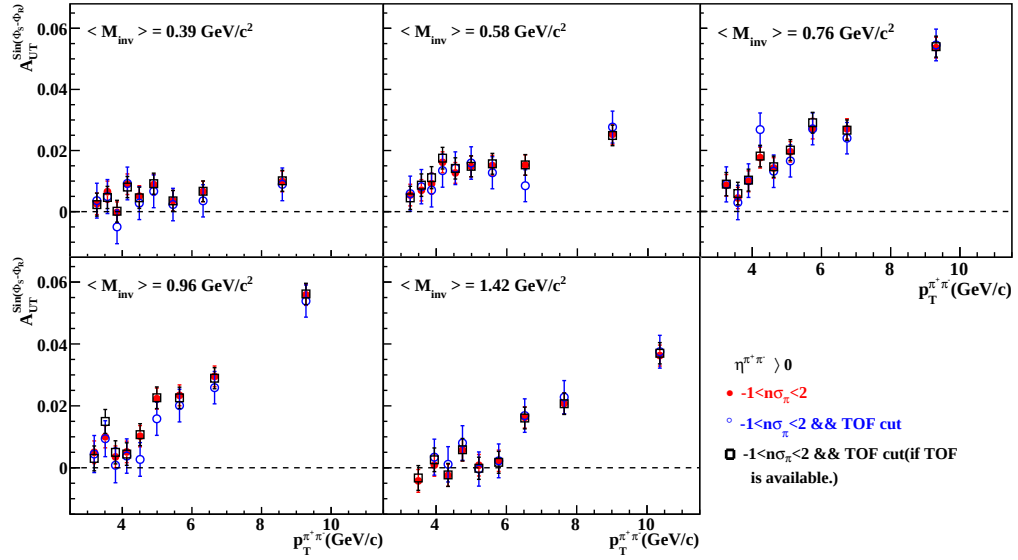


Figure A.8: $A_{UT}^{\sin(\phi_{RS})}$ vs $p_T^{\pi^+\pi^-}$ for varying pair purity samples. The purity is varied by applying PID cuts using TPC and TOF $n\sigma_\pi$.

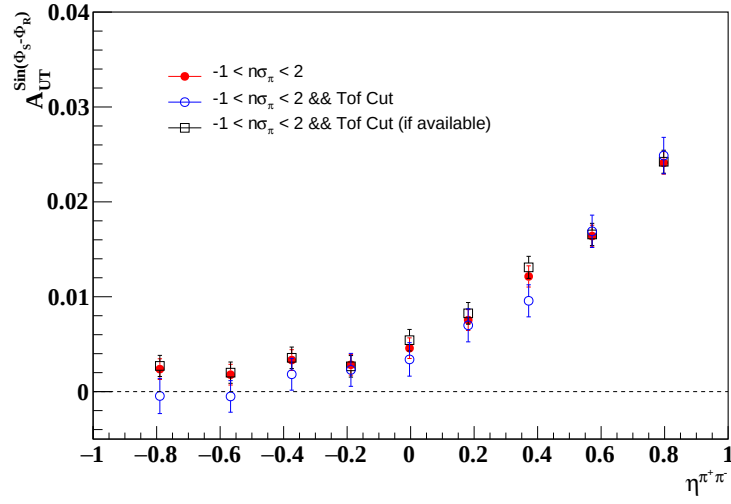


Figure A.9: $A_{UT}^{\sin(\phi_{RS})}$ in $\eta^{\pi^+\pi^-}$ bins for varying purity samples.

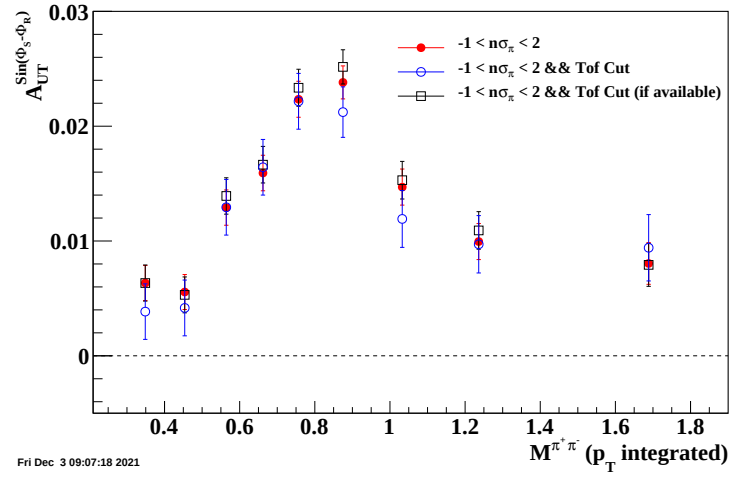


Figure A.10: $A_{UT}^{\sin(\phi_{RS})}$ in $M_{inv}^{\pi^+\pi^-}$ bins for varying purity samples.

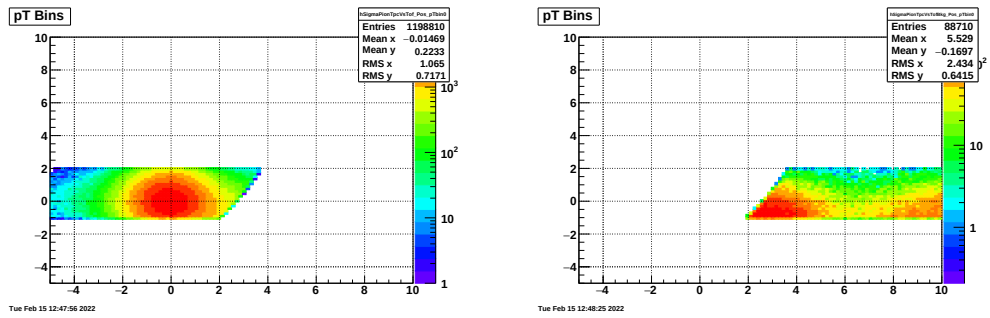


Figure A.11: Left: pion signal Right: Pion background.

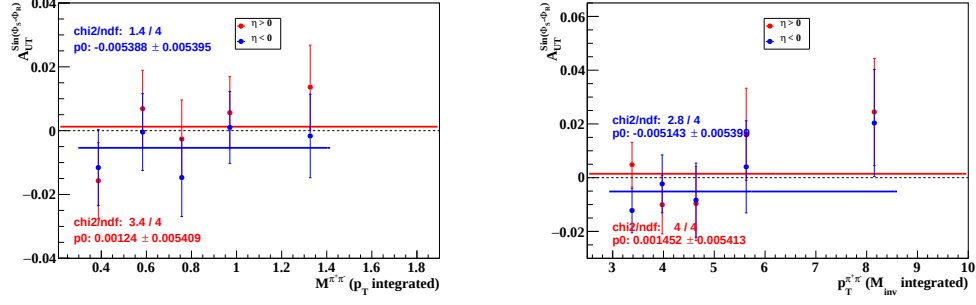


Figure A.12: Left: A_{UT}^{bkg} vs M_{inv} , p_T integrated. Right: A_{UT}^{bkg} vs p_T , M_{inv} integrated.

is displayed in figure A.14. This way of quantifying accurately reflects the background effects. The new PID systematic uncertainty is small within the statistical uncertainties. A similar level of improvement in the PID systematic uncertainty is observed in all kinematic bins. These improved results, which are not preliminary, will undergo final publication.

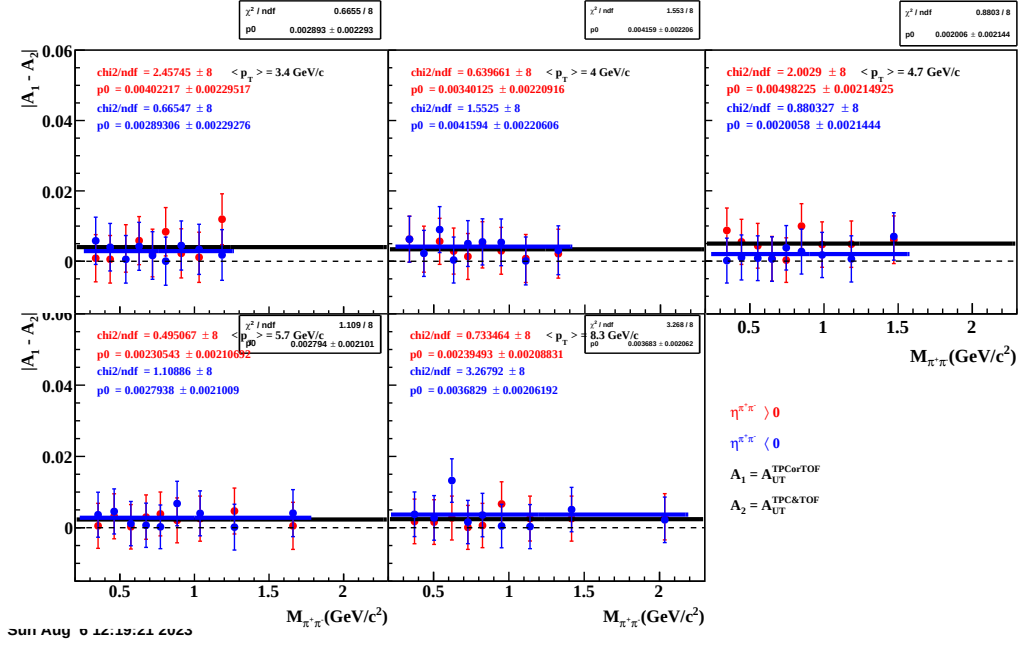


Figure A.13: Absolute difference in asymmetries between $S^{TPCorTOF}$ and $S^{TPC\&TOF}$ samples. The difference is assigned as a systematic uncertainty associated with the PID.

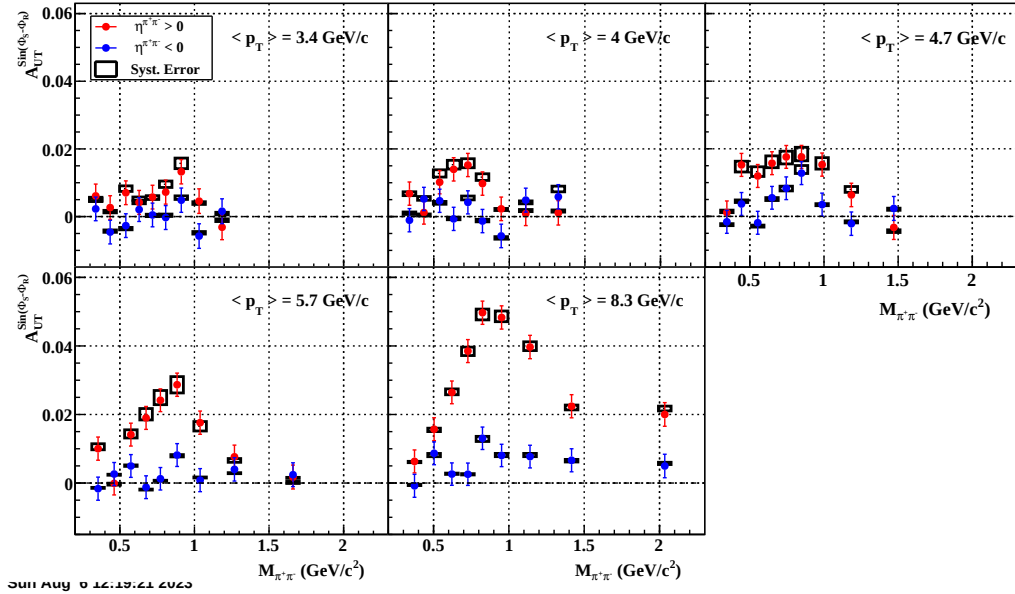


Figure A.14: Final $A_{UT}^{\sin(\phi_{RS})}$ as a function of $M_{inv}^{\pi^+\pi^-}$ differential in $p_T^{\pi^+\pi^-}$ and $\eta^{\pi^+\pi^-}$ from $S^{TPCorTOF}$ sample, with new PID systematic uncertainty.

A.2 Run 2015 Data And Embedding Comparison

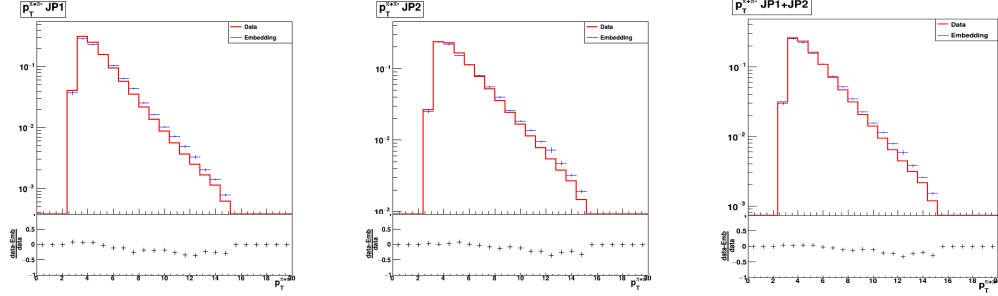


Figure A.15: $p_T^{\pi^+\pi^-}$ comparison for JP1, JP2, and combined JP from left to right respectively.

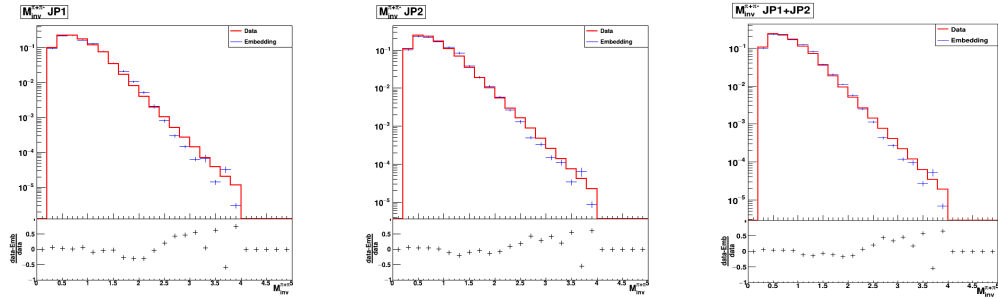


Figure A.16: $M_{inv}^{\pi^+\pi^-}$ comparison for JP1, JP2, and combined JP from left to right respectively.

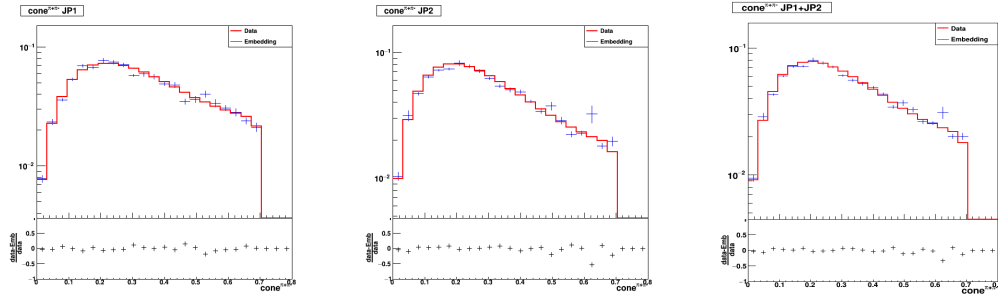


Figure A.17: Cone comparison for JP1, JP2, and combined JP from left to right respectively.

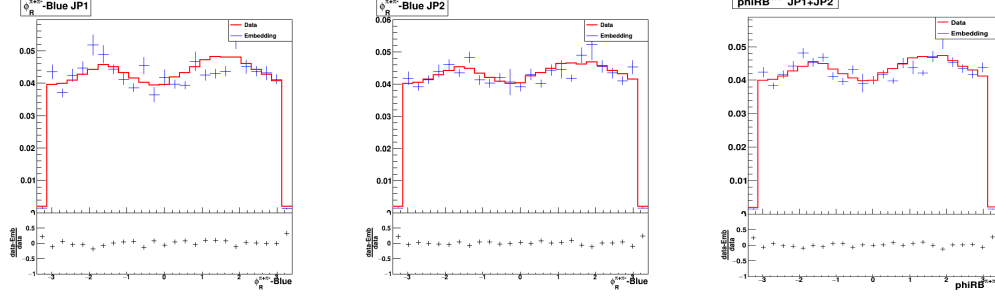


Figure A.18: $\phi_R(\text{Blue})$ comparison for JP1, JP2, and combined JP from left to right respectively.

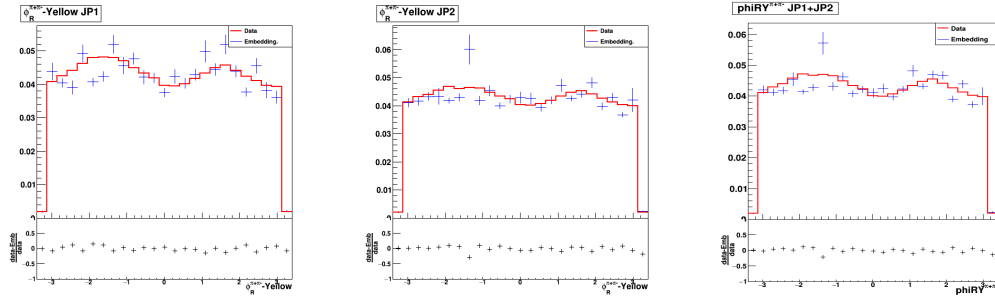


Figure A.19: $\phi_R(\text{Yellow})$ comparison for JP1, JP2, and combined JP from left to right respectively.

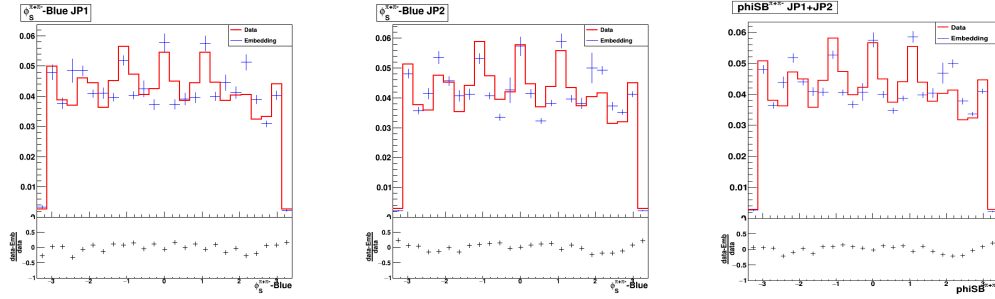


Figure A.20: $\phi_S(\text{Blue})$ comparison for JP1, JP2, and combined JP from left to right respectively.

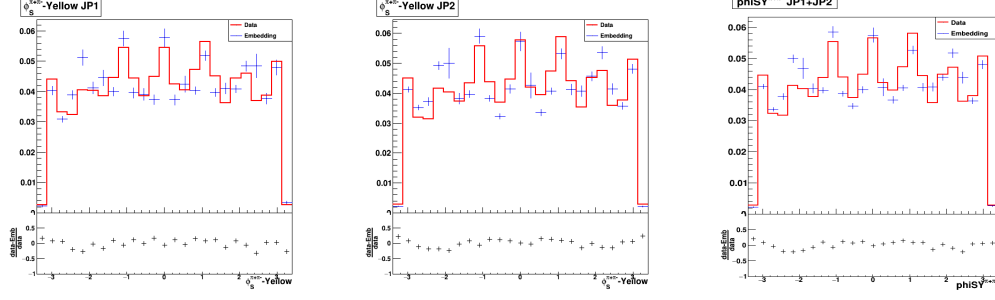


Figure A.21: $\phi_S(\text{Yellow})$ comparison for JP1, JP2, and combined JP from left to right respectively.

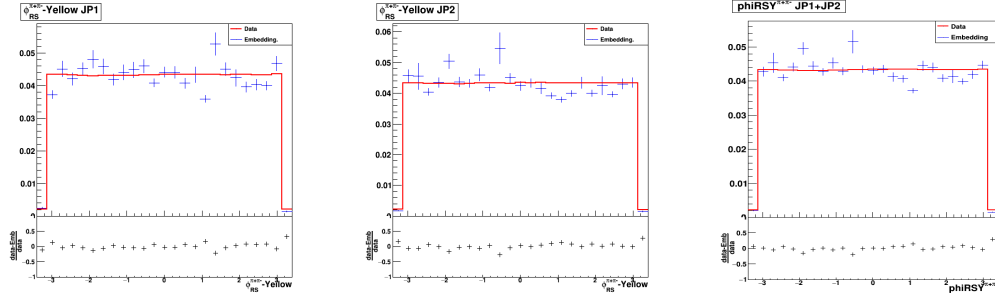


Figure A.22: $\phi_{RS}(\text{Yellow})$ comparison for JP1, JP2, and combined JP from left to right respectively.

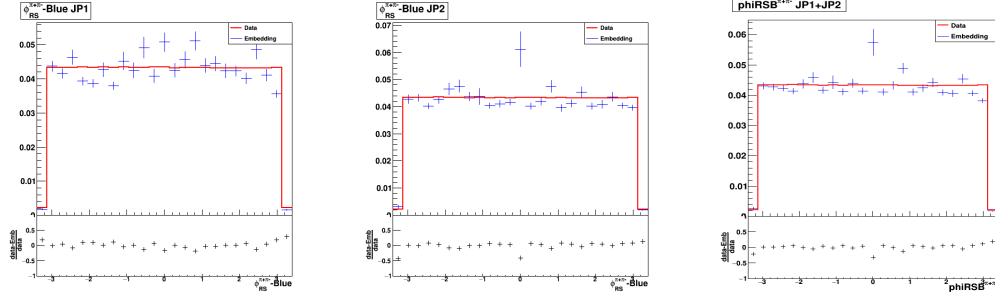


Figure A.23: $\phi_{RS}(\text{Blue})$ comparison for JP1, JP2, and combined JP from left to right respectively.

A.3 Run 15 Spin Patterns and Configurations

Blue	Yellow	8-bit	Decimal Eqv.
Up	Up	0011 0011	51
Down	Down	0101 0101	85
Up	Down	0011 0101	53
Down	Up	0101 0011	83

Table A.1: Beam spin configuration.

A.4 Run List

16065023 16065024 16065025 16065026 16065027 16065028 16065036 16065037 16065038
16065039 16065041 16065042 16065044 16065045 16065046 16065047 16065048 16065060
16065061 16065072 16066001 16066002 16066003 16066004 16066005 16066006 16066007
16066008 16066009 16066011 16066012 16066015 16066016 16066017 16066018 16066019
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16067015 16067016 16067017 16067019 16067020 16067021 16067022 16067039 16067040
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16068030 16068032 16068034 16068035 16068036 16068037 16068038 16068039 16068040
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16069005 16069006 16069007 16069008 16069009 16069010 16069011 16069012 16069016

16069049 16069053 16069054 16069055 16069060 16069062 16069063 16069064 16069065
16069066 16069067 16070003 16070004 16070005 16070006 16070008 16070009 16070010
16070012 16070013 16070014 16070015 16071016 16071018 16071021 16071022 16071023
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16093017 16093018

A.5 Asymmetry Results

A.5.1 A_{UT} vs $M_{inv}^{\pi^+\pi^-}$ (Multi-Binning)

$< M_{inv} >$	$\eta^{\pi^+\pi^-} > 0$			$\eta^{\pi^+\pi^-} < 0$		
	A_{UT}	σ_{stat}	σ_{sys}	A_{UT}	σ_{stat}	σ_{sys}
$< p_T > = 3.4 \text{ GeV}/c$						
0.3381	0.008601	0.003454	0.002086	0.002455	0.00346	0.0006034
0.435	0.003831	0.003464	0.0009292	-0.00192	0.003458	0.0005142
0.54	0.005288	0.003465	0.001282	-0.001011	0.003467	0.0003895
0.6291	0.001614	0.003491	0.0004979	0.001942	0.00348	0.0005192
0.7189	0.005459	0.00347	0.001324	0.002428	0.003467	0.0005992
0.8074	0.007763	0.003503	0.001883	0.0005015	0.003497	0.0003476
0.9117	0.01347	0.00356	0.003268	0.005114	0.003548	0.001161
1.031	0.003655	0.003592	0.0008865	-0.006578	0.003594	0.001399
1.185	0.0009538	0.003639	0.0004187	0.002352	0.003633	0.000595
$< p_T > = 4.0 \text{ GeV}/c$						
0.3395	0.007235	0.0034	0.001622	0.0004913	0.003399	0.0003631
0.4351	-0.0003393	0.0034	0.0002955	0.005318	0.003401	0.001177
0.5393	0.01156	0.003411	0.002591	0.005042	0.003393	0.001116
0.6321	0.01252	0.003408	0.002808	0.0005916	0.00339	0.0003679
0.726	0.01445	0.003392	0.00324	0.002034	0.00338	0.0005291
0.8226	0.01358	0.003416	0.003046	0.000416	0.003407	0.0003602
0.9465	0.004669	0.003449	0.001047	-0.00557	0.003436	0.001147
1.109	0.0008406	0.003504	0.0003435	0.006284	0.003501	0.001391
1.325	0.001404	0.003567	0.0004192	0.006004	0.003554	0.001329
$< p_T > = 4.7 \text{ GeV}/c$						
0.3451	0.00265	0.003362	0.0006668	-0.001786	0.003362	0.0004297
0.4436	0.01387	0.003362	0.003287	0.004655	0.003347	0.0009253

0.5534	0.01303	0.003351	0.003089	0.001872	0.003353	0.0004409
0.6495	0.01394	0.003343	0.003303	0.004585	0.003329	0.0009115
0.7454	0.01535	0.003332	0.003638	0.007228	0.003319	0.001437
0.8483	0.01562	0.003359	0.003702	0.01406	0.003359	0.002795
0.9874	0.01714	0.003391	0.004062	0.002982	0.00337	0.0006075
1.185	0.005015	0.003428	0.001189	-0.002351	0.003422	0.000513
1.471	-0.002538	0.003516	0.0006565	0.002507	0.003505	0.00054
$\langle p_T \rangle = 5.7 \text{ GeV}/c$						
0.3541	0.00741	0.003341	0.001744	-0.001636	0.003335	0.0003707
0.4615	-0.0007936	0.00332	0.0003263	0.003468	0.003326	0.0007419
0.5739	0.01199	0.003296	0.002823	0.007297	0.003287	0.001561
0.6756	0.01671	0.003302	0.003934	0.0006758	0.003287	0.0001979
0.7718	0.02513	0.003281	0.005915	8.487e-05	0.003238	0.0001371
0.8837	0.02893	0.003355	0.006811	0.007015	0.003292	0.001501
1.038	0.01923	0.003351	0.004525	0.001617	0.003327	0.0003669
1.268	0.006795	0.003384	0.001599	0.003172	0.003375	0.0006803
1.662	0.003547	0.003454	0.0008349	0.002205	0.003442	0.0004846
$\langle p_T \rangle = 8.3 \text{ GeV}/c$						
0.3743	0.007635	0.003314	0.002104	0.0002714	0.003314	0.0002048
0.5033	0.01572	0.003304	0.004332	0.007784	0.003292	0.002078
0.6216	0.02542	0.003277	0.007006	0.003097	0.003232	0.0008286
0.7279	0.04121	0.003323	0.01135	0.0009343	0.003221	0.0003069
0.8249	0.04738	0.003344	0.01305	0.01132	0.003235	0.003023
0.9517	0.04906	0.003365	0.01352	0.007349	0.00323	0.001962
1.14	0.03806	0.003366	0.01049	0.008324	0.003273	0.002223

1.416	0.02308	0.003339	0.006361	0.006407	0.003323	0.001711
2.036	0.02101	0.003415	0.005789	0.002998	0.003383	0.0008057

Table A.2: Table corresponding to Figure 4.30. ($\sigma_{sys} = \sqrt{\sigma_{pid}^2 + \sigma_{bias}^2}$)

A.5.2 A_{UT} vs $M_{inv}^{\pi^+\pi^-}$ (Integrated $p_T^{\pi^+\pi^-}$)

Radici et. al.			Run 15				Run 06		
$< M_{inv} >$	A_{UT}^{low}	A_{UT}^{high}	$< M_{inv} >$	A_{UT}	σ_{stat}	σ_{sys}	$< M_{inv} >$	A_{UT}	σ_{stat}
0.36	0.0027	0.0076	0.3487	0.006584	0.001514	0.001591	0.36	0.0054	0.01
0.5	0.0077	0.0132	0.4537	0.005064	0.001506	0.0009352	0.5	0.018	0.0068
0.69	0.0173	0.0286	0.5643	0.01267	0.001519	0.002576	0.69	0.023	0.0081
0.88	0.0262	0.0438	0.662	0.0143	0.001525	0.00291	0.88	0.07	0.013
1.19	0.02	0.036	0.7567	0.02268	0.001554	0.003966	1.19	0.039	0.02
			0.8749	0.02373	0.001423	0.00571			
			1.033	0.01585	0.00156	0.003521			
			1.236	0.01022	0.001555	0.003543			
			1.689	0.008694	0.0018	0.002546			

Table A.3: Table corresponding to Figure 4.32.

A.5.3 A_{UT} vs $p_T^{\pi^+\pi^-}$ (Multi-Binning)

$< p_T >$	$\eta^{\pi^+\pi^-} > 0$			$\eta^{\pi^+\pi^-} < 0$		
	A_{UT}	σ_{stat}	σ_{sys}	A_{UT}	σ_{stat}	σ_{sys}
$< M_{inv} > = 0.39 \text{ GeV}/c^2$						
3.268	0.003722	0.003496	0.001229	0.002673	0.003496	0.0009286
3.572	0.008756	0.003428	0.00289	-0.002344	0.003423	0.0008185
3.848	0.001468	0.003402	0.0005046	0.002822	0.003417	0.000977
4.146	0.00645	0.003333	0.002129	0.001989	0.003329	0.0007006
4.488	0.005387	0.003367	0.001778	0.001627	0.003362	0.0005837
4.902	0.008494	0.003343	0.002804	0.0008209	0.003342	0.0003358

5.456	0.0006703	0.003325	0.0002667	0.0006717	0.003325	0.0002944
6.32	0.007075	0.003311	0.002335	-0.002362	0.003318	0.0008234
8.595	0.009595	0.003353	0.003167	0.002882	0.003353	0.0009966
$< M_{inv} > = 0.58 \text{ GeV}/c^2$						
3.261	0.00477	0.003501	0.0008683	0.004915	0.003501	0.0008351
3.577	0.003524	0.003448	0.0006416	-0.002355	0.003445	0.0004151
3.865	0.00877	0.003426	0.001597	0.001316	0.003405	0.0002629
4.178	0.01643	0.003392	0.002992	0.00431	0.003371	0.0007323
4.54	0.01216	0.003353	0.002214	0.002613	0.003348	0.0004534
4.989	0.01447	0.003338	0.002634	0.003828	0.003327	0.0006503
5.59	0.01302	0.003314	0.002371	0.001889	0.003303	0.0003424
6.523	0.01449	0.003291	0.002638	0.009014	0.003284	0.001532
8.996	0.02443	0.003285	0.004446	0.005222	0.003251	0.0008872
$< M_{inv} > = 0.76 \text{ GeV}/c^2$						
3.249	0.007772	0.003514	0.001596	-0.002249	0.003506	0.0004979
3.583	0.004645	0.003453	0.0009538	0.007189	0.003456	0.001471
3.891	0.01268	0.003408	0.002603	-0.002832	0.003399	0.0005944
4.227	0.0167	0.003374	0.00343	0.004367	0.00336	0.0008934
4.618	0.01259	0.003337	0.002584	0.0114	0.003334	0.002332
5.101	0.01806	0.003316	0.003709	0.005925	0.003299	0.001212
5.744	0.02607	0.003313	0.005354	-0.002412	0.003267	0.0005176
6.737	0.02864	0.003294	0.00588	0.00372	0.003245	0.0007612
9.314	0.05319	0.003359	0.01092	0.007432	0.003203	0.00152
$< M_{inv} > = 0.96 \text{ GeV}/c^2$						
3.192	0.00497	0.00362	0.001208	-0.001234	0.00362	0.0003669

3.509	0.01031	0.003552	0.002505	-0.002235	0.003539	0.0006071
3.802	0.004301	0.003486	0.001045	0.001191	0.003474	0.0003537
4.126	0.007145	0.003439	0.001736	-0.00172	0.003427	0.0004782
4.51	0.01121	0.003398	0.002723	0.005186	0.003388	0.001373
4.991	0.0228	0.003378	0.00554	0.006219	0.003349	0.001647
5.642	0.02525	0.003349	0.006137	0.006295	0.003311	0.001667
6.65	0.02913	0.003323	0.007079	0.005573	0.003274	0.001476
9.273	0.05622	0.003383	0.01366	0.008352	0.003202	0.002212
$< M_{inv} > = 1.42 \text{ GeV}/c^2$						
3.491	-0.0007382	0.003638	0.0002754	0.004597	0.003633	0.001099
3.95	0.001759	0.00356	0.0004273	0.006834	0.003553	0.001633
4.34	-0.003868	0.003514	0.0008298	-0.001587	0.003506	0.0005652
4.751	0.005204	0.003469	0.001125	0.003925	0.003465	0.000938
5.217	0.003279	0.003434	0.0007115	0.0004268	0.003424	0.0004666
5.781	0.001148	0.00341	0.0003201	-0.0005202	0.003398	0.0004669
6.521	0.01736	0.003395	0.003752	0.002969	0.003366	0.0007407
7.644	0.01959	0.003375	0.004234	0.006852	0.003353	0.001638
10.36	0.03789	0.003363	0.008188	0.008832	0.003331	0.002111

Table A.4: Table corresponding to Figure 4.33.

A.5.4 A_{UT} vs $\eta^{\pi^+\pi^-}$

$< \eta^{\pi^+\pi^-} >$	$< x >$	$< z >$	A_{UT}	σ_{stat}	σ_{sys}
-0.7899	0.1341	0.4916	0.001988	0.00108	0.0005746
-0.5671	0.1175	0.4985	0.001756	0.001078	0.0004792

-0.3745	0.1165	0.4948	0.004088	0.001083	0.001109
-0.1881	0.1266	0.4692	0.002959	0.001079	0.0007964
-0.003494	0.1427	0.4504	0.004993	0.001075	0.00135
0.1814	0.1575	0.467	0.008043	0.001089	0.002197
0.3718	0.1707	0.4817	0.01298	0.00111	0.003609
0.5716	0.18	0.4997	0.0159	0.001125	0.0045
0.7972	0.2192	0.4805	0.02386	0.001178	0.007198

Table A.5: Table corresponding to Figure 4.29.

A.5.5 A_{UT} vs $M_{inv}^{\pi^+\pi^-}$ (Highest $p_T^{\pi^+\pi^-}$ Bin)

$\langle M_{inv}^{\pi^+\pi^-} \rangle$	$\langle x \rangle$	$\langle z \rangle$	A_{UT}	σ_{stat}	σ_{sys}
0.3743	0.2058	0.539	0.007635	0.003314	0.002104
0.5033	0.221	0.56	0.01572	0.003304	0.004332
0.6216	0.2032	0.56	0.02542	0.003277	0.007006
0.7279	0.1921	0.588	0.04121	0.003323	0.01135
0.8249	0.1861	0.6042	0.04738	0.003344	0.01305
0.9517	0.1955	0.5928	0.04906	0.003365	0.01352
1.14	0.2044	0.5998	0.03806	0.003366	0.01049
1.416	0.1964	0.6042	0.02308	0.003339	0.006361
2.036	0.1937	0.6096	0.02101	0.003415	0.005789

Table A.6: Table corresponding to Figure 4.31.

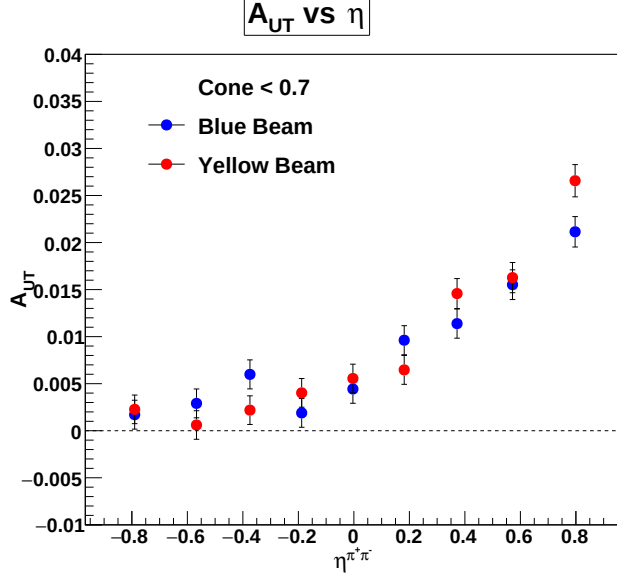


Figure A.24: A_{UT} vs $\eta^{\pi^+\pi^-}$ for blue and yellow beams.

A.5.6 Blue And Yellow Beam asymmetry

RHIC has both beams, yellow and blue, polarized. We have taken advantage of this by measuring asymmetries from both beams. Since the beam conditions are similar during collisions, we expect similarities in the asymmetry of both beams. This provides a systematic check on the asymmetry measurement.

Figures A.24-A.30 show asymmetries for the blue and yellow beams in the same kinematic bins. Within the statistical uncertainties, both beam asymmetries are consistent with each other, which serves as proof of the framework employed for the asymmetry extraction. Moreover, utilizing both beams doubles the statistics, providing higher statistical precision.

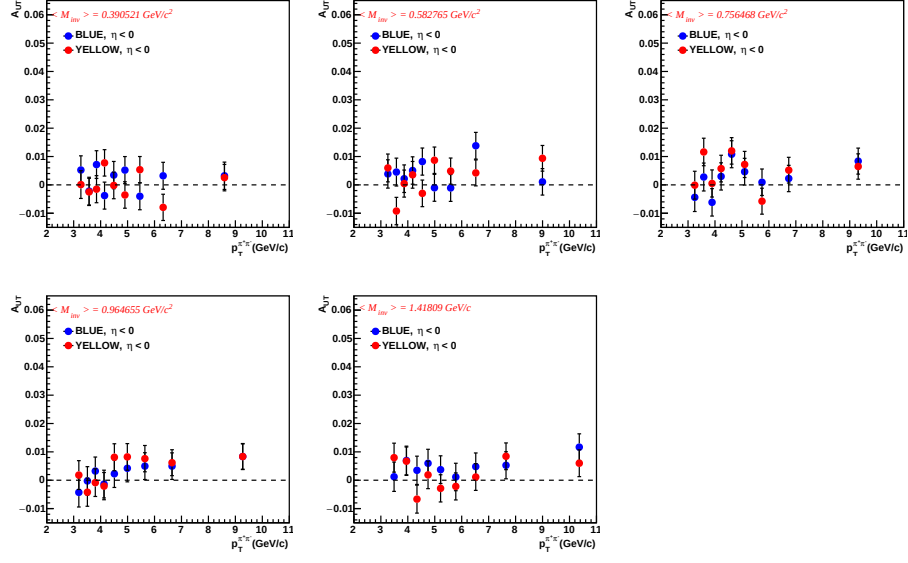


Figure A.25: A_{UT} vs $p_T^{\pi^+\pi^-}$, $\eta^{\pi^+\pi^-} < 0$ for blue and yellow beams.

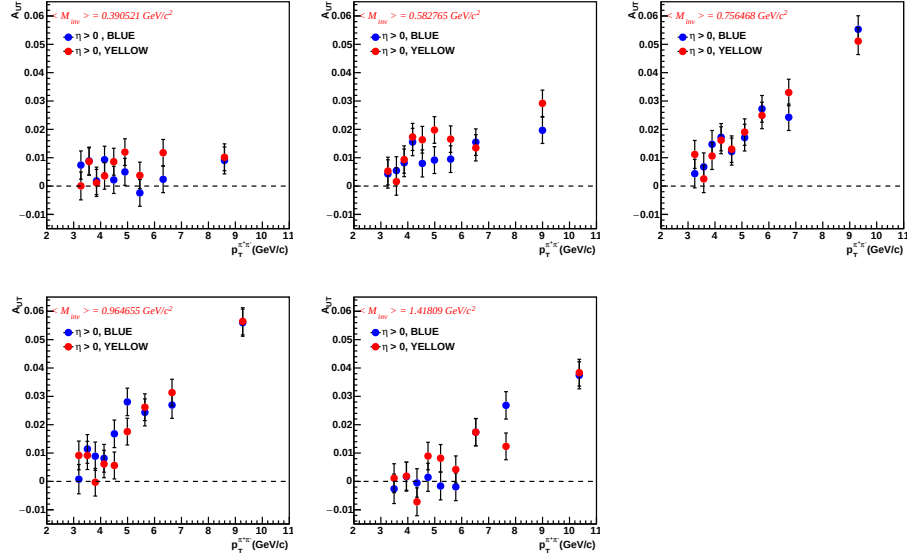


Figure A.26: A_{UT} vs $p_T^{\pi^+\pi^-}$, $\eta^{\pi^+\pi^-} > 0$

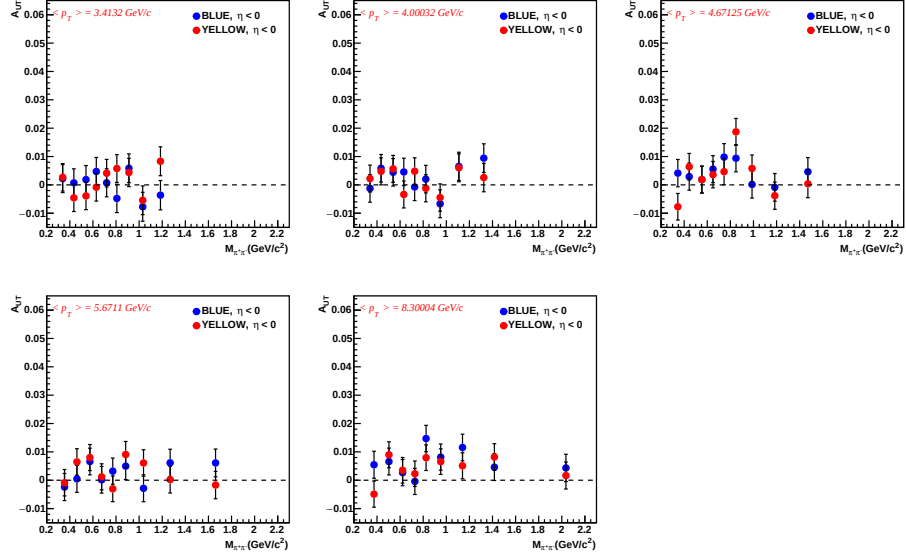


Figure A.27: A_{UT} vs $M_{inv}^{\pi^+\pi^-}$, $\eta^{\pi^+\pi^-} < 0$ for blue and yellow beams.

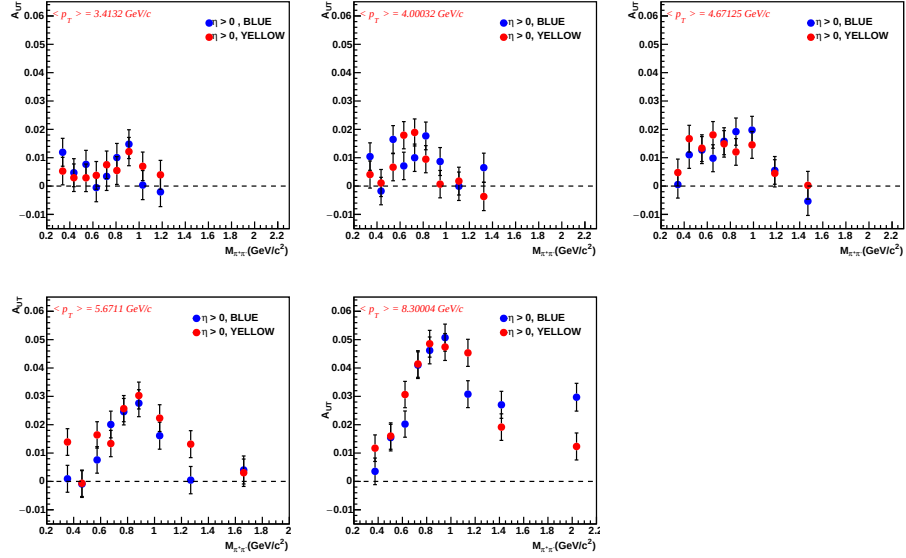


Figure A.28: A_{UT} vs $p_T^{\pi^+\pi^-}$, $\eta^{\pi^+\pi^-} > 0$ for blue and yellow beams.

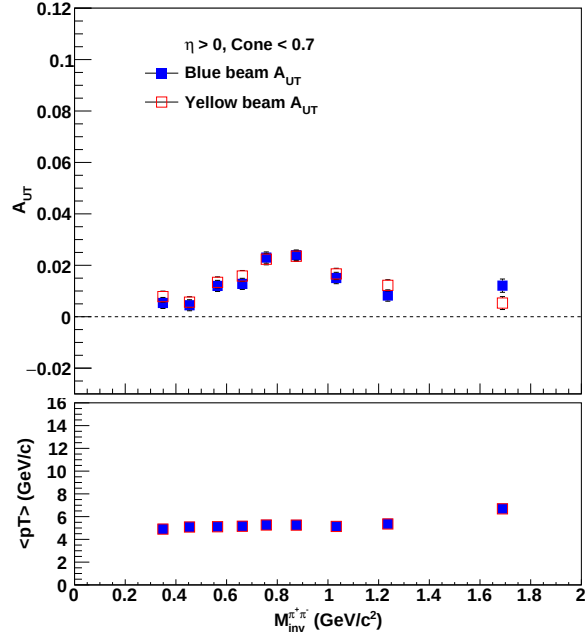


Figure A.29: A_{UT} vs $M_{inv}^{\pi^+\pi^-}$, $\eta^{\pi^+\pi^-} > 0$, integrated $p_T^{\pi^+\pi^-}$.

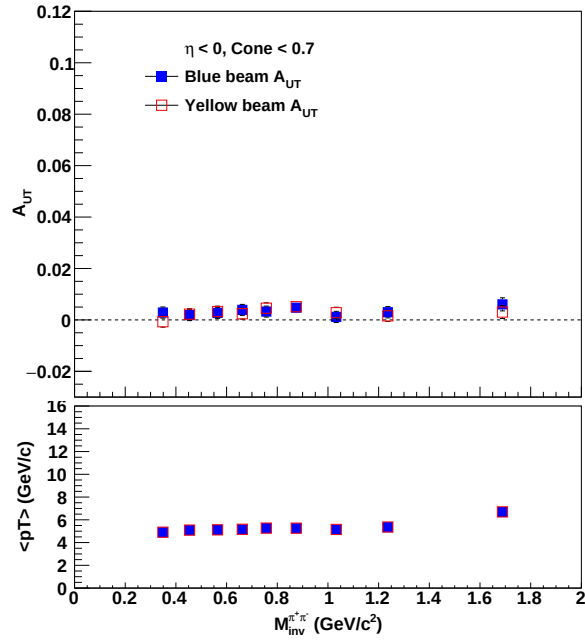


Figure A.30: A_{UT} vs $M_{inv}^{\pi^+\pi^-}$, $\eta^{\pi^+\pi^-} < 0$, integrated $p_T^{\pi^+\pi^-}$.

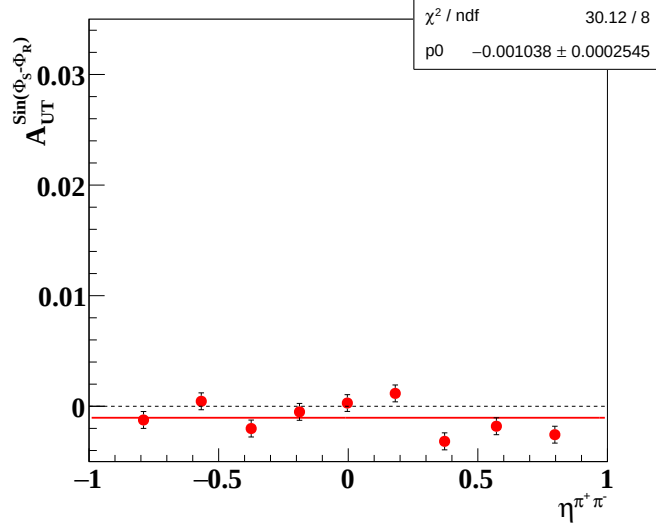


Figure A.31: Random asymmetry vs $\eta^{\pi^+\pi^-}$. The direction of $\mathbf{R}$ is randomized by randomizing charge ordering.

A.5.7 Random R Asymmetry

The relative difference between the momentum vectors of two pions, $\vec{R}(= 1/2(\vec{p}_{\pi^+} - \vec{p}_{\pi^-}))$, is crucial to access quark transverse spin. The charge order of the pions determines the direction of this vector, which needs to be fixed; otherwise, the vector direction gets randomized, leading to diluted asymmetry.

To test this argument, the asymmetry is produced by randomizing the direction of $\vec{R}$. In $\vec{R} = 1/2(\vec{p}_{\pi^+} - \vec{p}_{\pi^-})$, the constraint that the first vector should be π^+ and the second should be π^- is removed. In every pair, whichever pion comes first in the loop serves as the first hadron vector. Thus the pair could be either $\pi^+\pi^-$ or $\pi^-\pi^+$. The random asymmetries are shown in Figures A.31-A.33 in the $\eta^{\pi^+\pi^-} > 0$ region. All the asymmetries are consistent with zero. This confirms that the analysis framework is implemented and working correctly.

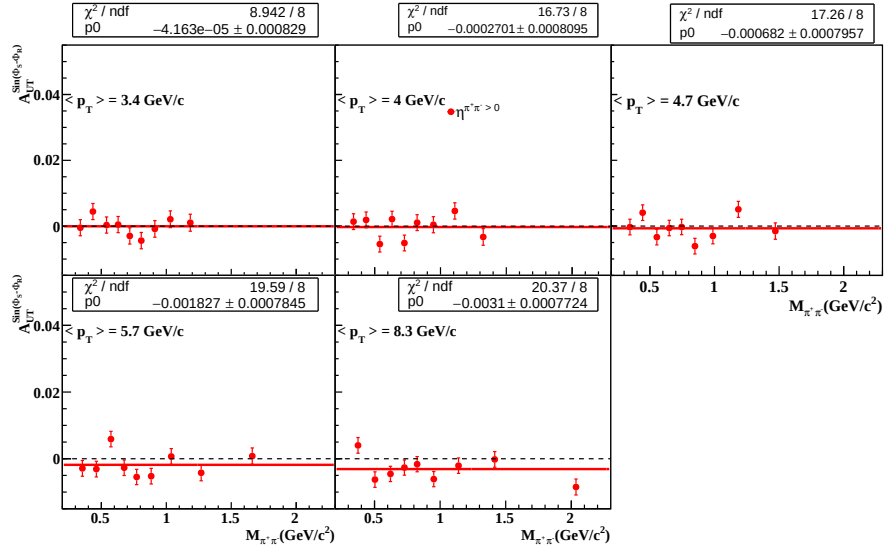


Figure A.32: Random asymmetry vs $M_{inv}^{\pi^+\pi^-}$. The direction of $\mathbf{R}$ is randomized by randomizing charge ordering.

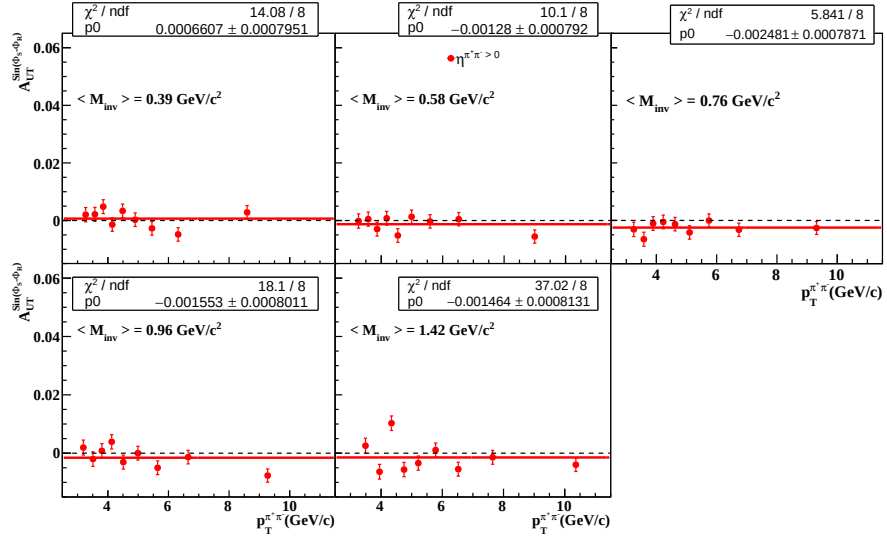
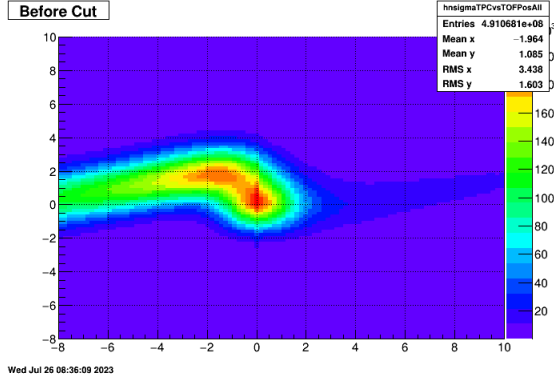


Figure A.33: Random asymmetry vs $p_T^{\pi^+\pi^-}$. The direction of $\mathbf{R}$ is randomized by randomizing charge ordering.

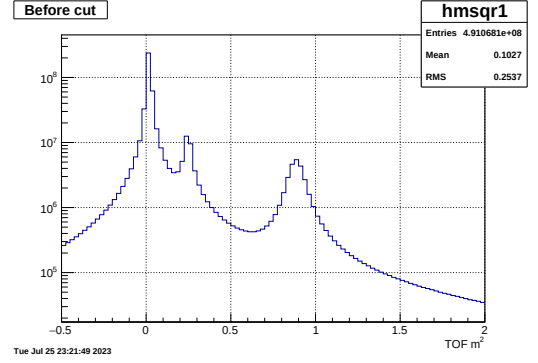
A.5.8 PID for K^+K^- IFF Projection

The PID for the K^+K^- pair is still very preliminary. A more stringent PID approach is employed for the kaon selection. The TPC and TOF PID information is utilized simultaneously to identify high-purity kaon samples, albeit at the cost of statistics. The track QA cuts are similar to those used in the $\pi^+\pi^-$ analysis, except for the p_T cut, which is reduced to $p_T > 0.2$ GeV/ c .

To isolate high-quality kaons cuts on $n\sigma_\pi^{TOF}$ and $n\sigma_\pi^{TPC}$ are implemented. Figure A.34a illustrates the nearly overlapping nature of kaons and pions, presenting challenges in kaon identification. The corresponding m^2 distribution from TOF, as shown in figure A.34b, clearly depicts the pion's dominance in the kaon signal region. A graphical cut on $n\sigma_\pi^{TPC}$ ($-1 < n\sigma_\pi^{TPC} < 2$) and $n\sigma_\pi^{TOF}$ ($-2 < n\sigma_\pi^{TOF} < 2$) is applied to mitigate pion and proton backgrounds, as depicted in figure A.35a. Subsequently, the TOF m^2 distribution (figure A.35b) shows a considerable reduction in pion dominance. The pion peak is visible on the left side, with an amplitude approximately 1.5 orders of magnitude lower than the kaon peak. An additional cut on TOF m^2 , $0.18 < m^2 < 0.32$, is imposed to refine the kaon signal further. Despite the preliminary fit in the m^2 distribution (the TOF m^2 distribution follows the Voigt profile), a collection of high-purity kaons is achieved after the PID selection. Notably, the PID approach is applied exclusively to the kaon projection, with the final analysis adhering to STAR's standard approach.

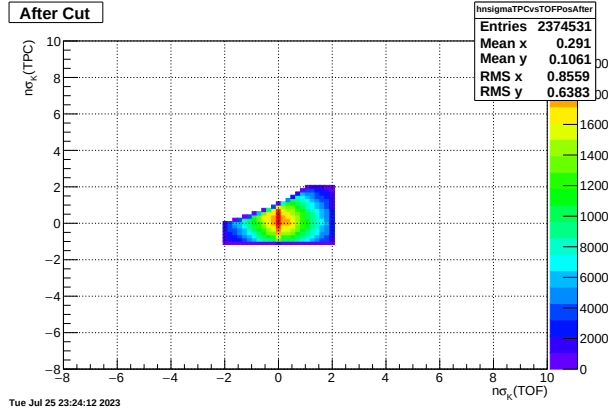


(a) $n\sigma_K^{TPC}$ vs $n\sigma_K^{TOF}$ before PID.

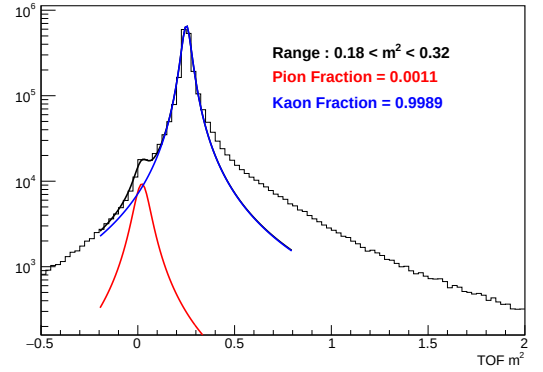


(b) TOF m^2 before PID.

Figure A.34: Kaon PID: before applying cuts.



(a) $n\sigma_K^{TPC}$ vs $n\sigma_K^{TOF}$ after PID.



(b) TOF m^2 after PID.

Figure A.35: Kaon PID: after the PID cuts.

APPENDIX B

RUN 2012 UNPOLARIZED

CROSS-SECTION

MEASUREMENT

B.1 Data-Embedding Comparison

B.1.1 Vertex Shape Correction

To ensure that the embedding sample matches with the data, it is necessary to match the z-vertices of the event. Once the z-vertex is matched, other distributions should also be matched accordingly. In Figure B.1(*left*), the z-vertices comparison between data and embedding is shown, where the embedding vertex distribution is shifted to the left. This mismatch cannot be ignored and requires correction. In physics analysis, this mismatch can be fixed by fitting the ratio between the two distributions (*bottom panel*) with the best polynomial fit. The fit equation, with the z-vertex as a variable, can then be used as a weight to the z-vertex to correct the mismatch. The

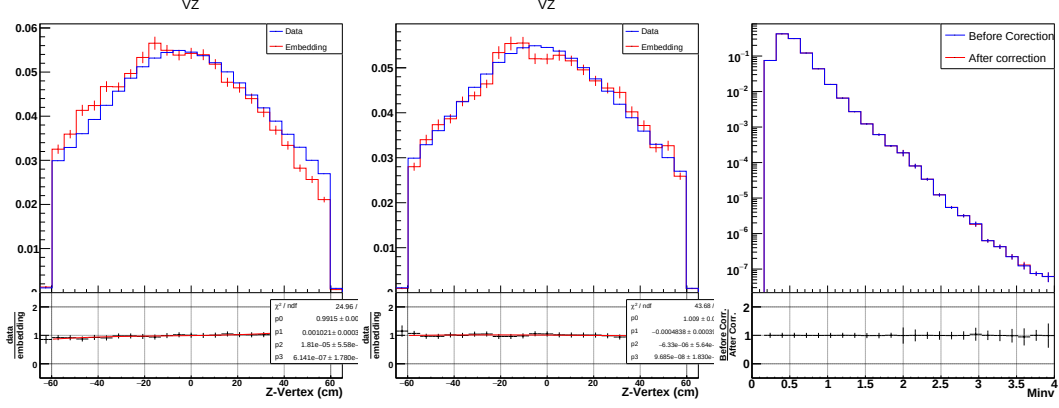


Figure B.1: Vertex shift (*left*), after vertex correction (*middle*), and impact of vertex correction in invariant mass distribution for JP0 trigger.

middle plot displays the comparison after the z-vertex shape correction. The impact of this shape correction is verified by comparing the same distribution before and after the correction, and finding that it has no impact on the rest of the distributions, as shown in the right plot.

In this analysis, only JP0, JP1, and JP2 events are used, whereas, in the data, the events are heavily prescaled during production. This means that only one event is kept for every event (*prescale*). In data, the triggering algorithm checks the highest threshold trigger, for example, JP2, followed by lower threshold triggers, for example, JP1, and JP0, in that order. An event meeting JP2 criteria will meet all the lower trigger categories. This may occur trigger overlapping and have a chance that the same event would be counted for each trigger. The prescale factor is applied at the production level to overcome trigger overlapping issues in the data. Prescale for the JP2 is always one, and the prescale value is larger for the lower threshold triggers.

However, the embedding sample takes everything, as the simulation has no triggering scheme. Trigger information can be accessed using the trigger simulator in the embedding sample. The same prescale factors should be applied to replicate the triggering scheme in the data and overcome trigger overlapping. Individual trigger

event distributions can be directly compared with the data; however, one should be careful while combining all triggers in the embedding. For combined triggers, the correlation between data and embedding can be established as follows:

$$H_{d,JP0} + H_{d,JP1} + H_{d,JP2} \leftrightarrow \frac{1}{p_{JP0}} H_{e,JP0} + \frac{1}{p_{JP1}} H_{e,JP0} + \frac{1}{p_{JP2}} H_{e,JP2} \quad (\text{B.1})$$

where, H is distributions from embedding(e) and data(d), and p is prescale factor for each individual triggers JP0, JP1, and JP2. The following section includes data-embedding comparison plots for all three and combined triggers. Embedding distributions match well with the data.

B.1.2 Data-Embedding Comparison plots

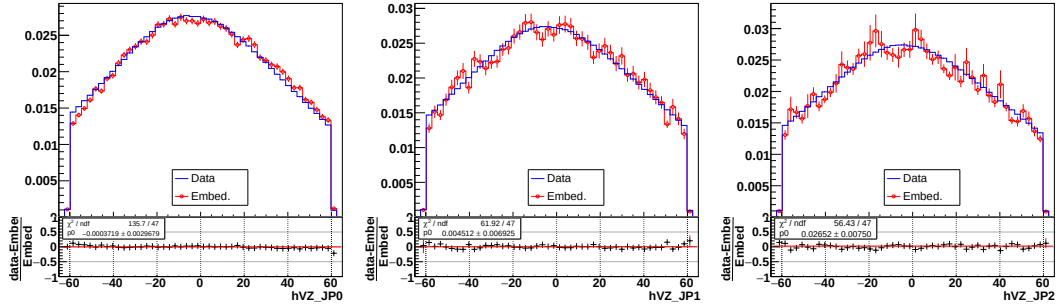


Figure B.2: Z-vertex distribution comparison for JP0, JP1, and JP2 from left to right respectively.

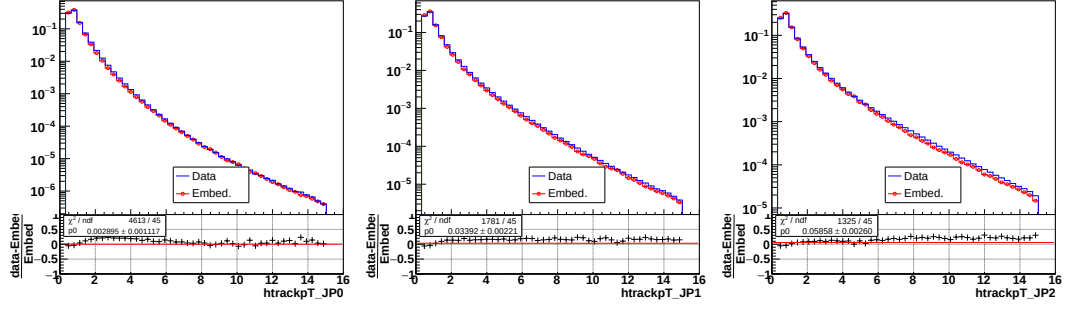


Figure B.3: Hadrons p_T distribution comparison for JP0, JP1, and JP2 from left to right respectively.

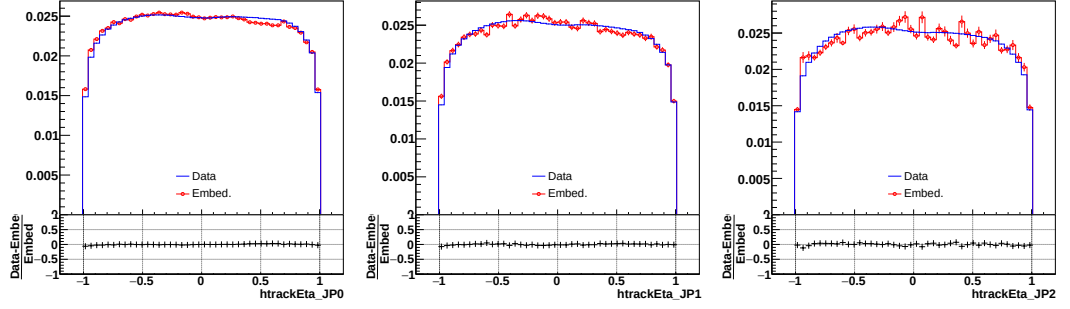


Figure B.4: Hadrons η distribution comparison for JP0, JP1, and JP2 from left to right respectively.

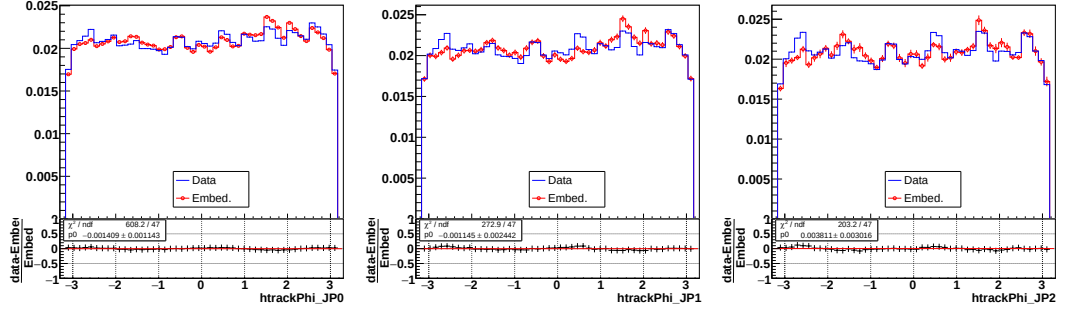


Figure B.5: Hadrons ϕ distribution comparison for JP0, JP1, and JP2 from left to right respectively.

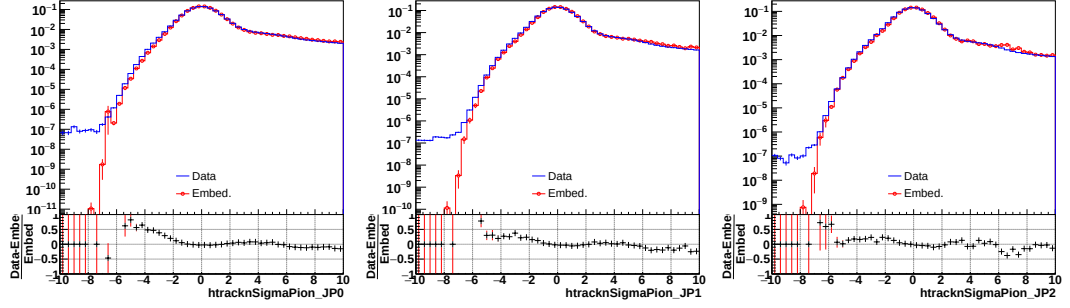


Figure B.6: Hadrons $n\sigma\pi$ distribution comparison for JP0, JP1, and JP2 from left to right respectively.

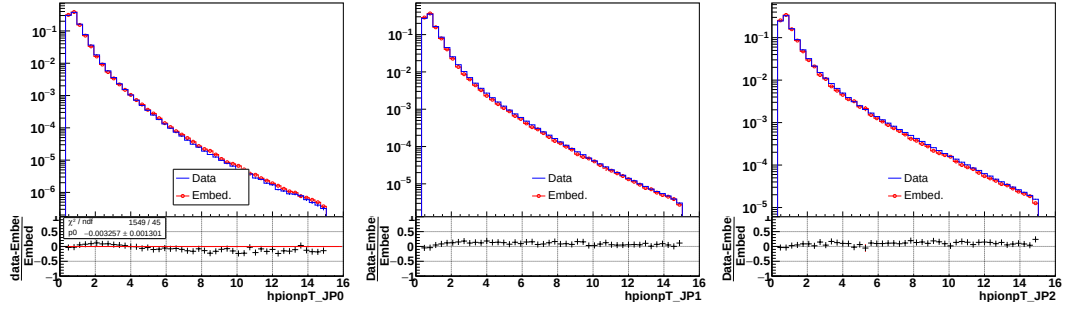


Figure B.7: Pion p_T distribution comparison for JP0, JP1, and JP2 from left to right respectively.

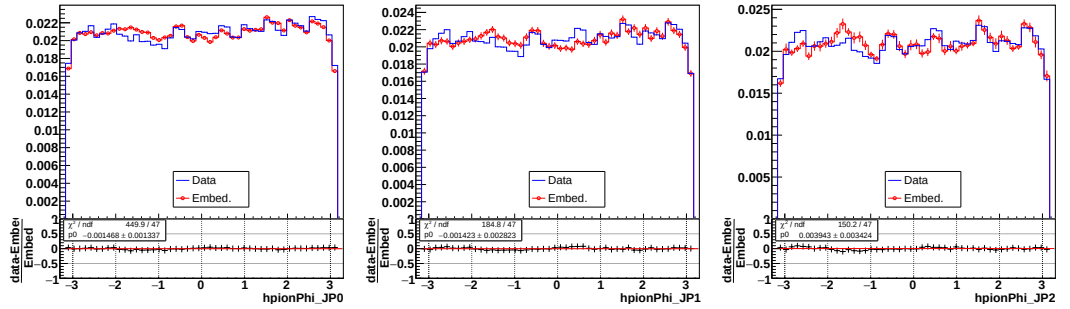


Figure B.8: Pion ϕ distribution comparison for JP0, JP1, and JP2 from left to right respectively.

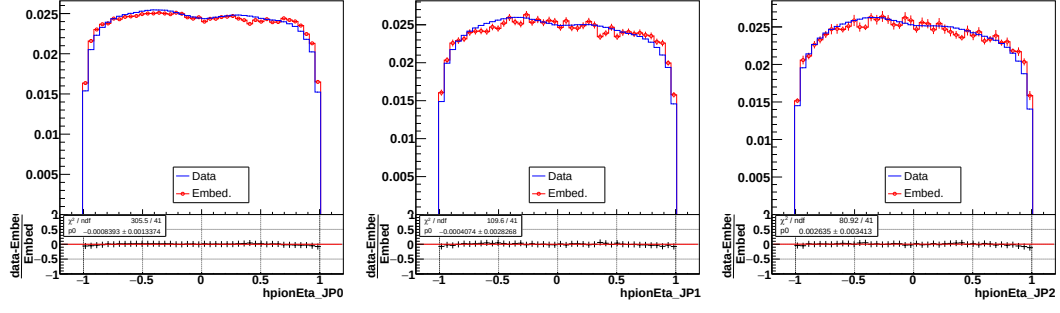


Figure B.9: Pion η distribution comparison for JP0, JP1, and JP2 from left to right respectively.

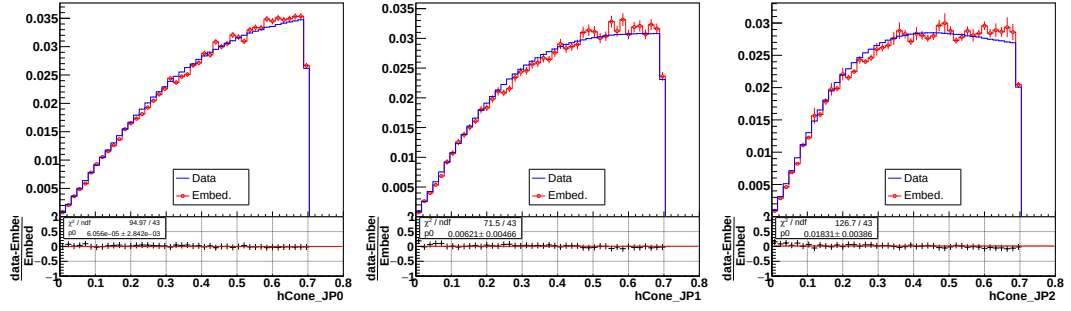


Figure B.10: $\pi^+\pi^-$ opening angle (in $\eta - \phi$ space) distribution comparison for JP0, JP1, and JP2 from left to right respectively.

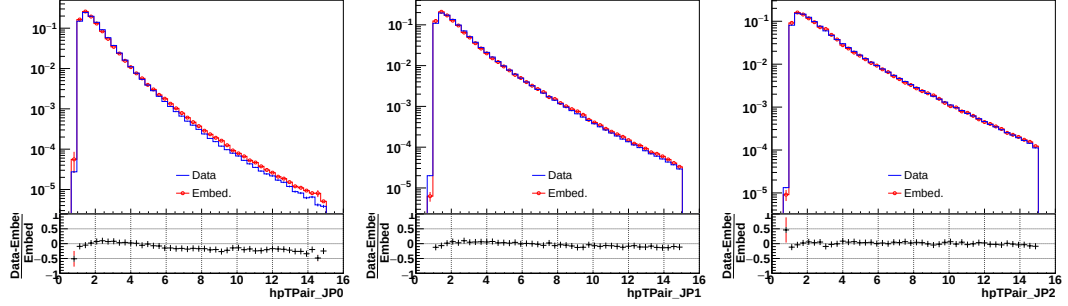


Figure B.11: $\pi^+\pi^-$ p_T distribution comparison for JP0, JP1, and JP2 from left to right respectively.

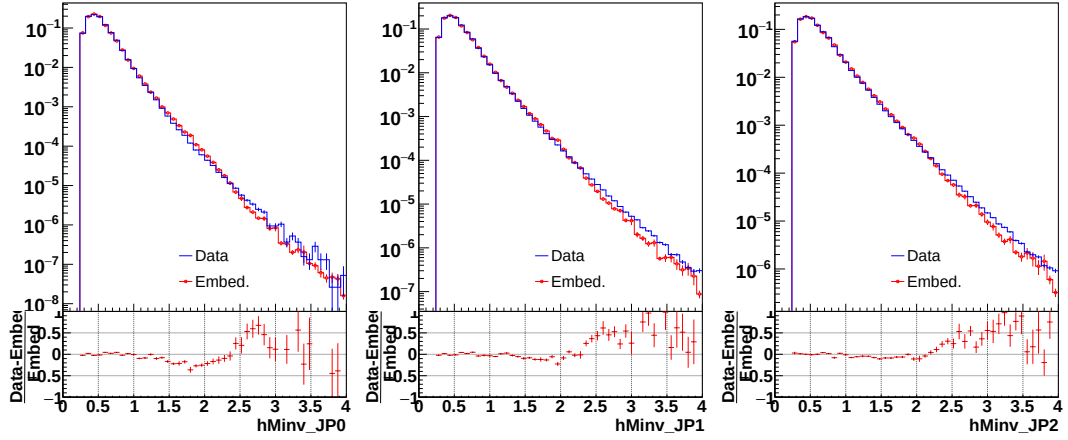


Figure B.12: $M_{inv}^{\pi^+\pi^-}$ distribution comparison for JP0, JP1, and JP2 from left to right respectively.

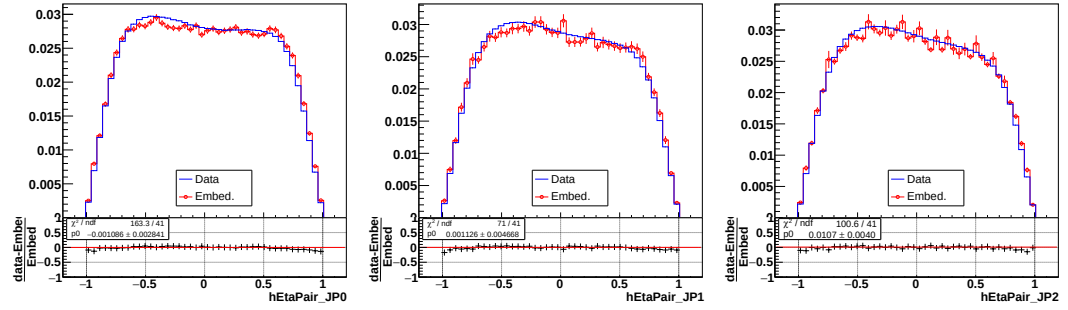


Figure B.13: $\eta^{\pi^+\pi^-}$ distribution comparison for JP0, JP1, and JP2 from left to right respectively.

B.2 Run List: 2012

13047003 13047004 13047022 13047023 13047024 13047026 13047027 13047028 13047029
13047030 13047031 13047032 13047033 13047034 13047035 13047036 13047037 13047039
13047041 13047042 13047043 13047044 13047122 13047123 13047124 13047126 13048009
13048010 13048011 13048012 13048013 13048014 13048015 13048016 13048017 13048018
13048019 13048030 13048031 13048032 13048040 13048041 13048042 13048043 13048044
13048045 13048046 13048049 13048050 13048051 13048052 13048053 13048087 13048088

13048089 13048090 13048091 13048092 13048093 13049004 13049005 13049006 13049007
13049031 13049032 13049035 13049039 13049041 13049042 13049044 13049045 13049046
13049047 13049048 13049049 13049050 13049072 13049073 13049080 13049081 13049082
13049086 13049087 13049088 13049089 13049093 13049094 13049096 13049098 13049099
13049101 13050001 13050006 13050007 13050009 13050011 13050012 13050015 13050016
13050020 13050022 13050023 13050025 13050026 13050027 13050028 13050029 13050031
13050032 13050033 13050036 13050037 13050038 13050039 13050041 13050042 13050043
13050044 13050046 13050047 13050049 13050050 13051006 13051007 13051008 13051009
13051010 13051011 13051012 13051015 13051016 13051017 13051019 13051020 13051021
13051022 13051023 13051024 13051026 13051028 13051068 13051069 13051070 13051071
13051072 13051073 13051074 13051080 13051081 13051083 13051085 13051086 13051087
13051088 13051092 13051093 13051095 13052001 13052002 13052003 13052004 13052005
13052009 13052010 13052011 13052012 13052013 13052014 13052015 13052016 13052017
13052018 13052019 13052020 13052021 13052022 13052036 13052037 13052039 13052042
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13052061 13052088 13053004 13053005 13053006 13053007 13053012 13053013 13053015
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13054063 13054064 13054065 13054066 13054068 13054069 13054070 13054084 13054085
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13069004 13069005 13069006 13069007 13069008 13069013 13069014 13069016 13069017
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13071006 13071008 13071009 13071010 13071011 13071012 13071064 13072001 13072002
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13072014 13072015 13072016 13072017 13072018 13072019 13072020

APPENDIX C

TPC SPACE CHARGE AND GRID LEAK CALIBRATION

The STAR TPC is responsible for the track reconstruction of charged particles. These reconstructed tracks determine various quantities, such as the collision vertex, particle momentum, and scattering angles. The TPC is a complex detector system with different calibrations that need to be done. One particular calibration charged to Temple University is the space charge and grid leak (SC&GL) calibration. SC&GL effects lead to distortions in the electric field and must be corrected to have accurate track and momentum reconstruction. Various sources of field distortions can impact the performance of the TPC, such as the electric field distortion caused by space charge (SC) and grid leak (GL).

C.1 Space Charge

As a charged particle moves through the TPC gas, it ionizes the gas in its path. The resulting electrons move toward the anode at $5.45 \text{ cm}/\mu\text{s}$, while the ions move toward

the cathode at a speed of approximately 100 cm/s. This causes a positive charge density in the drift space, known as the Space Charge (SC), which adds an electrical field transverse to the original uniform field along the z -direction. As a result, the drift velocity of the charged particles acquires non-zero x and y components, causing the hits to shift from their expected positions.

C.2 Grid Leak

During the analysis of residual TPC hit clusters from track fits, a discontinuity was observed in the residuals at the gap between the inner and outer readout wire chambers of the TPC. This suggests that the gated grid cannot prevent ions from flowing around the anode wire, resulting in a thin layer of ions reaching the TPC drift space, which is known as Grid Leak (GL). The GL adds up to the SC, contributing to the field distortion in the TPC.

C.3 What do SC and GL do to the tracks?

The SC and GL add an extra electric field. Drifting electrons experience a Lorentz force aligned along the direction of the electric field. However, an additional electric field in the drift space adds non-zero drift velocity along the x and y -axes. As a consequence, tracks get rotated azimuthally. This distortion changes the track properties and must be brought back to the original position before any of the track-based reconstruction is done. Simulation shows that the magnitude of the distortions is proportional to the accumulated space charge, ρ_{SC} ; however, the calibration is performed for the average charge density over the volume of the TPC chamber, $\langle \rho_{SC} / \epsilon_0 \rangle$. The correction for the distorted track by the SC and GL is performed by the SC and

GL calibration task force at STAR.

C.4 SC and GL Calibration

The direct measurement of the ρ_{SC} is not possible; however, it can be estimated using the distorted tracks with the help of simulation. Simulation shows the signed dca , sDCA, of the distorted tracks is proportional to ρ_{SC} in the drift space,

$$\rho_{SC}^{sim} \propto sDCA^{sim} \Rightarrow C^{sim} = \rho_{SC}^{sim} / sDCA^{sim} \quad (C.1)$$

Then with the real track ρ_{SC}^{obs} can be estimated as [121],

$$\rho_{SC}^{obs} = C^{sim} \cdot sDCA^{obs} = \rho_{SC}^{sim} \cdot \left(\frac{sDCA^{obs}}{sDCA^{sim}} \right) \quad (C.2)$$

Distortion due to the GL can be modeled in the same as in the SC, providing a map of cluster position corrections whose magnitude is proportional to the leaked charge, ρ_{leak} , which is found to be scaled with the collision rates. The effect for both the SC and GL is calibrated by finding a ratio between ρ_{leak} and ρ_{SC} , which removes the residual discontinuity while simultaneously zeroing the sDCA [121].

$$(\rho_{SC}^{obs} + \rho_{leak}^{obs}) = (\rho_{SC}^{sim} + \rho_{leak}^{sim}) \cdot (sDCA^{obs} / sDCA^{sim}) \quad (C.3)$$

$$\rho_{leak} \equiv D \cdot \rho_{SC} \quad (C.4)$$

The calibration is performed event by event using a self-correcting approach for each event for SC and GL simultaneously.

C.5 Calibrated Datasets During 2019 - 2023

Year	CM Energy(GeV)	Collision	Status
2019	200	AuAu	Completed
	19.6	AuAu	
	14.5	AuAu	
	7.7	AuAu	
2020	9.2	AuAu	Completed
2021	7.7	AuAu	Completed
	17.3	AuAu	Completed
	200	dAu	Completed
2022	508	pp	In progress

Table C.1: List of SC&GL calibrated datasets by Temple University group.